



OET 2.0 READING MATERIALS

(INCLUDING 2 OFFICIAL SAMPLES)

READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAMES:

Your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE: _____

Passport Photo



Fractures, dislocations and sprains: Texts

Text A

Fractures (buckle or break in the bone) often occur following direct or indirect injury, e.g. twisting, violence to bones. Clinically, fractures are either:

- closed, where the skin is intact, or
- compound, where there is a break in the overlying skin

Dislocation is where a bone is completely displaced from the joint. It often results from injuries away from the affected joint, e.g. elbow dislocation after falling on an outstretched hand.

Sprain is a partial disruption of a ligament or capsule of a joint.

Text B

Simple Fracture of Limbs

Immediate management:

- Halt any external haemorrhage by pressure bandage or direct pressure
- Immobilise the affected area
- Provide pain relief

Clinical assessment:

- Obtain complete patient history, including circumstances and method of injury
 - medication history – enquire about anticoagulant use, e.g. warfarin
- Perform standard clinical observations. Examine and record:
 - colour, warmth, movement, and sensation in hands and feet of injured limb(s)
- Perform physical examination
 - Examine:
 - all places where it is painful
 - any wounds or swelling
 - colour of the whole limb (especially paleness or blue colour)
 - the skin over the fracture
 - range of movement
 - joint function above and below the injury site

Check whether:

- the limb is out of shape – compare one side with the other
- the limb is warm
- the limb (if swollen) is throbbing or getting bigger
- peripheral pulses are palpable

Management:

- Splint the site of the fracture/dislocation using a plaster backslab to reduce pain
- Elevate the limb – a sling for arm injuries, a pillow for leg injuries
- If in doubt over an injury, treat as a fracture
- Administer analgesia to patients in severe pain. If not allergic, give morphine (preferable); if allergic to morphine, use fentanyl
- Consider compartment syndrome where pain is severe and unrelieved by splinting and elevation or two doses of analgesia
- X-ray if available



Text C

Drug Therapy Protocol:

Authorised Indigenous Health Worker (IHW) must consult Medical Officer (MO) or Nurse Practitioner (NP).
Scheduled Medicines Rural & Isolated Practice Registered Nurse may proceed.

Drug	Form	Strength	Route of administration	Recommended dosage	Duration
Morphine	Ampoule	10 mg/mL	IM/SC	Adult only: 0.1-0.2 mg/kg to a max. of 10 mg	Stat
			IV (IHW may not administer IV)	Adult only: Initial dose of 2 mg then 0.5-1 mg increments slowly, repeated every 3-5 minutes if required to a max. of 10 mg	Further doses on MO/NP order

Use the lower end of dose range in patients ≥ 70 years.

Provide Consumer Medicine Information: advise can cause nausea and vomiting, drowsiness.

Respiratory depression is rare – if it should occur, give naloxone.

Text D

Technique for plaster backslab for arm fractures – use same principle for leg fractures

1. Measure a length of non-compression cotton stockinette from half way up the middle finger to just below the elbow. Width should be 2–3 cm more than the width of the distal forearm.
2. Wrap cotton padding over top for the full length of the stockinette – 2 layers, 50% overlap.
3. Measure a length of plaster of Paris 1 cm shorter than the padding/stockinette at each end. Fold the roll in about ten layers to the same length.
4. Immerse the layered plaster in a bowl of room temperature water, holding on to each end. Gently squeeze out the excess water.
5. Ensure any jewellery is removed from the injured limb.
6. Lightly mould the slab to the contours of the arm and hand in a neutral position.
7. Do not apply pressure over bony prominences. Extra padding can be placed over bony prominences if applicable.
8. Wrap crepe bandage firmly around plaster backslab.

END OF PART A

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READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAMES:

Your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE: _____

Passport Photo

TIME: 15 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** or the **Text Booklet** until you are told to do so.

Write your answers on the spaces provided on this **Question Paper**.

You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this **Question Paper** and the **Text Booklet**.

DO NOT remove OET material from the test room.



Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate **Text Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

Fractures, dislocations and sprains: Questions

Questions 1-7

For each question, **1-7**, decide which text (**A**, **B**, **C** or **D**) the information comes from. You may use any letter more than once.

In which text can you find information about

- | | | |
|----------|--|-------|
| 1 | procedures for delivering pain relief? | _____ |
| 2 | the procedure to follow when splinting a fractured limb? | _____ |
| 3 | what to record when assessing a patient? | _____ |
| 4 | the terms used to describe different types of fractures? | _____ |
| 5 | the practitioners who administer analgesia? | _____ |
| 6 | what to look for when checking an injury? | _____ |
| 7 | how fractures can be caused? | _____ |

Questions 8-14

Answer each of the questions, **8-14**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- | | | |
|-----------|--|-------|
| 8 | What should be used to elevate a patient's fractured leg? | _____ |
| 9 | What is the maximum dose of morphine per kilo of a patient's weight that can be given using the intra-muscular (IM) route? | _____ |
| 10 | Which parts of a limb may need extra padding? | _____ |



- 11 What should be used to treat a patient who suffers respiratory depression?
-
- 12 What should be used to cover a freshly applied plaster backslab?
-
- 13 What analgesic should be given to a patient who is allergic to morphine?
-
- 14 What condition might a patient have if severe pain persists after splinting, elevation and repeated analgesia?
-

Questions 15-20

Complete each of the sentences, **15-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- 15 Falling on an outstretched hand is a typical cause of a _____ of the elbow.
- 16 Upper limb fractures should be elevated by means of a _____.
- 17 Make sure the patient isn't wearing any _____ on the part of the body where the plaster backslab is going to be placed.
- 18 Check to see whether swollen limbs are _____ or increasing in size.
- 19 In a plaster backslab, there is a layer of _____ closest to the skin.
- 20 Patients aged _____ and over shouldn't be given the higher dosages of pain relief.

END OF PART A
THIS QUESTION PAPER WILL BE COLLECTED



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READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAMES:

Your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE: _____

Passport Photo

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this **Question Paper**.

DO NOT remove OET material from the test room.

HOW TO ANSWER THE QUESTIONS:

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:

☐ A

☐ B

☐ C



Part B

In this part of the test, there are six short extracts relating to the work of health professionals. For **questions 1-6**, choose the answer (**A**, **B** or **C**) which you think fits best according to the text.

1. The manual informs us that the Blood Pressure Monitor

- Ⓐ is likely to interfere with the operation of other medical equipment.
- Ⓑ may not work correctly in close proximity to some other devices.
- Ⓒ should be considered safe to use in all hospital environments.

Instruction Manual: Digital Automatic Blood Pressure Monitor

Electromagnetic Compatibility (EMC)

With the increased use of portable electronic devices, medical equipment may be susceptible to electromagnetic interference. This may result in incorrect operation of the medical device and create a potentially unsafe situation. In order to regulate the requirements for EMC, with the aim of preventing unsafe product situations, the EN60601-1-2 standard defines the levels of immunity to electromagnetic interferences as well as maximum levels of electromagnetic emissions for medical devices. This medical device conforms to EN60601-1-2:2001 for both immunity and emissions. Nevertheless, care should be taken to avoid the use of the monitor within 7 metres of cellphones or other devices generating strong electrical or electromagnetic fields.



2. The notice is giving information about

- Ⓐ ways of checking that an NG tube has been placed correctly.
- Ⓑ how the use of NG feeding tubes is authorised.
- Ⓒ which staff should perform NG tube placement.

NG feeding tubes

Displacement of nasogastric (NG) feeding tubes can have serious implications if undetected. Incorrectly positioned tubes leave patients vulnerable to the risks of regurgitation and respiratory aspiration. It is crucial to differentiate between gastric and respiratory placement on initial insertion to prevent potentially fatal pulmonary complications. Insertion and care of an NG tube should therefore only be carried out by a registered doctor or nurse who has undergone theoretical and practical training and is deemed competent or is supervised by someone competent. Assistant practitioners and other unregistered staff must never insert NG tubes or be involved in the initial confirmation of safe NG tube position.



3. What must all staff involved in the transfusion process do?

- Ⓐ check that their existing training is still valid
- Ⓑ attend a course to learn about new procedures
- Ⓒ read a document that explains changes in policy

'Right Patient, Right Blood' Assessments

The administration of blood can have significant morbidity and mortality. Following the introduction of the 'Right Patient, Right Blood' safety policy, all staff involved in the transfusion process must be competency assessed. To ensure the safe administration of blood components to the intended patient, all staff must be aware of their responsibilities in line with professional standards.

Staff must ensure that if they take any part in the transfusion process, their competency assessment is updated every three years. All staff are responsible for ensuring that they attend the mandatory training identified for their roles. Relevant training courses are clearly identified in Appendix 1 of the Mandatory Training Matrix.



4. The guidelines establish that the healthcare professional should
- Ⓐ aim to make patients fully aware of their right to a chaperone.
 - Ⓑ evaluate the need for a chaperone on a case-by-case basis.
 - Ⓒ respect the wishes of the patient above all else.

Extract from 'Chaperones: Guidelines for Good Practice'

A patient may specifically request a chaperone or in certain circumstances may nominate one, but it will not always be the case that a chaperone is required. It is often a question of using professional judgement to assess an individual situation. If a chaperone is offered and declined, this must be clearly documented in the patient's record, along with any relevant discussion. The chaperone should only be present for the physical examination and should be in a position to see what the healthcare professional undertaking the examination/investigation is doing. The healthcare professional should wait until the chaperone has left the room/cubicle before discussion takes place on any aspect of the patient's care, unless the patient specifically requests the chaperone to remain.



5. The guidelines require those undertaking a clinical medication review to

- Ⓐ involve the patient in their decisions.
- Ⓑ consider the cost of any change in treatments.
- Ⓒ recommend other services as an alternative to medication.

Annual medication review

To give all patients an annual medication review is an ideal to strive for. In the meantime there is an argument for targeting all clinical medication reviews to those patients likely to benefit most.

Our guidelines state that 'at least a level 2 medication review will occur', i.e. the minimum standard is a treatment review of medicines with the full notes but not necessarily with the patient present. However, the guidelines go on to say that 'all patients should have the chance to raise questions and highlight problems about their medicines' and that 'any changes resulting from the review are agreed with the patient'.

It also states that GP practices are expected to

- minimise waste in prescribing and avoid ineffective treatments.
- engage effectively in the prevention of ill health.
- avoid the need for costly treatments by proactively managing patients to recovery through the whole care pathway.



6. The purpose of this email is to

- Ⓐ report on a rise in post-surgical complications.
- Ⓑ explain the background to a change in patient care.
- Ⓒ remind staff about procedures for administering drugs.

To: All staff

Subject: Advisory Email: Safe use of opioids

In August, an alert was issued on the safe use of opioids in hospitals. This reported the incidence of respiratory depression among post-surgical patients to an average 0.5% – thus for every 5,000 surgical patients, 25 will experience respiratory depression. Failure to recognise respiratory depression and institute timely intervention can lead to cardiopulmonary arrest, resulting in brain injury or death. A retrospective multi-centre study of 14,720 cardiopulmonary arrest cases showed that 44% were respiratory related and more than 35% occurred on the general care floor. It is therefore recommended that post-operative patients now have continuous monitoring, instead of spot checks, of both oxygenation and ventilation.



Part C

In this part of the test, there are two texts about different aspects of healthcare. For **questions 7-22**, choose the answer (**A, B, C** or **D**) which you think fits best according to the text.

Text 1: Sleep deprivation

Millions of people who suffer sleep problems also suffer myriad health burdens. In addition to emotional distress and cognitive impairments, these can include high blood pressure, obesity, and metabolic syndrome. 'In the studies we've done, almost every variable we measured was affected. There's not a system in the body that's not affected by sleep,' says University of Chicago sleep researcher Eve Van Cauter. 'Every time we sleep-deprive ourselves, things go wrong.'

A common refrain among sleep scientists about two decades ago was that sleep was performed by the brain in the interest of the brain. That wasn't a fully elaborated theory, but it wasn't wrong. Numerous recent studies have hinted at the purpose of sleep by confirming that neurological function and cognition are messed up during sleep loss, with the patient's reaction time, mood, and judgement all suffering if they are kept awake too long.

In 1997, Bob McCarley and colleagues at Harvard Medical School found that when they kept cats awake by playing with them, a compound known as adenosine increased in the basal forebrain as the sleepy felines stayed up longer, and slowly returned to normal levels when they were later allowed to sleep. McCarley's team also found that administering adenosine to the basal forebrain acted as a sedative, putting animals to sleep. It should come as no surprise then that caffeine, which blocks adenosine's receptor, keeps us awake. Teaming up with Basheer and others, McCarley later discovered that, as adenosine levels rise during sleep deprivation, so do concentrations of adenosine receptors, magnifying the molecule's sleep-inducing effect. 'The brain has cleverly designed a two-stage defence against the consequences of sleep loss,' McCarley says. Adenosine may underlie some of the cognitive deficits that result from sleep loss. McCarley and colleagues found that infusing adenosine into rats' basal forebrain impaired their performance on an attention test, similar to that seen in sleep-deprived humans. But adenosine levels are **by no means the be-all and end-all** of sleep deprivation's effects on the brain or the body.

Over a century of sleep research has revealed numerous undesirable outcomes from staying awake too long. In 1999, Van Cauter and colleagues had eleven men sleep in the university lab. For three nights, they spent eight hours in bed, then for six nights they were allowed only four hours (accruing what Van Cauter calls a sleep debt), and then for six nights they could sleep for up to twelve hours (sleep recovery). During sleep debt and recovery, researchers gave the participants a glucose tolerance test and found striking differences. While sleep deprived, the men's glucose metabolism resembled a pre-diabetic state. 'We knew it would be affected,' says Van Cauter. 'The big surprise was the effect being much greater than we thought.'



Subsequent studies also found insulin resistance increased during bouts of sleep restriction, and in 2012, Van Cauter's team observed impairments in insulin signalling in subjects' fat cells. Another recent study showed that sleep-restricted people will add 300 calories to their daily diet. Echoing Van Cauter's results, Basheer has found evidence that enforced lack of sleep sends the brain into a catabolic, or energy-consuming, state. This is because it degrades the energy molecule adenosine triphosphate (ATP) to produce adenosine monophosphate and this results in the activation of AMP kinase, an enzyme that boosts fatty acid synthesis and glucose utilization. 'The system sends a message that there's a need for more energy,' Basheer says. Whether this is indeed the mechanism underlying late-night binge-eating is still speculative.

Within the brain, scientists have glimpsed signs of physical damage from sleep loss, and the time-line for recovery, if any occurs, is unknown. Chiara Cirelli's team at the Madison School of Medicine in the USA found structural changes in the cortical neurons of mice when the animals are kept awake for long periods. Specifically, Cirelli and colleagues saw signs of mitochondrial activation – which makes sense, as 'neurons need more energy to stay awake,' she says – as well as unexpected changes, such as undigested cellular debris, signs of cellular aging that are unusual in the neurons of young, healthy mice. 'The number [of debris granules] was small, but it's worrisome because it's only four to five days' of sleep deprivation,' says Cirelli. After thirty-six hours of sleep recovery, a period during which she expected normalcy to resume, those changes remained.

Further insights could be drawn from the study of shift workers and insomniacs, who serve as natural experiments on how the human body reacts to losing out on such a basic life need for chronic periods. But with so much of our physiology affected, an effective therapy – other than sleep itself – is hard to imagine. 'People like to define a clear pathway of action for health conditions,' says Van Cauter. 'With sleep deprivation, everything you measure is affected and interacts synergistically to produce the effect.'



Text 1: Questions 7-14

7. In the first paragraph, the writer uses Eve Van Cauter's words to
- (A) explain the main causes of sleep deprivation.
 - (B) reinforce a view about the impact of sleep deprivation.
 - (C) question some research findings about sleep deprivation.
 - (D) describe the challenges involved in sleep deprivation research.
8. What do we learn about sleep in the second paragraph?
- (A) Scientific opinion about its function has changed in recent years.
 - (B) There is now more controversy about it than there was in the past.
 - (C) Researchers have tended to confirm earlier ideas about its purpose.
 - (D) Studies undertaken in the past have formed the basis of current research.
9. What particularly impressed Bob McCarley of Harvard Medical School?
- (A) the effectiveness of adenosine as a sedative
 - (B) the influence of caffeine on adenosine receptors
 - (C) the simultaneous production of adenosine and adenosine receptors
 - (D) the extent to which adenosine levels fall when subjects are allowed to sleep
10. In the third paragraph, what idea is emphasised by the phrase 'by no means the be-all and end-all'?
- (A) Sleep deprivation has consequences beyond its impact on adenosine levels.
 - (B) Adenosine levels are a significant factor in situations other than sleep deprivation.
 - (C) The role of adenosine as a response to sleep deprivation is not yet fully understood.
 - (D) The importance of the link between sleep deprivation and adenosine should not be underestimated.



11. What was significant about the findings in Van Cauter's experiment?
- (A) the rate at which the sleep-deprived men entered a pre-diabetic state
 - (B) the fact that sleep deprivation had an influence on the men's glucose levels
 - (C) the differences between individual men with regard to their glucose tolerance
 - (D) the extent of the contrast in the men's metabolic states between sleep debt and recovery
12. In the fifth paragraph, what does the word 'it' refer to?
- (A) an enzyme
 - (B) new evidence
 - (C) a catabolic state
 - (D) enforced lack of sleep
13. What aspect of her findings surprised Chiara Cirelli?
- (A) There was no reversal of a certain effect of sleep deprivation.
 - (B) The cortical neurons of the mice underwent structural changes.
 - (C) There was evidence of an increased need for energy in the brains of the mice.
 - (D) The neurological response to sleep deprivation only took a few hours to become apparent.
14. In the final paragraph, the quote from Van Cauter is used to suggest that
- (A) the goals of sleep deprivation research are sometimes unclear.
 - (B) it could be difficult to develop any treatment for sleep deprivation.
 - (C) opinions about the best way to deal with sleep deprivation are divided.
 - (D) there is still a great deal to be learnt about the effects of sleep deprivation.



Text 2: ADHD

The American Psychiatric Association (APA) recognised Attention Deficit Hyperactivity Disorder (ADHD) as a childhood disorder in the 1960s, but it wasn't until 1978 that the condition was formally recognised as afflicting adults. In recent years, the USA has seen a 40% rise in diagnoses of ADHD in children. It could be that the disorder is becoming more prevalent, or, as seems more plausible, doctors are making the diagnosis more frequently. The issue is complicated by the lack of any recognised neurological markers for ADHD. The APA relies instead on a set of behavioural patterns for diagnosis. It specifies that patients under 17 must display at least six symptoms of inattention and/or hyperactivity; adults need only display five.

ADHD can be a controversial condition. Dr Russell Barkley, Professor of Psychiatry at the University of Massachusetts insists; 'the science is overwhelming: it's a real disorder, which can be managed, in many cases, by using stimulant medication in combination with other treatments'. Dr Richard Saul, a behavioural neurologist with five decades of experience, disagrees; 'Many of us have difficulty with organization or details, a tendency to lose things, or to be forgetful or distracted. Under such subjective criteria, the entire population could potentially qualify. Although some patients might need stimulants to function well in daily life, the lumping together of many vague and subjective symptoms could be causing a national phenomenon of misdiagnosis and over-prescription of stimulants.'

A recent study found children in foster care three times more likely than others to be diagnosed with ADHD. Researchers also found that children with ADHD in foster care were more likely to have another disorder, such as depression or anxiety. This finding certainly reveals the need for medical and behavioural services for these children, but it could also prove the non-specific nature of the symptoms of ADHD: anxiety and depression, or an altered state, can easily be mistaken for manifestations of ADHD.

ADHD, the thinking goes, begins in childhood. In fact, in order to be diagnosed with it as an adult, a patient must demonstrate that they had traits of the condition in childhood. However, studies from the UK and Brazil, published in JAMA Psychiatry, are fuelling questions about the origins and trajectory of ADHD, suggesting not only that it can begin in adulthood, but that there may be two distinct syndromes: adult-onset ADHD and childhood ADHD.

They echo earlier research from New Zealand. However, an editorial by Dr Stephen Faraone in JAMA Psychiatry highlights potential flaws in the findings. Among them, underestimating the persistence of ADHD into adulthood and overestimating the prevalence of adult-onset ADHD. In Dr Faraone's words, 'the researchers found a group of people who had sub-threshold ADHD in their youth. There may have been signs that things weren't right, but not enough to go to a doctor. Perhaps these were smart kids with particularly supportive parents or teachers who helped them cope with attention problems. Such intellectual and social scaffolding would help in early life, but when the scaffolding is removed, full ADHD could develop'.



Until this century, adult ADHD was a seldom-diagnosed disorder. Nowadays however, it's common in mainstream medicine in the USA, a paradigm shift apparently driven by two factors: reworked – many say less stringent – diagnostic criteria, introduced by the APA in 2013, and marketing by manufacturers of ADHD medications. Some have suggested that this new, broader definition of ADHD was fuelled, at least in part, to broaden the market for medication. In many instances, the evidence proffered to expand the definitions came from studies funded in whole or part by manufacturers. And as the criteria for the condition loosened, reports emerged about clinicians involved in diagnosing ADHD receiving money from drug-makers.

This brings us to the issue of the addictive nature of ADHD medication. As Dr Saul asserts, 'addiction to stimulant medication isn't rare; it's common. Just observe the many patients periodically seeking an increased dosage as their powers of concentration diminish. This is because the body stops producing the appropriate levels of neurotransmitters that ADHD drugs replace – a trademark of addictive substances.' Much has been written about the staggering increase in opioid overdoses and abuse of prescription painkillers in the USA, but the abuse of drugs used to treat ADHD is no less a threat. While opioids are more lethal than prescription stimulants, there are parallels between the opioid epidemic and the increase in problems tied to stimulants. In the former, users switch from prescription narcotics to heroin and illicit fentanyl. With ADHD drugs, patients are switching from legally prescribed stimulants to illicit ones such as methamphetamine and cocaine. The medication is particularly prone to abuse because people feel it improves their lives. These drugs are antidepressants, aid weight-loss and improve confidence, and can be abused by students seeking to improve their focus or academic performance. So, more work needs to be done before we can settle the questions surrounding the diagnosis and treatment of ADHD.



Text 2: Questions 15-22

15. In the first paragraph, the writer questions whether
- (A) adult ADHD should have been recognised as a disorder at an earlier date.
 - (B) ADHD should be diagnosed in the same way for children and adults.
 - (C) ADHD can actually be indicated by neurological markers.
 - (D) cases of ADHD have genuinely increased in the USA.
16. What does Dr Saul object to?
- (A) the suggestion that people need stimulants to cope with everyday life
 - (B) the implication that everyone has some symptoms of ADHD
 - (C) the grouping of imprecise symptoms into a mental disorder
 - (D) the treatment for ADHD suggested by Dr Barkley
17. The writer regards the study of children in foster care as significant because it
- (A) highlights the difficulty of distinguishing ADHD from other conditions.
 - (B) focuses on children known to have complex mental disorders.
 - (C) suggests a link between ADHD and a child's upbringing.
 - (D) draws attention to the poor care given to such children.
18. In the fourth paragraph, the word 'They' refers to
- (A) syndromes.
 - (B) questions.
 - (C) studies.
 - (D) origins.



19. Dr Faraone suggests that the group of patients diagnosed with adult-onset ADHD
- (A) had teachers or parents who recognised the symptoms of ADHD.
 - (B) should have consulted a doctor at a younger age.
 - (C) had mild undiagnosed ADHD in childhood.
 - (D) were specially chosen by the researchers.
20. In the fifth paragraph, it is suggested that drug companies have
- (A) been overly aggressive in their marketing of ADHD medication.
 - (B) influenced research that led to the reworking of ADHD diagnostic criteria.
 - (C) attempted to change the rules about incentives for doctors who diagnose ADHD.
 - (D) encouraged the APA to rush through changes to the criteria for diagnosing ADHD.
21. In the final paragraph, the word 'trademark' refers to
- (A) a physiological reaction.
 - (B) a substitute medication.
 - (C) a need for research.
 - (D) a common request.
22. In the final paragraph, what does the writer imply about addiction to ADHD medication?
- (A) It is unlikely to turn into a problem on the scale of that caused by opioid abuse.
 - (B) The effects are more marked in certain sectors of the population.
 - (C) Insufficient attention seems to have been paid to it.
 - (D) The reasons for it are not yet fully understood.

END OF READING TEST
THIS BOOKLET WILL BE COLLECTED



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READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

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Passport Photo



Text A

Paracetamol: contraindications and interactions

4.4 Special warnings and precautions for use

Where analgesics are used long-term (>3 months) with administration every two days or more frequently, headache may develop or increase. Headache induced by overuse of analgesics (MOH medication-overuse headache) should not be treated by dose increase. In such cases, the use of analgesics should be discontinued in consultation with the doctor.

Care is advised in the administration of paracetamol to patients with alcohol dependency, severe renal or severe hepatic impairment. Other contraindications are: shock and acute inflammation of liver due to hepatitis C virus. The hazards of overdose are greater in those with non-cirrhotic alcoholic liver disease.

4.5 Interaction with other medicinal products and other forms of interaction

- Anticoagulants – the effect of warfarin and other coumarins may be enhanced by prolonged regular use of paracetamol with increased risk of bleeding. Occasional doses have no significant effect.
- Metoclopramide – may increase speed of absorption of paracetamol.
- Domperidone – may increase speed of absorption of paracetamol.
- Colestyramine – may reduce absorption if given within one hour of paracetamol.
- Imatinib – restriction or avoidance of concomitant regular paracetamol use should be taken with imatinib.

A total of 169 drugs (1042 brand and generic names) are known to interact with paracetamol.

14 major drug interactions (e.g. amyl nitrite)

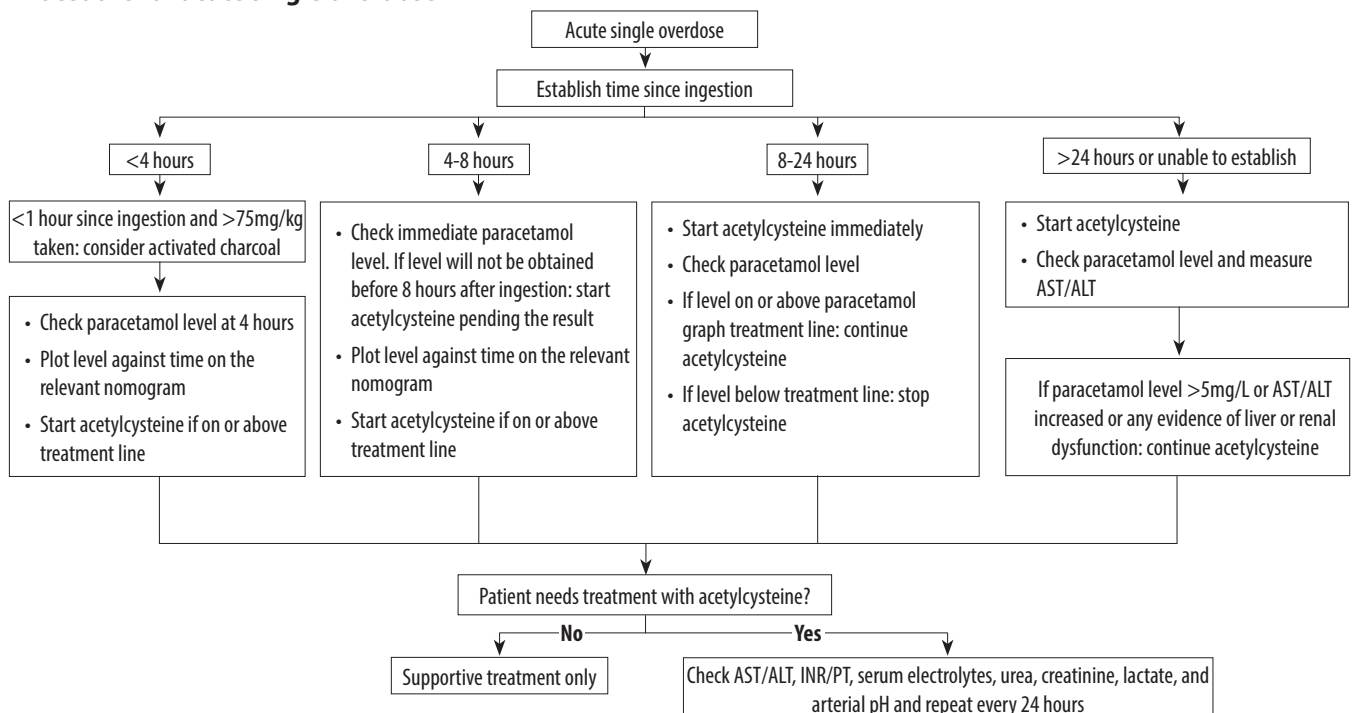
62 moderate drug interactions

93 minor drug interactions

A total of 118 brand names are known to have paracetamol in their formulation, e.g. Lemsip.

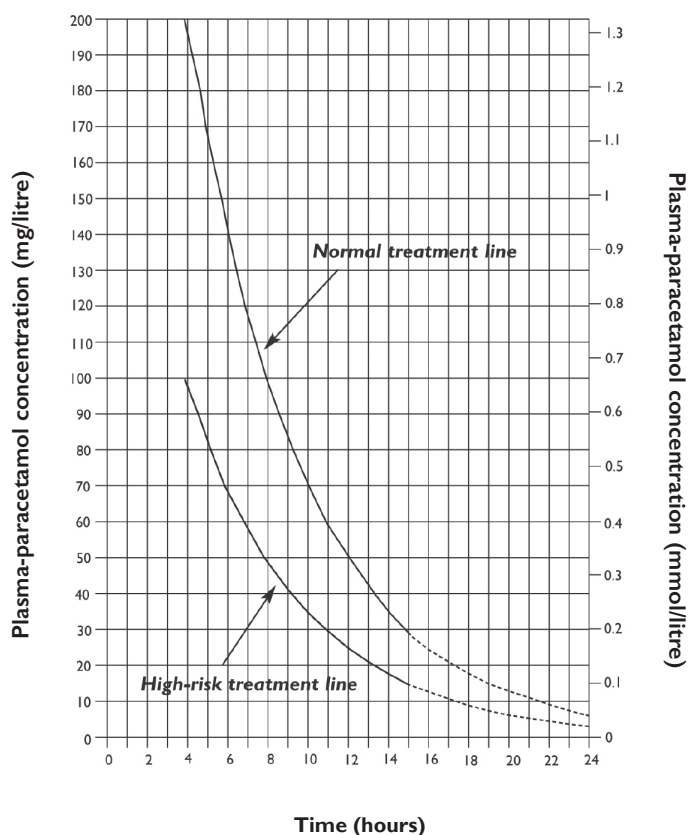
Text B

Procedure for acute single overdose



Text C

Paracetamol poisoning – Emergency treatment of poisoning



Patients whose plasma-paracetamol concentrations are above the **normal treatment line** should be treated with acetylcysteine by intravenous infusion (or, if acetylcysteine cannot be used, with methionine by mouth, provided the overdose has been taken **within 10-12 hours** and the patient is not vomiting).

Patients on enzyme-inducing drugs (e.g. carbamazepine, phenobarbital, phenytoin, primidone, rifampicin and St John's wort) or who are malnourished (e.g. in anorexia, in alcoholism, or those who are HIV positive) should be treated with acetylcysteine if their plasma-paracetamol concentration is above the **high-risk treatment line**.

Text D

Clinical Assessment

- Commonly, patients who have taken a paracetamol overdose are asymptomatic for the first 24 hours or just have nausea and vomiting
- Hepatic necrosis (elevated transaminases, right upper quadrant pain and jaundice) begins to develop after 24 hours and can progress to acute liver failure (ALF)
- Patients may also develop:
 - Encephalopathy
 - Oliguria
 - Hypoglycaemia
 - Renal failure – usually occurs around day three
 - Lactic acidosis

History

- Number of tablets, formulation, any concomitant tablets
- Time of overdose
- Suicide risk – was a note left?
- Any alcohol taken (acute alcohol ingestion will inhibit liver enzymes and may reduce the production of the toxin NAPQI, whereas chronic alcoholism may increase it)

END OF PART A
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READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAMES:

Your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE: _____

Passport Photo

TIME: 15 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** or the **Text Booklet** until you are told to do so.

Write your answers on the spaces provided on this **Question Paper**.

You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this **Question Paper** and the **Text Booklet**.

DO NOT remove OET material from the test room.



Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate **Text Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

Paracetamol overdose: Questions

Questions 1-7

For each question, **1-7**, decide which text (**A**, **B**, **C** or **D**) the information comes from. You may use any letter more than once.

In which text can you find information about

- 1 the various symptoms of patients who have taken too much paracetamol? _____
- 2 the precise levels of paracetamol in the blood which require urgent intervention? _____
- 3 the steps to be taken when treating a paracetamol overdose patient? _____
- 4 whether paracetamol overdose was intentional? _____
- 5 the number of products containing paracetamol? _____
- 6 what to do if there are no details available about the time of the overdose? _____
- 7 dealing with paracetamol overdose patients who have not received adequate nutrition? _____

Questions 8-13

Answer each of the questions, **8-13**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- 8 If paracetamol is used as a long-term painkiller, what symptom may get worse?

- 9 It may be dangerous to administer paracetamol to a patient with which viral condition?

- 10 What condition may develop in an overdose patient who presents with jaundice?



11 What condition may develop on the third day after an overdose?

12 What drug can be administered orally within 10 - 12 hours as an alternative to acetylcysteine?

13 What treatment can be used if a single overdose has occurred less than an hour ago?

Questions 14-20

Complete each of the sentences, **14-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

14 If a patient has taken metoclopramide alongside paracetamol, this may affect the

_____ of the paracetamol.

15 After 24 hours, an overdose patient may present with pain in the _____.

16 For the first 24 hours after overdosing, patients may only have such symptoms as

_____.

17 Acetylcysteine should be administered to patients with a paracetamol level above the high-risk treatment line who are taking any type of _____ medication.

18 A non-high-risk patient should be treated for paracetamol poisoning if their paracetamol level is above _____ mg/litre 8 hours after overdosing.

19 A high-risk patient who overdosed _____ hours ago should be given acetylcysteine if their paracetamol level is 25 mg/litre or higher.

20 If a patient does not require further acetylcysteine, they should be given treatment categorised as _____ only.

END OF PART A

THIS QUESTION PAPER WILL BE COLLECTED



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READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAMES:

Your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE: _____

Passport Photo

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this **Question Paper**.

DO NOT remove OET material from the test room.

HOW TO ANSWER THE QUESTIONS:

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:

☐ A

☐ B

☐ C



Part B

In this part of the test, there are six short extracts relating to the work of health professionals. For **questions 1-6**, choose the answer (**A**, **B** or **C**) which you think fits best according to the text.

1. This guideline extract says that the nurse in charge

- Ⓐ must supervise the opening of the controlled drug cupboard.
- Ⓑ should make sure that all ward cupboard keys are kept together.
- Ⓒ can delegate responsibility for the cupboard keys to another ward.

Medicine Cupboard Keys

The keys for the controlled drug cupboard are the responsibility of the nurse in charge. They may be passed to a registered nurse in order for them to carry out their duties and returned to the nurse in charge. If the keys for the controlled drug cupboard go missing, the locks must be changed and pharmacy informed and an incident form completed. The controlled drug cupboard keys should be kept separately from the main body of keys. Apart from in exceptional circumstances, the keys should not leave the ward or department. If necessary, the nurse in charge should arrange for the keys to be held in a neighbouring ward or department by the nurse in charge there.



2. When seeking consent for a post-mortem examination, it is necessary to

- Ⓐ give a valid reason for conducting it.
- Ⓑ allow all relatives the opportunity to decline it.
- Ⓒ only raise the subject after death has occurred.

Post-Mortem Consent

A senior member of the clinical team, preferably the Consultant in charge of the care, should raise the possibility of a post-mortem examination with the most appropriate person to give consent. The person consenting will need an explanation of the reasons for the post-mortem examination and what it hopes to achieve. The first approach should be made as soon as it is apparent that a post-mortem examination may be desirable, as there is no need to wait until the patient has died. Many relatives are more prepared for the consenting procedure if they have had time to think about it beforehand.



3. The purpose of these notes about an incinerator is to

- Ⓐ help maximise its efficiency.
- Ⓑ give guidance on certain safety procedures.
- Ⓒ recommend a procedure for waste separation.

Low-cost incinerator: General operating notes

3.2.1 Hospital waste management

Materials with high fuel values such as plastics, paper, card and dry textile will help maintain high incineration temperature. If possible, a good mix of waste materials should be added with each batch. This can best be achieved by having the various types of waste material loaded into separate bags at source, i.e. wards and laboratories, and clearly labelled. It is not recommended that the operator sorts and mixes waste prior to incineration as this is potentially hazardous. If possible, some plastic materials should be added with each batch of waste as this burns at high temperatures. However, care and judgement will be needed, as too much plastic will create dense dark smoke.



4. What does this manual tell us about spacer devices?

- Ⓐ Patients should try out a number of devices with their inhaler.
- Ⓑ They enable a patient to receive more of the prescribed medicine.
- Ⓒ Children should be given spacers which are smaller than those for adults.

Manual extract: Spacer devices for asthma patients

Spacer devices remove the need for co-ordination between actuation of a pressurized metered-dose inhaler and inhalation. In addition, the device allows more time for evaporation of the propellant so that a larger proportion of the particles can be inhaled and deposited in the lungs. Spacer devices are particularly useful for patients with poor inhalation technique, for children, for patients requiring higher doses, for nocturnal asthma, and for patients prone to candidiasis with inhaled corticosteroids. The size of the spacer is important, the larger spacers with a one-way valve being most effective. It is important to prescribe a spacer device that is compatible with the metered-dose inhaler. Spacer devices should not be regarded as interchangeable; patients should be advised not to switch between spacer devices.



5. The email is reminding staff that the

- Ⓐ benefits to patients of using bedrails can outweigh the dangers.
- Ⓑ number of bedrail-related accidents has reached unacceptable levels.
- Ⓒ patient's condition should be central to any decision about the use of bedrails.

To: All Staff

Subject: Use of bed rails

Please note the following.

Patients in hospital may be at risk of falling from bed for many reasons including poor mobility, dementia or delirium, visual impairment, and the effects of treatment or medication. Bedrails can be used as safety devices intended to reduce risk.

However, bedrails aren't appropriate for all patients, and their use involves risks. National data suggests around 1,250 patients injure themselves on bedrails annually, usually scrapes and bruises to their lower legs. Statistics show 44,000 reports of patient falls from bed annually resulting in 11 deaths, while deaths due to bedrail entrapment occur less than one every two years, and are avoidable if the relevant advice is followed. Staff should continue to take great care to avoid bedrail entrapment, but be aware that in hospital settings there may be a greater risk of harm to patients who fall out of bed.



6. What does this extract from a handbook tell us about analeptic drugs?

- Ⓐ They may be useful for patients who are not fully responsive.
- Ⓑ Injections of these drugs will limit the need for physiotherapy.
- Ⓒ Care should be taken if they are used over an extended period.

Analeptic drugs

Respiratory stimulants (analeptic drugs) have a limited place in the treatment of ventilatory failure in patients with chronic obstructive pulmonary disease. They are effective only when given by intravenous injection or infusion and have a short duration of action. Their use has largely been replaced by ventilatory support. However, occasionally when ventilatory support is contra-indicated and in patients with hypercapnic respiratory failure who are becoming drowsy or comatose, respiratory stimulants in the short term may arouse patients sufficiently to co-operate and clear their secretions.

Respiratory stimulants can also be harmful in respiratory failure since they stimulate non-respiratory as well as respiratory muscles. They should only be given under expert supervision in hospital and must be combined with active physiotherapy. At present, there is no oral respiratory stimulant available for long-term use in chronic respiratory failure.



Part C

In this part of the test, there are two texts about different aspects of healthcare. For **questions 7-22**, choose the answer (**A**, **B**, **C** or **D**) which you think fits best according to the text.

Text 1: Patient Safety

Highlighting a collaborative initiative to improve patient safety

In a well-documented case in November 2004, a female patient called Mary was admitted to a hospital in Seattle, USA, to receive treatment for a brain aneurysm. What followed was a tragedy, made worse by the fact that it needn't have occurred at all. The patient was mistakenly injected with the antiseptic chlorhexidine. It happened, the hospital says, because of 'confusion over the three identical stainless steel bowls in the procedure room containing clear liquids — chlorhexidine, contrast dye and saline solution'. Doctors tried amputating one of Mary's legs to save her life, but the damage to her organs was too great: she died 19 days later.

This and similar incidents are what inspired Professor Dixon-Woods of the University of Cambridge, UK, to set out on a mission: to improve patient safety. It is, she admits, going to be a challenge. Many different policies and approaches have been tried to date, but few with widespread success, and often with unintended consequences. Financial incentives are widely used, but recent evidence suggests that they have little effect. 'There's a danger that they tend to encourage **effort substitution**,' explains Dixon-Woods. In other words, people concentrate on the areas that are being incentivised, but neglect other areas. 'It's not even necessarily conscious neglect. People have only a limited amount of time, so it's inevitable they focus on areas that are measured and rewarded.'

In 2013, Dixon-Woods and colleagues published a study evaluating the use of surgical checklists introduced in hospitals to reduce complications and deaths during surgery. Her research found that that checklists may have little impact, and in some situations might even make things worse. 'The checklists sometimes introduced new risks. Nurses would use the lists as box-ticking exercises – they would tick the box to say the patient had had their antibiotics when there were no antibiotics in the hospital, for example.' They also reinforced the hierarchies – nurses had to try to get surgeons to do certain tasks, but the surgeons used the situation as an opportunity to display their power and refuse.

Dixon-Woods and her team spend time in hospitals to try to understand which systems are in place and how they are used. Not only does she find differences in approaches between hospitals, but also between units and even between shifts. 'Standardisation and harmonisation are two of the most urgent issues we have to tackle. Imagine if you have to learn each new system wherever you go or even whenever a new senior doctor is on the ward. This introduces massive risk.'



Dixon-Woods compares the issue of patient safety to that of climate change, in the sense that it is a 'problem of many hands', with many actors, each making a contribution towards the outcome, and there is difficulty in identifying where the responsibility for solving the problem lies. 'Many patient safety issues arise at the level of the system as a whole, but policies treat patient safety as an issue for each individual organisation.'

Nowhere is this more apparent than the issue of 'alarm fatigue', according to Dixon-Woods. Each bed in an intensive care unit typically generates 160 alarms per day, caused by machinery that is not integrated. 'You have to assemble all the kit around an intensive care bed manually,' she explains. 'It doesn't come built as one like an aircraft cockpit. This is not something a hospital can solve alone. It needs to be solved at the sector level.'

Dixon-Woods has turned to Professor Clarkson in Cambridge's Engineering Design Centre to help. 'Fundamentally, my work is about asking how we can make it better and what could possibly go wrong,' explains Clarkson. 'We need to look through the eyes of the healthcare providers to see the challenges and to understand where tools and techniques we use in engineering may be of value.' There is a difficulty, he concedes: 'There's no formal language of design in healthcare. Do we understand what the need is? Do we understand what the requirements are? Can we think of a range of concepts we might use and then design a solution and test it before we put it in place? We seldom see this in healthcare, and that's partly driven by culture and lack of training, but partly by lack of time.' Dixon-Woods agrees that healthcare can learn much from engineers. 'There has to be a way of getting our two sides talking,' she says. 'Only then will we be able to prevent tragedies like the death of Mary.'



Text 1: Questions 7-14

7. What point is made about the death of a female patient called Mary?
- (A) It was entirely preventable.
 - (B) Nobody was willing to accept the blame.
 - (C) Surgeons should have tried harder to save her life.
 - (D) It is the type of incident which is becoming increasingly common.
8. What is meant by the phrase '**effort substitution**' in the second paragraph?
- (A) Monetary resources are diverted unnecessarily.
 - (B) Time and energy is wasted on irrelevant matters.
 - (C) Staff focus their attention on a limited number of issues.
 - (D) People have to take on tasks which they are unfamiliar with.
9. By quoting Dixon-Woods in the second paragraph, the writer shows that the professor
- (A) understands why healthcare employees have to make certain choices.
 - (B) doubts whether reward schemes are likely to put patients at risk.
 - (C) believes staff should be paid a bonus for achieving goals.
 - (D) feels the people in question have made poor choices.
10. What point is made about checklists in the third paragraph?
- (A) Hospital staff sometimes forget to complete them.
 - (B) Nurses and surgeons are both reluctant to deal with them.
 - (C) They are an additional burden for over-worked nursing staff.
 - (D) The information recorded on them does not always reflect reality.



11. What problem is mentioned in the fourth paragraph?
- (A) failure to act promptly
 - (B) outdated procedures
 - (C) poor communication
 - (D) lack of consistency
12. What point about patient safety is the writer making by quoting Dixon-Woods' comparison with climate change?
- (A) The problem will worsen if it isn't dealt with soon.
 - (B) It isn't clear who ought to be tackling the situation.
 - (C) It is hard to know what the best course of action is.
 - (D) Many people refuse to acknowledge there is a problem.
13. The writer quotes Dixon-Woods' reference to intensive care beds in order to
- (A) present an alternative viewpoint.
 - (B) illustrate a fundamental obstacle.
 - (C) show the drawbacks of seemingly simple solutions.
 - (D) give a detailed example of how to deal with an issue.
14. What difference between healthcare and engineering is mentioned in the final paragraph?
- (A) the types of systems they use
 - (B) the way they exploit technology
 - (C) the nature of the difficulties they face
 - (D) the approach they take to deal with challenges



Text 2: Migraine – more than just a headache

When a news reporter in the US gave an unintelligible live TV commentary of an awards ceremony, she became an overnight internet sensation. As the paramedics attended, the worry was that she'd suffered a stroke live on air. Others wondered if she was drunk or on drugs. However, in interviews shortly after, she revealed, to general astonishment, that she'd simply been starting a migraine. The bizarre speech difficulties she experienced are an uncommon symptom of aura, the collective name for a range of neurological symptoms that may occur just before a migraine headache. Generally aura are visual – for example blind spots which increase in size, or have a flashing, zig-zagging or sparkling margin, but they can include other odd disturbances such as pins and needles, memory changes and even partial paralysis.

Migraine is often thought of as an occasional severe headache, but surely symptoms such as these should tell us there's more to it than meets the eye. In fact many scientists now consider it a serious neurological disorder. One area of research into migraine aura has looked at the phenomenon known as Cortical Spreading Depression (CSD) – a storm of neural activity that passes in a wave across the brain's surface. First seen in 1944 in the brain of a rabbit, it's now known that CSD can be triggered when the normal flow of electric currents within and around brain cells is somehow reversed. Nouchine Hadjikhani and her team at Harvard Medical School managed to record an episode of CSD in a brain scanner during migraine aura (in a visual region that responds to flickering motion), having found a patient who had the rare ability to be able to predict when an aura would occur. **This** confirmed a long-suspected link between CSD and the aura that often precedes migraine pain. Hadjikhani admits, however, that other work she has done suggests that CSD may occur all over the brain, often unnoticed, and may even happen in healthy brains. If so, aura may be the result of a person's brain being more sensitive to CSD than it should be.

Hadjikhani has also been looking at the structural and functional differences in the brains of migraine sufferers. She and her team found thickening of a region known as the somatosensory cortex, which maps our sense of touch in different parts of the body. They found the most significant changes in the region that relates to the head and face. 'Because sufferers return to normal following an attack, migraine has always been considered an episodic problem,' says Hadjikhani. 'But we found that if you have successive strikes of pain in the face area, it actually increases cortical thickness.'

Work with children is also providing some startling insights. A study by migraine expert Peter Goadsby, who splits his time between King's College London and the University of California, San Francisco, looked at the prevalence of migraine in mothers of babies with colic - the uncontrolled crying and fussiness often blamed on sensitive stomachs or reflux. He found that of 154 mothers whose babies were having a routine two-month check-up, the migraine sufferers were 2.6 times as likely to have a baby with colic. Goadsby believes it is possible that a baby with a tendency to migraine may not cope well with the barrage of sensory information they experience as their nervous system starts to mature, and the distress response could be what we call colic.



Linked to this idea, researchers are finding differences in the brain function of migraine sufferers, even between attacks. Marla Mickleborough, a vision specialist at the University of Saskatchewan in Saskatoon, Canada, found heightened sensitivity to visual stimuli in the supposedly 'normal' period between attacks. Usually the brain comes to recognise something repeating over and over again as unimportant and stops noticing it, but in people with migraine, the response doesn't diminish over time. 'They seem to be attending to things they should be ignoring,' she says.

Taken together this research is worrying and suggests that it's time for doctors to treat the condition more aggressively, and to find out more about each individual's triggers so as to stop attacks from happening. But there is a **silver lining**. The structural changes should not be likened to dementia, Alzheimer's disease or ageing, where brain tissue is lost or damaged irreparably. In migraine, the brain is compensating. Even if there's a genetic predisposition, research suggests it is the disease itself that is driving networks to an altered state. That would suggest that treatments that reduce the frequency or severity of migraine will probably be able to reverse some of the structural changes too. Treatments used to be all about reducing the immediate pain, but now it seems they might be able to achieve a great deal more.



Text 2: Questions 15-22

15. Why does the writer tell the story of the news reporter?
- (A) to explain the causes of migraine aura
 - (B) to address the fear surrounding migraine aura
 - (C) to illustrate the strange nature of migraine aura
 - (D) to clarify a misunderstanding about migraine aura
16. The research by Nouchine Hadjikhani into CSD
- (A) has less relevance than many believe.
 - (B) did not result in a definitive conclusion.
 - (C) was complicated by technical difficulties.
 - (D) overturned years of accepted knowledge.
17. What does the word '**This**' in the second paragraph refer to?
- (A) the theory that connects CSD and aura
 - (B) the part of the brain where auras take place
 - (C) the simultaneous occurrence of CSD and aura
 - (D) the ability to predict when an aura would happen
18. The implication of Hadjikhani's research into the somatosensory cortex is that
- (A) migraine could cause a structural change.
 - (B) a lasting treatment for migraine is possible.
 - (C) some diagnoses of migraine may be wrong.
 - (D) having one migraine is likely to lead to more.



19. What does the writer find surprising about Goadsby's research?
- (A) the idea that migraine may not run in families
 - (B) the fact that migraine is evident in infancy
 - (C) the link between childbirth and onset of migraine
 - (D) the suggestion that infant colic may be linked to migraine
20. According to Marla Mickleborough, what is unusual about the brain of migraine sufferers?
- (A) It fails to filter out irrelevant details.
 - (B) It struggles to interpret visual input.
 - (C) It is slow to respond to sudden changes.
 - (D) It does not pick up on important information.
21. The writer uses the phrase 'a silver lining' in the final paragraph to emphasise
- (A) the privileged position of some sufferers.
 - (B) a more positive aspect of the research.
 - (C) the way migraine affects older people.
 - (D) the value of publicising the research.
22. What does the writer suggest about the brain changes seen in migraine sufferers?
- (A) Some of them may be beneficial.
 - (B) They are unlikely to be permanent.
 - (C) Some of them make treatment unnecessary.
 - (D) They should still be seen as a cause for concern.

END OF READING TEST
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READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

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INSTRUCTIONS TO CANDIDATES:

DO NOT remove OET material from the test room.

TEXT A

Identifying opioid dependence

The International Classification of Disease, Tenth Edition [ICD-10] is a coding system created by the World Health Organization (WHO) to catalogue and name diseases, conditions, signs and symptoms.

The ICD-10 includes criteria to identify dependence. According to the ICD-10, opioid dependence is defined by the presence of three or more of the following features at any one time in the preceding year:

- a strong desire or sense of compulsion to take opioids
- difficulties in controlling opioid use
- a physiological withdrawal state
- tolerance of opioids
- progressive neglect of alternative interests or pleasures because of opioid use
- persisting with opioid use despite clear evidence of overtly harmful consequences.

There are other definitions of opioid dependence or 'use disorder' (e.g. the Diagnostic and Statistical Manual of Mental Disorders, 5th edition, [DSM-5]), but the central features are the same. Loss of control over use, continuing use despite harm, craving, compulsive use, physical tolerance and dependence remain key in identifying problems.

TEXT B

Why not just prescribe codeine or another opioid?

Now that analgesics containing codeine are no longer available OTC (over the counter), patients may request a prescription for codeine. It is important for GPs to explain that there is a lack of evidence demonstrating the long-term analgesic efficacy of codeine in treating chronic non-cancer pain. Long-term use of opioids has not been associated with sustained improvement in function or quality of life, and there are increasing concerns about the risk of harm.

GPs should explain that the risks associated with opioids include tolerance leading to dose escalation, overdose, falls, accidents and death. It should be emphasised that OTC codeine-containing analgesics were only intended for short-term use (one to three days) and that longer-term pain management requires a more detailed assessment of the patient's medical condition as well as clinical management.

New trials have shown that for acute pain, non-opioid combinations can be as effective as combination analgesics containing opioids such as codeine and oxycodone. If pain isn't managed with non-opioid medications then consider referring the patient to a pain specialist or pain clinic.

Patient resources for pain management are freely available online to all clinicians at websites such as:

- Pain Management Network in NSW - www.aci.health.nsw.gov.au/networks/pain-management
- Australian and New Zealand College of Anaesthetists Faculty of Pain Medicine - www.fpm.anzca.edu.au

TEXT C

TABLE 2. COMPARISON OF MEDICATION APPROACHES FOR SHORT-TERM MANAGEMENT OF OPIOID WITHDRAWAL (OVER SEVEN TO 10 DAYS)⁸⁻¹⁶

Medication	Regulatory requirements	Requires off-label use	Strength of evidence	Advantages	Disadvantages	Notes on dosage
Buprenorphine–naloxone (sublingual)	Permit before treatment; patient registered as drug dependent in some jurisdictions	Indicated for opioid dependence (off-label for pain)	Multiple well-conducted RCTs demonstrate efficacy and safety ⁸	A large evidence base (e.g. Cochrane reviews) shows it is the most effective option for opioid taper¹⁶	- More restrictive than other options (regulatory requirements and supervised dosing) - Indicated only for opioid dependence, not for chronic pain in the absence of opioid dependence	See sample withdrawal regimen (Box 8) or refer to state and national guidelines
Tramadol (oral)	Only those that apply to S4 medications	Yes (indicated for pain not opioid dependence)	Low – a small number of RCTs demonstrate efficacy and safety ^{9,12}	A small number of studies examined a one-week tramadol taper, with outcomes comparable with those of other opioid tapers and superior to clonidine	- May produce serotonergic side effects, known drug interactions, use with caution in the elderly - Use is off label - Risk of seizures, even at usual doses - Variable metabolism through CYP P450, similar to codeine	100 to 300 mg sustained-release formulation twice a day for one week - Supervised dosing and daily dispensing may be indicated
Buprenorphine (transdermal patch)	Permit before treatment if patient is drug dependent	Yes (indicated for pain not opioid dependence)	None – no RCTs or published cases	Good safety profile	Dose likely to be insufficient for patients with clear evidence of opioid dependence	5, 10 or 20 mcg weekly patches are available - A single patch should be sufficient for taper from oral codeine
Symptomatic medications	Only those relating to S4/OTC medications	No (if use is consistent with product indication)	Moderate to high – well-conducted RCTs demonstrate efficacy; however, poorer outcomes than buprenorphine or tramadol ¹⁵	- Fewer prescribing restrictions - Relatively safe in outpatient setting	- Shown to be less effective than buprenorphine and tramadol - Multiple medications can be confusing - Caution using sedative medications in outpatient setting - Clonidine can cause severe hypotension	See Table 1

Abbreviations: OTC = over the counter; RCT = randomised controlled trial.

TEXT D

Preparation for tapering

As soon as a valid indication for tapering of opioid analgesics is established, it is important to have a conversation with the patient to explain the process and develop a treatment agreement. This agreement could include:

- time frame for the agreement
- objectives of the taper
- frequency of dose reduction
- requirement for obtaining the prescriptions from a designated clinician
- scheduled appointments for regular review
- anticipated effects of the taper
- consent for urine drug screening
- possible consequences of failure to comply.

Before starting tapering, it needs to be clearly emphasised to the patient that reducing the dose of opioid analgesia will not necessarily equate to increased pain and that it will, in effect, lead to improved mood and functioning as well as a reduction in pain intensity. The prescriber should establish a therapeutic alliance with the patient and develop a shared and specific goal

READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

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INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** or the **Text Booklet** until you are told to do so.

Write your answers on the spaces provided on this **Question Paper**.

You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this Question Paper and the Text Booklet.

DO NOT remove **OET** material from the test room.

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate Text **Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

Managing Opioid Dependence: Questions

Questions 1-7

For each of the questions, 1-7, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1. What GPs should say to patients requesting codeine? -----
2. Basic indications of an opioid problem? -----
3. Different medications used for weaning patients off opioids? -----
4. Decisions to make before beginning treatment of dependence? -----
5. Defining features of a use disorder? -----
6. The development of a common goal for both prescriber and patient? -----
7. Sources of further information on pain management? -----

Questions 8-14

Answer each of the questions, **8-14**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8. What will reduce doses of opioids lead to a reduction of?

9. What is the most effective medication for tapering opioid dependence?

10. How long should over the counter codeine analgesics be used for?

11. When should doctors consider referring a patient to a pain expert or clinic?

12. What might a patient give permission to before starting treatment?

13. What might be increasingly neglected as a result of opioid use?

14. How many Buprenorphine patches are needed to taper from codeine tablets?

Questions 15-20

Complete each of the sentences, **15-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15. The use of Buprenorphine-naxolone requires a -----
before treatment.

16. The use of symptomatic medications for the treatment of opioid dependence has
been found to have ----- than tramadol.

17. Different definitions of opioid dependence share the same -----

18. Once it is decided that opioid taper is a suitable treatment the doctor and patient
should create a -----

19. Recent research indicates that ----- can
work as well as combination analgesics including codeine and oxycodone.

20. The ICD-10 defines a patient as dependent if they have -----
----- key symptoms simultaneously.

READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this Question Paper until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this Question Paper.

DO NOT remove **OET** material from the test room.

HOW TO ANSWER THE QUESTIONS

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:



In this part of the test, there are six short extracts relating to the work of health professionals. For questions 1-6, choose the answer (**A, B or C**) which you think fits best according to the text.

1. According to the guidelines nurses must

- Ⓐ Advise the practice as soon as they get to the next home visit
- Ⓑ Call the patient to confirm a time before they make a home visit
- Ⓒ Inform fellow staff members when they return from a home visit

Home visit guidelines

The nurse will complete all consultation notes in the patient's name home (unless not appropriate), prior to beginning the next consultation. With a focus on nurse safety, the nurse will call the practice at the end of each visit before progressing to the next home and will also communicate any unexpected circumstances that may delay arrival back at the practice (more than one hour).

Calling from the patient's home to make a review appointment with the GP is sufficient and can help minimise time making phone calls. On return to the practice the nurse will immediately advise staff members of their return. This time will be documented on the patient visit list, scanned and filled by administration staff

2. In progressive horizontal evacuation

- Ⓐ Patients are evacuated through fire proof barriers one floor at a time
- Ⓑ Patients who can't walk should not be moved until the fire is under control
- Ⓒ Patients are moved to fire proof areas on the same level to safely wait for help.

Progressive horizontal evacuation

The principle of progressive horizontal evacuation is that of moving occupants from an area affected by fire through a fire-resisting barrier to an adjoining area on the same level, designed to protect the occupants from the immediate dangers of fire and smoke (a refuge). The occupants may remain there until the fire is dealt with or await further assisted onward evacuation by staff to similar adjoining area or to the nearest stairway. Should it become necessary to evacuate on entire storey, this procedure should give sufficient time for non-ambulant and partially ambulant patients to be evacuated vertically to place of safety.

3. The main purpose of the extract is to

- Ⓐ Provide information of the legal requirements for disposing of animal waste.
- Ⓑ Describe rules for proper selling and export of animal products.
- Ⓒ Define the meaning of animal by-products for healthcare researchers.

Proper disposal of animal waste

Animal by-products from healthcare (for example research facilities) have specific legislative requirements for disposal and treatment. They are defined as “entire bodies or parts of animals or products of animal origin not intended for human consumption, including ova, embryos and semen.” The Animal By-Products Regulations are designed to prevent animal by-products from presenting a risk to animal or public health through the transmission of disease. This aim is achieved by rules for the collection, transport, storage, handling, processing and use or disposal of animal byproducts, and the placing on the market, export and transit of animal by-products and certain products derived from them.

4. According to the extract, what is the outcome of reusing medical equipment meant to be used once?

- Ⓐ The maker will take no legal responsibility for safety.
- Ⓑ Endoscopy units will save on equipment costs.
- Ⓒ There is a higher incidence of cross infection.

Cleaning and disinfection of endoscopes should be undertaken by trained staff in a dedicated room. Thorough cleaning with detergent remains the most important and first step in the process. Automated washer/disinfectors have become an essential part of the endoscopy unit. Machines must be reliable, effective, easy to use and should prevent atmospheric pollution by the disinfectant if an irritating agent is used. Troughs of disinfectant should not be used unless containment or exhaust ventilated facilities are provided.

Whenever possible, "single use" or autoclavable accessories should be used. The risk of transfer of infection from inadequately decontaminated reusable items must be weighed against the cost. Reusing accessories labelled for single use will transfer legal liability for the safe performance of the product from the manufacturer to the user or his/her employers and should be avoided unless Department of Health criteria are met.

5. According to the extract what is the purpose of the guidelines?

- Ⓐ To present statistics on the incidence of melanoma in Australia and New Zealand.
- Ⓑ To support the early detection of melanoma and select the best treatments.
- Ⓒ To explain the causes of melanoma in populations of Celtic origin.

Foreword

Australia and New Zealand have the highest incidence of melanoma in the world. Comprehensive, up-to-date, evidence-based national guidelines for its management are therefore of great importance. Both countries have populations of predominantly Celtic origin, and in the course of day-to-day life their citizens are inevitably subjected to high levels of solar UV exposure. These two factors are considered predominantly responsible for the very high incidence of melanoma (and other skin cancers) in the two nations. In Australia, melanoma is the third most common cancer in men and the fourth most common in women, with over 13, 000 new cases and over 1, 750 deaths each year.

The purpose of evidence-based clinical guidelines for the management of any medical condition is to achieve early diagnosis whenever possible, make doctors and patients aware of the most effective treatment options, and minimise the financial burden on the health system by documenting investigations and therapies that are inappropriate.

6. What should employees declare?

- Ⓐ Every item received from one donor.
- Ⓑ Each item from one donor valued at over \$50.
- Ⓒ Every item from one donor if the combined value is more than \$50.

Reporting of Gifts and Benefits

Employees must declare all non-token gifts which they are offered, regardless of whether or not those gifts are accepted. If multiple gifts, benefits or hospitality are received from the same donor by an employee and the cumulative value of these is more than \$50 then each individual gift, benefit or hospitality event must be declared. The Executive Director of Finance will be responsible for ensuring the gifts and benefits register is subject to annual review by the Audit Committee. The review should include analysis for repetitive trends or patterns which may cause concern and require corrective and preventive action. The Audit Committee will receive a report at least annually on the administration and quality control of the gifts, benefits and hospitality policy, processes and register.

In this part of the test, there are two texts about different aspects of healthcare. For questions 7-22, choose the answer (**A, B, C or D**) which you think fits best according to the text.

Text 1: The case for and against e-cigarettes

Electronic cigarettes first hit European and American markets in 2006 and 2007, and their popularity has been propelled by international trends favouring smoke-free environments. Sales reportedly have reached \$650 million a year in Europe and were estimated to reach \$3.6 billion in the US in 2018.

Although research on e-cigarettes is not extensive, a picture is beginning to emerge. Surveys suggest that the vast majority of those who use e-cigarettes treat them as smoking-cessation aides and self-report that they have been key to quitting. Data also indicate that e-cigarettes help to reduce tobacco cigarette consumption. A 2011 survey, based on a cohort of first-time e-cigarette purchasers, found that 66.8 percent reported reducing the number of cigarettes they smoked per day and after six months, 31 percent reported not smoking. These results compare favorably with nicotine replacement therapies (NRTs) like the patch and nicotine gum. Interestingly, a randomized controlled trial found that even e-cigarettes not containing nicotine were effective both in achieving a reduction of tobacco cigarette consumption and longer term abstinence, suggesting that “factors such as the rituals associated with cigarette handling and manipulation may also play an important role.” Some tobacco control advocates worry that they simply deliver an insufficient amount of nicotine to ultimately prove effective for cessation.

Nevertheless, the tobacco control community has embraced FDA approved treatments—NRTs, as well as the drugs bupropion and varenicline—that have relatively low success rates. In a commentary published in the Journal of the American Medical Association, smoking cessation experts Andrea Smith and Simon Chapman of the University of Sydney said that smoking cessation drugs fail most of those who try them. “Sadly, it remains the case that by far the most common outcome at 6 to 12 months after using such medication in real world settings is continuing smoking. Few, if any, other drugs with such records would ever be prescribed,” they wrote.

Amongst smokers not intending to quit, e-cigarettes—both with and without nicotine—substantially reduced consumption in a randomized controlled trial, not only resulting in decreased cigarette consumption but also in “enduring tobacco abstinence. ” In a second study from 2013, the authors reported that after 24 months, 12. 5 percent of smokers remained abstinent while another 27. 5 percent reduced their tobacco cigarette consumption by 50 percent. Finally, a third study commissioned in Australia has come to the same conclusion, though a high dropout rate (42 percent) makes these findings questionable.

Users widely perceive e-cigarettes to be less toxic. While the FDA has found trace elements of carcinogens, levels are comparable to those found in nicotine replacement therapies. Results from a laboratory study released in 2013 found that that while e-cigarettes do contain contaminants, the levels range from 9 to 450 times lower than in tobacco cigarette smoke. These are comparable with the trace amounts of toxic or carcinogenic substances found in medicinal nicotine inhalers. A prominent anti-tobacco advocate, Stanton Glantz, has warned of the need to protect people from secondhand emissions. While one laboratory study indicates that passive “vaping,” as smoking an e-cigarette is commonly known, releases volatile organic compounds and ultrafine particles into the indoor environment, it noted that the actual health impact is unknown and should remain a chief concern. A 2014 study concluded that e-cigarettes are a source of second hand exposure to nicotine but not to toxins. Nevertheless, bystanders are exposed to 10 times less nicotine exposure from e-cigarettes compared to tobacco cigarettes.

There are a number of interesting points of agreement among proponents and skeptics of e-cigarettes. First, all agree that regulation to ensure the quality of e-cigarettes should be uniform. Laboratory analyses have found sometimes wide variation across brands, in the level of carcinogens, the presence of contaminants, and the quality of nicotine. Second, proponents and detractors of e-cigarettes tend to agree that — considered only at the individual level—e-cigarettes are a safer alternative to tobacco cigarette consumption. The main concern is how e-cigarettes might shape tobacco use patterns at the population level. Proponents stress the evidence base that we have reviewed. Skeptics remain worried that e-cigarettes will

become “dual use” products. That is, smokers will use e-cigarettes, but will not reduce their smoking or quit.

Perhaps most troubling to public health officials is that e-cigarettes will “renormalize” smoking, subverting the cultural shift that has occurred over the past 50 years and transforming what has become a perverse habit into a pervasive social behaviour. In other words, the fear is that e-cigarettes will allow for re-entry of the tobacco cigarette into public view. This would unravel the gains created by smoke-free indoor (and, in some scientifically-unwarranted instances) outdoor environments. Careful epidemiological studies will be needed to determine whether the individual gains from e-cigarettes will be counteracted by population-level harms. For policy makers, the challenge is how to act in the face of uncertainty.

Text 1: Questions 7-14

7. What does the writer suggest about the research into e-cigarettes?

- Ⓐ Not enough research is being carried out.
- Ⓑ Early conclusions are appearing from the evidence.
- Ⓒ Too much of the available data is self-reported.
- Ⓓ An extensive picture of e-cigarette use has emerged.

8. What explanation does the writer offer for the effect of non-nicotine e-cigarettes?

- Ⓐ They deliver an insufficient volume of nicotine to help smoking cessation.
- Ⓑ They compare well with patches, nicotine gum and other NRT's.
- Ⓒ First time e-cigarette buyers tend to use them
- Ⓓ Behavioural elements are significant in quitting smoking.

9. What is the attitude of Andrea Smith and Simon Chapman to the use of smoking cessation drugs?

- Ⓐ They approve of and embrace these treatments.
- Ⓑ They consider them largely unsuccessful as treatments.
- Ⓒ They think they should be replaced with other treatments.
- Ⓓ They believe they should never be prescribed as treatment.

10. What problem with one of the studies is mentioned in paragraph 4?

- Ⓐ The research questions the study asked.
- Ⓑ The number of participants who left the study.
- Ⓒ The similarity of the conclusion to other studies.
- Ⓓ The study used e-cigarettes without nicotine.

11. What is **"these"** in paragraph 5 referring to?

- Ⓐ Laboratory study results
- Ⓑ Nicotine inhalers
- Ⓒ Contamination levels
- Ⓓ Tobacco cigarettes

12. Research mentioned in paragraph 5 suggests that

- Ⓐ E-cigarettes release dangerous toxins into the air.
- Ⓑ E-cigarettes should be banned from indoor environments.
- Ⓒ E-cigarettes are more toxic than nicotine replacement therapies
- Ⓓ Cigarettes present a far greater risk of secondhand exposure to toxins

13. The word **uniform** in paragraph 7 suggests that e-cigarettes should

- Ⓐ Be clearly regulated against.
- Ⓑ Only come in one brand.
- Ⓒ Be of a standard quality.
- Ⓓ Contain no contaminants.

14. What do both critics and supporters of e-cigarettes agree?

- Ⓐ Available research evidence must be reviewed.
- Ⓑ E-cigarette use may not result in quitting.
- Ⓒ Smoking tobacco is more dangerous than vaping.
- Ⓓ E-cigarettes are shaping the public's tobacco use.

Text 2: Vivisection

In 1875, Charles Dodgson, under his pseudonym Lewis Carroll, wrote a blistering attack on vivisection. He sent this to the governing body of Oxford University in an attempt to prevent the establishment of a physiology department. Today, despite the subsequent evolution of one of the most rigorous governmental regulatory systems in the world, little has changed. A report sponsored by the UK Royal Society, "The use of non-human primates in research", attempts to establish a sounder basis for the debate on animal research through an in-depth analysis of the scientific arguments for research on monkeys.

In the UK, no great apes have been used for research since 1986. Of the 3000 monkeys used in animal research every year, 75% are for toxicology studies by the pharmaceutical industry. Although expenditure on biomedical research has almost doubled over the past 10 years, the number of monkeys used for this purpose (about 300) has tended to fall. The report, which mainly discusses the use of monkeys in biomedical research, pays particular attention to the development of vaccines for AIDS, malaria, and tuberculosis, and to the nervous system and its disorders. The report assesses the impact of these issues on global health, together with potential approaches that might avoid the use of animals in research. Other research areas are also discussed, together with ethics, animal welfare, drug discovery, and toxicology.

The report concludes that in some cases there is a valid scientific argument for the use of monkeys in medical research. However, no blanket decisions can be made because of the speed of progress in biomedical science (particularly in molecular and cell biology) and because of the available non-invasive methods for study of the brain. Every case must be considered individually and supported by a fully informed assessment of the importance of the work and of alternatives to the use of animals.

Furthermore, the report asks for greater openness from medical and scientific journals about the amount of animal suffering that occurred in studies and for regular publication of the outcomes of animal research and toxicology studies. It calls for the development of a national strategic plan for animal research, including the

dissemination of information about alternative research methods to the use of animals, and the creation of centres of excellence for better care of animals and for training of scientists. Finally, it suggests some approaches towards a better-informed public debate on the future of animal research.

Although the report was received favourably by the mass media, animal-rights groups thought that it did not go far enough in setting priorities for development of alternatives to the use of animals. In fact, it investigates many of these approaches, including cell and molecular biology, use of transgenic mice (an alternative to use of primates), computer modelling, in-silico technology, stem cells, microdosing, and pharmacometabonomic phenotyping. However, the report concludes that although many of these techniques have great promise, they are at a stage of development that is too early for assessment of their true potential.

The controversy of animal research continues unabated. Shortly after publication of the report, two highly charged stories were published in the media. A study that used systematic reviews to compare treatment outcome from clinical trials of animals with those of human beings suggested that discordance in the results might have been due to bias, poor design, or inadequacies of animals for modelling of human disease. Although the study made some helpful suggestions for the future, its findings are not surprising. The imperfections of animals for study of human disease and of drug trials are documented widely.

The current furore about the UK Government's ban on human nuclear-transfer experiments involving animals should not surprise us either. This area of research had a bad start when this method of production of stem cells was labelled as therapeutic cloning, thus confusing it with reproductive cloning - **a problem** that, surely, licensing bodies and the scientific community should have anticipated. The possibilities that insufficient human eggs will be available, and that insertion of human nuclei into animal eggs might be necessary, have been discussed by the scientific community for several years, but have been aired rarely in public, leaving much room for confusion

Biomedical science is progressing so quickly that maintenance of an adequate level of public debate on ethical issues is difficult. Hopefully the sponsors of the recent report will now activate its recommendations, not least how better mechanisms can be developed to broaden and sustain interactions between science and the public. Although any form of debate will probably not satisfy the extremists of the antivivisection movement, the rest of society deserves to receive the information it needs to deal with these extremely difficult issues.

Text 2: Questions 15-22

15. How does the writer characterise Lewis Carroll's attitude to vivisection?

- Ⓐ He was in favour of clear regulations to control it.
- Ⓑ He felt the Royal Society should not support it.
- Ⓒ He was strongly opposed to it.
- Ⓓ He supported its use in physiology.

16. The word **rigorous** in paragraph 1 implies that the writer thinks UK vivisection laws are

- Ⓐ Strict and severe
- Ⓑ Careful and thorough
- Ⓒ Ambiguous and unhelpful
- Ⓓ Accurate and effective

17. What is the major focus of the report mentioned in paragraph 2?

- Ⓐ Animal experimentation in the pharmaceutical industry
- Ⓑ Recent increases in spending on Biomedical research
- Ⓒ Testing new treatments for serious disease on monkeys
- Ⓓ Possible alternatives to testing new drugs on animals

18. What is the main conclusion of the report?

- Ⓐ Scientific experimentation on monkeys is justified.
- Ⓑ Rapid development in biomedicine makes it hard to draw conclusions.
- Ⓒ Non-invasive techniques should be preferred in most cases.
- Ⓓ Research that requires monkeys should be evaluated independently.

19. What conclusion is drawn about alternative techniques to vivisection?

- Ⓐ Developing alternatives should be prioritised.
- Ⓑ Transgenic mice are a viable alternative to monkeys.
- Ⓒ Many alternative techniques are more promising than animal testing.
- Ⓓ They aren't well enough understood yet to adopt for research.

20. What does the writer claim about the use of animals in medical research?

- Ⓐ The limitations of using animals in research are well understood.
- Ⓑ Results from too many animal trials are biased.
- Ⓒ Human studies are known to be more reliable.
- Ⓓ Strong media reaction has kept up the controversy.

21. The phrase **a problem** in paragraph 6 refers to the

- Ⓐ Government licensing of animal experiments.
- Ⓑ Confusion between the names of two different methods.
- Ⓒ Chortage of human embryos available for experiments.
- Ⓓ Prohibition against human nuclear transfer in the UK.

22. The author thinks it is hard to keep the public adequately informed about this research because

- Ⓐ The report sponsors have not activated the recommendations.
- Ⓑ Of the rapid evolution of biomedical technologies.
- Ⓒ Scientists don't interact with the public enough.
- Ⓓ Extreme views from opponents cloud the debate.

READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT remove OET material from the test room.

TEXT A

The GP's role in the management of ADHD

It helps to remind patients that ADHD is not all bad. ADHD is associated with positive attributes such as being more spontaneous and adventurous. Some studies have indicated that people with ADHD may be better equipped for lateral thinking. It has been suggested that explorers or entrepreneurs are more likely to have ADHD.

In addition, GPs can reinforce the importance of developing healthy sleep–wake behaviours, obtaining adequate exercise and good nutrition. These are the building blocks on which other treatment is based. For patients who are taking stimulant medication, it is helpful if the GP continues to monitor their blood pressure, given that stimulant medication may cause elevation. Once a patient has been stabilised on medication for ADHD, the psychiatrist may refer the patient back to the GP for ongoing prescribing in line with state-based guidelines. However, in most states and territories, the GP is not granted permission to alter the dose.

TEXT B

ADHD: Overview

Contrary to common belief, ADHD is not just a disorder of childhood. At least 40 to 50% of children with ADHD will continue to meet criteria in adulthood, with ADHD affecting about one in 20 adults. ADHD can be masked by many comorbid disorders that GPs are typically good at recognising such as depression, anxiety and substance use. In patients with underlying ADHD, the attentional, hyperactive or organisational problems pre-date the comorbid disorders and are not episodic as the comorbid disorders may be. GPs are encouraged to ask whether the complaints are of recent onset or longstanding. Collateral history can be helpful for developing a timeline of symptoms (e. g. parent or partner interview). Diagnosis of underlying ADHD in these patients will significantly improve their treatment outcomes, general health and quality of life.

TEXT C

TABLE 2. MEDICATIONS FOR ATTENTION DEFICIT HYPERACTIVITY DISORDER AND TYPICAL DOSING

Medication	Initiation	Dose
Immediate-release methylphenidate	5 to 10 mg in the morning the first day; add a second dose of 5 to 10 mg at lunch time for a week; then add further increments weekly	Total dose typically varies between 10 mg/day and 60 mg/day Doses of more than 80 mg/day are uncommon (maximum recommended dose in the NICE guidelines is 100 mg/day) ¹¹ Transition to longer-acting formulations can occur after a month
Extended-release methylphenidate	18 or 36 mg/day taken once daily in the morning	Increase in 18 mg increments to a maximum of 72 mg/day Adjust dosage at weekly intervals
Long-acting methylphenidate	20 mg/day taken once daily in the morning	Adjust dose weekly in 10 mg increments Dose usually would not exceed 60 mg/day
Dexamfetamine	2.5 to 5.0 mg in the morning the first day; add a second dose of 2.5 to 5.0 mg at lunch time for a week; then add further increments weekly	Total dose typically varies between 5 mg/day and 30 mg/day Doses over 40 mg/day are uncommon (maximum recommended dose in the NICE guidelines is 60 mg/day) ¹¹
Lisdexamfetamine	30 mg in the morning the first day; increase up to 70 mg according to response	Dose range typically 30 to 70 mg/day
Atomoxetine	For those weighing less than 70 kg, start at 0.5 mg/kg taken once daily for three days then increase to 1.2 mg/kg once daily in the morning or as evenly divided doses in the morning and late afternoon/early evening. For those weighing more than 70 kg, start at 40 mg/day taken once daily for three days then increase to target dose of 80 mg ³	Target dose 80 mg/day Maximum dose 100 mg ³

TEXT D

Treatment of ADHD

It is very important that the dosage of medication is individually optimised. An analogy may be made with getting the right pair of glasses – you need the right prescription for your particular presentation with not too much correction and not too little. The optimal dose typically requires careful titration by a psychiatrist with ADHD expertise. Multiple follow-up appointments are usually required to maximise the treatment outcome. It is essential that the benefits of treatment outweigh any negative effects. Common side effects of stimulant medication may include:

- appetite suppression
- insomnia
- palpitations and increased heart rate
- feelings of anxiety
- dry mouth and sweating

READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** or the **Text Booklet** until you are told to do so.

Write your answers on the spaces provided on this **Question Paper**.

You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this Question Paper and the Text Booklet.

DO NOT remove **OET** material from the test room.

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate Text **Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

ADHD: Questions

Questions 1-7

For each of the questions, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1. different types of ADHD medication? -----
2. possible side effects of medication? -----
3. conditions which may be present alongside ADHD? -----
4. a doctor's control over a patient's medication? -----
5. positive perspectives on having ADHD? -----
6. when patients should take their ADHD medicine? -----
7. figuring out a patient's optimal dosage of medication? -----

Questions 8-14

Answer each of the questions, **8-14**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8. What is the maximum recommended dose of Dexamfetamine?

9. What is typically needed to get the best results from ADHD treatment?

10. How can GP's collect information about their patient's collateral history?

11. What causes symptoms such as palpitations and anxiety in some patients?

12. What proportion of children with ADHD will carry symptoms into adulthood?

13. What positive personality traits are sometimes associated with ADHD?

14. Which medication has dose recommendations related to patient weight?

Questions 15-20

Complete each of the sentences, **15-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15. Sleep, exercise and nutrition comprise the -----
of further ADHD treatment.
16. When diagnosing ADHD, it is important to ask if the issues arose recently or
are -----
17. It is possible to move to ----- after
one month of immediate-release methylphenidate.
18. Signs of ADHD can be disguised by ----- which
GPs are more likely to recognise.
19. GPs should regularly check the ----- of
patients prescribed stimulant medication.
20. Establishing the ideal dose of ADHD medication needs -----
----- by an expert psychiatrist.

READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this Question Paper until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this Question Paper.

DO NOT remove **OET** material from the test room.

HOW TO ANSWER THE QUESTIONS

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:



In this part of the test, there are six short extracts relating to the work of health professionals. For questions 1-6, choose the answer (**A, B or C**) which you think fits best according to the text.

1. According to the extract, to prevent the spread of infection, emergency department isolation rooms

- Ⓐ should be placed away from the main entry doors.
- Ⓑ are more numerous than those of other departments.
- Ⓒ ought to be situated near where people enter the unit.

DESIGN PRINCIPLES FOR ISOLATION ROOMS

The aim of environmental control in an isolation room is to control the airflow, thereby reducing the number of airborne infectious particles that may infect others within the environment.

This is achieved by:

- controlling the quality and quantity of intake and exhaust air;
- diluting infectious particles in large volumes of air;
- maintaining differential air pressures between adjacent areas; and
- designing patterns of airflow for particular clinical purposes.

The location and design of isolation rooms within a particular department or inpatient unit should ideally enable their separation from the rest of the unit. Multiple isolation rooms should be clustered and located away from the main entrance of the unit. An exception is an emergency department where it is recommended that designated isolation rooms be located near the entry to prevent spread of possible airborne infection throughout the unit.

2. What does staff need to be conscious of when working in Anterooms?

- Ⓐ Keeping used and unused medical clothing apart.
- Ⓑ Ensuring the ambient pressure in the room is a minimum 15 Pascal.
- Ⓒ Keeping the door closed at all times.

ANTEROOMS

Anterooms allow staff and visitors to change into, and dispose of, personal protective equipment used on entering and leaving rooms when caring for infectious patients. Clean and dirty workflows within this space should be considered so that separation is possible. Anterooms increase the effectiveness of isolation rooms by minimising the potential escape of airborne nuclei into a corridor area when the door is opened.

For Class N isolation rooms the pressure in the anteroom is lower than the adjacent ambient (corridor) pressure, and positive with respect to the isolation room. The pressure differential between rooms should be not less than 15 Pascal.

Anterooms are provided for Class N isolation rooms in intensive care units, emergency departments, birthing units, infectious diseases units, and for an agreed number of patient bedrooms within inpatient units accommodating patients with respiratory conditions.

3. What is the basic principle of flexible design?

- Ⓐ Creating systems which match current policy and can adjust to other possible guidelines.
- Ⓑ Designing healthcare facilities which strictly adhere to current policy.
- Ⓒ Changing healthcare policies regularly to match changes in the market place.

FLEXIBLE DESIGN

In healthcare, operational policies change frequently. The average cycle may be as little as five years. This may be the result of management change, government policy, and turnover of key staff or change in the marketplace. By contrast, major healthcare facilities are typically designed for 30 years, but may remain in use for more than 50 years. If a major hospital is designed very tightly around the operational policies of the day, or the opinion of a few individuals, who may leave at any time, then a significant investment may be at risk of early obsolescence. Flexible design refers to planning models that can not only adequately respond to contemporary operational policies but also have the inherent flexibility to adapt to a range of alternative, proven and forward-looking policies.

4. When prescribing antibiotics for a human bite, what should the medical professional remember?

- Ⓐ Not all patients should be given antibiotics given the nominal infection risk.
- Ⓑ The bacterium *Streptococcus* spp. is the most common in bite patients.
- Ⓒ *Eikenella corrodens* is not susceptible to several antibiotics often used for skin infections.

Human bites

Human bite injuries comprise clenched-fist injuries, sustained when a closed fist strikes the teeth of another person, and occlusive bites, resulting from direct closure of teeth on tissue. Clenched-fist injuries are more common than occlusive bites, particularly in men, with most human bites occurring on the hands. Human bites result in a greater infection and complication rate than animal bites. Cultures of human bites are typically polymicrobial. Mixed aerobic and anaerobic organisms are common, with the most common isolates including *Streptococcus* spp. and *Eikenella corrodens*, which occurs in up to one-third of isolates.

Some authors suggest that all patients with human bites should be commenced on antibiotic prophylaxis, given the high risk of infection. The choice of antibiotic therapy should cover *E. corrodens*, which is resistant to first-generation cephalosporins (such as cefalexin), flucloxacillin and clindamycin, antibiotics that are often used for skin and soft tissue infections.

5. The extract informs us that a model of care

- Ⓐ is only implemented at certain times and places.
- Ⓑ should include its own application and assessment.
- Ⓒ involves the development of a project management tool.

What is a MoC?

A “Model of Care” broadly defines the way health services are delivered. It outlines best practice care and services for a person, population group or patient cohort as they progress through the stages of a condition, injury or event. It aims to ensure people get the right care, at the right time, by the right team and in the right place.

When designing a new MoC, the aim is to bring about improvements in service delivery through effecting change. As such creating a MoC must be considered as a change management process. Development of a new MoC does not finish when the model is defined, it must also encompass implementation and evaluation of the model and the change management needed to make that happen. Developing a MoC is a project and as such should follow a project management methodology

6. What is the basic difference between delegation, referral, and handover?

- Ⓐ How many practitioners are involved in each part of the process.
- Ⓑ How much authority is attributed to each practitioner.
- Ⓒ How long each of the processes takes a practitioner to complete.

Delegation, referral and handover

Delegation involves one practitioner asking another person or member of staff to provide care on behalf of the delegating practitioner while that practitioner retains overall responsibility for the care of the patient or client.

Referral involves one practitioner sending a patient or client to obtain an opinion or treatment from another practitioner. Referral usually involves the transfer in part of responsibility for the care of the patient or client, usually for a defined time and a particular purpose, such as care that is outside the referring practitioner's expertise or scope of practice.

Handover is the process of transferring all responsibility to another practitioner

In this part of the test, there are two texts about different aspects of healthcare. For questions 7-22, choose the answer (A, B, C or D) which you think fits best according to the text.

Text 1: Personal devices and hearing loss

Most of us have experienced walking past someone and being able to hear every sound coming from their headphones. If you've ever wondered whether this could be damaging their hearing, the answer is yes. In the past, noise-induced hearing loss typically affected industrial workers, due to prolonged exposure to excessive levels of noise with limited or non-existent protective equipment. There are now strict limits on occupational noise exposure and many medico-legal claims have been filed as a result of regulation. The ubiquitous use of personal music players has, however, radically increased our recreational noise exposure, and research suggests there may be some cause for concern.

The problem is not just limited to children and teenagers either; adults listen to loud music too. According to the World Health Organization, hearing loss is already one of the leading causes of disability in adults globally, and noise-induced hearing loss is its second-largest cause. In Australia, hearing loss is a big public health issue, affecting one in six people and costing taxpayers over A\$12 billion annually for diagnosis, treatment, and rehabilitation.

When sounds enter our ear, they set in motion tiny frequency-specific hair cells within the cochlea, our hearing organ, which initiate the neural impulses which are perceived by us as sounds. Exposure to high levels of noise causes excessive wear and tear, leading to their damage or destruction. The process is usually gradual and progressive; as our cochlea struggles to pick up sounds from the damaged frequencies we begin to notice poorer hearing. Unfortunately, once the hair cells are gone, they don't grow back.

A number of US studies have shown the prevalence of noise-induced hearing loss in teenagers is increasing, and reports from Australia have suggested there's an increased prevalence of noise-induced hearing loss in young adults who use personal music players. This is a worrying trend considering the widespread usage

of these devices. Even a slight hearing loss can negatively affect a child's language development and academic achievement. This is of significant concern considering some studies have reported a 70% increased risk of hearing loss associated with use of personal music players in primary school-aged children.

Some smartphones and personal music players can reach up to 115 decibels, which is roughly equivalent to the sound of a chainsaw. Generally, 85 decibels and above is considered the level where noise exposure can cause permanent damage. Listening at this level for approximately eight hours is likely to result in permanent hearing loss. What's more, as the volume increases, the amount of time needed to cause permanent damage decreases. At 115 decibels, it can take less than a minute before permanent damage is done to your hearing.

In Australia a number of hearing education campaigns, such as Cheers for Ears, are teaching children and young adults about the damaging effects of excessive noise exposure from their personal music players with some encouraging results. Hopefully, this will lead to more responsible behaviour and prevent future cases of noise-induced hearing loss in young adults.

Currently, there are no maximum volume limits for the manufacturers of personal music players in Australia. This is in stark contrast to Europe, where action has been taken after it was estimated that 50 and 100 million Europeans were at risk of noise-induced hearing loss due to personal music players. Since 2009, the European Union has provided guidance to limit both the output and usage time of these devices. Considering the impact of hearing loss on individuals and its cost to society, it's unclear why Australia has not adopted similar guidelines. Some smartphones and music players allow you to set your own maximum volume limits. Limiting the output to 85 decibels is a great idea if you're a regular user and value preserving your hearing. Taking breaks to avoid continued noise exposure will also help reduce your risk of damaging your hearing.

Losing your hearing at any age will have a huge impact on your life, so you should do what you can to preserve it. Hearing loss has often been referred to as a "silent epidemic", but in this case it is definitely avoidable.

Text 1: Questions 7-14

7. The writer suggests that the risks from exposure to excessive industrial noise

- Ⓐ Have become better regulated over time.
- Ⓑ Have increased with the spread of new media devices.
- Ⓒ Were limited or non-existent in the past.
- Ⓓ Are something most people have experienced.

8. The word '**ubiquitous**' in paragraph 1 suggests that use of personal media players is

- Ⓐ Getting out of control.
- Ⓑ Radically increasing.
- Ⓒ Extremely common.
- Ⓓ A serious health risk.

9. In the second paragraph, the writer aims to emphasise the

- Ⓐ Impacts of hearing loss on young people.
- Ⓑ Significant global effect of noise related hearing loss.
- Ⓒ WHO's statistical information on hearing loss.
- Ⓓ Huge cost of hearing loss treatment in Australia.

10. What does the word **'their'** in paragraph 3 refer to?

- Ⓐ Smart phones and music players.
- Ⓑ People with hearing loss.
- Ⓒ Neural impulses entering our ear.
- Ⓓ Tiny hair cells in the ear.

11. What does the research mentioned in paragraph four show?

- Ⓐ A higher prevalence of personal music devices in primary schools.
- Ⓑ The negative impact of device related hearing loss on academic and linguistic skills.
- Ⓒ An increasing number of teens and young adults suffering noise related hearing loss.
- Ⓓ The widespread trend for increased use of personal music devices.

12. In paragraph 5, the writer suggests that

- Ⓐ Chainsaws and smartphones are negatively impacting the public's hearing.
- Ⓑ Listening to music on a Smartphone will damage your hearing.
- Ⓒ Smartphones are designed to play music at dangerously high volumes.
- Ⓓ More rules should be in place to control how loud smartphones can go.

13. Why does the writer mention the Australian education programs in paragraph 6?

- Ⓐ To encourage schools to adopt the Cheers for Ears program.
- Ⓑ To suggest that education could lead to safer behaviour in young people
- Ⓒ To criticise governments for not educating youths on the danger of excessive noise.
- Ⓓ To highlight a successful solution to the issue of hearing loss in young people.

14. What is the writer's attitude to the lack of manufacturing guidelines for music devices in Australia?

- Ⓐ There is no clear reason why Australia has not created guidelines.
- Ⓑ The implementation of guidelines in Australia is unnecessary.
- Ⓒ Guidelines probably won't be created in Australia.
- Ⓓ It will be difficult to create guidelines in Australia.

Text 2: What is herd immunity?

A recent outbreak of chickenpox is a reminder that even in countries where immunisation rates are high; children and adults are still at risk of vaccine-preventable diseases. Outbreaks occur from time to time for two main reasons. The first is that vaccines don't always provide complete protection against disease and, over time, vaccine protection tends to diminish. The second is that not everyone in the population is vaccinated. This can be for medical reasons, by choice, or because of difficulty accessing medical services. When enough unprotected people come together, infections can spread rapidly. This is particularly the case in settings such as schools where large numbers of children spend long periods of time together.

When a high proportion of a community is immune it becomes hard for diseases to spread from person to person. This phenomenon is known as herd immunity. Herd immunity protects people indirectly by reducing their chances of coming into contact with an infection. By decreasing the number of people who are susceptible to infection, vaccination can starve an infectious disease outbreak in the same way that firebreaks can starve a bushfire: by reducing the fuel it needs to keep spreading. If the immune proportion is high enough, outbreaks can be prevented and a disease can even be eliminated from the local environment. Protection of "the herd" is achieved when immunity reaches a value known as the "critical vaccination threshold". This value varies from disease to disease and takes into account how contagious a disease is and how effective the vaccine against it is.

For a disease outbreak to "grow", each infected person needs to pass their disease on to more than one other person, in the same way that we think about population growth more generally. If individuals manage only to "reproduce" themselves once in the infectious process, a full-blown outbreak won't occur. For example, on average someone with influenza infects up to two of the people they come into contact with. If one of those individuals was already fully protected by vaccination, then only one of them could catch the flu. By immunising half of the population, we could **stop flu in its tracks**.

On the other hand, a person with chickenpox might infect five to ten people if everyone were susceptible. This effectively means that we need to vaccinate around nine out of every ten people (90% of the population) to prevent outbreaks from occurring. As mentioned earlier, vaccines vary in their ability to prevent infection completely, particularly with the passing of time. Many vaccines require several “booster” doses for this reason. When vaccine protection is not guaranteed, the number of people who need to be vaccinated to achieve herd immunity and prevent an outbreak is higher. Chickenpox vaccine is one such example: infections can occur in people who have been vaccinated. However, such cases are typically less severe than in unimmunized children, with fewer spots and a milder symptom course.

In Australia, overall vaccine coverage rates are high enough to control the spread of many infectious diseases. Coverage shows considerable geographic variation, though, with some communities recording vaccination levels of less than 85%. In these communities, the conditions necessary for herd immunity may not be met. That means localised outbreaks are possible among the unvaccinated and those for whom vaccination did not provide full protection. In the Netherlands, for example, high national measles vaccine uptake was not enough to prevent a very large measles outbreak (more than 2,600 cases) in orthodox Protestant communities opposed to vaccination.

Australia's National Immunisation Strategy specifically focuses on achieving high vaccine uptake within small geographic areas, rather than just focusing on a national average. Although uptake of chickenpox vaccine in Australia was lower than other infant vaccines, coverage is now comparable.

Media attention has emphasised those who choose not to vaccinate their children due to perceived risks associated with vaccination. However, while the number of registered conscientious objectors to vaccination has increased slightly over time, these account for only a small fraction of children. A recent study found only 16% of incompletely immunised children had a mother who disagreed with vaccination. Other factors associated with under vaccination included low levels of social contact, large family size and not using formal childcare.

Tailoring services to meet the needs of all parents requires a better understanding of how families use health services, and of the barriers that prevent them from immunising. To ensure herd immunity can help protect all children from preventable disease, it's vital to maintain community confidence in vaccination. It's equally important the other barriers that prevent children from being vaccinated are identified, understood and addressed.

Text 2: Questions 15-22

15. According to the writer what causes occasional outbreaks of preventable diseases?

- Ⓐ A high prevalence of disease.
- Ⓑ Limited access to vaccination.
- Ⓒ A low prevalence of vaccination.
- Ⓓ Attitudes towards vaccination

16. Why does the writer mention bushfires in paragraph 2?

- Ⓐ To emphasise the effectiveness of herd immunity.
- Ⓑ To describe a method for eliminating disease.
- Ⓒ To warn of the risks of vaccination.
- Ⓓ To highlight the severity of the flu.

17. The phrase "**stop flu in its tracks**" in paragraph 3 refers to the

- Ⓐ Prevention of flu spreading.
- Ⓑ Eradication of the flu virus.
- Ⓒ Minimisation of flu victims.
- Ⓓ Reduction in severity of flu symptoms.

18. Information in paragraph 4 implies that

- Ⓐ The chickenpox vaccine is highly unreliable.
- Ⓑ Chickenpox is more contagious than the flu.
- Ⓒ Booster vaccines should be given in schools.
- Ⓓ Outbreaks of chickenpox are on the rise.

19. In paragraph 5, the writer emphasises the importance of

- Ⓐ How geographical variation contributes to outbreaks.
- Ⓑ Differences in global vaccination guidelines.
- Ⓒ The influence of religious beliefs on vaccination.
- Ⓓ Enforcing high vaccine coverage rates.

20. Why does the writer mention Australia's National Immunisation Strategy?

- Ⓐ To serve as a counter argument
- Ⓑ To engage Australian readers.
- Ⓒ To reinforce a previous point.
- Ⓓ To introduce a new topic.

21. The research quoted in paragraph 7 reinforces that

- Ⓐ The media presents vaccination negatively.
- Ⓑ Many factors contribute to under vaccination.
- Ⓒ Parental objections account for most unvaccinated children.
- Ⓓ The number of conscientious objectors has increased over time.

22. In the final paragraph, the writer focuses on

- Ⓐ The importance of widespread faith in vaccination.
- Ⓑ The difficulty of tailoring health services to all parents.
- Ⓒ The identification of barriers to overcoming under vaccination.
- Ⓓ The different kinds of preventable disease that need to be overcome.

READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT remove OET material from the test room.

TEXT A

Iron deficiency and iron deficiency anaemia are common. The serum ferritin level is the most useful indicator of iron deficiency, but interpretation can be complex. Identifying the cause of iron deficiency is crucial. Oral iron supplements are effective first-line treatment. Intravenous iron infusions, if required, are safe, effective and practical.

Key Points

- Measurement of the serum ferritin level is the most useful diagnostic assay for detecting iron deficiency, but interpretation may be difficult in patients with comorbidities.
- Identifying the cause of iron deficiency is crucial; referral to a gastroenterologist is often required.
- Faecal occult blood testing is not recommended in the evaluation of iron deficiency; a negative result does not impact on the diagnostic evaluation.
- Oral iron is an effective first-line treatment, and simple strategies can facilitate patient tolerance.
- For patients who cannot tolerate oral therapy or require more rapid correction of iron deficiency, intravenous iron infusions are safe, effective and practical, given the short infusion times of available formulations.
- Intramuscular iron is no longer recommended for patients of any age.

TEXT B

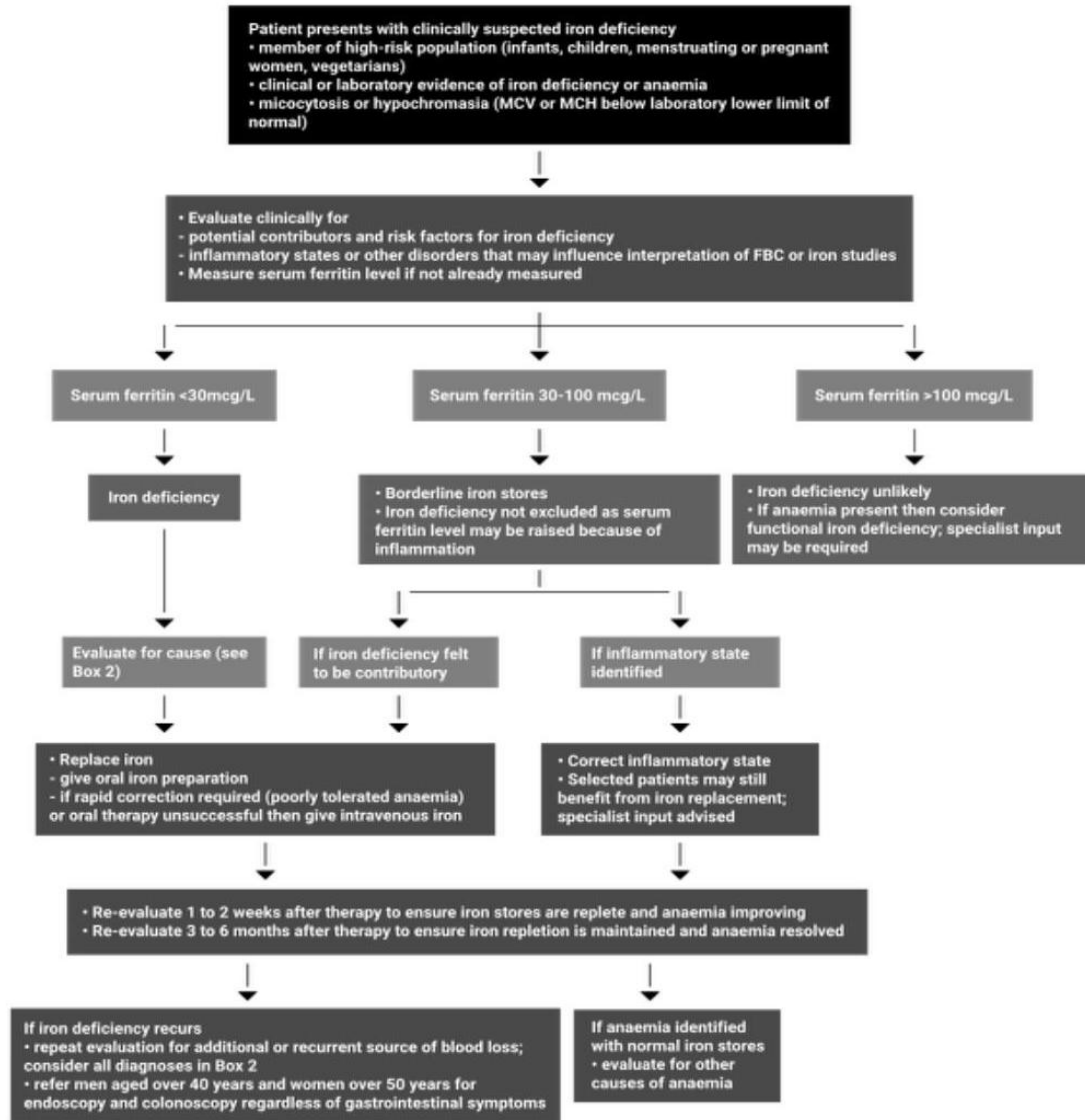
Treatment of infants and children

Although iron deficiency in children cannot be corrected solely by dietary change, dietary advice should be given to parents and carers. Cows' milk is low in iron compared with breast milk and infant formula, and enteropathy caused by hypersensitivity to cows' milk protein can lead to occult gastrointestinal blood loss. Excess cows' milk intake (in lieu of iron-rich solid foods) is the most common cause of iron deficiency in young children. Other risk factors for dietary iron deficiency include late introduction of or insufficient iron-rich foods, prolonged exclusive breastfeeding and early introduction of cows' milk.

Adult doses of iron can be toxic to children, and paediatric-specific protocols on iron supplementation should be followed. The usual paediatric oral iron dosage is 3 to 6mg/kg elemental iron daily. If oral iron is ineffective or not tolerated then consider other causes of anaemia, referral to a specialist paediatrician and use of IV iron.

TEXT C

AN ALGORITHM FOR THE IDENTIFICATION AND MANAGEMENT OF ADULTS WITH IRON DEFICIENCY



TEXT D

INTRAVENOUS PREPARATIONS FOR IRON REPLACEMENT				
Form of Iron	Presentation	Maximum dose per administration	Dosing frequency	Rate of administration
Ferric carboxymaltose	500 mg/10 mL vial or 100 mg/2 mL vial	1000 mg (or 20 mg/kg)	Maximum dose once per week, or 200 mg three times per week	IV injection or infusion 100-200mg: 3 minutes 200-500 mg: 6 minutes 500-1000 mg: 15 minutes
Iron polymaltose	100 mg/2 mL ampoule	2500 mg	Not applicable as entire dose can be delivered in single administration	IV infusion: first 50 mL infused slowly (20 to 40 mL/h); if tolerated then rate can be increased to 120 mL/h*
Iron sucrose	100 mg/5 mL ampoule	100 mg	Maximum three times per week	IV infusion 100 mg over 15 minutes

*Iron polymaltose can also be administered by the intramuscular route. Different maximum doses and dosing frequencies apply.

READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** or the **Text Booklet** until you are told to do so.

Write your answers on the spaces provided on this **Question Paper**.

You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this Question Paper and the Text Booklet.

DO NOT remove **OET** material from the test room.

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate Text **Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

Iron deficiency: Questions

Questions 1-7

For each of the questions, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

- | | |
|---|-------|
| 1. considerations when treating children with iron deficiency? | ----- |
| 2. essential steps for identifying iron deficiency? | ----- |
| 3. evaluating iron deficiency by testing for blood in stool? | ----- |
| 4. risk factors associated with dietary iron deficiency? | ----- |
| 5. different types of iron solutions? | ----- |
| 6. a treatment for iron deficiency that is no longer supported? | ----- |
| 7. appropriate dosage when administering IV iron infusions? | ----- |

Questions 8-14

Answer each of the questions, **8-14**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8. What level of serum ferritin leads to a diagnosis of iron deficiency?

9. What is the most likely cause of iron deficiency in children?

10. Which form of iron can also be injected into the muscle?

11. What should a clinician do if iron stores are normal and anaemia is still present?

12. How long after iron replacement therapy should a patient be re-tested?

13. Which form of iron is presented in a vial?

14. What is the first type of treatment iron deficient patients are typically given?

Questions 15-20

Complete each of the sentences, **15-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15. In comparison to breast milk and infant formula, cows' milk is -----
16. Special procedures should be used because ----- may be
poisonous for children.
17. Men over 40 and women over 50 with a recurring iron deficiency should have an -

18. Iron sucrose can be given to a patient no more than -----
19. Although serum ferritin level is a good indication of deficiency, interpreting the
results is sometimes difficult -----
20. IV iron infusions are a safe alternative when patients are unable to -----

READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this Question Paper until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this Question Paper.

DO NOT remove **OET** material from the test room.

HOW TO ANSWER THE QUESTIONS

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:



In this part of the test, there are six short extracts relating to the work of health professionals. For questions 1-6, choose the answer (**A, B or C**) which you think fits best according to the text.

1. The code of conduct applies to

- Ⓐ doctors friending patients on Facebook.
- Ⓑ privacy settings when using social media.
- Ⓒ electronic and face to face communication.

Professional obligations

The Code of conduct contains guidance about the required standards of professional behaviour, which apply to registered health practitioners whether they are interacting in person or online. The Code of conduct also articulates standards of professional conduct in relation to privacy and confidentiality of patient information, including when using social media. For example, posting unauthorised photographs of patients in any medium is a breach of the patient's privacy and confidentiality, including on a personal Facebook site or group, even if the privacy settings are set at the highest setting (such as for a closed, 'invisible' group).

2. Why does dysphagia often require complex management?

- Ⓐ Because it negatively influences the cardiac system.
- Ⓑ Because it is difficult contrast complex and non-complex cases.
- Ⓒ Because it seldom occurs without other symptoms.

6.1 General principles

Dysphagia management may be complex and is often multi-factorial in nature. The speech pathologist's understanding of human physiology is critical. The swallowing system works with the respiratory system. The respiratory system is in turn influenced by the cardiac system, and the cardiac system is affected by the renal system. Due to the physiological complexities of the human body, few clients present with dysphagia in isolation.

6.2 Complex vs. non-complex cases

Broadly the differentiation between complex and non-complex cases relates to an appreciation of client safety and reduction in risk of harm. All clinicians, including new graduates, should have sufficient skills to appropriately assess and manage non-complex cases. Where a complex client presents, the skills of an advanced clinician are required. Supervision and mentoring should be sought for newly graduated clinicians or those with insufficient experience to manage complex cases.

3. The main point of the extract is

- Ⓐ how to find documents about infection control in Australia.
- Ⓑ that dental practices must have a guide for infection control.
- Ⓒ that dental infection control protocols must be updated.

1. Documentation

1.1 Every place where dental care is provided must have the following documents in either hard copy or electronic form (the latter includes guaranteed Internet access). Every working dental practitioner and all staff must have access to:

- a) A manual setting out the infection control protocols and procedures used in that practice, which is based on the documents listed at sections 1.1(b), (c) and (d) of these guidelines and with reference to the concepts in current practice noted in the documents listed under References in these guidelines.
- b) The current Australian Dental Association Guidelines for Infection Control (available at: <http://www.ada.org.au>)

4. Negative effects from prescription drugs are often

- Ⓐ avoidable in young people.
- Ⓑ unpredictable in the elderly.
- Ⓒ caused by miscommunication.

Reasons for Drug-Related Problems: Manual for Geriatrics Specialists

Adverse drug effects can occur in any patient, but certain characteristics of the elderly make them more susceptible. For example, the elderly often take many drugs (polypharmacy) and have age-related changes in pharmacodynamics and pharmacokinetics; both increase the risk of adverse effects.

At any age, adverse drug effects may occur when drugs are prescribed and taken appropriately; e.g. , new-onset allergic reactions are not predictable or preventable. However, adverse effects are thought to be preventable in almost 90% of cases in the elderly (compared with only 24% in younger patients). Certain drug classes are commonly involved: antipsychotics, antidepressants, and sedative-hypnotics.

In the elderly, a number of common reasons for adverse drug effects, ineffectiveness, or both are preventable. Many of these reasons involve inadequate communication with patients or between health care practitioners (particularly during health care transitions).

5. The guideline tries to use terminology that

- Ⓐ presents value-free information about different social groups.
- Ⓑ distinguishes disadvantaged groups from the traditional majority.
- Ⓒ clarifies the proportion of each race, gender and culture.

Terminology

Terminology in this guideline is a difficult issue since the choice of terminology used to distinguish groups of persons can be personal and contentious, especially when the groups represent differences in race, gender, sexual orientation, culture or other characteristics. Throughout the development of this guideline the panel endeavoured to maintain neutral and non-judgmental terminology wherever possible. Terms such as “minority”, “visible minority”, “non-visible minority” and “language minority” are used in some areas; when doing so the panel refers solely to their proportionate numbers within the larger population and infers no value on the term to imply less importance or less power. In some of the recommendations the term “under-represented groups” is used, again, to refer solely to the disproportionate representation of some citizens in those settings in comparison to the traditional majority.

6. What is the purpose of this extract?

- Ⓐ To illustrate situations where patients may find it difficult to give negative feedback.
- Ⓑ To argue that hospital brochures should be provided in many languages.
- Ⓒ To provide guidance to people who are victims of discrimination.

Special needs

Special measures may be needed to ensure everyone in your client base is aware of your consumer feedback policy and is comfortable with raising their concerns. For example, should you provide brochures in a language other than English?

Some people are less likely to complain for cultural reasons. For example, some Aboriginal people may be culturally less inclined to complain, particularly to non-Aboriginal people. People with certain conditions such as hepatitis C or a mental illness, may have concerns about discrimination that will make them less likely to speak up if they are not satisfied or if something is wrong.

In this part of the test, there are two texts about different aspects of healthcare. For questions 7-22, choose the answer (**A, B, C or D**) which you think fits best according to the text.

Text 1: Difficult-to-treat depression

Depression remains a leading cause of distress and disability worldwide. In one country's survey of health and wellbeing in 2007, 7.2% of people surveyed had experienced a mood (affective) disorder in the previous 12 months. Those affected reported a mean of 11.7 disability days when they were "completely unable to carry out or had to cut down on their usual activities owing to their health" in the previous 4 weeks. There was also evidence of substantial under-treatment: amazingly only 35% of people with a mental health problem had a mental health consultation during the previous 12 months. Three-quarters of those seeking help saw a general practitioner (GP). In the 2015–16 follow-up survey, not much had changed. Again, there was evidence of substantial unmet need, and again GPs were the health professionals most likely to be providing care.

While GPs have many skills in the assessment and treatment of depression, they are often faced with people with depression who simply do not get better, despite the use of proven psychological or pharmacological therapies. GPs are well placed in one regard, as they often have a longitudinal knowledge of the patient, understand his or her circumstances, stressors and supports, and can marshal this knowledge into a coherent and comprehensive management plan. Of course, GPs should not soldier on alone if they feel the patient is not getting better.

In trying to understand what happens when GPs feel "stuck" while treating someone with depression, a qualitative study was undertaken that aimed to gauge the response of GPs to the term "difficult-to-treat depression". It was found that, while there was confusion around the exact meaning of the term, GPs could relate to it as broadly encompassing a range of individuals and presentations. More specific terms such as "treatment-resistant depression" are generally reserved for a subgroup of people with difficult-to-treat depression that has failed to respond to treatment, with particular management implications.

One scenario in which depression can be difficult to treat is in the context of physical illness. Depression is often expressed via physical symptoms, however it is also true is that people with chronic physical ailments are at high risk of depression. Functional pain syndromes where the origin and cause of the pain are unclear, are particularly tricky, as complaints of pain require the clinician to accept them as “legitimate”, even if there is no obvious physical cause. The use of analgesics can create its own problems, including dependence. Patients with comorbid chronic pain and depression require careful and sensitive management and a long-term commitment from the GP to ensure consistency of care and support.

It is often difficult to tackle the topic of depression co-occurring with borderline personality disorder (BPD). People with BPD have, as part of the core disorder, a perturbation of affect associated with marked variability of mood. This can be very difficult for the patient to deal with and can feed self-injurious and other harmful behaviour. Use of mentalisation-based techniques is gaining support, and psychological treatments such as dialectical behaviour therapy **form the cornerstone** of care. Use of medications tends to be secondary, and prescription needs to be judicious and carefully targeted at particular symptoms. GPs can play a very important role in helping people with BPD, but should not “go it alone”, instead ensuring sufficient support for themselves as well as the patient.

Another particularly problematic and well-known form of depression is that which occurs in the context of bipolar disorder. Firm data on how best to manage bipolar depression is surprisingly lacking. It is clear that treatments such as unopposed antidepressants can make matters a lot worse, with the potential for induction of mania and mood cycle acceleration. However, certain medications (notably, some mood stabilisers and atypical antipsychotics) can alleviate much of the suffering associated with bipolar depression. Specialist psychiatric input is often required to achieve the best pharmacological approach. For people with bipolar disorder, psychological techniques and long-term planning can help prevent relapse. Family education and support is also an important consideration.

Text 1: Questions 7-14

7. In the first paragraph, what point does the writer make about the treatment of depression?

- Ⓐ 75% of depression sufferers visit their GP for treatment.
- Ⓑ Have GPs struggle to meet the needs of patients with depression
- Ⓒ Treatment for depression takes an average of 11.7 days a month.
- Ⓓ Most people with depression symptoms never receive help.

8. In the second paragraph, the writer suggests that GPs

- Ⓐ are in a good position to conduct long term studies on their patients.
- Ⓑ lack training in the treatment and assessment of depression.
- Ⓒ should seek help when treatment plans are ineffective.
- Ⓓ sometimes struggle to create coherent management plans.

9. What do the results of the study described in the third paragraph suggest?

- Ⓐ GPs prefer the term “treatment resistant depression” to “difficult-to-treat depression”.
- Ⓑ Patients with “difficult-to-treat depression” sometimes get “stuck” in treatment.
- Ⓒ The term “difficult-to-treat depression” lacks a precise definition.
- Ⓓ There is an identifiable sub-group of patients with “difficult-to-treat depression”.

10. paragraph 4 suggest that

- Ⓐ prescribing analgesics is unadvisable when treating patients with depression.
- Ⓑ the co-occurrence of depression with chronic conditions makes it harder to treat.
- Ⓒ patients with depression may have undiagnosed chronic physical ailments.
- Ⓓ doctors should be more careful when accepting pain complaints as legitimate.

11. According to paragraph 5, people with BPD have

- Ⓐ depression occurring as a result of the disorder.
- Ⓑ noticeable mood changes which are central to their disorder.
- Ⓒ a tendency to have accidents and injure themselves.
- Ⓓ problems tackling the topic of their depression.

12. In paragraph 5, what does the phrase **'form the cornerstone'** mean regarding BPD treatment?

- Ⓐ Psychological therapies are generally the basis of treatment.
- Ⓑ There is more evidence for using mentalisation than dialectical behaviour therapy.
- Ⓒ Dialectical behaviour therapy is the optimum treatment for depression.
- Ⓓ In some unusual cases prescribing medication is the preferred therapy.

13. In paragraph 6, what does the writer suggest about research into bipolar depression management?

- Ⓐ There is enough data to establish the best way to manage bipolar depression.
- Ⓑ Research hasn't provided the evidence for an ideal management plan yet.
- Ⓒ A lack of patients with the condition makes it difficult to collect data on its management.
- Ⓓ Too few studies have investigated the most effective ways to manage this condition.

14. In paragraph 6, what does the writer suggest about the use of medications when treating bipolar depression?

- Ⓐ There is evidence for the positive and negative results of different medications.
- Ⓑ Medications typically make matters worse rather than better.
- Ⓒ Medication can help prevent long term relapse when combined with family education.
- Ⓓ Specialist psychiatrists should prescribe medication for bipolar disorder rather than GPs.

Text 2: Are the best hospitals managed by doctors?

Doctors were once viewed as ill-prepared for leadership roles because their selection and training led them to become “heroic lone healers. ” However, the emphasis on patient-centered care and efficiency in the delivery of clinical outcomes means that physicians are now being prepared for leadership. The Mayo Clinic is America’s best hospital, according to the 2016 US News and World Report (USNWR) ranking. Cleveland Clinic comes in second. The CEOs of both — John Noseworthy and Delos “Toby” Cosgrove — are highly skilled physicians. In fact, both institutions have been physician-led since their inception around a century ago. Might there be a general message here?

A study published in 2011 examined CEOs in the top-100 hospitals in USNWR in three key medical specialties: cancer, digestive disorders, and cardiovascular care. A simple question was asked: are hospitals ranked more highly when they are led by medically trained doctors or non-MD professional managers? The analysis showed that hospital quality scores are approximately 25% higher in physician-run hospitals than in manager-run hospitals. Of course, this does not prove that doctors make better leaders, though the results are surely consistent with that claim.

Other studies find a similar correlation. Research by Bloom, Sadun, and Van Reenen revealed how important good management practices are to hospital performance. However, they also found that it is the proportion of managers with a clinical degree that had the largest positive effect; in other words, the separation of clinical and managerial knowledge inside hospitals was associated with more negative management outcomes. Finally, support for the idea that physician-leaders are advantaged in healthcare is consistent with observations from many other sectors. Domain experts-“expert leaders” (like physicians in hospitals) — have been linked with better organizational performance in settings as diverse as universities, where scholar-leaders enhance the research output of their organizations, to basketball teams, where former All-Star players turned coaches are **disproportionately linked** to NBA success.

What are the attributes of physician-leaders that might account for this association with enhanced organizational performance? When asked this question, Dr. Toby

Cosgrove, CEO of Cleveland Clinic, responded without hesitation, “credibility ... peer-to-peer credibility. ” In other words, when an outstanding physician heads a major hospital, it signals that they have **“walked the walk”**. The Mayo website notes that it is physician-led because, “This helps ensure a continued focus on our primary value, the needs of the patient come first. ” Having spent their careers looking through a patient-focused lens, physicians moving into executive positions might be expected to bring a patient-focused strategy.

In a recent study that matched random samples of U. S. and UK employees with employers, we found that having a boss who is an expert in the core business is associated with high levels of employee job satisfaction and low intentions of quitting. Similarly, physician-leaders may know how to raise the job satisfaction of other clinicians, thereby contributing to enhanced organizational performance. If a manager understands, through their own experience, what is needed to complete a job to the highest standard, then they may be more likely to create the right work environment, set appropriate goals and accurately evaluate others’ contributions.

Finally, we might expect a highly talented physician to know what “good” looks like when hiring other physicians. Cosgrove suggests that physician-leaders are also more likely to tolerate innovative ideas like the first coronary artery bypass, performed by René Favaloro at the Cleveland Clinic in the late ‘60s. Cosgrove believes that the Cleveland Clinic unlocks talent by giving safe space to people with extraordinary ideas and importantly, that leadership tolerates appropriate failure, which is a natural part of scientific endeavour and progress.

The Cleveland Clinic has also been training physicians to lead for many years. For example, a cohort-based annual course, “Leading in Health Care, ” began in the early 1990s and has invited nominated, high-potential physicians (and more recently nurses and administrators) to engage in 10 days of offsite training in leadership competencies which fall outside the domain of traditional medical training. Core to the curriculum is emotional intelligence (with 360-degree feedback and executive coaching), teambuilding, conflict resolution, and situational leadership. The course culminates in a team-based innovation project presented to hospital leadership. 61% of the proposed innovation projects have had a positive institutional impact.

Moreover, in ten years of follow-up after the initial course, 48% of the physician participants have been promoted to leadership positions at Cleveland Clinic.

Text 2: Questions 15-22

15. In paragraph 1, why does the writer mention the Mayo and Cleveland Clinics?

- Ⓐ To highlight that they are the two highest ranked hospitals on the USNWR
- Ⓑ To introduce research into hospital management based in these clinics
- Ⓒ To provide examples to support the idea that doctors make good leaders
- Ⓓ To reinforce the idea that doctors should become hospital CEOs

16. What is the writer's opinion about the findings of the study mentioned in paragraph 2?

- Ⓐ They show quite clearly that doctors make better hospital managers.
- Ⓑ They show a loose connection between doctor-leaders and better management.
- Ⓒ They confirm that the top-100 hospitals on the USNWR ought to be physician-run.
- Ⓓ They are inconclusive because the data is insufficient.

17. Why does the writer mention the research study in paragraph 3?

- Ⓐ To contrast the findings with the study mentioned in paragraph 2
- Ⓑ To provide the opposite point of view to his own position
- Ⓒ To support his main argument with further evidence
- Ⓓ To show that other researchers support him

18. In paragraph 3, the phrase 'disproportionately linked' suggests

- Ⓐ all-star coaches have a superior understanding of the game.
- Ⓑ former star players become comparatively better coaches.
- Ⓒ teams coached by former all-stars consistently outperform other teams.
- Ⓓ to be a successful basketball coach you need to have played at a high level.

19. In the fourth paragraph, what does the phrase "walked the walk," imply about physician- leaders?

- Ⓐ They have earned credibility through experience.
- Ⓑ They have ascended the ranks of their workplace.
- Ⓒ They appropriately incentivise employees.
- Ⓓ They share the same concerns as other doctors.

20. In paragraph 6, the writer suggests that leaders promote employee satisfaction because

- Ⓐ they are often cooperative.
- Ⓑ they tend to give employees positive evaluations.
- Ⓒ they encourage their employees not to leave their jobs.
- Ⓓ they understand their employees' jobs deeply.

21. In the seventh paragraph, why is the first coronary artery bypass operation mentioned?

- Ⓐ To demonstrate the achievements of the Cleveland clinic
- Ⓑ To present René Favaloro as an exemplar of a 'good' doctor
- Ⓒ To provide an example of an encouraging medical innovation
- Ⓓ To show how failure naturally contributes to scientific progress

22. In paragraph 8, what was the outcome of the course "Leading in Health Care"?

- Ⓐ The Cleveland Clinic promoted almost half of the participants.
- Ⓑ 61% of innovation projects lead to participants being promoted.
- Ⓒ Some participants took up leadership roles outside the medical domain.
- Ⓓ A culmination of more team-based innovations.

READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT remove OET material from the test room.

TEXT A

Terminology Cognitive difficulties

Cognitive changes are normal for almost all people as they age, and assessment should focus on differentiating the normal changes of ageing from abnormal cognitive functioning. While concerns about memory are common in older patients, when patients complain of memory problems, they could be referring to difficulties in a number of possible cognitive domains. Although learning and memory is often the most salient of these domains, the problems could also be in:

- attention (ability to sustain or shift focus),
- language (naming, producing words, comprehension, grammar or syntax),
- perceptual and motor skills (construction, visual perception),
- executive function (decision making, mental flexibility), or
- social cognition. It is thus often more appropriate to refer to cognitive rather than memory complaints or deficits.

TEXT B

Pharmacological treatments

There are currently no evidence-based recommendations on medications to treat mild cognitive impairment (MCI). If dementia is suspected then specialist referral is recommended for confirmation of the diagnosis. If Alzheimer's disease is confirmed then pharmacological treatment can be considered (e.g. acetylcholinesterase inhibitors such as donepezil, galantamine or rivastigmine).

A psychiatric or psychogeriatric referral should be considered for:

- patients who do not respond to first- or second-line treatment
- patients with atypical mental health presentations
- patients with significant psychiatric histories, including complicated depression and/or anxiety or comorbid severe mental illnesses such as schizophrenia and bipolar affective disorder.

Follow up

If the diagnosis remains unclear after a detailed assessment then provide general advice and watchfully wait. All patients should have a cognitive review with a screening instrument every 12 months or sooner if deterioration is detected by the patient or their family. Risk factors for progression of MCI to dementia include older age, less education, stroke, diabetes and hypertension. Patients who are younger more educated with higher baseline cognitive function and no amnesia symptoms are more likely to revert from MCI to normal cognition. Even after 10 years, between 40 and 70% of patients with MCI may not have developed dementia.

TEXT C		
Domain	Examples of skills	Warning signs and questions
Learning and memory	Short-term recall Semantic and autobiographical Long- term memory Implicit learning	Have you noticed that you have been talking to someone and soon after forget the conversation? Have you had difficulty remembering the names of people you have just met? Have you had trouble keeping track of dates and appointments? Have you had any difficulty remembering events from your past? Have you had difficulty doing activities previously thought as automatic, like driving or typing? [To informant] Has he or she been repeating him or herself lately?
Language	Object naming Word finding Receptive language	Have you noticed any word-finding difficulties? [To informant] Has he or she had more difficulty understanding you lately?
Executive function	Planning , decision making, working, memory, flexibility	Have you had more difficulty managing your finances lately? [To informant] Have you noticed difficulties with his or her capacity to plan activities or make decisions?
Perceptual motor function	Visual perception Perceptual motor co-ordination	Have you had trouble using day-to-day objects, such as phone or cutlery? Have there been new driving difficulties such as staying in the lane?
Complex attention	Sustained attention Selective attention	Are you having difficulty following what's going on around you? [To informant] Have you noticed that he or she is more easily distracted?
Social cognition	Recognition of emotions Appropriateness of behaviour to social norms	[To informant] Has he or she been behaving inappropriately in social situations? Is he or she able to recognise social cues? Is she or she able to motivate him or herself?

TEXT D
Dementia, now also referred to as ‘major neurocognitive disorder’ in the Diagnostic and Statistical Manual of Mental Disorders (DSM-5), is defined by the presence of substantial cognitive decline from a previous level of functioning to the degree that the individual’s ability to live independently is compromised owing to the cognitive deficits. Dementia is a syndrome with many possible causes, with Alzheimer’s disease being the most common in older people. It is generally of gradual onset with a chronic course, although there are exceptions. Dementia must be distinguished from delirium (acute confusional state), which by definition is of acute or recent onset and associated with loss of awareness of surroundings, a global disturbance in cognition, changes in perception and the sleep–wake cycle, and other features.

READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** or the **Text Booklet** until you are told to do so.

Write your answers on the spaces provided on this **Question Paper**.

You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this Question Paper and the Text Booklet.

DO NOT remove **OET** material from the test room.

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate Text **Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

Evaluating cognitive function : Questions

Questions 1-7

For each of the questions, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1. What to ask patients when evaluating cognitive functioning? -----
2. Possible choices for pharmaceutical treatments? -----
3. The best way to describe patient symptoms? -----
4. The defining features of dementia? -----
5. The proper focus of cognitive assessment? -----
6. Different types of mental processing? -----
7. What to do when a diagnosis is remains uncertain? -----

Questions 8-14

Answer each of the questions, **8-14**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8. What is the other name for dementia?

9. What is the most common cause of dementia in older people?

10. Which domain of cognition is the skill of planning associated with?

11. What is the most appropriate way to confirm a diagnosis of dementia?

12. What is recommended for patients when standard treatments are unsuccessful?

13. What is often the most noticeable of the many cognitive domains?

14. How often should a patient be cognitively screened if they are not getting worse?

Questions 15-20

Complete each of the sentences, **15-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15. Dementia differs in important ways from -----
which, for example, has a sudden onset.

16. The DSM-5 defines dementia as substantial cognitive decline that
compromises the individual's-----

17. There are ----- medications
for MCI that are recommended based on available research.

18. Many symptoms described as problems with memory are probably better
described as ----- complaints.

19. Social cognition includes the ability to follow accepted social rules and the ----

20. To assess perceptual motor functioning doctors can ask if patients have had
difficulty using ----- objects like knives and forks.

READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this Question Paper until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this Question Paper.

DO NOT remove **OET** material from the test room.

HOW TO ANSWER THE QUESTIONS

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:



In this part of the test, there are six short extracts relating to the work of health professionals. For questions 1-6, choose the answer (**A, B or C**) which you think fits best according to the text.

1. The purpose of this memo is to

- Ⓐ provide staff information on appropriate methods.
- Ⓑ notify staff of a possible change in standard procedure.
- Ⓒ remind staff of the importance of following best practice.

Memo to: Department physicians and clinical staff

Subject: Aseptic technique

Aseptic technique protects patients during invasive clinical procedures by employing infection control measures that minimise, as far as practicably possible, the presence of pathogenic organisms. Good aseptic technique procedures help prevent and control healthcare associated infections and must be preserved. As you are aware, the aim of every procedure should be to maintain asepsis at all times by protecting the key parts and key sites from contact contamination by microorganisms. This can be achieved through correct hand hygiene, a non-touch technique, glove use and ensuring asepsis and sterility of equipment. While the principles of aseptic technique remain constant for all procedures, the level of practice will change depending upon a standard risk assessment

2. The guidelines require those administering thrombolysis to

- Ⓐ explore other options before proceeding.
- Ⓑ contact the coronary care unit prior to transfer.
- Ⓒ ensure support staffs are readily available.

6.2 Thrombolysis for STEMI patients

Primary percutaneous coronary intervention (PCI) is the treatment of choice for patients presenting with an acute ST Elevation Myocardial Infarction (STEMI). However, if it is not possible to transfer the patient to the cardiac catheter laboratory immediately, for whatever reason, then the need for thrombolysis to be given should be considered. The admitting team must ask the primary PCI operator if they are able to achieve the arrival in hospital to first balloon inflation target of 120 minutes. If not, then thrombolysis will be given on the advice of the primary PCI operator without delay. Support for this may be given by Coronary Care Unit (CCU) staff/Chest Pain Nurses depending on the patient's location. Transfer the patient with resuscitation equipment to CCU immediately after thrombolysis is administered.

3. The guidelines specify that those performing an MRI on patients with implants or foreign bodies

- Ⓐ should abandon the scan if unsure of the device.
- Ⓑ have the final say in whether to scan a patient.
- Ⓒ use a lower field strength for conditional items.

Guidelines for the management of implants and foreign bodies during MRI scans

Implantable devices or other foreign bodies may contraindicate MRI scanning and/or cause significant image artefacts. There is a growing number of medical devices and implants that are classified as 'MRI conditional', placing the responsibility for safety on the operator. It should be stressed that safety at defined field strength or for a specific MRI system is no guarantee of safety at higher (or lower) field strength, or a different MRI system at the same field strength. If there is any doubt as to the nature of a device then a scan should only proceed after a careful assessment of the potential risks and benefits of the scan with the device in situ. The MRI Safety Expert can assist with identifying and quantifying the risks, but the decision to scan is a clinical one.

4. The manual informs us that the AP14 syringe pump

- Ⓐ should be disconnected in times of power outage.
- Ⓑ facilitates easy cleaning by its smooth exterior.
- Ⓒ has a unique patient transportation feature.

Manual extract: Operation of AP14 Manual Syringe Pump

Pump Application

The AP 14 syringe pump is simple to operate, reliable and is of general application. It is suitable for various types of single-use syringes. BOLUS function enables quick and repeated delivery of bolus doses to the patient, with accurately established volume and within a specified infusion time. The pump can operate without connection to the mains, as it is automatically supplied by the internal battery in cases, e. g. of mains failure. It also enables to continue the infusion when the patient is being transported from one area of the hospital to another. Simple casing, without any parts protruding from the front panel, facilitates maintenance and disinfection.

5. The notice on indwelling urinary catheters provides information about

- Ⓐ the order for correct insertion.
- Ⓑ optimal positioning of the patient
- Ⓒ how best to avoid harming patients.

Indwelling urinary catheters

Urethral, prostate or bladder neck injury resulting in false tracts, strictures and bleeding are related to traumatic urethral insertion. Traumatic injury is less likely to occur with appropriate catheter selection, lubrication, correct patient positioning and insertion into a full bladder. Retention balloons should only be inflated inside the bladder, which is indicated by urine return with IUC inserted to the hilt. If there is any uncertainty regarding catheter placement, the balloon should not be inflated. If the patient experiences pain with inflation, deflate the balloon immediately and reassess IUC position as this may indicate the catheter is outside the bladder. IUCs should be used with caution in patients at risk of self-extraction, such as those with dementia or who are delirious. When available, ultrasonography is recommended to evaluate bladder volumes and guide SPC insertions.

6. This extract from a handbook says that patients with delirium experience

- Ⓐ a similar cognitive decline as with dementia.
- Ⓑ a loss of interest during conversations.
- Ⓒ influences that may trigger the disorder.

Delirium is an acute deterioration in cognition, often with altered arousal (drowsiness, stupor, or hyperactivity) and psychotic features (e.g. paranoia). The main cognitive deficit in delirium is 'inattention', e.g. the patient is distractible, cannot consistently follow commands, and loses the thread during a verbal exchange. Delirium and dementia commonly co-exist, however, with the latter there is a much slower deterioration in thinking, perceiving and understanding, and inattention is much less prominent. Because delirium is usually due to an interaction between multiple predisposing and precipitating factors, management should be aimed at not just finding and treating the assumed cause, but also optimising all aspects of care.

In this part of the test, there are two texts about different aspects of healthcare. For questions 7-22, choose the answer (**A, B, C or D**) which you think fits best according to the text.

Text 1: Shedding Light on Complex Regional Pain Syndrome (CRPS)

Eleven years ago, Debbie had a routine bunion operation that changed her life. Instead of finding relief, her pain grew worse, until it was excruciating and constant. “I became disabled and had to stop working. My foot is permanently in an air cast and I walk with a cane. Most of the time the pain is a 10 out of 10,” she says. Debbie’s surgeon sent her to a pain specialist, who recommended a psychiatrist. “I knew the pain wasn’t in my head,” she says, but the medical community didn’t believe her. It wasn’t until she met neurologist Anne Louise Oaklander that she finally received a diagnosis: Complex Regional Pain Syndrome, or CRPS.

CRPS is a chronic pain condition that develops following trauma to a limb, such as surgery or a fracture. As Debbie learned, “this is a very controversial condition that not a lot of doctors understand,” says Oaklander. “Historically, the field of medicine has been very sceptical of patients with CRPS. On top of their illness, patients have had to navigate a medical system that is suspicious of them and hasn’t had effective treatment to offer. It adds insult to injury.” But those who treat CRPS are hopeful the **tide is turning**. Recent attempts to better comprehend CRPS have produced consensus guidelines for which patient outcomes should be included in future research, as well as internationally agreed-upon diagnostic criteria. Investigators are also learning more about the causes of CRPS from laboratory studies.

CRPS starts off with a surprising amount of pain that doesn’t match the initial trauma. In the first few months, instead of the expected healing, patients describe an increase in pain levels. They often report that a cast on the affected limb feels excessively tight and the sensation that the limb might “explode,” says Candy McCabe, a CRPS clinician and researcher at the University of the West of England, Bristol, UK. The limb often swells, changes colour to red or purple, and is perceived by the patient as either very cold or very hot. Changes in hair and nail growth, and sweating are also common. Research from Oaklander’s lab has identified persistent problems with certain neurons in patients’ injured limbs. These nerve cells carry pain

messages, but also control the small blood vessels and sweat glands, explaining why patients have a constellation of symptoms in addition to chronic nerve pain.

Many patients report that within a few days or weeks the limb feels completely alien, and of a very different size and shape than it really is. Many also describe very negative feelings toward the limb and a strong desire to have it amputated. “In CRPS, the brain’s perception of the limb changes pretty quickly,” McCabe says. The good news is that, while in some cases CRPS becomes persistent, about 75% of people get better without intervention, by six months to a year. “Getting a CRPS diagnosis does not necessarily equate to a lifetime of disability,” she emphasises.

While the features mentioned above describe the “average” CRPS patient, not everyone experiences the disease in the same way. Beyond differences in the length and severity of the syndrome, different patients report different symptoms as the most prominent and bothersome. For some, movement problems cause the most difficulty, while for others, the pain they experience may **take centre stage**. “The presentation of CRPS is variable within a common picture, but unfortunately we don’t yet know why these different scenarios happen,” says McCabe.

As reflected in the original name for CRPS, Reflex Sympathetic Dystrophy, one of the earliest ideas about the biological underpinnings of the condition is the presence of dysfunction of the sympathetic nervous system, the network of neurons that governs the body’s automatic “fight or flight” response. Currently, researchers believe that such alterations are important in the initial generation and acute phase of CRPS. For example, studies suggest that in the tibial fracture model, sympathetic neurons release an immune system protein called interleukin-6 that stimulates inflammation and pain. Andreas Goebel, a clinician and pain researcher at the University of Liverpool, UK has identified a number of autoantibodies, which are immune system proteins directed against a person’s own tissues or organs, in the blood of people with chronic CRPS.

The first CRPS trial is underway, to evaluate the efficacy and safety of neridronate, a new bisphosphonate, which is a class of drugs already widely used to prevent and treat osteoporosis. This is a placebo-controlled clinical trial and has completed enrolment of 230 patients at 59 sites in the US and Europe. Debbie is one of the trial

participants, and has received several intravenous infusions. Neither she nor Oaklander are aware as yet if she received neridronate or a placebo. "If this trial finds neridronate to be safe and effective and receives approval to be marketed for CRPS, it will be historic", says Oaklander. "It's only when there's an approved drug that there's advertising, which increases public awareness, and spurs doctors to learn more," she adds. "I felt it was important to participate in this trial because it makes a statement to the world that CRPS is a real medical disease that deserves high quality trials. This could be a landmark trial."

Text 1: Questions 7-14

7. In the first paragraph, the writer uses Debbie's case to convey
- Ⓐ The dangers of having even minor surgery.
 - Ⓑ A lack of awareness of CRPS among the medical fraternity.
 - Ⓒ The psychological causes of pain experienced by CRPS sufferers.
 - Ⓓ That specialist attention is warranted in such instances
8. What is meant by the phrase the **tide is turning** in the second paragraph?
- Ⓐ Doctors now believe in the existence of CRPS.
 - Ⓑ Beneficial treatment is now more readily available.
 - Ⓒ Recent investigations are indicating a cure is in sight.
 - Ⓓ The medical community's understanding is beginning to shift.
9. Evidence mentioned in the third paragraph has revealed
- Ⓐ Possible reasons for the multitude of symptoms experienced.
 - Ⓑ Better post-operative care of limbs is needed.
 - Ⓒ Temperature patterns remain consistent throughout cases.
 - Ⓓ Further research is required into the possible causes of pain.

10. What do we learn about CRPS in the fourth paragraph?

- Ⓐ Patients respond very differently to available treatment.
- Ⓑ Professional diagnosis is necessary to see improvements.
- Ⓒ Profound psychological impacts are often reported.
- Ⓓ Amputation should only be performed when all else has failed.

11. In the fifth paragraph, what point is made about the symptoms of CRPS?

- Ⓐ The length and severity of CRPS are quite consistent.
- Ⓑ Pain is the dominant symptom for CRPS sufferers.
- Ⓒ CRPS presents itself in a diverse number of ways.
- Ⓓ The average CRPS patient is very well understood.

12. What point is made about the sympathetic nervous system in the sixth paragraph?

- Ⓐ It only affects CRPS in the very early stages.
- Ⓑ It causes CRPS following a fractured tibia.
- Ⓒ It has a critical role from the outset of CRPS.
- Ⓓ It has less influence on CRPS than initially believed.

13. Anne Louise Oaklander values the trials highly because?

- Ⓐ Of the inclusion of the recently created neridronate.
- Ⓑ They may help validate the authenticity of CRPS.
- Ⓒ She gets to be a part of ground-breaking research.
- Ⓓ It is the first time a cohort of this size has been used.

14. The final paragraph mentions that confirmation has yet to be received regarding

- Ⓐ Whether Debbie was given the neridronate infusion.
- Ⓑ The final number of participant enrolments for the trial.
- Ⓒ Having the backing of the entire medical community.
- Ⓓ The approval for public advertising campaigns.

Text 2: Antibiotic Resistance now a global threat to public health

In 1945, Alexander Fleming, the man who discovered the first antibiotic said in his Nobel Prize acceptance speech, "The time may come when penicillin can be bought by anyone in the shops. Then there is the danger that the ignorant may easily under dose themselves and by exposing their microbes to non-lethal quantities of the drug, making them resistant." A recent report from the Centres for Disease Control and Prevention (CDC) revealed that more than 2 million people in the US alone become ill every year as a result of antibiotic-resistant infections, and 23,000 die from such infections.

The World Health Organization (WHO) has recently published their first global report on the issue, looking at data from 114 countries. WHO focused on determining the rate of antibiotic resistance to seven bacteria responsible for many common infections, including pneumonia, diarrhoea, urinary tract infections, gonorrhoea and sepsis. Their findings were worrying. The report revealed that resistance to common bacteria has reached "alarming" levels in many parts of the world, with some areas already out of treatment options for common infections. For example, they found resistance to carbapenem antibiotics used to tackle *Klebsiella pneumoniae* - the bacteria responsible for hospital-acquired infections such as pneumonia and infections in newborns - has spread to all parts of the globe.

Dr Keiji Fukuda, WHO's assistant director-general for health security, said of the report's findings: "Effective antibiotics have been **one of the pillars** of recent generations, and unless we take significant actions to improve efforts to prevent infections and also change how we produce, prescribe and use antibiotics, the world will lose more and more of these global public health goods that allow us to live longer, healthier lives, and the implications will be devastating. We're heading for a post-antibiotic era effectively wiping out what is a marvel of modern medicine."

Bacteria have shown the ability to become resistant to an antibiotic with great speed. "It's true that they've saved millions of lives over the years, and there's also undoubtedly a growing worldwide need. But their use at any time in any setting puts biological pressure on bacteria that promotes the development of resistance. That's where the blame lies, and only the medical officer assumes this responsibility," says

Dr Steve Solomon, Director of the CDC's Office of Antimicrobial Resistance. "When antibiotics are needed to prevent or treat disease, they should always be used. But research has shown that as much as 50% of the time, antibiotics are prescribed when they're not needed or they're dispensed incorrectly, such as when a patient is given the wrong dose. Whether it's a lack of experience or knowledge, or just the easier option, I really can't say."

Dr Charles Penn, coordinator of antimicrobial resistance at WHO, takes a slightly different viewpoint from his peers. "One of many reasons why antibiotic use is so high is that there is a poor understanding of the differences between bacteria, viruses and other pathogens, and also of the value of antibiotics," he said. "Too many antibiotics are prescribed for viral infections such as colds, flu and diarrhoea. Unfortunately, these public misconceptions are often perpetuated by marketing and advertising campaigns through the use of generic terms such as 'germs' and 'bugs.' It's difficult to try and narrow down the blame to a single origin."

Dr Penn noted that reliance on antibiotics for modern medical benefits has contributed to drug resistance. "Surgery, cancer treatment, intensive care, transplant surgery, even simple wound management would all become much riskier, more difficult options if we could not use antibiotics to prevent infection, or treat infections if they occurred," he said. "Similarly, we now take it for granted that many infections are treatable with antibiotics, such as tonsillitis, gonorrhoea and bacterial pneumonia. But some of these are now becoming untreatable." Add to this the excessive and incorrect use of antibiotics in food-producing animals since resistant bacteria can be transmitted to humans through the food we eat, and you literally have a recipe for disaster.

Dr Penn goes on to say, "Although many warnings about resistance were issued, physicians, that is to say prescribers, became somewhat complacent about preserving the effectiveness of antibiotics - new drugs always seemed to be available. However, the pipeline for discovery of new antibiotics has diminished in the past 30 years and has now run dry." He noted, however, that health care providers have now started to become more vigilant in prescribing antibiotics. "Greater awareness of the urgency of the problem has given new impetus to careful

stewardship of existing antibiotics. Medical practitioners are now **heeding the warning** that the pioneer of the antibiotic gave all those years ago."

Text 2: Questions 15-22

15. The writer quotes Alexander Fleming in the first paragraph to

- Ⓐ Emphasise the impressive history of antibiotics.
- Ⓑ Reveal the ease at which people may purchase antibiotics.
- Ⓒ Compare current usage of medication to an earlier time.
- Ⓓ Show that his prediction of antibiotic resistance was accurate.

16. In the second paragraph, what does the writer find particularly worrisome?

- Ⓐ One particular antibiotic no longer provides resistance anywhere.
- Ⓑ New born are quickly becoming resistant to all antibiotics.
- Ⓒ Resistance is at an all-time low for the most common infections.
- Ⓓ Although treatment is available globally it is largely ineffective.

17. What is meant by one of the pillars in the third paragraph?

- Ⓐ An innovation that changed the healthcare industry.
- Ⓑ A permanent fixture in the field of medicine
- Ⓒ An essential component of the medical system
- Ⓓ A remedy that is among the greatest inventions

18.According to Dr Steve Solomon, what is ultimately responsible for antibiotic resistance?

- Ⓐ Their everyday use for common diseases.
- Ⓑ The prescriber's lack of experience.
- Ⓒ The increase in global demand.
- Ⓓ The medical professional's misuse.

19.In the fifth paragraph, Dr Charles Penn argues that when it comes to antibiotic resistance

- Ⓐ Increasing their cost would deter overuse.
- Ⓑ The general public should be held responsible.
- Ⓒ Mass media plays a crucial role in its demand.
- Ⓓ More understanding is needed to overcome it.

20.In paragraph 6, Dr Penn gives examples of our dependence on antibiotics to

- Ⓐ Stress that substitute medications are needed.
- Ⓑ Justify the need to change our habits.
- Ⓒ Show that it's too late to reverse the damage.
- Ⓓ Highlight our lack of appreciation for current treatments.

21. In the final paragraph, Dr Penn makes the point that medical practitioners

- Ⓐ Have depleted the supply of antibiotics through overuse.
- Ⓑ Were reluctant to take advice regarding antibiotics.
- Ⓒ Once believed there was an endless supply of antibiotics.
- Ⓓ Are yet to understand the damage caused by their actions.

22. In the final paragraph, the phrase heeding the warning refers to

- Ⓐ Prescribers being attentive to the problem.
- Ⓑ Doctors now issuing warnings to patients.
- Ⓒ The medical community regretting their carelessness.
- Ⓓ Practitioners looking ahead to a brighter future.

READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT remove OET material from the test room.

TEXT A

INVESTIGATING ACUTE TRANSFUSION REACTIONS

1. Immediately report all acute transfusion reactions with the exceptions of mild hypersensitivity and non-haemolytic febrile transfusion reactions, to the appropriate departments.
2. Record the following information on the patient's notes:
 - Type of transfusion reaction
 - Length of time after the start of the transfusion and when the reaction occurred
 - Volume, type and pack numbers of the blood components transfused
3. Take the samples and send them to the appropriate laboratory
 - Immediate post-transfusion blood samples from a vein in the opposite arm:
 - Group & Antibody Screen
 - Direct Antiglobulin Test
 - Blood unit and giving set should contain residues of the transfused donor blood
4. Take the following samples and send them to the Haematology/ Clinical Chemistry Laboratory for:
 - Full blood count
 - Urea
 - Coagulation screen
 - Electrolytes
 - Creatinine
 - Blood culture in an appropriate blood culture bottle
5. Complete a transfusion reaction report form.
6. Record the results of the investigations in the patient's records for future follow-up, if required.

TEXT B

immediate Reaction - Life Threatening Situations

- Maintain airway and give high concentration oxygen by mask
- STOP the transfusion. Replace the giving set and keep the IV line open
- Manage as anaphylaxis protocol and ensure help is coming: stridor, wheeze and hypotension require treatment with oxygen and adrenaline. Critical Care admission.
- Notify consultant haematologist and Hospital Transfusion Laboratory immediately.
- Send the blood unit with the giving set, freshly collected blood samples with appropriate request form to the Hospital Transfusion Laboratory for investigations.
- Check a fresh urine sample visually for signs of haemoglobinuria.
- Commence urine collection (24 hours) and record all intake and output. Maintain fluid balance.
- Assess for bleeding from puncture sites or wounds.
- Reassess: 1. treat bronchospasm and shock as per protocol. 2. Acute renal failure or hyperkalaemia may require urgent renal replacement therapy

TEXT C			
Relevant effects	Drugs and doses		Notes
	Name	Route & Dosage	
	Oxygen	60 -100 %	1 st line
Bronchodilator Vasopressor	Adrenaline	500 micrograms im repeated after 5 mins if no better, or worse	1 st line
Expand blood volume	0.9% - saline, Gelufusine	If patient hypotensive, 20 ml/kg over 5 minutes	1 st line
Reduce fever and inflammatory response	Paracetamol	Oral or rectal 10mg/kg	2 nd line avoid aspirin containing products if patient has low platelet count
Inhibits histamine mediated responses	Chlorphenamine (chlorpheniramine)	IV 0.1 mg/ kg	2 nd line
Inhibits immune mediated bronchospasm	Salbutamol Aminophylline	By 5ml nebuliser Use under expert guidance	2 nd line
Vasopressor Bronchodilator	Adrenaline 6mg in 100ml 5% dextrose (6%)	5-10 ml / hr	Use only under expert guidance

TEXT D			
Guidelines for recognition and management of acute transfusion reactions			
Category	Signs	Symptoms	Possible causes
Category 1 Mild	Localised cutaneous reactions: • Urticaria • Rash • Mild fever	Pruritus	Hypersensitivity Febrile non-hemolytic transfusion reactions: • Antibodies to white blood cells, platelets • Antibodies to proteins, including IgA
Category 2 Moderately severe	Flushing Urticaria Rash Fever Restlessness Tachypnoea Tachycardia	Anxiety Pruritus Palpitations Mild dyspnea Headache	Hypersensitivity (moderate- severe) Febrile non-haemolytic transfusion reactions: • Antibodies to white blood cells, platelets • Antibodies to proteins, including IgA Possible contamination with pyrogens and/or bacteria
Category 3 Life Threatening	Rigors Fever Restlessness Hypotension (fall of >20% in systolic BP) Tachypnoea+++ Tachycardia (rise of >20% in heart rate) Haemoglobinuria Unexplained bleeding (DIC)	Anxiety Chest pain Pain near infusion site Respiratory distress/ shortness of breath Loin /back pain Headache	Acute intravascular hemolysis Bacterial contamination and Septic shock Fluid overload Anaphylaxis Transfusion related acute lung injury (TRALI) Transfusion associated Graft versus Host Dyspnoea disease (TA-GvHD)

READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** or the **Text Booklet** until you are told to do so.

Write your answers on the spaces provided on this **Question Paper**.

You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this Question Paper and the Text Booklet.

DO NOT remove **OET** material from the test room.

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate Text **Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

Transfusion reaction : Questions

Questions 1-7

For each of the questions, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1. the correct route for the administration of chlorphenamine? -----
2. the likely cause of rigors and fever? -----
3. the best way to describe patient symptoms? -----
4. initial steps to take when treating a critically ill patient? -----
5. the various symptoms of patients who have had a transfusion reaction? -----
6. where to document the findings of the appropriate investigations? -----
7. the effects of various medications for managing patient's symptoms? -----

Questions 8-13

Answer each of the questions, **8-13**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8. For how long should a patient's urine be collected and documented?

9. What should be used to appropriately transport a blood culture?

10. How long should 0.9% saline be given if the patient is hypotensive?

11. What type of admission is warranted for a patient experiencing stridor?

12. What might a category 3 patient show more than a twenty percent drop in?

13. What is best avoided if the patient has a low platelet count?

Questions 14-20

Complete each of the sentences, **14-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

14. A vein from the ----- should be used for sample collection if a reaction occurs following transfusion.
15. If a patient experiences pain close to the site of infusion, it's likely to be classified a ----- reaction.
16. A nebuliser should be used to administer ----- at 5mg.
17. An assessment for bleeding from ----- should be conducted in an emergency situation.
18. There is no need to report ----- transfusion reactions if they do indeed occur.
19. Visual confirmation is sufficient to check for ----- in a patient's recent urine sample.
20. A patient may be considered ----- if they experience pruritus accompanied by a headache.

READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this Question Paper until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this Question Paper.

DO NOT remove **OET** material from the test room.

HOW TO ANSWER THE QUESTIONS

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:



In this part of the test, there are six short extracts relating to the work of health professionals. For questions 1-6, choose the answer (**A, B or C**) which you think fits best according to the text.

1. The guidelines for infection control require dentists to

- Ⓐ strictly abide by the rules set out within the document.
- Ⓑ use their own judgment when putting the strategies into practice.
- Ⓒ follow the example of well-established dental clinics.

1.12 Guidelines for Infection Control for Dental Practitioners

The routine work practices outlined here are designed to reduce the number of infectious agents in the dental practice environment; prevent or reduce the likelihood of transmission of these infectious agents from one person or item/location to another; and make items and areas as free as possible from infectious agents. It is important to acknowledge that professional discernment is essential in determining the application of these guidelines to the situation of the individual dental practice environment. Individual dental practices must have their own infection control procedures in place, which are tailored to their particular daily routines. Professional awareness is critical when applying these guidelines to the particular circumstances of each individual dental practice. Each dental practitioner is responsible for implementing these guidelines in their clinical practice and for ensuring their clinical support staffs are familiar with and able to apply them.

2. The email informs physiotherapists that

- Ⓐ the option of consent ultimately lies with the patient.
- Ⓑ information provided by the patient is confidential.
- Ⓒ patient consent forms are a legally binding document..

To: All physiotherapists

From: Ken Macarthur, Head Physiotherapist

Subject: Patient consent forms

This is a courtesy email reminding all staff that it is standard practice to not only provide the patient consent forms, but to also verbally go through all aspects of the form with the patient prior to the commencement of treatment. The purpose of this is to inform the patient of their rights and how we address the issue of a collaborative decision making and informed consent between physiotherapist and patient.

The patient's condition and options for treatment must be discussed so they are appropriately informed and are in a position to make decisions relating to their treatment. They must also be informed that they may choose to consent or refuse any form of treatment for any reason including religious or personal grounds. Once they have given consent, they may withdraw that consent at any time.

3. What does the policy for manual handling equipment tell employers?

- Ⓐ All areas of the hospital should be fitted with overhead tracking.
- Ⓑ Assistance devices should be used over physically handling the patient.
- Ⓒ Patients have the final decision on how they should be assisted.

Policy for manual handling equipment

The provision of ceiling hoist technology and air assisted patient lifting equipment should be considered as the first line handling aid by employers as significant evidence exists that their use reduces operator and patient injuries. Overhead tracking should be installed in all new or refurbished facilities. This should cover beds as a minimum, but should extend to ensuites and other areas of the facility where patients are likely to require assistance. Once an assessment has been made that equipment should be used for safe patient handling then equipment should be made available and used, even in situations where the patient and/or family's preference is for it not to be used.

4. The purpose of the notice is to explain to occupational therapists that

- Ⓐ confirmation of equipment is subject to availability at the time of request.
- Ⓑ mattresses are of standard size so may not be suitable for all bed types.
- Ⓒ patient factors must be considered prior to lodging a request form.

Equipment Request Form (ERF) for Pressure Care Mattresses

It is the responsibility of the occupational therapist attending to the individual patient to submit an Equipment Request Form (ERF) based on equipment eligibility criteria. A pressure mattress may be appropriate when someone is at risk of a pressure injury as evidenced by documented sound clinical reasoning and their pressure injury risk is unlikely to significantly change. Environmental and equipment considerations must be confirmed such as that a patient's weight is within the safe workload of the equipment requested. The size of the mattress must also be compatible with other bed equipment and accessories and the patient has been informed regarding the contraindications of placing items (e.g. continence products, sheepskins, electric blankets, ill-fitting bed sheets) on top of the mattress. Only after this confirmation should an ERF be submitted.

5. The memo about use of smart phones during surgery tells staff that

- Ⓐ their use may be a violation of patient confidentiality.
- Ⓑ they are to be used only by the surgeon
- Ⓒ they can potentially lead to patient harm.

Memo: Restricted use of smart phones during surgery.

As smart phone technology has become increasingly common, it is now cause for concern when used within the operating rooms, especially as a major source of distraction. For this reason, the use of smart phones within the operating rooms will now be restricted.

The undisciplined use of smart phones - whether for telephone, email or data communication, and whether by the surgeon or other members of the surgical team may compromise patient care. Whenever possible, members of the operating suite team should only engage in urgent outside communication during surgery. Personal and routine calls should be minimised and be kept as brief as possible. Incoming calls should be forwarded to voicemail or to the reception desk to be communicated promptly. Any use of a device or its accessories must not compromise the integrity of the sterile field and special care should be taken to avoid sensitive communications within the hearing of awake or sedated patients.

6. The main point of the extract on subcutaneous cannula is to explain

- Ⓐ the versatility of their design and function.
- Ⓑ that they must only be used by registered nurses.
- Ⓒ the need for a backup cannula in case of malfunction.

Subcutaneous cannula

A subcutaneous cannula is a small plastic tube designed to carry medication into a person's body. One end, inserted by a registered nurse, sits just under the person's skin. The other end divides into two parts and is shaped like a Y. One part of the Y-arm can be connected to a syringe driver or infusion pump; the other can be used for subcutaneous injections. The nurse may insert a second cannula in a different part of the body. This is in case the original cannula stops working and ensures that there will be no delays in giving medications to the person you are caring for. It can be especially useful if the original cannula stops working at night when nurses may not readily available or have the same level of support as during the day.

In this part of the test, there are two texts about different aspects of healthcare. For questions 7-22, choose the answer (**A, B, C or D**) which you think fits best according to the text.

Text 1: Witnessed resuscitation attempts - a question of support.

The idea of supporting relatives who witness resuscitation is nothing new, with research and reports going back to the 1980s. In 1996, the Research Councils UK (RCUK) published a booklet called Should Relatives Witness Resuscitation? Since then, practice has moved on, but many of its core elements are still considered valid today. It was suggested that family members who witness the resuscitation process may have a healthier bereavement, as they will find it easier to come to terms with the reality of their relative's death, and may feel reassured that everything possible has been done. It acknowledged that the reality of CPR may be distressing, but argued that it is "more distressing for a relative to be separated from their family member" at this critical time.

In the latest edition of its Advanced Life Support manual, the RCUK **remains adamant** that "many relatives want the opportunity to be present during the attempted resuscitation of their loved one." But do they have the right to demand it? 'The resuscitation team and the nurse caring for the patient have the responsibility of deciding whether to offer relatives the opportunity to witness a resuscitation attempt' says Judith Goldman, clinician and researcher at the University of Michigan, USA. 'Sometimes resuscitation teams may decide not to offer relatives the option of witnessing resuscitation; but this should never be based on their own anxieties rather than on evidence-based practice'.

When a patient is admitted to intensive care the question may be asked by the medical team whether the patient would want CPR. This would also provide an opportunity for witnessed resuscitation to be discussed with patients and relatives upon admission. 'The subject would have to be approached sensitively, but ascertaining patients' and/or relatives' wishes before an admission to intensive care would certainly help' says Frank Lang, researcher for the European Resuscitation Council. 'Recent studies show both public support for witnessed resuscitation and a

desire to be included in the resuscitation process and of those who have had this experience; over 90% would wish to do so again” he says.

‘Still, the decision regarding whether to be present during resuscitation should be left to the individual person because it’s certainly not for everyone,’ he adds. ‘Medical teams also need to gauge whether witnessed resuscitation would have benefits for the patient and/or the relatives, which can only be done through a holistic assessment of the specific situation at the time. It needs to remain a personal approach’ he says. What this way of thinking suggests is that regardless of research, witnessing resuscitation can be traumatic for all involved, particularly for family members, so it seems appropriate that health professionals explain everything that is happening. Even more so that a member of the team, ideally the nurse caring for the patient in cardiac arrest, be designated for that role and remain with the family during the whole process.

‘Nurses need to discuss the wishes of the patient and/or relatives as soon as possible to act in the best interests of both while remaining non-judgemental whatever the relatives decide, whether they choose to be present or not, and support them in making the decision’ says Judith Goldman. ‘Once it has been established that relatives want to be present, the nurse should inform the resuscitation team leader, seek their approval and ask them when the relatives should enter the resuscitation area. The team who are providing direct care retains the option to request that the family be escorted away from the bedside and/or out of the room if deemed appropriate’, she says.

Such decisions to request family removal are not taken lightly. ‘There are the more obvious occasions that family members must be removed, for instance, if they disrupt the work of the resuscitation team either through excessive grief, loss of self-control, exhibit violent or aggressive behaviour or try to become physically involved in the CPR attempt’ she says. ‘But the team also need to consider times when during a resuscitation attempt all members of staff are fully occupied and there is no one available to stay with the family. This is especially hard for them to take.’

If the family do remain present, and regardless of patient outcome, providing assistance is crucial for families to get through such a stressful and shocking event.

Frank Lang recommends that 'the nurse who is directing the family should point them towards all or any available support service within the hospital as well as towards professional bereavement counselling outside of the hospital. The latter provides distance from the scene and can help with symptoms of post-traumatic stress disorder.' Throughout any decision-making, however, it is clear that the patient's welfare, privacy and dignity must remain the utmost priority of the resuscitation team.

Text 1: Questions 7-14

7. In the first paragraph, the writer quotes the RCUK in order to

- Ⓐ stress the significance of family involvement in resuscitation attempts.
- Ⓑ show the significant benefits of family presence during resuscitation.
- Ⓒ highlight that many now consider witnessed resuscitation outdated.
- Ⓓ demonstrate that being witness to a resuscitation attempt is traumatic

8. In the second paragraph, Judith Goldman says that witnessed resuscitation

- Ⓐ should not be the sole decision of the resuscitation team.
- Ⓑ needs to be made available to all families.
- Ⓒ must not be denied because of personal feelings.
- Ⓓ is requested by a large number of relatives.

9. In the second paragraph, the phrase '**remains adamant**' is used to

- Ⓐ argue that relatives should have the ultimate decision.
- Ⓑ show that the opinion of the RCUK has not changed.
- Ⓒ express that greater understanding is needed from staff.
- Ⓓ emphasise RCUK's opposition to excluding family.

10. In the third paragraph, Frank Lang suggests that patients and family members

- Ⓐ would struggle to comprehend the process of CPR.
- Ⓑ require follow up support from resuscitation teams.
- Ⓒ have a good understanding of witnessed resuscitation.
- Ⓓ would benefit from early consultation with staff.

11. In paragraph four, the writer believes that a team member present at resuscitation attempts

- Ⓐ should provide the family with constant reassurance.
- Ⓑ will find the experience as stressful as family members.
- Ⓒ should focus on the patient rather than the relatives.
- Ⓓ needs to explain the process to each individual family member.

12. What does Judith Goldman regard as important during resuscitation?

- Ⓐ establishing that the resuscitation team are in charge.
- Ⓑ that relatives are instructed on whether to be present or not.
- Ⓒ the point at which family members enter or leave the scene.
- Ⓓ remaining courteous when requesting relatives to leave.

13.In the sixth paragraph, Judith Goldman suggests that families who wish to be present

- Ⓐ must understand that extra staff may not always be available.
- Ⓑ at times struggle to understand why they cannot enter.
- Ⓒ prefer to remain with the allocated member of staff.
- Ⓓ are sometimes concerned about witnessing the resuscitation.

14.In the final paragraph, Frank Lang insists that despite the outcome of the resuscitation attempt, families

- Ⓐ are required to seek counselling as soon as appropriate.
- Ⓑ should utilise the hospital network before outside assistance.
- Ⓒ sometimes regret their decision to remain present.
- Ⓓ will still often struggle to overcome the experience.

Text 2: A smoker's right to surgery

Smokers who do not try or do not succeed in quitting should not be offered a wide range of elective surgical procedures, according to an editorial published in The Medical Journal of Australia. The authors acknowledge this would be a controversial, overtly discriminatory approach, but they say it is also evidence-based. Dr Matthew Peters and colleagues from Concord Repatriation General Hospital say smokers who undergo surgery have substantially higher risks, poorer surgical outcomes and therefore consume more healthcare resources than non-smokers. Surprisingly, these new concerns are not based on cardiac and respiratory risks, but increased wound infection.

"A randomised study examining smoking cessation intervention before joint replacement surgery, saw wound infection rates reduced from 27 per cent in continuing smokers to zero in those who quit smoking," Dr Peters said. "Almost 8 per cent of breast reconstruction patients who smoke experience abdominal wall site necrosis, compared with 1 per cent of non-smokers. These results are obviously significant." He believes that it's much better that the prioritisation occurs on the basis of good evidence rather than on a whim or some political influence. "If there was a health care system that had everything patients need and want immediately, there wouldn't be a problem. But we don't have that and as far as I'm aware no country truly does. You have to determine priorities," Peters says.

However, not everyone agrees. Professor Andrew Coats, dean of the University of Sydney's faculty of medicine believes this is not accepted medical treatment. "You do not arrange patients based on them being more deserving or less deserving. You give treatment based on need and how a person will benefit. It's the urgency of that need that's the main factor." Coats says lifestyle factors should only affect treatment in very limited circumstances. "If, because of lifestyle factors, a treatment is not likely to work or it will be harmful, then obviously it should not proceed. But we don't take these factors into account in prioritising; that would be the end of the healthcare system as we know it." He says if a doctor believes a patient could give up smoking and therefore reduce complication rates, they should encourage the patient to quit,

but he says you cannot withhold an operation as punishment for not giving up. "Many people are not able to give up cigarettes. It is a real chemical condition."

Dr Mike Kramer, the Royal College of Surgeons representative agrees that smokers need to be treated differently. "You need to take risk into account. The risks of procedure versus the benefits, and that is affected by the smoking status of the patient," he says. Kramer, a cardiothoracic surgeon, says complications associated with smoking are so significant he will delay an operation for the removal of a lung cancer so a patient can stop smoking for a minimum of four weeks before an operation. "This is not a moral judgement or an ethical judgement. It is a pure clinical judgement for the benefits of a patients outcome," he says.

There is also the heavy burden of financial pressure that must be considered when dealing with the limited health dollar. Reverend Norman Ford, the director of the Caroline Chisholm Centre for Health Ethics, says while there should be no blanket ban or refusal for any surgery, the allocation of public health funds needs to be taken into account. "Why should non-smokers fork out for smokers?" Ford says the additional costs of wound infection complications should be calculated and smokers who refuse to quit before surgery should pay the additional expense if wound infections occur. "If they give up smoking they should be treated the same as non-smokers. If they don't give up smoking they should pay the difference," he says. "You've got to motivate them to stop smoking and the pocket is a great motivator - if they've got it. So their ability to pay should be means tested."

The essence of this argument comes down to the question of whether people who are knowingly doing things that may be harmful to their health are entitled to health care. Surgery is routinely performed on diabetics, who also are at risk of increased postoperative complications. If surgery can be denied to smokers, or even delayed, should the same treatment, or lack thereof be given diabetics with poor glycaemic control because they don't comply with diet or medications? Refusing to operate on smokers could land us on a very slippery slope, eventually allowing surgeons to choose to operate only on low risk patients. Perhaps it would be more prudent for physicians to educate their patients about the risks of smoking, as well as other risk factors, prior to surgery and entitle patients to make an informed decision about their healthcare.

Text 2: Questions 15-22

15. What possible reason does the writer give for refusing current smokers the opportunity for surgery?

- Ⓐ the negative effects seen in systematic research.
- Ⓑ the overall increased costs to the hospital system.
- Ⓒ the known impact on the patient's heart and lungs.
- Ⓓ the higher possibility of post-operative infection.

16. In the second paragraph, Dr Peters says that prioritising patients

- Ⓐ is unfortunately necessary.
- Ⓑ is less expensive in the long run.
- Ⓒ should start at a government level..
- Ⓓ has been shown to reduce harmful outcomes.

17. In the second paragraph, the writer uses the term '**on a whim**' to show Dr Peters' belief that

- Ⓐ further research should be carried out.
- Ⓑ current healthcare systems are not adequate.
- Ⓒ the findings of recent research are remarkable.
- Ⓓ careful consideration is extremely important.

18. In the third paragraph, Professor Coates says that treatment should be provided

- Ⓐ to all patients based on a system of merit..
- Ⓑ according to the necessity of the individual patient.
- Ⓒ regardless of a patient's lifestyle factors.
- Ⓓ once a patient has reduced their intake of cigarettes.

19. What does Dr Mike Kramer regard as a significant factor when treating a smoker?

- Ⓐ the length of time a patient has refrained from smoking.
- Ⓑ providing an unbiased assessment of each individual.
- Ⓒ considering the ethical implications of each case.
- Ⓓ the patient's attitude towards smoking cessation.

20. In the fifth paragraph, Reverend Norman Ford says that when considering the financial burden of healthcare

- Ⓐ smokers should fund their own operations.
- Ⓑ more public funding is needed to help smokers quit.
- Ⓒ making a smoker pay incentivises change.
- Ⓓ patients who smoke should not be held accountable.

21. In fifth paragraph, what opinion is highlighted by the phrase 'fork out'?

- Ⓐ Patients that continue to smoke should still have rights..
- Ⓑ Those that don't smoke have less complications.
- Ⓒ The public should not bear the cost of smokers' healthcare.
- Ⓓ Non-smokers are less of a burden on public funding.

22. In the final paragraph, the writer argues that treating smokers differently

- Ⓐ is fair as other patients haven't made such poor lifestyle choices.
- Ⓑ could in turn lead to poor decisions concerning other patients.
- Ⓒ may ultimately cause such patients to avoid having health checks.
- Ⓓ may lead surgeons to discriminate against patients with diabetes.

READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT remove OET material from the test room.

TEXT A

Anaemia is defined as an overall decrease in red blood cell mass. There are many varying causes of anaemia, which all present with some general symptoms. Anaemia results in a lack of red blood cells in the blood. Because it is the haemoglobin in red blood cells that carries oxygen from the lungs to the rest of the body, a decrease in red blood cells results in less oxygen going into the tissues. This causes a state known as hypoxia, or reduced oxygen in body tissues.

The common symptoms of all anaemias are those of hypoxia:

- Weakness, fatigue, difficult or laboured breathing
- Pale skin
- Headache and light-headedness
- Chest pain (if the patient already has a disease of the arteries supplying the heart)

TEXT B

There are many classification systems to differentiate anaemias. The most commonly used is based on the size of the red blood cell. Anaemias with red blood cells that are smaller than normal are known as microcytic anaemias. If the anaemia has normally sized red blood cells, it is referred to as a normocytic anaemia. Finally, if the red blood cells are too big, it is known as a macrocytic anaemia. Normocytic anaemias are further broken up into whether or not there is an increased number of young red blood cells (a.k.a. reticulocytes), which is an indication if the bone marrow is working properly—for example, if the red blood cells are being destroyed (haemolysis), there should be higher reticulocytes because there is no effect on the bone marrow's ability to produce new cells..

TEXT C

While there are many different causes of anaemia, laboratory studies and unique features of the patient can be used to help differentiate between various aetiologies.

Laboratory studies used to diagnose anaemia include:

- Haemoglobin (Hb)—a measure of the protein that transports oxygen in the red blood cell
- Haematocrit (Hct)—a measure of the percentage of red blood cells in the blood
- Red blood cell amount (erythrocyte count)—a measure of the number of red blood cells in the blood

A general diagnosis of anaemia can be determined by the following values:

- Haemoglobin level
 - Males: less than 13.5 g/dL
 - Females: less than 12.5 g/dL (women have a generally lower haemoglobin because of blood loss during the monthly menstrual cycle)
- Haematocrit
 - Males: less than 45% red blood cells
 - Females: less than 37% red blood cells (women have a generally lower haematocrit because of blood loss during the monthly menstrual cycle)
- Red blood cell amount
 - Male: less than 4.7 million cells/mL
 - Female: less than 4.2 million cells/mL (women have a generally lower red blood cell amount because of blood loss during the monthly menstrual cycle)

While these laboratory tests are good estimates of the red blood cell mass, they are not perfect. Red blood cell mass is very difficult to measure, and therefore these laboratory tests are used together to assess whether or not someone has anaemia.

TEXT D

Management of Acute Asthma

- The treatment of anaemia depends heavily on the type of anaemia that the patient is experiencing. However, there are several overarching goals of treatment.
- If possible, treat the underlying cause of the red blood cell loss. For example, if the patient has anaemia because of blood loss, give a blood transfusion.
- Identify and treat any complications that have occurred because of the anaemia.
- Educate the patient on how to manage their anaemia. For example, a patient with anaemia because of iron deficiency can supplement their treatment with iron rich foods, such as leafy green vegetables.
- Alternatively, a patient with anaemia caused by vitamin deficiency should be advised to increase their intake of folic acid and B-12. Note that patients who follow vegetarian or vegan diets may struggle to meet B-12 requirements, so eating fortified foods and using supplements should be advised.

READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** or the **Text Booklet** until you are told to do so.

Write your answers on the spaces provided on this **Question Paper**.

You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this Question Paper and the Text Booklet.

DO NOT remove **OET** material from the test room.

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate Text **Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

Anemia: Questions

Questions 1-6

For each of the questions, 1-6, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1. treating patients with anaemia?
2. the symptoms of hypoxia?
3. methods used to identify anaemic patients?
4. the different types of anaemia?
5. the levels of haemoglobin in a woman with anaemia?
6. how red blood cell size affects anaemia?

Questions 7-14

Answer each of the questions, **7-14**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

7. What should vegan patients with vitamin deficiency anaemia be encouraged to add to their diets?
.....

8. If there are a decreased number of young red blood cells, what type of anaemia is being dealt with?

9. How will a patient's breathing sound when experiencing a significant reduction of oxygen in the body's tissues?

10. A male with anaemia must have less than what percentage of red blood cells?

11. What is an increase in the number of reticulocytes an indication of?

12. What reduces the amount of red blood cells in some patients?

13. What should be treated in anaemic patients, after identifying the cause?

14. How are the different types of anaemia most commonly distinguished?

Questions 15-20

Complete each of the sentences, **15-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15. Anaemia caused by -----
should be treated with a blood transfusion.
16. Patients suffering from hypoxia and chest pain are likely to also have a -----

17. If ----- is functioning properly, high reticulocyte anaemia is likely to be present.
18. A number of tests may be necessary to diagnose anaemia, due to the difficulties involved in measuring -----
-----.
19. Patients with anaemia caused by -----
----- should be instructed to adjust their diet.
20. When identifying the type of aetiology, ----- of the patient should be considered, in addition to laboratory studies.

READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this Question Paper until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this Question Paper.

DO NOT remove **OET** material from the test room.

HOW TO ANSWER THE QUESTIONS

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:



In this part of the test, there are six short extracts relating to the work of health professionals. For questions 1-6, choose the answer (**A, B or C**) which you think fits best according to the text.

1. The notice reminds staff that patients who are dying

- Ⓐ Will need to be prescribed anti-emetics.
- Ⓑ Might not need to continue with certain medication.
- Ⓒ Should be encouraged to discuss their condition with loved ones.

End-of-Life Decision Making

Remember the five priorities when caring for a dying patient:

1. Recognise that the end of life may be approaching.
2. Communicate with patients, families, carers and staff.
3. Involve patients and those close to them in decisionmaking.
4. Support the needs of families and carers.
5. Develop an individualised plan of care for the patient.

An end-of-life care plan must ensure the physical, psychological, social and spiritual comfort of the patient, and should strive for the best possible quality of life for the patient's remaining time. This includes prescribing anticipatory medications which can be given as required, falling under the following categories which staff are encouraged to remember as the 'Four As': Analgesia (pain relief), Anxiolytics (anti-anxiety), Anti-emetics (for nausea and vomiting), and Anti-secretory (for respiratory and airway secretions). Any unnecessary medications, such as long-term diabetes control and blood pressure medications can be stopped. A Do-Not-Resuscitate (DNACPR) decision also needs to be made.

2. The guidelines inform us that multiple anaesthetics can be used

- Ⓐ To increase the numbing effects.
- Ⓑ To prevent bleeding throughout the procedure.
- Ⓒ To more accurately control how long it will last.

Anaesthesia use at Harlow Dental Centre

At this practice, preference is given to the use of local anaesthetics in combination with conscious sedation.

Many local anaesthetics may be used in order to reversibly block specific pain pathways and/or cause paralysis of muscles. The most commonly used local anaesthetic at the centre is lidocaine - remember that the half-life of lidocaine in the body is about 1.5 to 2 hours. Other local anaesthetic agents include articaine, bupivacaine, prilocaine and mepivacaine. Often, a combination of local anaesthetics may be used, sometimes with adrenaline or another vasoconstrictor to modulate the metabolism of the local anaesthetic and control local bleeding.

Sedation during procedures should mostly be limited to conscious sedation. Benzodiazepines enhance the effect of neurotransmitter gamma-aminobutyric acid (GABA) at the GABAA receptor. This results in a sedative, hypnotic, anxiolytic, anticonvulsant and muscle relaxant properties.

3. The purpose of this memo is to explain

- Ⓐ How to treat multi-resistant pathogens.
- Ⓑ The causes of bacterial infections.
- Ⓒ When to prescribe antibiotics.

For the attention of all medical staff:

Microbial resistance to antibiotics is on the rise and infection with multi-resistant pathogens, such as *Clostridium difficile* and MRSA amongst others, is becoming more common. Patients receiving antibiotics are at increased risk of such infections. As such, please be aware of our antimicrobial prescribing guidelines, which ensure that antibiotics are only prescribed with clear, clinical justification; evidence of infection; and/or guaranteed medical benefit. It is recommended that specimens should be cultured and results obtained before commencing treatment with antibiotics, thus only prescribing the therapy to which the microbe is sensitive. Prescription of broad-spectrum antibiotics should be avoided where possible, as these not only damage the normal bacteria of the human body, but also increase microbial exposure to antimicrobial medications, increasing their potential for developing resistance. Review narrow-spectrum antibiotic prescriptions within 5 days, and broad-spectrum prescriptions within 48 hours.

4. The purpose of the notice is to explain to occupational therapists that

- Ⓐ The disorder is more difficult to identify in patients with ADHD.
- Ⓑ Most children with autism are diagnosed before the age of three.
- Ⓒ Young people with autism are more likely to suffer from other conditions.

Autism in Young People

More than 1% of the UK population has an autism spectrum disorder. Signs can vary widely between individuals and at different stages of an individual's development. When children present with other conditions such as ADHD (attention deficit hyperactivity disorder) or other learning difficulties, autism spectrum disorders often go undiagnosed.

In children with autism spectrum disorders, symptoms are present before three years of age but diagnosis can be made after this age too. Individuals with autism spectrum disorder tend to have issues with social interaction and communication, including difficulty with eye contact, facial expressions, body language and gestures. Often, children with autism spectrum disorders may lack awareness or interest in other children and tend to play alone.

The causes of autism spectrum disorder are unknown but are linked to several complex genetic and environmental interactions.

5. The memo reminds all staff to avoid

- Ⓐ Challenging a patient's criticisms.
- Ⓑ Handling grievances of a sensitive nature.
- Ⓒ Recording complaints that is not legitimate.

Subject: Fielding Patient Complaints

For the attention of all hospital staff:

At County Green Hospital, we endeavour to provide our patients and families with the highest quality of services. Unfortunately, there may be times where performance does not meet expectation. We routinely survey our patients on how we can do better, but members of the treatment team may also be approached with patient feedback, so all employees must be aware of the correct procedure for handling patient complaints. The first step is to listen to what patients have to say and document details appropriately. Whether or not you feel there is a legitimate grievance, it is important to keep a record for later examination. While listening to the complaint, the employee should validate the patient or family member's experience. This does not mean there needs be agreement about the nature of the complaint, but that the employee demonstrates a clear understanding of why the patient or family member might be feeling this way.

6. The Patients with delirium are more likely to recover quickly if

- Ⓐ Kept in a darkened environment.
- Ⓑ Staff changes are kept to a minimum.
- Ⓒ Treatment ensures they receive adequate rest.

Diagnostic Criteria for Delirium

Delirium affects up to 87% of patients in intensive care and is particularly common among the elderly. Delirium can have serious adverse effects and even lead to mortality and must therefore be treated as a medical emergency.

All hospital staff must know how to prevent, detect, and rapidly assess and treat delirium on the hospital wards. Risk factors for developing delirium include: change of environment, loss of vision/hearing aids, inappropriate noise or lighting, sleep deprivation, severe pain, dehydration, drug withdrawal, infections of any kind, recent surgery, and old age. For patients at risk of delirium, think of the mnemonic DELIRIUM which indicates the common causes: Drugs or Dehydration, Electrolyte Imbalance, Level of pain, Infection or Inflammation (such as post-surgery), Respiratory failure, Impaction of faeces (severe constipation), Urinary retention, Metabolic disorder (such as liver or renal failure).

Management requires re-orientation of the patient to where they are and who everybody around them is, as well as re-assurance and a non-confrontational, empathetic approach towards agitated and distressed patients. Please refrain from changing the staff of the medical team responsible for a delirious patient's care, in order to ensure consistency for the patient. Avoid unfamiliar noises, equipment and staff in the immediate vicinity of the patient, and facilitate visits from family and friends as much as possible.

In this part of the test, there are two texts about different aspects of healthcare. For questions 7-22, choose the answer (**A, B, C or D**) which you think fits best according to the text.

Text 1: Work-Related Stress & Medical Errors

Stress is a term that crops up all too often in modern conversation, used to describe every unfortunate circumstance, every out-of-sequence event, and every foot out of line. What is stress? Most definitions of stress cover any internal or external stimulus which results in a negative response or disturbance in one's physical, social or mental wellbeing. Unfortunately, stress is common, and it can be devastating to people's lives and health when it is maintained over long periods of time, and when it gains the capacity to overwhelm one's coping abilities and mechanisms.

In the medical profession, daily stress is almost guaranteed. Recently, changes to many healthcare workers' contracts in the UK have resulted in longer and more antisocial working hours, as well as an increased workload, greater bed crises in hospitals and larger budget cuts, so stress levels amongst UK healthcare professionals are on the rise. A 1996 questionnaire study in the Lancet reported that 27% of doctors in the UK believed that the stress they experienced was triggered by poor management, low job satisfaction, financial concerns, and patients' suffering, amongst other factors.

Over two decades later, these problems still exist; some healthcare workers argue that conditions have actually deteriorated. A 2013 report by the British Medical Association stated that over 50% of UK doctors had experienced an increase in work-related stress over the preceding year, in addition to an increase in the complexity of their work. 25% of junior doctors in hospitals also reported a reduced quality of care for patients due to high levels of stress and the pressures put on individual members of staff, with levels of stress exacerbated by longer working hours. In many healthcare jobs, stress is the **elephant in the room**, particularly with junior staff, who may feel unable to voice concerns about their workload. Unfortunately, however, these factors have the potential to lead to medical mistakes, which could be detrimental to patient lives. In such a circumstance, who is really to blame? The overworked medical staff, or the poor management of modern hospitals?

We do not need to look far to examine the effect that stress can have on doctors today. In 2015, Dr Hadiza Bawa-Garba was found guilty of manslaughter after failing to provide life-saving treatment to a patient when needed, resulting in the unfortunate death of a six-year-old child, Jack Adcock. In 2018 this experienced senior paediatrician with a previously unblemished record was struck off the medical register, unable to ever practice again as a doctor. The case of Dr Bawa-Garba infuriated many in the medical profession, as fingers were pointed at an overworked doctor working under immense pressure who was blamed for gross negligence. But who is the truly negligent one in our current healthcare system? While the death of young Jack is extremely saddening, it is important to explore the circumstances around his death in order to prevent such tragedies from reoccurring. On the day of the incident, Dr Bawa-Garba was covering her own workload as well as that of two senior colleagues who were away, across six wards, spanning four floors, with malfunctioning IT software and out-of-order results systems. Did Dr Bawa-Garba make detrimental mistakes? Yes. But one must ask, are we creating a recipe for disaster when we require our medical staff to work under such immense pressures? Could this be one tragic event of many waiting to happen? Such mistakes ruin lives.

Studies have shown that the most common cause of medical errors is the use of heuristics in medical decision-making, leading to bias. Heuristics are shortcuts taken to reach decisions quickly, based on previous patterns of disease and similar cases seen by the doctor. Mistakes are more likely when such shortcuts are used by junior doctors who lack the experience necessary to make such fast decisions accurately. Tversky and Kahneman outlined seven types of heuristics in their 1974 article: Availability heuristics are based on how easy specific diagnoses are to recall, resulting in over-diagnosis of rare but memorable conditions; Representativeness heuristics are based on similarity of patient presentations to previous typical cases, leading to delayed or missed diagnoses in atypical or non-characteristic patients; Anchoring heuristics occur when a diagnosis is based on one piece of information only, leading to rapid conclusions which lack evidence and early diagnosis without consideration of all available information; Confirmation bias occurs when a diagnosis is based on a preconceived idea, where the doctor pays attention to the information that supports their theory, and evidence which challenges the diagnosis is consciously or subconsciously ignored; Commissioning bias where a doctor acts too

soon rather than waiting to gather and review all the information first; Gambler's Fallacy which is where consecutive patients have the same diagnosis and so the doctor assumes a similar patient who follows must also have the same diagnosis; Fundamental Attribution Error which is the tendency to blame patients rather than their circumstances for their poor health.

Research shows that the best way to avoid medical errors in diagnosis is to consider several hypotheses, known as "differential diagnoses", and investigate them all equally until the one with the most supporting evidence is found and agreed upon. Use of heuristics and the resultant flawed decision-making could be prevented by reducing work stresses and pressures on medical professionals. One way to achieve this would be to reduce working hours and shift durations in order to prevent sleep deprivation in medical staff, which is known to hinder focus, thus creating a safer medical environment for both staff and patients.

Text 1: Questions 7-14

7. The first paragraph explains that stress

- Ⓐ Is usually caused by a factor than cannot be controlled.
- Ⓑ Is interpreted in various ways by different people.
- Ⓒ Is unusual when it lasts for an extended time.
- Ⓓ Generally impacts people's behaviour.

8. In the second paragraph, doctors are said to claim that stress

- Ⓐ Is often improperly managed by chronic sufferers.
- Ⓑ Could be improved by increasing the welfare budget.
- Ⓒ Generally resulted in their having to work longer hours.
- Ⓓ Was caused by a number of issues including money worries.

9. The writer uses the phrase '**the elephant in the room**' to emphasise the fact that

- Ⓐ Levels of stress experienced by staff have declined.
- Ⓑ Senior staff generally experiences less stress than their juniors.
- Ⓒ Many healthcare professionals do not discuss the stress they experience.
- Ⓓ Junior doctors have reported a lower quality personal life as a result of stress.

10. Why does the writer comment on Dr Hadiza Bawa-Garba and her patient Jack?

- Ⓐ to suggest that doctors are more likely to make significant errors when stressed
- Ⓑ to outline a scenario where a doctor's concerns about stress were ignored
- Ⓒ to demonstrate that stress in healthcare professionals is unacceptable
- Ⓓ to emphasise the impact the death of a patient can have on stress

11. The writer suggests that Jack Adcock's death was partly caused by

- Ⓐ Technology that was out of date and faulty.
- Ⓑ A hospital ward overcrowded with patients.
- Ⓒ An insufficient number of nursing team staff.
- Ⓓ A lack of experience among the clinical team.

12. Why might doctors who use heuristics be at a greater risk of making clinical errors?

- Ⓐ Heuristics are more likely to be used by junior doctors
- Ⓑ Doctors might take too long to complete their tasks
- Ⓒ Doctors might skip over the relevant information
- Ⓓ The different types of heuristics are confused

13. The writer claims that confirmation bias might cause doctors to ignore relevant information if

- Ⓐ They have recently treated a patient with the same condition.
- Ⓑ They are very familiar with the evidence being presented.
- Ⓒ The patient displays extreme symptoms.
- Ⓓ It does not support their existing theory.

14. What does the word **them** refer to in the final paragraph?

- Ⓐ The team of healthcare staff
- Ⓑ A variety of possible causes
- Ⓒ The mistakes in patient care
- Ⓓ A number of different texts

Text 2: Electroconvulsive therapy (ECT)

Electrodes Wires Bite Blocks For many these terms bring to mind a sinister mental asylum and the foreboding image of a patient about to suffer a tortuous electric shock. Literature written in the 20th century did much to criticise this practice, with writers frequently describing electroconvulsive therapy (ECT) as a form of torture, reserved for the most vulnerable members of society. Interestingly enough, ECT has actually been used in the healthcare field for hundreds of years. Before the advent of effective antipsychotic medications, a wide variety of therapies were trialled for serious mental illnesses. One of these involved the therapeutic use of inducing seizures in patients. As early as Benjamin Franklin's (1705 – 1790) time, an electrostatic machine could be used to cure someone of 'hysterical fits'.

Through the 19th century, British asylums began to employ electroconvulsive therapy in a widespread effort to cure diseases of the mind. In the early 20th century, a neuropsychiatrist by the name of Ladislav J. Meduna promoted the idea that schizophrenia and epilepsy were antagonistic disorders, and that precipitating seizures could serve as a potential treatment of schizophrenia. There were several methods used to induce seizures, including insulin coma, seizure-inducing medications (metrazol), and most famously, ECT.

While many of these practices are now seen as barbaric, there were very few options for psychiatric treatment before the development of antipsychotics, mood stabilisers, and anti-depressants. With the rise of these new treatment options came an increase in the public awareness of the often inhuman conditions of electroshock. The revelations resulted in widespread backlash, and the use of ECT therapy began to swiftly decline. However, in the later part of the 20th century, after much debate and research, the National Institute of Mental Health in the US came to a consensus that ECT was both safe and effective when proper guidelines were implemented. In the US today, ECT treatment is routinely covered by insurance for severe and treatment-resistant forms of mental illness.

The exact mechanism of action for ECT is unknown, but there are several hypotheses: Firstly, increased release of monoamine neurotransmitters such as dopamine, serotonin, and norepinephrine; secondly, enhanced transmission of

monoamine neurotransmitters between synapses; thirdly, release of hypothalamus or pituitary gland hormones and fourthly, anticonvulsant effect. ECT has several indications, the most notable being refractory major depression, catatonia, persistent suicidality, and bipolar disorder. It is also used in pregnancy as it is effective and does not have the teratogenic effects of some other psychiatric medications. While there are no absolute contraindications, it goes without saying that when using ECT, the risks involved will carry more weight with certain patients. Those with unstable cardiovascular conditions, those who have recently suffered a stroke, and those with increased intracranial pressure, severe pulmonary conditions, or a high risk in anaesthesia may not be suitable candidates for ECT. To further explore the appropriateness of using of ECT on specific patients, consider the following case study.

The patient, let's call her Dana, is a 35 year old female who has a history of schizophrenia. She was taken to the hospital by ambulance because her parents found her motionless in her bed, staring blankly, not responding to external stimuli, and not eating or drinking for two days. The psychiatrist caring for her is understandably concerned, because this represents symptoms of catatonia. If Dana does not eat or drink, she may develop life-threatening nutritional deficiencies and electrolyte imbalances. If she does not move, Dana may end up developing a blood clot that could result in a fatal pulmonary embolism. The first-line treatment is benzodiazepines, but in this particular case, there is no improvement in her condition. The psychiatrist decides that that ECT is the next best option. There is the issue of informed consent. Legal jurisdiction handles this differently throughout the world, but if a patient lacks capacity or is too ill to provide consent, a court must provide substitute consent to ensure adequate legal oversight. Once this happens, Dana is medically screened and prepped for treatment.

A course of ECT treatments does not have a standard regimen. Generally, most patients require between six to twelve treatments, but the actual endpoint is determined by the level of improvement. ECT is often given two to three times a week, usually on a Monday/Wednesday/Friday schedule with psychiatric symptoms and testing carried out on a regular basis to monitor progress. Dana starts Monday by being NPO (nothing by mouth) except for any necessary medications. This

reduces the chance for aspiration under anaesthesia during the seizure. She will be taken down to the ECT suite where an anaesthesiologist, psychiatrist, and nurse will greet her. She will be placed in a supine position with EEG monitoring to determine the quality of the seizure given. She will have electrodes placed on her head bitemporally, bifrontally, or unilaterally on the right. In this case, given her life-threatening catatonia, we will use the bitemporal position. The anesthesiologist will then induce anaesthesia, first preoxygenating the patient, then administering anticholinergic agent to reduce oral secretions, anaesthetic medication, muscle relaxation medication, and any cardiovascular prophylaxis as needed.

Once the patient is sufficiently sedated, a brief (0.5 to 2.0 milliseconds) electrical pulse will be introduced at a level determined to reliably cause a seizure. A therapeutic ECT seizure should last at least 15 seconds but no more than 180 seconds. Dana will be monitored for thirty to sixty minutes once this has finished, to ensure her recovery. The goal is for further treatments to reduce her symptoms and enable her to eat, drink, communicate, and move again. Of course, there are adverse effects that must be considered. Anaesthesia can cause nausea, aspiration pneumonia, dental and tongue injuries. The seizure itself can cause cardiovascular issues, and fractures in patients with osteoporosis, and can temporarily impair cognition and memory. It is advised that patients do not make any major or financial decisions during or after ECT treatment, and patients must refrain from driving until a few weeks after the last session. For most patients, one treatment may be all that is needed. For some, continuation of ECT as a single session every couple of weeks may help to prevent relapse. Maintenance treatment for patients with chronically recurring psychiatric illness may also be appropriate. The scheduling of these sessions generally depends on the patient's needs and episodes, sometimes even going on indefinitely. In Dana's case, a few treatments are all that is needed to resolve her catatonia and soon she will be healthy enough to be discharged home with outpatient follow-up for her mental health management.

Text 2: Questions 15-22

15. In the first paragraph, the writer mentions the role of 20th century literature in

- Ⓐ Informing patients of the side effects of antipsychotic medication.
- Ⓑ Preventing the mistreatment of defenseless people.
- Ⓒ Increasing the number of patients receiving ECT.
- Ⓓ Promoting a negative image of ECT.

16. What do we learn about schizophrenia in the second paragraph?

- Ⓐ It was less prevalent in patients who experienced seizures.
- Ⓑ It had a significant impact on the treatment of epilepsy.
- Ⓒ Many asylums in the UK were not prepared to treat it.
- Ⓓ The medication metrazol could be used to induce it.

17. What did the US National Institute of Mental Health decide in the 20th century?

- Ⓐ Practitioners must follow identical treatment plans when using ECT.
- Ⓑ Patients should be given the right to refuse ECT treatment.
- Ⓒ ECT should only be used as a treatment in severe cases.
- Ⓓ ECT was accepted as a safe treatment for patients.

18. In the fourth paragraph, what idea does the writer emphasise with the phrase **'it goes without saying'**?

- Ⓐ Some women find ECT treatments successful while carrying a child.
- Ⓑ It is well known that some patients will not respond well to ECT.
- Ⓒ Few patients realise that they could benefit from ECT therapy.
- Ⓓ The risks associated with ECT are rarely discussed.

19. In the case study, the psychiatrist decides to use ECT on Dana

- Ⓐ Despite Dana's parents' concerns about this type of procedure.
- Ⓑ Because the patient expresses a preference for this treatment.
- Ⓒ After treatment with benzodiazepines proves ineffective.
- Ⓓ As she has developed an electrolyte imbalance.

20. In the sixth paragraph, why isn't Dana given food before her ECT treatment?

- Ⓐ To lower the likelihood of anaesthesia-related aspiration
- Ⓑ To reduce the likelihood of vomiting during treatment
- Ⓒ As medication can interfere with the treatment
- Ⓓ As the catatonic state makes eating difficult

21. In the seventh paragraph, what does the word **'this'** refer to?

- Ⓐ A treatment plan
- Ⓑ A seizure caused by ECT
- Ⓒ An abnormal reaction to medication
- Ⓓ An improvement to the patient's condition

22. In the final paragraph, the writer suggests that Dana's treatment

- Ⓐ Was complete after only one ECT session.
- Ⓑ Will ultimately cure her catatonia using only ECT sessions.
- Ⓒ Will continue for a number of weeks before improvement can be seen.
- Ⓓ Will consist of two ECT sessions each week for the foreseeable future.

READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT remove OET material from the test room.

TEXT A**Establishing the severity of an acute asthma attack**

	Moderate asthma	Severe asthma	Life-threatening asthma
Adults	Measure PEF and arterial saturation		
	PEF > 50-75% predicted	PEF 33-50% predicted	PEF < 33% predicted
	SpO ₂ ≥ 92% PEF 50-75% predicted • no features of acute severe asthma	SpO ₂ ≥ 92% PEF < 50 % predicted • RR ≥ 25/ min • HR ≥ 110/ min • Difficulty talking	SpO ₂ ≥ 92% Silent chest • Cyanosis • Poor respiratory effort • Arrhythmia • Hypotension • Exhaustion • Altered consciousness

Asthma sufferers of any severity may also experience the following:

- shortness of breath
- coughing
- tightness or pain in the chest
- a whistling sound when exhaling

TEXT B

Lung Function Tests in Asthma

Asthma tests should be undertaken to diagnose and aid management of the condition. This is particularly important in asthma, because it presents slightly differently with each patient. Spirometry is the most important test, however several different types of test are available:

- **Peak expiratory flow rate (PEFR):** this is the maximum flow rate during exhalation, after full lung inflation. Diurnal variation in PEFR is a good measure of asthma and useful to the long-term management of patients and the response to treatment. Monitor PEFR over 2-4 weeks in adults if there is uncertainty about diagnosis. It is measured with a peak flow meter - a small, handheld device - into which the patient blows, giving a reading in l/min.
- **Spirometry:** measures volume and flow of air that can be exhaled or inhaled during normal breathing. Asthma can be diagnosed with a >15% improvement in FEV1 or PEFT following bronchodilator inhalation. Alternatively, consider FEV1/FVC < 70% as a positive result for obstructive airway disease. A spirometry test usually takes less than 10 minutes, but will last about 30 minutes if it includes reversibility testing.
 - **Direct bronchial challenge test with histamine or methacholine:** in this test, patients breathe in a bronchoconstrictor. The degree of narrowing can be quantified by spirometry. Asthmatics will react to lower doses, due to existing airway hyperactivity.
 - **Exercise tests:** these are often used for the diagnosis of asthma in children. The child should run 6 minutes (on a treadmill or other) at a workload sufficient to increase their heart rate > 160/min. Spirometry is used before and after the exercise - an FEV1 decrease > 10% indicates exercise-induced asthma.
- **Allergy testing:** can be useful if year-round allergies trigger a patient's asthma. This will be recommended if inhaled corticosteroids are not controlling symptoms. Three different tests are used to measure the patient's reaction to allergens: nitric oxide testing, sputum eosinophils and blood eosinophils.

TEXT C

Patients with asthma of any severity may find their attacks panic-inducing. Remember that the patient's struggle to breathe can cause stress, panic and a feeling of helplessness. There is a strong link between people who suffer from asthma and those who experience panic attacks. Staff must keep this in mind when treating patients with asthma, as some sufferers will require additional emotional support.

Patients may find breathing exercises beneficial. Advise patients to practice daily, to allow these exercises to become habitual. When experiencing an attack, patients should make a conscious effort to relax their muscles and maintain steady breathing. Advise patients to breathe deeply in through the nose and out through the mouth.

Smokers are at a higher risk of developing both panic attacks and asthma. In addition, smoking can irritate the airways in patients with asthma, causing neutrophilic inflammation, and exacerbating breathing problems in those with asthma. Ensure that patients who smoke are fully aware of the risks of smoking with asthma.

TEXT D

Management of Acute Asthma

Rapid treatment and reassessment is of paramount importance. It is sometimes difficult to assess severity. Maintaining a calm atmosphere is helpful to resolving an acute asthmatic attack.

Assess the severity of the attack

1. Check peak expiratory flow (PEF)
2. Is the patient able to speak?
3. Check respiratory rate (RR)
4. Check pulse rate
5. Check O₂ saturation

If life-threatening or severe: warn ICU



Immediate Treatment

1. Maintain O₂ saturation with oxygen (94-98%)
2. Salbutamol 5mg with O₂ (nebulised)
3. Ipratropium 5mg every 6 hours if severe
4. Prednisolone 40-50mg PO or Hydrocortisone 100mg IV



Every 15 minutes: re-assess

1. PEF < 75%: salbutamol repeated every 15-30 minutes, or 10mg every hour continuously. If not yet given, add ipratropium.
2. Monitor ECG and check for arrhythmias
3. Magnesium Sulfate (MgSO₄), 1.2-2g IV over 20 minutes is an option in severe cases not responding to therapy



No improvement

1. Refer to ICU for ventilator support
2. Escalation of medical therapy
3. Check for:
 - PEF deteriorating
 - hypoxia
 - hypercapnia
 - ABG: low pH or high H⁺
 - Exhaustion
 - Drowsiness and confusion
 - Respiratory arrest



Improvement within 15 – 30 minutes

1. Continue salbutamol every 4-6 hours
2. Check peak PEF and O₂ saturation
3. Prednisolone 40-50 mg PO OD for 5-7 days
4. If PEF > 75% an hour after treatment, consider discharge with follow-up

READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** or the **Text Booklet** until you are told to do so.

Write your answers on the spaces provided on this **Question Paper**.

You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this Question Paper and the Text Booklet.

DO NOT remove **OET** material from the test room.

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate Text **Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

Asthma : Questions

Questions 1-7

For each of the questions, 1-7, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1. the different types of dengue virus? -----
2. how fever presents in patients? -----
3. how dengue fever is transmitted? -----
4. the stages at which to conduct tests for dengue fever? -----
5. monitoring and assessing a patient's condition? -----
6. what advice to give patients to avoid mosquito bites? -----
7. advice for patients regarding medication? -----

Questions 8-14

Answer each of the questions, **8-14**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8. How long after being bitten by an infected mosquito does high fever occur?

9. What might patients with dengue fever complain of?

10. Which test should only be ordered 5 days after symptoms appear?

11. What other test is also useful when checking for dengue fever?

12. Who is at risk of seizures during the febrile stage of dengue?

13. What takes places in the most lethal cases of dengue?

14. How long does the most serious stage of dengue last?

Questions 15-20

Complete each of the sentences, **15-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15. Dengue fever does not spread -----

16. In many ----- dengue infections cause almost no symptoms.

17. Within three days of symptoms beginning a PCR or -----
----- can be ordered.

18. To avoid haemorrhage patients mustn't take anti-inflammatory medications or

19. Advise patients be cared for by someone at home in -----
----- accommodation.

20. Patients must be made aware of the need to check their -----
-----.

READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this Question Paper until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this Question Paper.

DO NOT remove **OET** material from the test room.

HOW TO ANSWER THE QUESTIONS

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:



In this part of the test, there are six short extracts relating to the work of health professionals. For questions 1-6, choose the answer (**A, B or C**) which you think fits best according to the text.

1. Why is epinephrine added to Lidocaine injections?

- Ⓐ To numb the area.
- Ⓑ To prolong the effects.
- Ⓒ To reduce patient discomfort.

Preparation of Injection

Lidocaine is a local anesthetic that is often injected subcutaneously before minor medical procedures such as laceration repair, excisional biopsy, and hormone implantation. A key step to prepare for this procedure is clearing a suitable workspace and obtaining any necessary supplies. First, be sure to check with your provider about the concentration and mixture of Lidocaine to be used. Epinephrine is often included to constrict local blood vessels for longer duration, but can increase the risk of causing ischemia in areas with poor blood supply (fingers, ears, toes). Sodium bicarbonate can also be added to avoid pain during injection due to Lidocaine's acidic pH. Be sure to obtain the proper sized needle and syringe, which will be dependent on the location of the injection and the size of the area requiring anesthesia, respectively.

2. The policy document on collateral information offers advice to staff about how to

- Ⓐ Gather information from colleagues about specific patients.
- Ⓑ Collect information about patients from their friends and relatives.
- Ⓒ Inform patients and their carers about recent diagnoses over the phone.

Policy Reminder: Collecting Collateral Information

Collateral information is an important factor in determining appropriate disposition for psychiatric patients in the Emergency Department. Often, patients with psychiatric complaints are unable to accurately or thoroughly describe their medical history, baseline condition, or events leading up to their arrival at the hospital. Thus, it becomes imperative to contact those who might know the patient best or were in the patient's company prior to their arrival. Contact information can be obtained from the patient themselves, persons accompanying the patient, or the medical record. When initiating contact, confirm the other person's identity before revealing the patient's name or the reason you are speaking with them. If you reach voicemail and the answering machine does not clearly identify the person you are looking for, do not reveal any information about the patient – simply state your name, number, position, and whom you are requesting a callback from.

3. When dealing with patients with symptoms of peripheral arterial disease, staff should

- Ⓐ Look for signs of swelling in the upper body.
- Ⓑ Confirm that the patient has a history of poor diet.
- Ⓒ Identify the cause through physical examination and tests.

Assessing and Managing Peripheral Arterial Disease

Staff should assess patients who have symptoms suggestive of peripheral arterial disease or diabetes with non-healing wounds for the presence of peripheral arterial disease. Ask about the presence of intermittent claudication and critical limb ischaemia. Examine the lower limbs for evidence of critical limb ischaemia. Imaging is possible for patients with peripheral arterial disease: duplex ultrasound is the first-line imaging technique. If patients require additional imaging, contrast-enhanced magnetic resonance angiography is used. If this is contraindicated or not possible, use computed tomography angiography instead. Lifestyle changes are the first-line treatment for peripheral arterial disease, this includes: smoking cessation, better control of diabetes, better management of hypertension, management of high cholesterol, in combination with antiplatelet drugs. Finally, regular exercise has shown to beneficially revascularise tissues in those with claudication. Examine pulses in the lower limbs: femoral, popliteal and feet. Measure the ankle brachial pressure index.

4. The guidelines on alcohol withdrawal treatment informs healthcare professionals about

- Ⓐ Determining the quantity of medication required.
- Ⓑ Reducing the dosage as the symptoms improve.
- Ⓒ Various types of drugs to prescribe to patients.

Guidelines: Alcohol Withdrawal Treatment

Alcohol withdrawal can present as a life-threatening emergency and requires treatment at a hospital. Providers use algorithms to determine when and how much medication to administer for a safe and optimal outcome. A key component of this assessment is determining the severity of alcohol withdrawal using the Clinical Institute Withdrawal Assessment for Alcohol (CIWAAr). The scale contains 10 subjective and objective items that can be queried and scored in minutes. Symptoms asked about include nausea, vomiting, tremors, sweating, anxiety, agitation, tactile/auditory/visual disturbances, headache, and cognitive dysfunction. Every hospital has different cutoffs for treatment, but as a general rule, treatment with benzodiazepines begin starting at a score 8–10, with higher scoring indicating increasing amount and frequency of medication.

5. The memo reminds all staff to avoid

- Ⓐ Identify the cause of the agitation.
- Ⓑ Handling grievances of a sensitive nature.
- Ⓒ Recording complaints that is not legitimate.

For the attention of all staff:

RE: AGITATED PATIENTS

Agitated patients are a common occurrence in the Emergency Department. There are many reasons for agitation, ranging from medical conditions, substance intoxication, psychiatric illness, and distressing circumstances. While both physical and chemical restraints are available to providers, these are items of last resort as their use creates significant risk to the patient, staff, and other persons in the area. Verbal de-escalation is a proven, effective technique that can be used to calm a patient down and promote a safe treatment environment. When de-escalating, designate one person to speak for the group. Agitated patients can be easily confused by multiple speakers and a unified message must be presented. Respect personal space to prevent the patient from feeling 'trapped' and maintain sufficient distance to avoid any resultant physical aggression. Remember to introduce yourself and your role on the treatment team to the patient. Use their name and orient them to their surroundings and why they are here in the hospital

6. The guidelines advise that patients with heart problems

- Ⓐ May need to avoid ibuprofen.
- Ⓑ Should be given lidocaine for pain relief.
- Ⓒ Must receive a lower dose of acetaminophen.

Extract from Appropriate Treatment for Pain

Pain is one of the most common complaints that will be brought to a physician's attention. This section will cover treatment of mild to moderate pain without the use of opioids. More severe pain may require judicious use of short-acting opioid medications or a consult to pain medicine. For most patients, the first line medications for pain are acetaminophen and ibuprofen. Maximum daily dosage of acetaminophen is suggested to be 4 grams, reduced to under 2 grams for patients with liver issues such as a cirrhosis. Ibuprofen is particularly effective in patients whose pain is caused by inflammation, though caution is urged in elderly patients, patients with diagnosed bleeding issues (especially gastrointestinal bleeds), or any cardiac issues. Maximum daily dosage suggested is 2.4 grams. A combination of acetaminophen and ibuprofen can be used if either one used alone is not sufficient. For more localised pain relief, consider using lidocaine dermal patches over non-broken areas of skin.

In this part of the test, there are two texts about different aspects of healthcare. For questions 7-22, choose the answer (**A, B, C or D**) which you think fits best according to the text.

Text 1: Delivering Serious News

Delivering serious news to patients and relatives: its many healthcare professionals' most dreaded task. Unfortunately, it's not something that can be avoided, and it's something that must be done right. Patients and relatives need our guidance and support, particularly when the prognosis is serious. In this article, we use the phrase '**serious news**' or 'life-altering news' rather than choosing a term with negative connotations, such as 'bad news', for example, as it helps to reframe the discussion. If you discuss 'serious news' with a patient, they can decide how to respond, whereas giving a patient 'bad news', may prevent them from being able to accept the news in a more constructive light. Studies show the vast majority of patients would prefer to be informed of a life altering diagnosis, rather than remain in ignorance. However, the amount of information they wish to receive can vary, with most wanting to know details concerning the different treatment options, and the effectiveness of proposed treatments, while they may want to hear less about the specific details of their prognosis. According to statistics, in western cultures, the majority of patients may not wish to know certain details, such as life expectancy. Healthcare workers may also find families asking that diagnoses be kept from the patient, or that patients prefer to have care wholly managed by their family, rather than themselves.

One model for delivering serious news is called SPIKES, developed by Walter Baile and initially used for discussions with cancer patients. The first step in SPIKES is setting up the interview. A quiet private area such as an exam room or family meeting room is an ideal setting. The patient should be able to choose family members or friends to be present for support. **For those** who don't speak fluent English, a hospital-contracted medical interpreter should be used. The healthcare professional should be prepared to answer difficult queries about prognosis, treatment, and overall plan going forward, but also know when to refer to a specialist for more esoteric information. If there is a multi-disciplinary approach, every team member should be on the same page with regards to the care plan to avoid confusion.

The second item in SPIKES is the patient's perception. Last week, I asked a patient, let's call him Harry, if he understood his current condition. Of course, he said he did, but when he came to explain it to me, I saw that there were many gaps in his knowledge that needed to be addressed. A good way to assess the patient's understanding is to ask what the patient already knows about their condition and what they have been told so far. Make sure to assess the level of their understanding, as well as their awareness of the basic facts. This will allow you to assess their level of background knowledge, their current knowledge, and where to begin your own discussion.

The third item in SPIKES is the patient's invitation for discussion. Different patients desire different levels of information about their condition. Some of the more technical-minded or younger patients may want to know their diagnosis, prognosis, treatments, course of illness, etc. Others, including older patients, may simply wish to know the diagnosis and accept the recommendations of the treatment team as being in their best interests. Before beginning to discuss their condition, you might find it helpful to ask "Would you like me to discuss all the information we know about your condition or just certain parts? What would you like us to tell your family?"

The fourth item in SPIKES is giving knowledge to the patient. You should be direct, but avoid being unfeeling or blunt when you discuss their condition, and utilise non-technical terms in small chunks. Prognosis and course of illness should be realistic, but also convey hope and planning for the future. An appropriate opening for our patient would be, "I'm afraid, we have some serious news about the CT scan that was performed. It showed that the cancer in your liver has spread to your spine." Take note of how the words 'hepatocellular carcinoma' and 'metastasis' were rephrased into layman's terms.

The fifth item In SPIKES is addressing the patient's emotions. You should identify the emotion the patient is experiencing, the reasoning, and provide support during this difficult time. Don't try to change the patient's emotions, just help them to express how they feel. For example, in a patient who is dysphoric and crying, you can offer a tissue box and physical support if appropriate. You might say something like, "I know these results weren't what you wanted to hear. I wish we had better news for you." Other responses can range from asking the patient to elaborate on

their reaction, “Can you tell me what you’re worried about?” to validating their concerns, “I can understand why you felt that way. Many of her patients have had similar reactions.”

The sixth item in SPIKES is strategy and summary. Patients who receive serious news will often feel that they are in over their head, so you should make sure that they leave with a clear plan for the future. This will help them to feel less anxious and more hopeful. Patients should know what options are available for them and what follow-up is planned. You should also recheck that they understand what has just been discussed and have had all their questions answered. A good opening statement could be, “I understand this is a lot to take in, but you have several options available. A decision does not need to be made now, but we would like to refer you to an oncologist and follow-up with us in a week to discuss your next steps.” Giving serious news is one of the most difficult parts of being a healthcare professional. However, with careful planning and an effective protocol, patients can leave feeling well-informed and in control of their own outcome.

Text 1: Questions 7-14

7. Why does the writer prefer the term **'serious news'**?

- Ⓐ It enables doctors to avoid unnecessary conversations.
- Ⓑ It avoids influencing the patient's emotional response.
- Ⓒ It helps patients to better understand their condition.
- Ⓓ It offers a more specific definition of the information.

8. The writer's purpose in the second paragraph is to highlight

- Ⓐ The treatment options available to most patients.
- Ⓑ The difficulty of knowing what a patient wants to be told.
- Ⓒ The trends concerning what patients and relatives want to hear.
- Ⓓ The different topics that healthcare workers should cover with patients.

9. What does the word **'those'** refer to?

- Ⓐ Healthcare staff
- Ⓑ Treatment experts
- Ⓒ Language translators
- Ⓓ Patients and relatives

10. In the fourth paragraph, the writer mentions the patient, Harry, in order to explain that

- Ⓐ Patients are often reluctant to ask for help.
- Ⓑ Patients may not be aware of their ignorance.
- Ⓒ Healthcare professionals often find it hard to relate to patients.
- Ⓓ Healthcare professionals may not always explain things effectively.

11. The writer suggests that older patients may be more likely to

- Ⓐ Require more information.
- Ⓑ Limit their family's involvement.
- Ⓒ Accept the staff's suggested plan.
- Ⓓ Inquire further about their treatment plans.

12. In the sixth paragraph, the writer offers an example to emphasise that when explaining information professionals should

- Ⓐ Avoid using complex medical language.
- Ⓑ Prevent patients from becoming upset.
- Ⓒ Discuss how the illness was identified.
- Ⓓ Repeat information multiple times.

13. The seventh paragraph focuses on

- Ⓐ Ensuring the patient understands how to react.
- Ⓑ Helping the patient to feel more positive.
- Ⓒ Comparing different patient responses
- Ⓓ Empathising with the patient's reaction.

14. The expression 'in over their head' is used to stress that patients might

- Ⓐ Find the information overwhelming.
- Ⓑ Struggle to remember information.
- Ⓒ Make a choice about their treatment quickly.
- Ⓓ Have difficulty understanding their prognosis.

TEXT 2: TREATING OPIUM ADDICTION

In the United States alone, there are around 115 deaths caused by opioid addiction every day. The addiction impacts individuals rapidly and drastically, damages families, and costs the US huge amounts of money: the total economic burden of prescription opioid abuse is estimated to be \$78.5 billion a year, while the economic burden of non-prescription opioid abuse simply cannot be calculated. Measures are constantly being improved to prevent patients from developing opioid addictions to begin with, but it is also imperative that we continue to provide treatment for those already in the thrall of opioid addiction.

Jane's story is one heard over and over again in opioid addiction clinics. When she was 20, she had a bad automobile accident that required two surgeries. She was soon home from the hospital but her residual pain meant she was prescribed scheduled opiates. Jane's body soon became tolerant of the dosage; however, and she needed higher and higher doses in order to achieve the same pain relieving effect. She eventually reached a level that her physician felt uncomfortable prescribing. Unable to find another prescriber in time, Jane turned to alternative sources of narcotics. Unfortunately, when purchased on the street, these pills are exorbitantly expensive and increasingly hard to come by in an era of prescription monitoring throughout the United States. Heroin is much cheaper and, when delivered by IV, produces a much more potent high and greater pain relief.

Eventually, after destroying relationships with her loved ones, bankrupting her savings, and **hitting rock-bottom**, Jane turned to a local opioid addiction clinic for help. At the clinic, they put her on Methadone, a long-acting opioid agonist that is standard for addiction treatment. It binds to the mu-opioid receptors, prevents withdrawal symptoms, reduces cravings, and can also provide a level of pain relief. Of course, as an opioid agonist, methadone serves as a substitute for the primary addiction, meaning many of the issues associated with long-term opioid usage remain. Patients must often begin treatment with daily visits, which can be disruptive. Fortunately for Jane, these visits are her first steps towards putting her life back together. As Jane's road to recovery is likely to be long and fraught with difficulty, many doctors are led to wonder: does she have any other options?

One of the increasingly popular alternatives to methadone is buprenorphine, a partial mu-opioid agonist. Aside from its unique mechanism of action (MOA), there are two major differences when compared to Methadone: first, it can be administered as oral tablets, sublingual/buccal films, and a long-acting implant, second, It can be prescribed month-to-month from a clinician's office directly to a local pharmacy. These factors make it much easier to use in the community, and are ideal for patients who cannot visit a methadone clinic every day.

To initiate buprenorphine, a patient must already be in a mild state of withdrawal due to the high affinity for the mu-opioid receptor displacing other opioids. **This means that** patients generally transition best from a short-acting opioid like heroin or oxycodone rather than a long-acting opioid agonist like Methadone, given the length of time needed until mild withdrawal occurs. As Jane had been using opioids for a long time prior to her admission, however, she was better suited to treatment with Methadone, as there is no ceiling effect to this drug, and Jane had developed a high tolerance to opioids. Buprenorphine, being a partial agonist, has a maximum level of effect which it cannot be increased beyond. For this reason, buprenorphine can be used as a maintenance therapy in some patients, but it can also be tapered down over time. This allows patients to resume their normal lives with minimal interruptions and avoid relapse through pharmacological blocking.

Alongside treatment with medication, patients recovering from opioid addiction must also deal with recovery at a mental level. As with many healing processes, the first stage is acceptance. Jane was not able to seek the treatment she needed until she had nowhere else to hide. Once everything was lost, she couldn't deny that she was in trouble anymore, so she came to the clinic. Many patients suffering from opioid addictions are reluctant to admit that they are addicted, and reluctant to ask for help. Patients are often worried about being judged, being treated like a criminal, and meeting with disapproval from the healthcare professionals who must treat them.

When patients do seek aid, healthcare professionals need to help them to build a support network around themselves, so that they are protected when they feel the need to relapse. Opioid addicts are likely to have burned bridges with friends and family who have not enabled their addiction, so patients beginning recovery may not have positive role models to support and influence their recovery. Talking therapies,

such as cognitive behavioural therapy (CBT) can be offered to recovering patients experiencing anxiety or depression, though patients may find it more useful to join local confidential support groups, such as Narcotics Anonymous, as they can discuss recovery with those who have first-hand experience. Though Jane was hesitant to discuss her experiences with anyone when she was first admitted to the clinic for treatment, she has since gone on to attend weekly sessions at Narcotics Anonymous, where she not only listens to others share their stories of recovery, but where she also is beginning to tell her own.

Text 2: Questions 15-22

15. In the first paragraph, the writer highlights that opioid addiction in the US

- Ⓐ Has been gradually increasing for a number of years.
- Ⓑ Is largely influenced by the illegal sale of drugs.
- Ⓒ Causes more deaths than any other addiction.
- Ⓓ Has a significant financial and social impact.

16. In the second paragraph, the writer outlines Jane's case in order to emphasise that

- Ⓐ Opioid addiction is increasingly rare.
- Ⓑ It can be remarkably easy for a patient to become addicted.
- Ⓒ In some cases, heroin is less harmful to addicts than opioids.
- Ⓓ Healthcare professionals must take responsibility for opioid addiction.

17. The writer uses the phrase **'hitting rock bottom'** about the patient Jane in order to describe

- Ⓐ How her addiction led to the most distressing point in her life.
- Ⓑ Her sudden awareness that she had to recover.
- Ⓒ The large tolerance she developed for opioids.
- Ⓓ The physical pain she felt at that time.

18. In the fourth paragraph, the writer suggests that buprenorphine may be preferable because

- Ⓐ It is less addictive than alternatives.
- Ⓑ It can be easier for patients to access.
- Ⓒ It does not interfere with other treatments.
- Ⓓ It can be picked up more often than other medications.

19. What does '**this means that**' refer to?

- Ⓐ The effectiveness of buprenorphine when combating opioid displacement.
- Ⓑ The requirement for the medication to be reserved for heroin addicts.
- Ⓒ The need for patients to have begun to experience withdrawals.
- Ⓓ The impact of mu-opioids on recovered opioid addicts.

20. In the fifth paragraph, the writer suggests that Jane was prescribed methadone, rather than buprenorphine because

- Ⓐ Buprenorphine is too similar to heroin.
- Ⓑ The effects of methadone last for longer.
- Ⓒ She was dependent on high doses of opioids.
- Ⓓ It is more readily available at addiction clinics.

21. According to the seventh paragraph, why do patients often delay seeking treatment for opioid addiction?

- Ⓐ They are unwilling to face the damage they have caused.
- Ⓑ They do not realise they are addicted until it's too late.
- Ⓒ They think that they can recover without help.
- Ⓓ They do not want to be labelled as an addict.

22. In the final paragraph, the writer suggests that recovering addicts may prefer to discuss their experiences with

- Ⓐ Those who have experienced addiction.
- Ⓑ People who are not aware of their history.
- Ⓒ Healthcare professionals.
- Ⓓ Friends and family.

READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT remove OET material from the test room.

TEXT A

History

Hashimoto's thyroiditis is an autoimmune condition in which the body perceives its own tissue as foreign. It is the leading cause of hypothyroidism (underactive thyroid) in the Western World. Common, early presenting symptoms of hypothyroidism, such as fatigue, constipation, dry skin, and weight gain, are nonspecific.

Physical Examination

Physical findings are variable and depend on the extent of hypothyroidism and other factors such as age. Findings include the following:

- Puffy face
- Cold, dry skin, which may be rough and scaly - Skin may appear yellow but does not involve the sclera, which distinguishes it from the yellowing of jaundice due to hypercarotenemia
- Peripheral oedema of hands and feet, typically non-pitting
- Thickened and brittle nails (may appear ridged)
- Hair loss involving the scalp, the lateral third of the eyebrows, and possibly skin, and facial hair
- Elevated blood pressure (typically diastolic hypertension) - Most often, blood pressure is normal or even low
- Diminished deep tendon reflexes and the classic prolonged relaxation phase, most notable and initially described at the Achilles tendon (although it may be present in other deep tendon reflexes as well)
- The thyroid gland is typically enlarged, firm, and rubbery, without any tenderness or bruit; it may be normal in size or not palpable at all.
- Voice hoarseness
- Slow speech
- Impairment in memory function

TEXT B

Testing Recommendations

Serum TSH Test

In the presence of suggestive symptoms and physical findings, a serum TSH (thyroid stimulating hormone) test is needed for the diagnosis of primary hypothyroidism, and it serves to assess the functional status of the thyroid. This should be followed up periodically to monitor for symptoms of hypothyroidism and to detect any rise in TSH or cholesterol levels. Checks can usually be performed every 6-12 months.

Free T4 test

A free T4 is usually needed to correctly interpret the TSH in some clinical settings.

T3 test

A low T3 level and a high reverse T3 level may be of additional help in the diagnosis of nonthyroidal illness.

Ultrasonography

This is useful for assessing thyroid size, echotexture, and, most importantly, whether thyroid nodules are present. Ultrasonographic study aids in confirming the presence of a thyroid nodule, in defining a nodule as solid or cystic, and in defining features suggestive of malignancy, such as irregular margins, a poorly defined halo, microcalcification, and increased vascularity on Doppler interrogation.

TEXT C

Thyroid levels chart

Condition	TSH	T4	T3	Notes
Lab error	High	High	High	Can also occur with TSH-secreting tumour (very rare)
Hashimotos's thyroiditis	High	Low	Low	Thyroid peroxidase antibodies high in 90 % of Hashimotos's
Sub-clinical (mild)	High	Normal	Normal	May also occur in patients with hypothyroidism not receiving adequate thyroxine replacement.

For treatment of Hashimoto's thyroiditis synthetic T4 or thyroxine (levothyroxine) should be prescribed at the correct dosage.

Average full replacement dose: 1.7 mcg/ kg/day (e.g, 100 to 125 mcg/day for 70 kg adult) orally.

Older patients may require less than 1 mcg / kg/day.

Doses greater than 200 mcg / day orally are seldom required.

TEXT D

For treatment of Hashimotos's thyroiditis synthetic T4 or thyroxine (levothyroxine) should be prescribed at the correct dosage.

Average full replacement dose: 1.7 mcg/kg/day (e.g., 100 to 125 mcg /day for a 70kg adult orally.

Older patients may require less than 1 mcg/kg/day.

Doses greater than 200mcg / day orally are seldom required

Because of the generally permanent and often progressive nature of Hashimoto's thyroiditis, it is usually necessary for treatment to continue throughout the patient's lifetime. Medicine dose requirement may have to be adjusted periodically.

READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

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Write your answers on the spaces provided on this **Question Paper**.

You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this Question Paper and the Text Booklet.

DO NOT remove **OET** material from the test room.

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate Text **Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

VITAMIN C: Questions

Questions 1-7

For each of the questions, 1-7, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1. signs that indicate characteristics of hypothyroidism? -----
2. the use of ultrasound scans when assessing thyroid function? -----
3. reference ranges for thyroid function tests? -----
4. hormones which determine the status of a patient's thyroid? -----
5. treatment options for Hashimoto's thyroiditis? -----
6. different types of thyroid conditions? -----
7. dosage rates for thyroid replacement medication? -----

Questions 8-14

Answer each of the questions, **8-14**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8. What can the face of a patient suffering with Hashimoto's thyroiditis look like?

9. If a Hashimoto's patient has high blood pressure, what does it usually indicate?

10. Which test is often required to understand TSH results?

11. How often should the TSH level be checked if hypothyroidism is present?

12. What medication is used to treat Hashimoto's thyroiditis?

13. How long will treatment for Hashimoto's thyroiditis typically last?

14. What can occur if a patient isn't receiving enough medication?

Questions 15-20

Complete each of the sentences, **15-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15. A patient's ----- can affect the type of symptoms they display.

16. Nonthyroidal illness can be detected via a -----

17. Check for a ----- if a patient returns a high result in all tests.

18. Irregular margins in a patient with nodules may indicate -----

19. The average medication dose for a 70kg adult is -----
----- a day.

20. The amount of medication a patient receives may need to be -----
----- from time to time.

READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this Question Paper until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this Question Paper.

DO NOT remove **OET** material from the test room.

HOW TO ANSWER THE QUESTIONS

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:



In this part of the test, there are six short extracts relating to the work of health professionals. For questions 1-6, choose the answer (**A, B or C**) which you think fits best according to the text.

1. The instructions state that disinfectant should be used

- Ⓐ Via the process of mechanical cleaning.
- Ⓑ In combination with physical cleaning.
- Ⓒ Only at the time of manual cleaning.

Routine environmental cleaning

Use of disinfectants

In acute –care settings where there is uncertainty about the nature of soiling on the surface (e.g. blood or body fluid contamination versus routine dust or dirt) or the presence of MROs (including *C.difficile*) or other infectious agents requiring transmission –based precautions (e.g. pulmonary tuberculosis) is known or suspected, surfaces should be physically cleaned with a detergent solution, followed or combined with a TGA-registered disinfectant.

This process must involve either

- A physical clean using detergent followed by a chemical disinfectant (2-step clean) i.e. clean with detergent and chemical disinfectant; or
- Physical clean using a detergent and chemical disinfectant (2-in-clean) i.e. a combined detergent/ disinfectant wipe or solution could be used if this process involves mechanical/manual cleaning.
- Physical (manual or mechanical) cleaning is the most important step in cleaning. Sole reliance on a disinfectant without mechanical/manual cleaning is therefore not recommended.

2. The email informs physiotherapists that

- Ⓐ Advise them on the importance of clear documentation.
- Ⓑ Warn them about the dangers of incorrect diagnoses.
- Ⓒ Inform staff of their obligations to a patient's family.

To: All staff

Subject: Advisory Email: Doctor found guilty in duty of care case

Recently, a jury returned a verdict of \$ 7.5 million against a pediatrician who neglected to tell a mother or other doctors that her daughter had a dangerous heart condition which later killed her.

The circumstances surrounding this incident highlight a breach of a doctor's duty of care to his or her patient where there was an omission to maintain an accurate record of treatment provided to the patient and subsequent failure to diagnose a dangerous cardiac condition.

It is therefore timely to remind all staff that inaccuracies or improper record keeping may have devastating consequences to the patient and all concerned in the care of the patient.

3. The guidelines are providing information about

- Ⓐ When a basal test should be conducted.
- Ⓑ The steps for conducting a basal test.
- Ⓒ Who should perform a basal test.

Management of adults using insulin Pump Therapy

In preparation for surgery, it is necessary to establish a consistent stable blood glucose concentration in the fated state, requiring a 'basal test' to be performed. The early involvement of the diabetes team in people on an insulin pump preoperatively, is paramount to facilitate this optimization of glycaemic management. The overnight basal assessment can be performed at least a few days or weeks before surgery, to allow sufficient time for input from the diabetes specialist, if necessary, to adjust and confirm the new settings by repeating the basal assessment. The timing of the assessment can be extended to cover the proposed timings of surgery. For example, if surgery is planned for 9am then the basal assessment could cover the overnight period, up until late morning.

4. The safety update states that during the past 12 months

- Ⓐ At least 1,000 errors were reported.
- Ⓑ An average of 3.1 errors was reported.
- Ⓒ More errors than last year were reported.

Medication Safety Update

Medication given to patients in hospital can be the most important part of treatment. However, medication is not without risk and occasionally medications can cause harm. Harm associated with medication remains the second most common type of incident in hospital, and some harm caused by medicines is due to errors that are preventable.

We aim to prevent harm by:

- Understanding what contributes to these errors
- Taking action
- Sharing this information with the community and health professionals

During the past 12 months there were 3.1 reported medication errors per 1,000 bed days, an increase from the previous 12 months, attributable mainly to an increase in hospital staff reporting medication incidents, in line with an increase in reported incidents is continually encouraged, and all staff are encouraged to report all incidents in order to help identify way to improve medication safety.

5. The policy extract states that doctors and specialists who visit the hospital must

- Ⓐ Wait for the emergency officer to provide advice on safety procedures.
- Ⓑ Follow the instructions of the area warden or emergency officer.
- Ⓒ Listen to the emergency officer until the area warden arrives

Emergency Response Procedures

We are committed to ensuring a safe environment and appropriate emergency preparedness for our visiting doctors.

In the event of an internal emergency, the person in charge has legal authority to direct all parties. Initially, the person in charge will be the area warden who will be identified by a yellow hat or vest. (once the emergency officer arrives on the scene they will support the area warden. The emergency officer is identified by a white hat or vest.)

Everybody on site, no matter how senior, is legally required to follow the directions of the area warden or emergency officer. Therefore, the primary role of visiting specialists in an internal emergency is to follow the instructions of those in charge.

6. This email suggests some staff have been looking at patient records

- Ⓐ Outside of normal work hours.
- Ⓑ When they didn't relate to them.
- Ⓒ Without their manager's approval.

To: All staff

Subject: important: New patient privacy safeguards

Ensuring the privacy of our patient is crucial, which is why we have taken steps to strengthen protection of patient health data.

Accessing patient records for reasons unrelated to your job is a violation of our privacy policies. Staff should only access patient information necessary to properly do their jobs. To better identify any inappropriate access of patient data, we have installed sophisticated new monitoring software.

The new software will use patient data, human resources data, and artificial intelligence to detect suspicious activity or unusual patterns in the health system's electronic health records. Such activity will trigger an alert, which will be sent to the Privacy Office who will then investigate the matter further. Staff found to have inappropriately accessed a patient record or other data may be disciplined up to and including termination.

In this part of the test, there are two texts about different aspects of healthcare. For questions 7-22, choose the answer (**A, B, C or D**) which you think fits best according to the text.

Text 1: Coffee

People who drink three to four cups of coffee a day are more likely to see health benefits than problems, experiencing lower risks of premature death and heart disease than those who abstain, scientists have said. The research also found coffee consumption was linked to lower risks of type-2 diabetes, dementia, and several cancers, including prostate, skin and liver cancer. The greatest benefit was seen for liver conditions such as cirrhosis of the liver. The largest reduction in relative risk of premature death is seen in people consuming three cups a day, compared with non-coffee drinkers. Drinking more than three cups a day was not linked to harm, but the beneficial effects were less pronounced. To better understand its effects on health, Robin Poole, a public health specialist at Britain's University of Southampton, led a research team in review of 201 studies based on observational research and 17 studies based on clinical trial across all countries and all settings. "Coffee drinking appears safe within usual patterns of consumption", Pool's team concluded in their research, published recently in the British Medical Journal.

In a linked editorial, Professor Eliseo Guallar from the John Hopkins Bloomberg School of Public Health In Maryland wrote that "coffee is safe, but hold the cake". He argued that the latest study showed that "coffee consumption seems generally safe", but added: "Coffee is often consumed with products rich in refined sugars and unhealthy fats, and these may independently contribute to adverse health outcomes". Does coffee to chronic disease and reduce mortality? We simply do not know. Should doctors recommend drinking coffee to prevent disease? Should people start drinking coffee for health reasons? The answer to both questions is no." Poole's team noted that because their review included mainly observational data, no firm conclusions could be drawn about cause and effect. But they said their findings support other recent reviews and studies of coffee intake.

Italian coffee scientist Dr Luciano Navarini seems to agree. "I believe that coffee can be a dietary bad guy only if it is consumed in high doses, when caffeine is present,"

he said. “But as far as I know, healthy adult coffee drinkers normally keep to a fixed number of cups, and they exceed that amount only in very special situations, when it is necessary to stay alert for some reason. Scientific literature indicates that moderate coffee consumption seems to be a good habit rather than a dangerous vice.” Navarini works for Italian coffee giant Illycaffè, so perhaps he may be expected to say that. Increasingly, however, independent as well as industry – aligned scientists are crowding into coffee research.

Professor Elvira Gonzalez de Mejia of Illinois University is one of the world’s leading experts on coffee and health. In a 2014 overview of the subject, she stated that available data supported “the view that habitual coffee consumption has several health benefits, including lower risks of Parkinson’s and Alzheimer’s disease, a favourable effect on liver function, a possible role in weight loss, and a decreased risk of developing certain cancers”. She added that the evidence for coffee in managing type 2 diabetes and cardiovascular disease was **“largely reassuring”** and that consumption of two to four cups a day “is not associated with adverse effects”.

Professor de Mejia also stressed the need for further research. “More consistent human studies are needed” she said. “Standardized coffee samples must be prepared and tested in human studies. The reproducibility of such studies will certainly help answer questions about coffee consumption”. She also sounded a note of caution regarding possible outcomes of the barrage of ongoing research into the therapeutic potential of coffee’s myriad components. Isolating individual compounds and fashioning them into health products- as is currently happening in the medical marijuana industry, for instance- could lead to problems. “it is better to recommend ‘whole foods’ rather than isolated compounds”, she said. “There is always the risk of using mega-doses of individual compounds, which may bring some risks”

Dr Simon Drew, of the Florey Department of Neuroscience and Mental Health at the University of Melbourne, takes a different view. Recently he worked with Navarini and others on a study into how roasting coffee beans changed their antioxidant structure. “Willow bark was used as an analgesic for millennia before

scientists finally extracted salicin from it, discovery that led to aspirin,” he said. “Likewise, something beneficial may be hiding within your daily brew.

Drew’s research is not aimed directly at investigating health claims made for coffee. Instead, he and Navarini set out to map exactly how antioxidant types and levels were affected by various roasting, storage and brewing techniques. The results may assist other scientists trying to understand how coffee-derived antioxidants behave in the body. Like de Mejia, Drew is cautiously optimistic about coffee’s health potential, but is reserving judgement until more evidence comes to light. “In terms of antioxidant intake alone, the jury is probably still out” he said. “But there’s much we don’t know about other potential benefits. Coffee is a melting pot of chemical compounds and roasting process leads to many new ones”.

Text 1: Questions 7-14

7. In the first paragraph, what do we learn about drinking coffee?

- Ⓐ Having more than four cups a day can be harmful.
- Ⓑ It is healthier than scientists had previously believed.
- Ⓒ Beneficial effects reduce if you drink four or more cups a day.
- Ⓓ People who drink coffee usually live longer than those who don't.

8. What concern does Professor Eliseo Guallar have about coffee consumption?

- Ⓐ No clear conclusions can be made about the benefits of drinking coffee.
- Ⓑ There may be health risks attributable to products taken with coffee.
- Ⓒ Not enough doctors are recommending coffee's health benefits.
- Ⓓ There haven't been enough studies conducted about coffee.

9. In the second paragraph, the word 'they' refers to

- Ⓐ Observational data.
- Ⓑ People drinking coffee.
- Ⓒ Recent reviews.
- Ⓓ Poole's team.

10. Dr Luciano Navarini's views could be considered controversial because he

- Ⓐ Agrees with Eliseo Guallar.
- Ⓑ Is employed by a coffee producer.
- Ⓒ Thinks coffee is good for you.
- Ⓓ Suggests people drink more coffee when tired.

11. The phrase '**largely reassuring**' in the fourth paragraph expresses the idea that

- Ⓐ There is an important link between some diseases and regular coffee consumption.
- Ⓑ Drinking coffee daily appeared beneficial for certain types of diseases.
- Ⓒ There are no negative effects associated with drinking coffee.
- Ⓓ Routinely drinking coffee makes people feel better.

12. What does Elvira Gonzalez de Mejia suggest about ongoing research?

- Ⓐ Separating coffee's components for its potential healing benefits could be harmful.
- Ⓑ There are too many studies taking place, which is causing problems in the industry.
- Ⓒ It's better to have coffee as a whole food in itself, but to avoid drinking too much.
- Ⓓ There are similarities between research into coffee and medical marijuana.

13. In the sixth paragraph, Dr Simon Drew indicates that

- Ⓐ may contain components that can be used medicinally.
- Ⓑ could contain similar properties to those found in aspirin.
- Ⓒ is beneficial for people who suffer from headaches.
- Ⓓ produces antioxidants when the beans are roasted.

14. The writer uses Dr Drew's comments in the final paragraph to express the idea that

- Ⓐ there currently isn't enough optimism about coffee.
- Ⓑ brewing coffee doesn't produce enough beneficial chemicals.
- Ⓒ roasting coffee produces too many chemical compounds.
- Ⓓ further research into coffee still needs to be done.

Text 2: Industry-funded Nutrition Research

With so much inaccurate information about diet and health out there, the last thing you would believe is misleading- and the one we have trust in most- are research studies. People rely on unbiased research to find out important statistics about all facets of nutrition, from sugar intake and food supplements, to genetically modified food and cereals. However, recent research suggests there is bias in industry-funded research studies, the full extent of which is still unknown. “We’re starting a whole program of work in this area. The first thing we did was a review of all studies that have looked at the association of industry sponsorship with the outcomes of nutrition studies,” said Lisa Bero, professor, chair of Medicines use and health outcomes and head of the bias node at the Charles Perkins at the University of Sydney.

Bero and her team reviewed 775 reports in the medical literature, narrowing down to 12 relevant reports, to determine whether nutrition studies funded by the food industry were “associated with outcomes favourable to the sponsor”. “It was a little surprising because most of the studies only looked at the conclusion of the research. By that I mean the author’s interpretation. So, if it were industry sponsored, they were more likely to have a conclusion that favoured the industry sponsor,” Bero said. “What we found is that only three of the studies looked at the actual results or data. That’s something we’re really interested in doing in the future.

Bero’s investigation has confirmed that researchers know little about the influence of corporate sponsors on nutrition studies. However, the research community **has been at odds** over the amount of damage funding bias is bringing to people’s understanding of food and health, and what should be done to stop it. On one end, experts have been highly critical of industry-funded studies and say that type of research should almost always be avoided. On the other end, researchers who accept corporate funds say funding is a very small part of larger issue, if even a issue at all.

Dr. Marion Nestle, nutrition and food studies professor at New York University, falls in the first group. “I worry a lot about the effects of industry sponsorship on public belief in the credibility of nutrition science,” says Nestle, the author of *Soda Politics: Taking on Big Soda (and Winning)*. “Just because a claim supposedly backed by

“clinical studies” doesn’t mean it can be trusted. Even if the research is scientifically sound” Nestle said. “Ultimately the basis for many corporate – sponsored researches is marketing, not just public health. And if there is no scientific basis for the research, companies can make one up”

Dr. David Katz, nutritionist and the founding director of Yale University’s Prevention Research Center, says there is a difference between conflict of interest and confluence of interest (when funders’ vested interests are in line with public’s interest). According to him, it’s a mistake for people to just assume that corporate bias always falls under the former: Instead of the rush to judgement, nutritionists should focus on raising industry standards and improving elusive nutrition research. “I think that’s what we should be working on”, Katz said. “Where do we draw the line? What are the things that are required to avoid conflict of interest, and to ensure that research is reliable, objective, impartial, and responsible no matter who funds it?” An all-out ban on corporations in research would leave people in need, says Katz, who uses the impact of corporate-led research on the pharmaceutical industry as an example. “Without corporations, many common medicines and pharmaceutical products people use, such as antibiotics and cancer-treating drugs, wouldn’t exist. If food companies want to use research to make a better product and improve consumers’ health at the same time, we should let them.”

Advocates of industry funding, like Dr Andrew Brown, a scientist and researcher at the Nutrition and Obesity Research Center & office of energetic at the University Of Alabama at Birmingham, argue that corporate funding is not really the issue on which to focus. “We like to say that with science, there’s only three things that matter: the data, the way the data were collected, and the logic connecting the data to the conclusions,” Brown said. Brown echoes Katz in saying there are bigger fish to fry in nutrition research. Instead of immediately labeling corporate-sponsored studies as problematic, Brown argues critics should tackle the larger issue of conflict of interest in nutrition research, finding far-reaching ways to hold researchers accountable for their scientific methods, not who they allow to fund them.

Text 2: Questions 15-22

15. The writer says that research into nutrition and health requires

- Ⓐ Further study.
- Ⓑ Innovative ideas.
- Ⓒ Additional resources.
- Ⓓ Greater independence.

16. In the second paragraph, what surprised Lisa Bero and her team when conducting their review?

- Ⓐ The positivity of the nutrition industry.
- Ⓑ The quality of the reports they had access to.
- Ⓒ The emphasis on opinions over facts.
- Ⓓ The dependence on industry sponsorship.

17. The phrase '**has been at odds**' is used to underline

- Ⓐ Mistrust around sponsorship.
- Ⓑ A desire to end scientific subsidies.
- Ⓒ Disagreement amongst researchers.
- Ⓓ Public misunderstanding about nutrition.

18. What does Dr Marion Nestle believe about nutrition and food studies?

- Ⓐ Companies sometimes fabricate their own research.
- Ⓑ The public doesn't trust most companies.
- Ⓒ A lot of company sponsorship comes from the drinks sector.
- Ⓓ Many companies use health as a way to advertise their products.

19. Dr David Katz expresses the opinion that when it comes to nutrition research

- Ⓐ More studies need to be undertaken.
- Ⓑ There are too many opinions confusing the issue.
- Ⓒ Nutritionists don't really know what a healthy diet looks like.
- Ⓓ It shouldn't matter who pays for it as long as it is in the public's interest.

20. What is Dr Katz worried about?

- Ⓐ The exclusion of companies from research at the expense of public health.
- Ⓑ The lack of development in many products used by the general public.
- Ⓒ The involvement of pharmaceutical companies in research areas.
- Ⓓ The declining quality of several types of medication.

21. In the final paragraph, Dr Andrew Brown makes the point that

- Ⓐ science is mainly interested in the process of gathering information.
- Ⓑ there aren't a lot of sensible solutions being offered at the moment.
- Ⓒ researchers should be made more responsible for their studies.
- Ⓓ opponents of industry-funded research aren't helping the issue.

22. Dr Brown uses the phrase '**bigger fish to fry**' to highlight

- Ⓐ There are more important issues than who pays for research.
- Ⓑ The scientific community and the business sector can't agree.
- Ⓒ Science isn't interested in the significance of any new findings.
- Ⓓ The problems associated with nutrition studies are getting worse.

READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT remove OET material from the test room.

TEXT A

Vitamin C Deficiency

Scurvy is a life-threatening condition due to dietary vitamin C deficiency. Those affected are mostly refugees or victims of famine, alcoholics, older people, fad dieters, or children with autism or idiosyncratic behavioural abnormalities. Diagnosis is often delayed due to incomplete review of dietary history.

Vitamin C deficiency may result from a diet deficient in fresh fruits and vegetables. Also, cooking can destroy some of the vitamin C in food.

The following conditions can significantly increase the body's requirements for vitamin C and the risk of vitamin C deficiency:

- Pregnancy
- Breastfeeding
- Disorders that cause a high fever or inflammation
- Diarrhoea that lasts a long time
- Surgery
- Burns
- Smoking, which increases the vitamin C requirement by 30%

TEXT B

The recommended daily intake of vitamin C varies by age, gender, pregnancy, lactation and smoking status.

Patient Group	Adequate intake or AI (milligrams/day)*		Tolerable upper intake level *** (milligrams/day)
0-6 months	40		not determined
7-12 months	50		not determined
	Recommended Daily Allowance (RDA)** (milligrams/day)		
1-3 years	15		400
4-8 years	25		650
	Men	Women	
9-13 years	45	45	1200
14-18 years	75 Smoker: add 35	Not pregnant: 65 Pregnant: 80 Lactating: 115 Smoker: add 35	1800
19 years and older	90 Smoker: add 35	Not pregnant: 75 Pregnant: 85 Lactating: 120 Smoker: add 35	2000

*Adequate intake is the mean intake of healthy breastfed children.

**RDA meets the needs of 97% to 98% of individuals.

*** Tolerable upper intake level is the maximum daily dose thought to pose no adverse risk.

TEXT C

Symptoms

The symptoms of scurvy develop only after a few months of deficiency.

Adults feel tired, weak, and irritable. They may lose weight and have vague muscle and joint aches.

Bleeding may occur under the skin (particularly around hair follicles or as bruises), around the gums, and into the joints. The gums become swollen, purple, and spongy. The teeth eventually loosen. The hair becomes dry and brittle, and the skin becomes dry, rough, and scaly. Fluid may accumulate in the legs. Anaemia may develop. Infections may develop, and wounds do not heal.

Infants may be irritable, have pain when they move, and lose their appetite. Infants do not gain weight as they normally do. In infants and children, bone growth is impaired and bleeding and anaemia may occur.

Normal examination has also been reported, presumably when symptoms have developed in the setting of very low but not critical body stores.

Examination

Although no consistent order of presenting signs is established, the earliest signs of scurvy are often gingival abnormalities and a comprehensive examination of the mouth when scurvy is recommended in patients presenting relatively early.

If the test is available, measuring the vitamin C level in blood can help establish diagnosis.

Blood tests to check for anaemia.

In children, x-rays to check for impaired bone growth.

Treatment

For scurvy in adults, ascorbic acid 100 to 500 mg orally twice daily must be given for 1 to 2 weeks, until signs disappear, followed by a nutritious diet supplying 1 to 2 times the daily recommended intake of fresh fruits and vegetables.

In scurvy, therapeutic doses of ascorbic acid restore the functions of vitamin C in a few days. The symptoms and signs usually disappear over 1 to 2 weeks. Chronic gingivitis with extensive subcutaneous haemorrhage persists longer.

TEXT D

Prior to discharge:

- Tell patients to drink plenty of fluids and get plenty of rest.
- Tell patients to take antipyretics to control their temperature. Children with dengue are at risk for febrile seizures during the febrile phase of illness.
- Warn patients to avoid aspirin and anti-inflammatory medications because they increase the risk of haemorrhage.
- Monitor your patients' hydration status during the febrile phase of illness. Educate patients and parents about the signs of dehydration and have them monitor their urine output.
- Assess hemodynamic status frequently by checking the patient's heart rate, capillary refill, pulse pressure, blood pressure, and urine output. If patients cannot tolerate fluids orally, they may need IV fluids.
- Perform hemodynamic assessments, baseline hematocrit testing, and platelet counts.
- Continue to monitor your patients closely during defervescence. The critical phase of dengue begins with defervescence and lasts 24–48 hours.

READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** or the **Text Booklet** until you are told to do so.

Write your answers on the spaces provided on this **Question Paper**.

You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this Question Paper and the Text Booklet.

DO NOT remove **OET** material from the test room.

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate Text **Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

VITAMIN C: Questions

Questions 1-7

For each of the questions, 1-7, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1. the types of people usually affected by scurvy?
2. the physical effects of scurvy?
3. tests that can be conducted to check for scurvy?
4. adequate intake totals for vitamin C each day?
5. the effects of taking high doses of vitamin C?
6. conditions that increase a person's need for vitamin C?
7. recovery time for a patient suffering from scurvy?

Questions 8-14

Answer each of the questions, **8-14**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8. What is frequently seen as an early indication of scurvy?
.....

9. What can occur in the gums, joints and under the skin of a patient with scurvy?

10. How many extra milligrams each day of vitamin C does a smoker require?

11. What increases the need for vitamin C by 30%?

12. What is the maximum amount of vitamin C per day that should be given to infants?

13. What do high levels of vitamin C protect cells from?

14. Which condition is most likely to benefit from higher levels of vitamin C?

Questions 15-20

Complete each of the sentences, **15-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15. Scurvy takes only a ----- of deficiency to develop.

16. In infants and children, ----- and anaemia may be present.

17. Incomplete review of dietary history frequently results in diagnosis being -----

18. RDA sufficiently meets the vitamin C requirements in ----- of patients.

19. 75mg of vitamin C daily is recommended for women who are -----.

20. It takes only a few days' worth of ----- for the normal functions of vitamin C to return.

READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this Question Paper until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this Question Paper.

DO NOT remove **OET** material from the test room.

HOW TO ANSWER THE QUESTIONS

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:



In this part of the test, there are six short extracts relating to the work of health professionals. For questions 1-6, choose the answer (**A, B or C**) which you think fits best according to the text.

1. The guidelines on paediatric procedural sedation suggest that

- Ⓐ Patients who are sick should not receive sedation.
- Ⓑ Sedation is likely to be more difficult in elderly patients.
- Ⓒ The approach to sedation changes depending on the patient.

Extract from guidelines: Paediatric Procedural Sedation

Clinicians who administer procedural sedation to patients must have a thorough understanding of the actions of the medication being administered, including modifications for age, concurrent drug therapy and disease processes. Knowledge of the correct procedures is also required to safely administer sedation.

Patient selection is a major factor in achieving safe and successful procedural sedation. The approaches described in this Guide are intended for use with patients who are generally healthy or have only mild systemic disease.

More severely ill patients, those with complex medical problems and infants under 12 months of age or less than 10 kilograms, should not be sedated outside of the operating theatre. Children, who are very anxious prior to the procedure, need special consideration and may be more suitable for general anaesthesia in an operating theatre.

2. Under what circumstances should a doctor pass on confidential information given by a patient?

- Ⓐ When the patient's treatment might otherwise suffer.
- Ⓑ When not disclosing information could cause further harm.
- Ⓒ When it could be considered breaking the law if they didn't.

Confidentiality- reporting gunshot and knife wounds

Trust is an essential part of the doctor patient relationship and confidentiality is central to this. Patients may avoid seeking medical help, or may under-report symptoms, if they think that their personal information will be disclosed by doctors without consent, or without the chance to have some control over the timing or amount of information shared.

Doctors owe a duty of confidentiality to their patients, but they also have a wider duty to protect and promote the health of patients and the public. If you consider that failure to disclose information would leave individuals or society exposed to a risk so serious that it outweighs the patient's and the public interest in maintaining confidentiality, you should disclose relevant information promptly to an appropriate person or authority.

3. This memo is providing information about

Ⓐ Who should wear PPE.

Ⓑ When to wear PPE.

Ⓒ Why to wear PPE.

Memo: Decision-making about Personal Protective equipment (PPE)

PPE is designed and issued for a particular in a protected environment and should not be worn outside that area. Protective clothing provided for staff in areas where there is high risk of contamination (e.g. operating suite/room) must be removed before leaving the area. Even where there is a lower risk of contamination, clothing that has been in contact with patients should not be worn outside the patient-care area. Inappropriate wearing of PPE (e.g. wearing operating suite/room attire in the public areas of a hospital or wearing such attire outside the facility) may also to a public perception of poor practice within the facility.

4. The Aboriginal and Torres Strait Islander Liaison Service assists by

- Ⓐ Establishing stronger connections between people.
- Ⓑ Ensuring improved healthcare for everyone.
- Ⓒ Educating patients about their rights.

Hospital Liaison Officers

The Aboriginal and Torres Strait Islander Liaison Service acts as a cultural link between health professionals, identified Aboriginal and Torres Strait Islander patients, and patient's families.

The service, and liaison officers, assists in breaking down any perceived barriers of communication so that Aboriginal and Torres Strait Islander patients and/or their families have a better understanding of their hospitalisation and treatment.

Patients and their families are supported by:

- Providing information, emotional and cultural support for Aboriginal and Torres Strait Islander patients, and patient's families to assist in delivering services.
- Coordination of patient travel and accommodation.
- Mediation and advocacy for referrals to social workers and support services when required.
- Consultation with hospital staff seeking further information on patient/family history or discussing any special needs of Aboriginal and Torres Strait Islander people.

Facilitating referrals to other facilities and community-based services. Providing support and practical assistance to significant others and/or family members.

5. The purpose of the safety notice about sharps injuries is to

- Ⓐ Praise staff who has been following sharps protocols.
- Ⓑ Reduce the frequency of future sharps harm to staff.
- Ⓒ Remind staff about the risks of working with sharps.

Hospital Bulletin Board Safety Alert Communication

Situation

- Last week, there were 4 reported sharps injuries at the Medical Centre in a 6 day period.
- In comparison, there were 18 sharps injuries in the previous calendar year, about 1-2 per month.

Background

- Although we have gone many weeks without an employee lost time injury, sharps injuries can be potentially serious events.

Assessment

- A through cause analysis investigation completed with staff involved, their managers and the occupational health and safety team, revealed that each of these injuries was preventable.

Recommendation

- Verify and validate: verify that features available on sharps have fully engaged before transporting or discarding a needle. Complete a visual check and listen for the 'click'.
- Star: make sure your fingers are not in the path of a needle when holding soft tissue for injection.
- Seek help: if required, obtain assistance from a co-worker before injecting a patient.

6. This email to staff indicates that older patients

- Ⓐ are being prescribed too many potentially dangerous drugs.
- Ⓑ are being unfairly targeted by pharmaceutical companies.
- Ⓒ are being burdened by the cost of expensive medication.

To : All staff

Subject: Medication use in older persons

Frequently prescribed medicines in older people include those with anticholinergis and sedative effects. These medicines are used in adult to treat medical conditions that often occur later in life, such as urinary incontinence, sleep and pain disorders, dementia and mental illness.

In many instances the benefits of theses medicines do not justify the risk of harm for older adults. The use of these medicines is associated with adverse effects including (but not limited to): impairment of physical and cognitive function, sedation, falls and fractures, and an increased risk of mortality. Their use in older people is also associated with economic costs such as an increase risk of hospitalisation.

In this part of the test, there are two texts about different aspects of healthcare. For questions 7-22, choose the answer (**A, B, C or D**) which you think fits best according to the text.

Text 1: Migraines

Migraines are often misunderstood, or dismissed as “just a headache.” Yet they have the capacity to disrupt a person’s life, relationships, and sense of well-being. A study from Thomas Jefferson University in Philadelphia found that chronic migraine sufferers experience as much social stigma as people with epilepsy—a disease that produces far more obvious and dramatic symptoms. Some of that stigma is external—for example, getting treated differently by friends or colleagues. “Migraines are the unseen and undocumented pain that takes them away from work,” says Dr. R. Joshua Wootton, of pain psychology at the Arnold Pain Management Center, and assistant professor of anesthesia at Harvard Medical School. “There’s no empirical test for migraine yet. That’s why people who report these problems with chronic pain are often not believed or are thought to be exaggerating in the work environment’.

Effective migraine treatments are available—but many migraine sufferers don’t take advantage of them, either because they don’t seek help or they mistakenly believe they’re just suffering from regular headaches. “I think 80% of all migraine sufferers can be effectively helped, but only about a quarter of them are effectively helped at the present time,” says Dr. Egilius Spierings, associate clinical professor of neurology at Harvard Medical School. The gold standard for migraine relief is a class of drugs called triptans. When taken at the first twinge of a migraine, triptans can relieve pain, nausea, and light sensitivity. “These medications have been on the market for about 20 years now,” Dr. Spierings says. “They are generally very safe and well tolerated, and also very effective.”

Despite being the seventh leading cause of time spent disabled worldwide, migraine “has received relatively little attention as a major public health issue,” Dr. Andrew Charles, a California neurologist, wrote recently in *The New England Journal of Medicine*. It can begin in childhood, becoming more common in adolescence and peaking in prevalence at ages 35 to 39. While the focus has long been on head pain, migraines are not just pains in the head. They are a body-wide disorder that recent research has shown results from “an abnormal state of the nervous system involving multiple parts of the brain,” said Dr. Charles, of the U.C.L.A. Goldberg Migraine Program at the David Geffen School of Medicine in Los Angeles. He hopes the journal article will educate practicing physicians, who learn little about migraines in medical school.

Before it was possible to study brain function through a functional M.R.I. or PET scan, migraines were thought to be caused by swollen, throbbing blood vessels in the scalp, usually affecting one side of the head. This classic migraine symptom prompted the use of medications that narrow blood vessels, drugs that help only some patients and are not safe for people with underlying heart disease. Neurologists who specialise in migraine research and treatment now approach migraine as a brain-based disorder, with symptoms and signs that can start a day or more before the onset of head pain and persist for hours or days after the pain subsides. Based on the new understanding, there are now potent and less disruptive treatments already available or awaiting approval. However, to be most effective, the new therapies may require patients to recognise and respond to the warning signs of a migraine in its so-called prodromal phase — when symptoms like yawning, irritability, fatigue, food cravings and sensitivity to light and sound occur a day or two before the headache.

Even with current remedies, people typically wait until they have a **full-blown** headache to start treatment, which limits its effectiveness, Dr. Charles said. His advice to patients: Learn to recognise your early symptoms signaling the onset of an attack and start treatment right away before the pain sets in. Conditions that can trigger a migraine in susceptible people include skipped meals, irregular intake of caffeine, erratic sleep habits and stress. Accordingly, Dr. Charles suggests practicing consistent dietary, sleep, caffeine and exercise habits to limit the frequency of migraines. Keeping a migraine diary that includes your stress level and what you've eaten and drunk can also help identify triggers.

But they aren't just a physical condition. Living with chronic pain, or the constant worry that **they** may strike at any moment, can take an emotional toll, too. Migraines have been linked to an increased risk of depression. A study presented at the American Academy of Neurology's annual meeting found that women with a history of migraines are 41 % more likely to be depressed than those without the condition. "When you can't find effective ways to manage your migraines that frequently results in feeling helpless, hopeless and as if everyone is against you," Dr Wootton says. If you're having these feelings, it can be helpful to see a psychiatrist or psychologist, particularly at a center that specialises in pain management. "If you have considerable anxiety and/or depression, addressing those issues is important because they can negatively affect migraine. They also make it much more difficult to cope with a condition like migraine.'

Text 1: Questions 7-14

7. The writer makes the comparison between migraines and epilepsy to show

- Ⓐ How the sufferers of both conditions feel a lot of shame.
- Ⓑ How people suffering from these conditions have social problems.
- Ⓒ How both conditions affect the amount of work a person is able to do.
- Ⓓ How friends and colleagues find it hard to trust people with these conditions

8. In the second paragraph, Dr Egilius Spierings says he believes

- Ⓐ There aren't enough migraine medications, but those that do exist work well.
- Ⓑ At least 80% of people with migraines are helped by medication.
- Ⓒ Only 25% of migraine sufferers currently take medication.
- Ⓓ Not enough migraine sufferers take medication.

9. What does Dr Andrew Charles hope will change as a result of his journal article?

- Ⓐ More doctors will understand that migraines are more than just head pain.
- Ⓑ More doctors will realise that migraines can be a life-long problem.
- Ⓒ More doctors will learn about migraines in medical school.
- Ⓓ More doctors will read about the issue of migraines.

10. What point does the writer make in the fourth paragraph?

- Ⓐ In the past, there was no way to accurately test patients for migraine.
- Ⓑ The triggers for migraine are more complex than was originally believed.
- Ⓒ Medications that narrow blood vessels in migraine patients are no longer useful.
- Ⓓ Enlarged blood vessels in a person's scalp are now seen on both sides of the head

11. In the fourth paragraph, the writer suggests patients should

- Ⓐ Try a range of improved therapies.
- Ⓑ Seek specialised treatment earlier.
- Ⓒ Start taking new types of medication.
- Ⓓ Become more aware of their triggers.

12. The use of the adjective 'full-blown' indicates

- Ⓐ The treatment of the headache.
- Ⓑ The duration of the headache.
- Ⓒ The severity of the headache.
- Ⓓ The location of the headache.

13. What does the word 'they' in the final paragraph refer to?

- Ⓐ Triggers.
- Ⓑ Migraines.
- Ⓒ Pains and worries.
- Ⓓ Physical conditions.

14. According to Dr Wootton, if a person is suffering with migraines

- Ⓐ They can feel quite alone.
- Ⓑ It can be very difficult to recover.
- Ⓒ They may develop pain in other areas.
- Ⓓ They should see a mental health professional.

Text 2: Water Consumption

How much fluid should you drink each day for good health? Eight glasses a day has been the widely circulated advice. But recently, two large studies have suggested that's probably overkill. It turns out that under normal circumstances, you get most of the liquid you need each day from what you routinely eat and drink, including coffee, tea, soft drinks, and even some alcoholic drinks.

So where did this notion of 'eight glasses a day' come from? In 1945, the Food and Nutrition Board of the United States National Research Council wrote: 'A suitable allowance of water for adults is 2.5 liters daily in most instances. Most of this quantity is in prepared foods: But in the 1990's Dr Heinz Valtin undertook a comprehensive investigation into the myths surrounding water consumption in humans. He found healthy people who drank more water didn't have a higher 'output of stool', and that there was no scientific evidence high fluid intake could relieve constipation.

And what of the belief that thirst is not a good indicator of a need to drink? Valtin states that while 'a rise in plasma osmolality' (which is an internal chemical change) of less than two per cent can elicit thirst, dehydration is defined as a rise of at least five per cent. This is a complicated way of saying you get thirsty before your body starts to dehydrate, so thirst is a good guide.

Australia's current dietary guidelines don't recommend a specific amount of water, but simply recommend we 'drink plenty of water'. "How much water each one of us needs depends on a range of factors," said CSIRO dietitian Pennie Taylor. "This can include our gender, bodyweight and how much physical activity we do: 'The guidelines also encourage drinking water over juices, soft drinks, cordials or the like. Also, pregnant or breastfeeding women (who require more fluid), people who live or

work in extremely hot climates, and people with high protein diets (the kidneys may need more fluid to help process the increased amount of protein) are encouraged to drink more water. It's on hot days that most of us notice we're thirstier than normal. This is because we're sweating more, and we lose fluid through sweat. "We can lose between 1 to 3 per cent of our fluid quite easily," Ms Taylor said.

Associate Professor Ben Desbrow from Griffith University agrees. "Those who work or exercise in hot climates lose the most fluid — up to 2.5 liters of sweat in an hour in extreme circumstances. You need to replace those fluids pretty quickly; otherwise it's going to fairly rapidly have an effect on your subsequent performance." Your body will give you some pretty clear signs that you're not getting dehydrated. So keep an eye out for symptoms such as a dry mouth, headache and feeling dizzy. Also pay attention to your toilet habits, the colour of your urine and how frequently you go to the toilet. It is true that 'copious and clear' is a good indicator of healthy wee. But 'clear' does not mean colourless. The depth of colour in urine will vary, what you need to look out for is cloudiness — that's the indicator of a problem. "Your kidneys do a great job in fluid regulation, so frequency of urination and colour of urination are your two best guides," Associate Professor Desbrow said.

What about the idea that a person may be drinking too much water. There is a thirst control centres in our brain that controls water intake, says Dr Michael McKinley, Senior Fellow at Florey Neuroscience Institute. When we drink water, this part of our brain stops us feeling thirsty long before the water has been fully absorbed into the bloodstream. "Usually if we take in too much water, it'll suddenly feel like hard work to drink," he said. However, in some circumstances when people drink a large volume of water, they can over-ride the thirst control centre in the brain. When this happens, their sodium levels can drop too low. This can lead to a condition known as

hyponatremia, where the body also starts to retain the excess water. “Normally if we drink too much water, our kidneys would excrete it [as urine],” Dr McKinley said. But sometimes, factors like heat, physical stress or certain drugs can switch off the hormonal signal that causes the kidneys to excrete excess water. Then there is a double whammy effect. Not only have you drunk a lot of water, but you start to hang onto all the water in your body. Drinking more just makes things worse. “This is when things can get dangerous,” Dr McKinley said.

Text 2: Questions 15-22

15. Drinking eight glasses of water a day is described as '**overkill**' because

- Ⓐ Staying hydrated isn't as important as we used to believe.
- Ⓑ It is now considered unnecessary to drink so much.
- Ⓒ The benefits have been known for many years.
- Ⓓ There are better ways to stay hydrated.

16. What do we learn about water consumption in the second paragraph?

- Ⓐ There was no scientific evidence to support past ideas about water.
- Ⓑ Healthy people don't need to drink as much water as unhealthy people.
- Ⓒ Not everything we previously believed about drinking water was correct.
- Ⓓ No one knows the origins of how we came to drink the amount we do.

17. In the third paragraph, the word '**this**' refers to

- Ⓐ The chemical changes that occur within a person's body.
- Ⓑ The lack of rules about how much water to drink.
- Ⓒ The idea that thirst has no connection to dehydration.
- Ⓓ The reasons why a person gets thirsty.

18. Why don't Australia's dietary guidelines state a specific amount of water to drink?

- Ⓐ Because everyone differs physically.
- Ⓑ Because there are too many factors to consider.
- Ⓒ Because drinking any amount of water has benefits.
- Ⓓ Because there is no current agreement among dieticians.

19. In the fifth paragraph, Associate Professor Ben Desbrow says he believes fluid loss

- Ⓐ Happens at a very fast rate.
- Ⓑ Is a sign of hydration issues.
- Ⓒ Can result in physical decline.
- Ⓓ Doesn't occur in cool climates.

20. When commenting on urine Professor Desbrow suggests

- Ⓐ Variations in colour are uncommon.
- Ⓑ It should always be clear or colourless.
- Ⓒ Frequency is a good indication of a problem.
- Ⓓ If it isn't clear, there may be something wrong

21. What idea does Dr McKinley express in the final paragraph?

- Ⓐ Drinking a lot of water can suppress a person's appetite.
- Ⓑ Not having enough water can affect concentration levels.
- Ⓒ Our bodies usually tell us when we've had enough water.
- Ⓓ Processing large amounts of water can stress our arteries.

22. Dr McKinley expresses concern about people who

- Ⓐ Drink dangerous quantities of water.
- Ⓑ Have trouble passing surplus urine.
- Ⓒ Consumes small amounts of salt.
- Ⓓ Take pills in extreme weather.

READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT remove OET material from the test room.

TEXT A

Dengue: virus, fever and mosquitoes

Dengue fever is a viral disease spread only by certain mosquitoes – mostly *Aedes aegypti* or “dengue mosquitoes” which are common in tropical areas around the world.

There are four types of the dengue virus that cause dengue fever – Dengue Type 1, 2, 3 and 4. People become immune to a particular type of dengue virus once they’ve had it, but can still get sick from the other types of dengue if exposed. Catching different types of dengue, an even year apart, increases the risk of developing severe dengue. Severe dengue causes bleeding and shock, and can be life threatening.

Dengue mosquitoes only live and breed around humans and buildings, and not in bush or rural areas. They bite during the day – mainly mornings and evenings. Dengue mosquitoes are not born with dengue virus in them, but if one bites a sick person having the virus in their blood, that mosquito can pass it on to another human after about a week. This time gap for the virus to multiply in the mosquito means that only elderly female mosquitoes transmit dengue fever. The mosquitoes remain infectious for life, and can infect several people. Dengue does not spread directly from person to person

TEXT B

Signs and Symptoms

Classic dengue fever, or “break bone fever,” is characterised by acute onset of high fever 3–14 days after the bite of an infected mosquito. Symptoms include frontal headache, retro-orbital pain, myalgias, arthralgias, hemorrhagic manifestations, rash, and low white blood cell count. The patient also may complain of weight loss and nausea. Acute symptoms, when present, usually last about 1 week, but weakness, malaise, and weight loss may persist for several weeks. A high proportion of dengue infections produce no symptoms or minimal symptoms, especially in children and those with no previous history of having a dengue infection.

TEXT C

Steps to take when seeing a suspected case of dengue fever

Step 1: Notify your nearest Public Health Unit immediately upon clinical suspicion.

Step 2: Take a comprehensive travel history and determine whether the case was acquired overseas or locally.

Step 3: Note the date of onset of symptoms to identify the correct diagnostic test, as suitable laboratory tests depend on when the blood sample is collected during the illness.

- Another useful test is full blood count. Cases often have leucopenia and/or thrombocytopenia.

The table below shows which test to order at which stage of illness:

TEST TYPE	PCR	NSI ELISA	IgM	IgG
Days after onset of symptoms	0-5 days	0-9 days	From day 5 onwards	From day 8 onwards

Stage 4: Provide personal protection advice.

- The patient should stay in screened accommodation and have someone stay home to look after them.
- The patient should use personal insect repellent particularly during daylight hours to avoid mosquito bites.
- All household members should use personal insect repellent during daylight hours.
- Advise family members or associates of the case who develop a fever to present immediately for diagnosis.

TEXT D

Prior to discharge:

- Tell patients to drink plenty of fluids and get plenty of rest.
- Tell patients to take antipyretics to control their temperature. Children with dengue are at risk for febrile seizures during the febrile phase of illness.
- Warn patients to avoid aspirin and anti-inflammatory medications because they increase the risk of haemorrhage.
- Monitor your patients' hydration status during the febrile phase of illness. Educate patients and parents about the signs of dehydration and have them monitor their urine output.
- Assess hemodynamic status frequently by checking the patient's heart rate, capillary refill, pulse pressure, blood pressure, and urine output. If patients cannot tolerate fluids orally, they may need IV fluids.
- Perform hemodynamic assessments, baseline hematocrit testing, and platelet counts.
- Continue to monitor your patients closely during defervescence. The critical phase of dengue begins with defervescence and lasts 24–48 hours.

READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** or the **Text Booklet** until you are told to do so.

Write your answers on the spaces provided on this **Question Paper**.

You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this Question Paper and the Text Booklet.

DO NOT remove **OET** material from the test room.

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate Text **Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

DENGUE FEVER : Questions

Questions 1-7

For each of the questions, 1-7, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1. the different types of dengue virus? -----
2. how fever presents in patients? -----
3. how dengue fever is transmitted? -----
4. the stages at which to conduct tests for dengue fever? -----
5. monitoring and assessing a patient's condition? -----
6. what advice to give patients to avoid mosquito bites? -----
7. advice for patients regarding medication? -----

Questions 8-14

Answer each of the questions, **8-14**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8. How long after being bitten by an infected mosquito does high fever occur?

9. What might patients with dengue fever complain of?

10. Which test should only be ordered 5 days after symptoms appear?

11. What other test is also useful when checking for dengue fever?

12. Who is at risk of seizures during the febrile stage of dengue?

13. What takes places in the most lethal cases of dengue?

14. How long does the most serious stage of dengue last?

Questions 15-20

Complete each of the sentences, **15-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15. Dengue fever does not spread -----

16. In many ----- dengue infections cause almost no symptoms.

17. Within three days of symptoms beginning a PCR or -----
----- can be ordered.

18. To avoid haemorrhage patients mustn't take anti-inflammatory medications or

19. Advise patients be cared for by someone at home in -----
----- accommodation.

20. Patients must be made aware of the need to check their -----
-----.

READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this Question Paper until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this Question Paper.

DO NOT remove **OET** material from the test room.

HOW TO ANSWER THE QUESTIONS

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:



In this part of the test, there are six short extracts relating to the work of health professionals. For questions 1-6, choose the answer (**A, B or C**) which you think fits best according to the text.

1. Which type of hazard does the workplace extract relate to?

Ⓐ Chemical agents.

Ⓑ Biological agents.

Ⓒ Physical agents.

Extract from Workplace Policy Document: Hazard Assessment

Hazards to Look for When Inspecting Hospitals

Examples of typical hazards include bacteria, viruses, fungi, and other living organisms that can cause acute and chronic infections by entering the body through ingestion, inhalation or breaks in the skin. They also include exposure to blood or other body fluids or to clients or patients with infectious diseases (e.g., MRSA, staph, HIV, HBV, HCV, influenza, tuberculosis). Hospital workers can be exposed to blood borne pathogens from blood and other potentially infectious materials if not following universal precautions.

2. The policy extract is explaining

- Ⓐ Why to make a LAM submission.
- Ⓑ How to make a LAM submission.
- Ⓒ Where to make a LAM submission.

Requesting a change to the list of approved medicines (LAM)

Any hospital staff member can request a change to the list of approved medicines (LAM). It is expected that applications for changes will include input from a senior prescriber. Changes should be requested by completing either the standard or, in limited circumstances, a minor submission form. A standard submission form is available online or from your local pharmacy department. A minor submission form can be obtained through contacting the relevant Secretariat. Staffs are also encouraged to flag potential issues regarding the use of medicines or pharmaceuticals in writing, with evidence attached. Requests from pharmaceutical manufacturers or their agents will not be accepted.

3. What point do the guidelines make about leadership for doctors?

- Ⓐ The role of a doctor should go beyond practising medicine.
- Ⓑ Doctors are the most important clinician in a health care setting.
- Ⓒ There could be harsh penalties for doctors who don't improve their skills.

Guidelines: Leadership and Management for all Doctors

This guidance sets out the wider management and leadership responsibilities of doctors in the workplace. The principles in this guidance apply to all doctors, whether they work directly with patients or have a formal management role.

Being a good doctor means more than simply being a good clinician. In their day-to-day role doctors can provide leadership to their colleagues and vision for the organisations in which they work and for the profession as a whole. However, unless doctors are willing to contribute to improving the quality of services and to speak up when things are wrong, patient care is likely to suffer. You must be prepared to explain and justify your decisions and actions. Serious or persistent failure to follow this guidance will put your registration, and so you're right to practice medicine, at risk.

4. The purpose of this memo to staff is to

- Ⓐ State the potential risks to patients who smoke electronic cigarettes.
- Ⓑ Provide information about the substances used in electronic cigarettes.
- Ⓒ Advise that no position has yet been reached about electronic cigarettes

Memo to staff: Electronic Cigarettes

Electronic cigarettes (e-cigarettes) are battery operated devices that heat a liquid (called 'e-liquid') to produce a vapour that users inhale. Although the composition of this liquid varies, it typically contains a range of chemicals, including solvents and flavouring agents, and may or may not contain nicotine.

Electronic cigarettes are a topic of contention among public health and tobacco-control advocates, some of whom argue they don't pose the same dangers to smokers as traditional cigarettes. Others, however, argue that electronic cigarettes should not be promoted as a lower threat option for smokers when their long-term safety is unknown.

5. As a result of an update in favour of patient-centeredness what is going to happen?

- Ⓐ There will be a greater focus on hospital wait times.
- Ⓑ More staff will be required to undertake training.
- Ⓒ New standards of practice will be developed.

Patient-centered Interdisciplinary Goal Setting in Rehabilitation Services

Although goal setting is fundamental to rehabilitation practice and optimal patient outcomes, it typically varies in the practices taught across different health professions, and the preparedness of rehabilitation clinicians to undertake it. Patient-centeredness has been shown to improve patient care experiences and create value for public services through increasing the quality and safety of health care.

Hospital pressures to facilitate discharge and decrease length of stay have been identified by staff as barriers to implementing patient-centered goal setting practice. This has resulted in goal setting often being hospital driven rather than patient driven. Furthermore, staff has recently expressed a lack of strategies or tools to implement patient-centered principles in care processes such as goal setting. There is therefore a need to enable rehabilitation services to improve goal setting models and patient engagement in health care related goals and decisions.

6. According to the procedure, when inserting a catheter clinicians should

- Ⓐ Only use a catheter once.
- Ⓑ Carefully follow all guidelines.
- Ⓒ Ensure the patient isn't left alone.

Catheter Insertion Procedure

Where possible, use a two clinician buddy system to carry out the procedure. The patient's ethical, religious and cultural beliefs and personal history should be considered when appointing clinicians to perform a catheterisation. A chaperone may also be required to observe the procedure.

It is recommended that the patient's genital area be washed with soap and water prior to catheterisation. If unable to insert a catheter after two attempts (includes changing to a different catheter size), seek further assistance from a senior clinician. A new catheter should be used for each attempt.

In this part of the test, there are two texts about different aspects of healthcare. For questions 7-22, choose the answer (**A, B, C or D**) which you think fits best according to the text.

Text 1: Restless Leg Syndrome

The relatively common neurologic movement disorder known as restless legs syndrome (RLS) is poorly understood. Patients describe uncomfortable sensations in their legs that often are worse at night, having a huge impact on their ability to sleep and overall quality of life. Experts now speculate that patients with RLS, like those with unrelieved chronic pain, might feel hopeless, leading to suicidal thoughts and actions. The risk correlates with history of depression, and is independent of the severity of restless leg symptoms and demographic factors.

Although evaluating restless legs syndrome and finding effective treatments is challenging, a recent study suggests that it's important to assess not only the impact of RLS on the patient's life, but also the presence of suicidal thoughts. People with severe RLS are more likely to plan and attempt suicide than people without it, even after controlling for depression, according to new research. "Lifetime suicidal ideation and attempts are very prevalent among people with restless legs syndrome and seem to be independent of demographic factors and depression and seem to be associated with severity of restless legs," said Brian Koo, MD, director of the Yale Center for Restless Legs Syndrome, Yale University, New Haven, Connecticut.

Until now, RLS and its potential relationship with suicide had not been studied in any depth. The Yale team investigated the frequency of lifetime suicidal behaviour in 198 patients with severe RLS and 164 controls. All participants completed the Suicidal Behaviour Questionnaire-revised (SBQ-R) and the Brief Lifetime Depression Scale. RLS and controls were similar in age (mean age, 51), income, and gender. Compared with controls, patients with RLS were more often white (96% vs 88%), less often had higher education (84% vs 96%), were more often married (72% vs 60%), and were less often employed or retired (80% vs 90%). Significantly more patients with RLS than controls were at high suicide risk (SBQ-R score ≥ 7) and had lifetime suicidal thoughts or behaviour, independent of depression history.

“Mood and anxiety disorders are highly comorbid in RLS patients,” noted John W. Winkelman, MD, PhD, from Harvard Medical School and Massachusetts General Hospital, Boston. “My feeling is that the suicidal ideation, or even plan or intent, and even some who have followed through, is the same thing you see in patients with chronic pain. In many respects, RLS is a chronic pain disorder. And if you have chronic pain, for which you feel there is no appropriate treatment and your physician may not understand what you have, or may not know how to treat it appropriately, it can lead you to feeling hopeless, and I think pain and hopelessness can lead to those kinds of thoughts,” Winkelman said.

One such case is Lisa, a 45-year-old married woman who came to see a psychiatrist initially for depressive symptoms. During the initial evaluation, she complained of difficulty in falling asleep and other depressive symptoms such as low mood, difficulty with concentration, poor appetite, and low energy along with daytime fatigue. Depression was diagnosed. A selective serotonin reuptake inhibitor (SSRI) was prescribed on an as-needed basis, and the patient was advised to take a nightly dose of diphenhydramine to help her sleep. Three days later—after staying up nearly all night—Lisa called her doctor in despair and complained of worsening insomnia. On more detailed questioning about the insomnia, Lisa revealed that for the past 2 years, she has experienced leg discomfort when she gets into bed. She is so uncomfortable that she needs to walk or ride on her exercise bike past 2 or 3 am until the discomfort subsides. While not painful, this leg discomfort sometimes prevents her from relaxing and watching television because she just “has to move” her legs.

Lisa describes a deep uncomfortable sensation that feels like “bugs crawling in her legs:” she also reveals that her mother used to suffer from similar night-time leg restlessness. Lisa’s leg discomfort became more intense and was lasting most of the night. After secondary causes of RLS, such as iron deficiency anaemia, pregnancy, uraemia, and neuropathy were ruled out, SSRI and diphenhydramine therapy were stopped. Low-dose dopamine agonist therapy was started, after which the symptoms subsided. However, despite resolution of the RLS symptoms, her depressive symptoms continued. This only serves to further reinforce the need to investigate

and treat any associated mood or anxiety disorders in conjunction with RLS symptoms.

Text 1: Questions 7-14

7. The writer suggests that restless legs syndrome (RLS)

- Ⓐ Is impossible to cure.
- Ⓑ Could lead to depression.
- Ⓒ Doesn't occur during the day.
- Ⓓ May relate to pain management.

8. Dr Brian Koo suggests it's important for clinicians to treat any suicidal thoughts because

- Ⓐ Older people are more likely to suffer from RLS.
- Ⓑ The effects of RLS can be better identified.
- Ⓒ It makes managing RLS much easier.
- Ⓓ RLS is a mental health condition.

9. What did the Yale team learn from their investigations?

- Ⓐ Some people in the control group had previously suffered from RLS.
- Ⓑ The likelihood of someone developing RLS depends on various factors.
- Ⓒ Answers to the questionnaires didn't provide a lot of useful data about RLS.
- Ⓓ A person with RLS is more likely to attempt suicide than someone without it.

10. The expression 'followed through' refers to

- Ⓐ RLS patients who have attempted suicide.
- Ⓑ The relationship between RLS and pain.
- Ⓒ A time when RLS has been resolved.
- Ⓓ Management of RLS by the doctor.

11. John Winkelman's comments in the fourth paragraph show his

- Ⓐ Concern that a lot of doctors have never heard of RLS.
- Ⓑ Belief that RLS relates to many other health conditions.
- Ⓒ Frustration that too many people with RLS commit suicide.
- Ⓓ Understanding of the situation facing a lot of RLS sufferers.

12. The case involving Lisa highlights that

- Ⓐ Some patients don't follow the recommended advice for RLS.
- Ⓑ Regular exercise is recommended for people with RLS.
- Ⓒ Sleep problems and exhaustion could indicate RLS.
- Ⓓ Medication is important in the treatment of RLS.

13. In the final paragraph, the writer suggests Lisa's treatment was changed because

- Ⓐ A new diagnosis was made.
- Ⓑ She no longer had depression.
- Ⓒ SSRI medication wasn't working for her.
- Ⓓ She developed a range of new symptoms.

14. What does the word 'this' in the final paragraph refer to?

- Ⓐ Low-dose dopamine agonist therapy.
- Ⓑ The differences between therapies.
- Ⓒ The end of her RLS symptoms.
- Ⓓ Lisa's unresolved depression.

Text 2: Statins- How Safe Are They?

Heart disease is the leading cause of death in the U.S and statins are a commonly prescribed medicine that helps to lower harmful levels of LDL cholesterol in the blood and mitigate the risks of cardiovascular disease, including heart attack and stroke. Trials have consistently demonstrated a clear correlation between reducing LDL cholesterol with statins and a decrease in cardiovascular risk. So it may appear puzzling that uncertainty over statins still remains.

As the body of evidence evaluating statins has expanded, so too have the indications for the drug. Guidelines released in 2013 by the American College of Cardiology (ACC) and the American Heart Association (AHA) recommended that statin therapy might be beneficial for people with cardiovascular disease, people who have high LDL cholesterol levels, people aged 40 to 75 years with diabetes and high LDL levels and people aged 40 to 75 years without diabetes, but with high LDL cholesterol levels and a predicted 10-year risk of cardiovascular disease of 7.5 percent or higher. However, experts questioned the 2013 guidelines, arguing that a 7.5 percent threshold seemed too low.

In 2015, two research teams examined the 7.5 percent threshold and published their findings. The first paper, led by Dr. Udo Hoffmann at Massachusetts General Hospital and Harvard Medical School – both in Boston - found that compared with guidelines published in 2004, the 2013 guidelines were more accurate at identifying individuals at a greater risk of cardiovascular disease. They estimated that by adopting the 2013 guidelines, between 41,000 and 63,000 cardiovascular events would be prevented over 10 years compared with previous guidelines. The second paper, led by Drs. Ankur Pandya and Thomas A. Gaziano at the Harvard T.H. Chan School of Public Health - also in Boston - assessed the cost-effectiveness of the 10-year cardiovascular disease threshold. The researchers concluded that the risk threshold of 7.5 percent or higher had an acceptable cost-effectiveness profile.

As a result of the expansion of the groups reported to benefit from statins, suspicions have been raised about the pharmaceutical industry and of the prescribing healthcare professionals. **Alarm bells started ringing** that people were being overmedicated and put at risk of adverse effects. Statins are generally considered to

be safe and well tolerated. However, as with any medication, statins may have negative effects in some people. "We know that statins can prevent a significant number of heart attacks and strokes. We know there is a small increase in the risk of diabetes, and at high doses there is a very small increase in myopathy, but overall the benefits greatly outweigh the harms," says Peter Sever, professor of clinical pharmacology and therapeutics at Imperial College London. "Widespread claims of high rates of statin intolerance still prevent too many people from taking an affordable, safe, and potentially life-saving medication."

Some people, however, believe heart disease is better treated by other means, such as diet. A study found those who had a diet rich in vegetables, nuts, fish and oils, such as a Mediterranean-style diet were a third less likely to die early, compared with those who ate larger quantities of red meat, such as beef, and butter. Sir David Nicholson, former chief executive of the National Health Service (NHS) in the UK, entered the debate over statins when he said he had stopped taking them as part of his medication for diabetes. "If a lifestyle change works then why would you take the statin? The trouble is that they give you a statin straightaway, so you don't know what is working," he said.

While a heart-healthy diet, regular physical activity, and maintaining a healthy weight are all components that may help to reduce cholesterol and lower the risk of heart disease and stroke, certain factors are unable to be influenced - such as genetics. In some people, lifestyle changes alone are not enough to lower cholesterol. According to a study published in the Journal of the American Medical Association, from 1969 to 2013, deaths from heart disease fell by 68 percent, and there were 77 percent fewer deaths from stroke. There may be a link between the rise in statin use and the fall of deaths connected to cardiovascular disease. However, the progress made could be attributed to the "cumulative effect of better prevention, diagnosis, and treatment," says Wayne D. Rosamond, Ph.D., professor of epidemiology at the University of North Carolina in Chapel Hill.

The mounting research appears to overturn debate around statins and aims to reassure doctors and patients that the risks of not taking statins - heart attack or stroke - far outweigh concerns about side effects associated with the drug. Serious side effects are rare, and study authors seem to agree that the substantial proven

benefits of statins have been **compromised** by “serious misrepresentations of the evidence for its safety.”

Text 2: Questions 15-22

15. The writer suggests that uncertainty over the use of statins is puzzling because

- Ⓐ No other medication is used as often to treat cardiovascular disease.
- Ⓑ Heart disease kills large numbers of people in the United States.
- Ⓒ Extensive studies have been conducted about their use.
- Ⓓ They are so effective in lowering LDL cholesterol.

16. In the second paragraph, what do we learn about the guidelines released in 2013?

- Ⓐ They were seen as worse than the previous guidelines.
- Ⓑ They recommended the use of statins for anyone with high LDL levels.
- Ⓒ They contained a lot of advice that health professionals didn't agree with.
- Ⓓ They suggested a connection between heart disease and other conditions.

17. The research papers written in 2015 concluded that the 7.5 percent threshold would

- Ⓐ Focus more on patient health than the previous guidelines.
- Ⓑ Result in lower treatment costs for most patients.
- Ⓒ Reduce the amount of cardiovascular disease.
- Ⓓ Take many years to implement.

18. The writer uses the phrase 'alarm bells started ringing' to indicate

- Ⓐ Some health professionals have been overprescribing statins.
- Ⓑ The numbers of people taking statins has grown too quickly.
- Ⓒ There are too many risks associated with taking statins.
- Ⓓ Research into the use of statins has cost too much.

19. What concerns does Peter Sever have about statins in the fourth paragraph?

- Ⓐ They aren't being promoted as widely as they should be.
- Ⓑ They are linked to several other health conditions.
- Ⓒ They are too expensive for some patients.
- Ⓓ They aren't being used enough.

20. Sir David Nicholson's comments show that he believes statins

- Ⓐ Should only be prescribed after other options have been tried.
- Ⓑ Aren't as effective as diet in improving a person's health.
- Ⓒ Only work after you have been taking them for a while.
- Ⓓ Don't work as an effective treatment for diabetes.

21. In the sixth paragraph, Wayne D. Rosamond attributes a reduction in deaths from heart attack and stroke to

- Ⓐ A combination of different factors that work together.
- Ⓑ The rise in medications that treat heart disease.
- Ⓒ A person's family history and background.
- Ⓓ Improved diet and regular exercise.

22. The benefits of statins are described as having been 'compromised' because

- Ⓐ Their benefits are too few in number.
- Ⓑ A lot more research needs to be done.
- Ⓒ There is still a lot of debate around their use.
- Ⓓ Too many lies have been told about their effects.

READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

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TEXT A

Pain assessment is a broad concept involving clinical judgment based on observation of the type, significance and context of the individual's pain experience. There are challenges in assessing paediatric pain, none more so than in the pre-verbal and developmentally disabled child. Therefore physiological and behavioural tools are used in place of the self-report of pain. However in children with developmental disabilities there can be incorrect assumptions and there is a risk of under-treating pain. It is important to take behavioral cues identified by parents and caregivers to improve pain assessment in these children.

When assessing a child's level of pain careful consideration needs to be given to their:

- Pain history
- Location of pain
- Intensity of pain
- Cognitive development and understanding of pain
- Environment (eg: hospital)
- Anxiety level
- Cause of pain (eg: post –operative)

When is pain assessment required?

- Pain scores should be documented for all children at least once per shift
- Children with pain should have pain scores documented more frequently.
- Children who are receiving oral analgesia should have pain scores documented at least 4 hourly during waking hours.
- Children on complex analgesia such as intravenous opioid and/ or ketamine, epidurals or regional analgesia should have hourly pain and sedation scores documented
- Assess and document pain before and after analgesia, and document effect.
- Assess and document pain on activity such as physiotherapy.

TEXT B

Pain Assessment Tools

Tools used for pain assessment have been selected on their validity, reliability and usability and are recognised by pain specialists to be clinically effective in assessing acute pain. All values are documented on the clinical observation chart as the 5th vital sign.

Three ways of measuring pain:

- Self report – what the child says (the gold standard)
- Behavioural – how the child behaves
- Physiological – clinical observations

Physiological indicators

Physiological indicators in isolation cannot be used as a measurement for pain. A tool that incorporates physical, behavioural and self report is preferred when possible. However, in certain circumstance (for example, the ventilated and sedated child) physiological indicators of pain can be helpful to determine a patient's experience of pain.

These include:

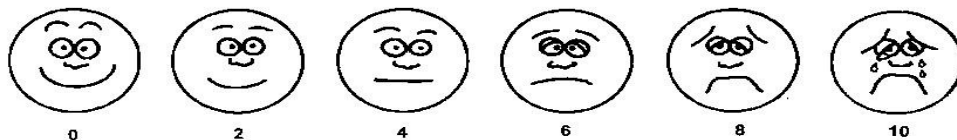
- heart rate may increase
- respiratory rate and pattern may shift from normal i.e.: increase, decrease or change pattern
- blood pressure may increase
- oxygen saturation may decrease

TEXT C

Wong-Baker faces pain scale

The Wong-Baker faces pain scale uses self report of pain to assess a patient's experience of pain. It can be used in children aged between 3 and 18 years of age, depending upon their cognitive ability. Explain to the patient that each face helps us understand how much pain they have, and how this makes them feel. Face 0 is very happy because he doesn't hurt at all (i.e. has no pain). Face 2 hurts just a little bit. Face 4 hurts a little more. Face 6 hurts even more. Face 8 hurts a whole lot. Face 10 hurts as much as you can imagine, although you don't have to be crying to feel this bad. Ask the person to choose the face that best describes how he is feeling.

Wong-Baker FACES Pain Rating Scale



TEXT D

FLACC: Face, Legs, Activity, Cry and Consolability

The FLACC is a pain assessment tool that uses that patient's behaviour to assess their pain experience. It can be used for children aged between 2 months and 18 years of age, and up to 18 years of age in children with cognitive impairment and/or developmental disability.

Each category (Face, Legs etc) is scored on a 0-2 scale, which results in a total pain score between 0 and 10. The person assessing the child should observe them briefly and then score each category according to the description supplied.

	0	1	2
Face	No particular expression or smile	Occasional grimace or frown, withdrawn, disinterested	Frequent to constant frown, clenched jaw, quivering chin
Legs	Normal position or relaxed	Uneasy , restless, tense	Kicking, or legs drawn up
Activity	Lying quietly, normal position, moves easily	Squirming, shifting, back and forth, tense	Arched, rigid or jerking
Cry	No cry (awake or asleep)	Moans or whimpers, occasional complaints	Crying steadily, screams or sobs, frequent complaints
Consolability	Content , relaxed	Reassured by occasional touching, hugging or "talking to". Distractable	Difficult to console or comfort

READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** or the **Text Booklet** until you are told to do so.

Write your answers on the spaces provided on this **Question Paper**.

You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this Question Paper and the Text Booklet.

DO NOT remove **OET** material from the test room.

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate Text **Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

Assessing pain: Questions

Questions 1-7

For each of the questions, **1-7**, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1. about necessary considerations when assessing a child's pain level -----
2. the methods used to measure pain -----
3. specific factors to note when assessing a child's pain -----
4. uses behavioural responses to assess pain level. -----
5. the rating scale to use for self report when assessing pain. -----
6. how to assess a patient's pain level via their facial expression? -----
7. signs of pain to be aware of in a patient who is under sedation? -----

Questions 8-13

Answer each of the questions, **8-13**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8. Which pain assessment tools should be used in children who are developmentally disabled or too young to speak?

9. What is the maximum age for behavioural assessment of pain in an intellectually disabled patient?

10. What risk should you be aware of when assessing pain in children with disabilities?

11. Who can help provide better assessment of pain in disabled children?

12. Where should pain values be documented?

13. How often should all children have their pain scores recorded?

Questions 14-20

Complete each of the sentences, **14-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

14. Self reporting is considered to be -----
when measure pain in children.

15. You should avoid using -----
----- observations on their own as a way of measuring pain.

16. You should ensure the patient understands they need to choose the -----
----- that most accurately represents
their mood when self reporting.

17. Don't have patients self report if they lack the required -----
----- ability

18. Make sure to allocate a score of between -----
----- for each sub-group when evaluating the behavioural responses of a
child.

19. The maximum possible pain score should be given to a patient if you notice it
is hard to ----- them.

20. Pain should be assessed and recorded -----
----- analgesia.

READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this Question Paper until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this Question Paper.

DO NOT remove **OET** material from the test room.

HOW TO ANSWER THE QUESTIONS

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:



In this part of the test, there are six short extracts relating to the work of health professionals. For questions 1-6, choose the answer (**A, B or C**) which you think fits best according to the text.

1. The special access scheme should be used when

- Ⓐ The patient requires specialised treatment.
- Ⓑ All other possibilities have been exhausted.
- Ⓒ The standard of the product has been verified.

Extract from policy document

Overview of medicinal cannabis and the special access scheme (SAS)

Medicines (products) that are included in the Australian Register of Therapeutic Goods (ARTG) have been evaluated to ensure they meet strict standards of safety, quality and effectiveness. For this reason, it is expected that medical practitioners (prescribers) will have considered all clinically appropriate treatment options that are included in the ARTG before applying to access an unapproved medicinal cannabis product under the SAS.

Where products in the ARTG are found to not be clinically suitable, the SAS provides a pathway for prescribers to access unapproved products for individual patients on a case-by-case basis. However, it is important to note that there can be no guarantee of the quality, safety and effectiveness of unapproved products accessed through SAS and therefore the prescriber and patient (via informed consent) accept responsibility for any adverse consequences of treatment.

2. What needs to be considered when recommending the use of cough and cold medicines in children?

- Ⓐ The possible dangers.
- Ⓑ The low success rate.
- Ⓒ The age of the child.

Cough and cold medicines for children

Health professionals are advised of the following:

No changes have been made to the scheduling of over-the-counter (OTC) cough and cold medicines and a prescription is not required. However, a recommendation for treatment with these medicines in a child under 6 years of age constitutes off-label use. These recommendations for treatment should not be made without serious consideration of the risks and benefits. There is no robust evidence of efficacy for these medicines in children and there are a number of safety concerns. While the safety concerns are lower in children aged 6 - 11 years than in children aged less than 6 years, they should be taken into consideration when advising parents on the management of coughs and colds.

3. The memo tells us that the new form

- Ⓐ Replaces all previous airway forms.
- Ⓑ Is designed to reduce airway incidences.
- Ⓒ Can be used in a variety of airway situations.

Guidelines: Leadership and Management for all Doctors

This guidance sets out the wider management and leadership responsibilities of doctors in the workplace. The principles in this guidance apply to all doctors, whether they work directly with patients or have a formal management role.

Being a good doctor means more than simply being a good clinician. In their day-to-day role doctors can provide leadership to their colleagues and vision for the organisations in which they work and for the profession as a whole. However, unless doctors are willing to contribute to improving the quality of services and to speak up when things are wrong, patient care is likely to suffer. You must be prepared to explain and justify your decisions and actions. Serious or persistent failure to follow this guidance will put your registration, and so you're right to practice medicine, at risk.

4. The guidelines inform us that pregnancy testing

- Ⓐ Is compulsory for any woman who suspects she may be pregnant.
- Ⓑ May result in scheduled treatment being postponed or cancelled.
- Ⓒ Should be conducted in accordance with established procedures.

Extract from guidelines: Anaesthesia and pregnancy testing

All women of childbearing age should be sensitively questioned on the day of surgery as to whether there is a possibility that they could be pregnant. Women who could possibly be pregnant should be informed of the risks and a pregnancy test should be discussed.

Pregnancy tests should only be carried out on women who may be pregnant with their consent, and any relevant discussions should be documented in the clinical notes. There should be locally agreed policies on the administration and checking of pregnancy tests prior to surgery.

5. The policy recommends that vitamin K be given to infants

- Ⓐ By a trained health professional.
- Ⓑ Within the first month of birth
- Ⓒ Only if they are healthy.

Vitamin K administration to newborn infants

Healthy newborn infants should receive vitamin K either by intramuscular injection of 1 mg (0.1 mL) of Konakion MM at birth (this is the preferred route for reliability of administration and level of compliance), or as three 2 mg (0.2 mL) oral doses of Konakion MM, given at birth, at the time of newborn screening (usually at three to five days of age) and in the fourth week. The last dose is not required in infants predominantly formula fed. It is imperative that the third dose is given no later than four weeks after birth as the effect of earlier doses decreases after this time. Undertaking this form of prophylaxis requires that the parent accepts responsibility and that clinician's support and advise them in the administration of the third dose. If the infant vomits or regurgitates the formulation within one hour of administration, the oral dose should be repeated.

6. The update on cosmetic and discretionary surgery informs us

- Ⓐ Who is eligible to provide surgery.
- Ⓑ When surgery should be performed.
- Ⓒ The process for approval of surgery.

Information Update: Cosmetic & Discretionary Surgery

Surgery should meet an identified clinical need to improve the physical health of the patient. The approval of the Local Health District/Network Program Director of Surgery, in consultation with senior management should be sought by the referring doctor before cosmetic and discretionary procedures are undertaken in any public hospital facility.

The referring doctor should document on the Request for Admission form, at the time a patient is referred, objective medical criteria supporting the decision for surgery for all procedures that may be considered cosmetic or discretionary. This requirement supports appropriate documentation of clinical decision-making and the review procedure. The patient should be advised when the Recommendation for Admission is going through the approval process.

In this part of the test, there are two texts about different aspects of healthcare. For questions 7-22, choose the answer (**A, B, C or D**) which you think fits best according to the text.

Text 1: Healthy Ageing

Having friends, and participating socially, can help to soften the stresses of life and reduce feelings of helplessness. The very large INTERHEART case control study across 52, countries found that the presence of psychosocial stressors was associated with increased risk of acute myocardial infarction. This was still significant after adjusting for other cardiovascular risk factors. The effect of stress was independent of socio-economic status and smoking, and occurred across all geographic regions and age groups, and in both men and women. The authors concluded that approaches aimed at modifying psychosocial stress should be developed.

In a US study, loneliness was prospectively associated with increased risk of incident coronary heart disease, after controlling for multiple confounding factors. A study of Thailand rural elders found that social support buffered the impact of disability and reduced the risk of depression. Studies also show that helping others helps older people to adjust to their own decline in function and health. Recently Holt-Lunstad et al undertook a helpful review and appraisal of the many studies that have examined the influence of social isolation. They concluded that people with adequate social relationships have a 50% greater likelihood of survival compared to those with poor or insufficient social relationships. The effect is so great that it is comparable with quitting smoking and exceeds the better-publicised risk factors such as obesity and lack of physical exercise. We should also note, however, that relationships that are unhappy with conflict or excessive demands can increase risk of depression or angina.

Many recent studies have added to our understanding of the pathways through which social isolation influences physiological mechanisms to cause disease. Grant et al explored the effect of social isolation on the body's ability to recover from physiological responses to stress. Men and women who were socially isolated had slower post-task recovery of systolic blood pressure and greater cortisol output over

the day; the men also had a higher cholesterol response to stress. In a European study, salivary cortisol responses to mental stress were associated with coronary artery calcification in healthy men and women. In a US cross-sectional study, level of social integration was associated with fibrinogen concentration in elderly men. In another US study social integration was found to modify physiologic pathways influenced by stress, such as blood pressure, reducing risk of cardiovascular disease.

Friendships, helping others, and social participation increase self-esteem and well-being. Older people are then more likely to be motivated to change behaviours that jeopardise their health, such as smoking and drinking, and to maintain their healthier behaviours. They are more likely to seek health care, and to have better self-care in the management of their conditions. The 'Package of Essential Non-communicable Disease Interventions for Primary Health Care in Low-Resource Settings' was developed by WHO to improve access to cost effective interventions in resource constrained settings. It addresses the risk factors of smoking, diets high in fat and salt, lack of physical activity, and high alcohol intake. The **'package'** does not include strategies to encourage social participation.

Simple messages are effective for advocacy purposes. So the choice by WHO of the 4 x 4 concept (4 diseases; 4 shared risk factors) is understandable. However, this sharp focus on the 'lifestyle' risks tends to keep other significant factors, such as social participation, **in the shadows**. The focus on 'lifestyle' factors can also suggest that individuals are responsible for their own behaviours and the illnesses that result from them. Yet the evidence that low birth weight and genetic factors influence vulnerability to diabetes and cardiovascular disease is very relevant in low income settings.

There are a number of possible reasons why the potential of encouraging social participation has not been recognised and promoted within the WHO package for addressing non-communicable diseases (NCDs). Understandably the WHO has a commitment to evidence-based policy making, which tends to prioritise results of randomised controlled trials. Much of the evidence for the influence of social participation on development of NCDs, and understanding of the mechanisms for this association, are relatively recent, so there have been few trials of interventions.

It is also difficult to standardise such interventions for trials, and they are likely to be context-dependent. However, because an intervention has not been trialled does not mean that it may not be effective — there is a difference between ‘no evidence’ and ‘evidence of no effect’.

Text 1: Questions 7-14

7. The writer says the INTERHEART study found that stress

- Ⓐ Is conditional on how much money a person earns.
- Ⓑ Can be affected by a person's level of social integration.
- Ⓒ Affects people from some nations more than others.
- Ⓓ Is worse in people who already have health concerns.

8. In the second paragraph, the writer cites several studies to suggest

- Ⓐ The quality of a person's relationships is integral to their health.
- Ⓑ Being on your own as you age may lead to premature death.
- Ⓒ Some diseases are more affected by loneliness than others
- Ⓓ There are many contributing factors that lead to poor health.

9. What does the phrase '**the effect**' refer to?

- Ⓐ Other issues that reduce mortality rates.
- Ⓑ A person's chance of staying healthy.
- Ⓒ The influence of being on your own.
- Ⓓ Having enough people in your life.

10. What point does the writer make in the third paragraph?

- Ⓐ The causes of social isolation differ between the US and Europe.
- Ⓑ More research has been done into social isolation than social integration.
- Ⓒ Social integration assists in minimising heart disease by altering nerve impulses.
- Ⓓ Social isolation affects women while social integration has a greater effect on men.

11. The writer suggests the 'package' developed by the WHO

- Ⓐ Has a strong focus on saving money.
- Ⓑ Encourages elderly people to take responsibility for their lives.
- Ⓒ Fails to acknowledge dangers beyond lifestyle choices and eating habits.
- Ⓓ Carefully balances a range of issues that impact on an individual's well-being.

12. The phrase 'in the shadows' suggests that social participation

- Ⓐ Is difficult to understand.
- Ⓑ Has become hidden.
- Ⓒ Is largely unknown.
- Ⓓ Has been ignored.

13. In the fifth paragraph, the writer expresses the view that

- Ⓐ The approach by the WHO is too simplistic.
- Ⓑ People in low-income areas have worse health.
- Ⓒ Lifestyle factors are not the only cause of illness.
- Ⓓ Irresponsible behaviour causes most health problems.

14. The writer suggests that social participation is not encouraged in the WHO package because

- Ⓐ Studies of this kind would be challenging to complete.
- Ⓑ It is a new area of research with limited data to support it.
- Ⓒ They have an obligation to focus on areas that promote profits.
- Ⓓ There is no indication it would make any difference to a person's health.

Text 2: No Scientific Proof That Multivitamins Promote Heart Health

Taking multivitamin/mineral (MVM) supplements does not prevent myocardial infarction, stroke, or death from a cardiovascular cause, according to a comprehensive meta-analytic review of relevant research. “The take-home message is simple: there is no scientific evidence that MVM supplements promote cardiovascular health,” said lead author Joonseok Kim, MD, University of Alabama at Birmingham. “We hope that our paper helps to settle the controversy on MVM use for CVD prevention. It has been exceptionally difficult to convince people, including nutritional researchers, to acknowledge that multivitamin and mineral supplements don’t prevent cardiovascular diseases.”

“Vitamin and mineral supplements are taken by nearly half of US adults, yet few benefits have been documented,” said JoAnn Manson, MD, who was not involved in the study and is chief of preventive medicine, Brigham and Women’s Hospital and professor of medicine, Harvard Medical School, both in Boston, Massachusetts. “Regarding multivitamins and cardiovascular disease, specifically, neither observational studies nor randomised clinical trials have demonstrated clear benefits for primary or secondary prevention,” Manson said. “Importantly, clinicians should emphasise with their patients that multivitamin supplements will never be a substitute for a healthful, balanced way of eating, which have many beneficial components for vascular health. Additionally, micronutrients in food are typically better absorbed by the body than those from supplements,” she advised.

Kim and colleagues did a systematic review and meta-analysis of 18 studies with more than 2 million adults (mean age, 57.8 years) with mean follow-up of 11.6 years. Eleven studies were from the United States, 4 from Europe, and 3 from Japan. Only 5 studies specified the dose and type of MVM supplement. Overall, there was no association between MVM supplement use and cardiovascular disease (CVD) mortality, the investigators report. There was also no link between MVM supplements and CVD or coronary heart disease (CHD) mortality in pre-specified subgroups categorised by mean follow-up; mean age; period of MVM use; sex; type of population; exclusion of patients with history of CHD; and adjustment for diet, adjustment for smoking, adjustment for physical activity, and study site. MVM supplement use did appear to be associated with a lower risk for CHD incidence.

However, this association did not remain significant in the pooled subgroup analysis of randomised controlled trials.

Alyson Haslam, MD, and Vinay Prasad, MD, both from Oregon Health & Science University in Portland, note that practices in biomedicine are often adopted because they “appeal to our hopes and there is biologic plausibility. In the case of multivitamins, it is logical that some vitamins may reduce cardiovascular events because they are anti-inflammatory or more broadly improve health and well-being. Yet, in this case, it appears they do not, and as such, multivitamins for cardiovascular disease joins the list of plausible but failed practices in cardiology,” they conclude.

Although multivitamins tend to be moderately dosed and are likely to be safer than mega-doses of individual dietary supplements, “they are not completely free of risk in all patients,” Dr Manson said. For example, dietary supplements may interact with some medications, such as vitamin K and warfarin; interfere with the measurement of some clinical laboratory tests, such as biotin and troponin levels; they also have side effects, such as gastrointestinal symptoms, for some patients, she explained. “Thus, routine multivitamin supplementation is not recommended for the general population, but a targeted approach is appropriate for certain life stages and high-risk groups.” Some examples of relevant life stages include pregnancy, where supplementation with folic acid/prenatal vitamins is of benefit, and in midlife or older adults, some of whom may benefit from supplemental vitamin B12, vitamin D, and/or calcium. High-risk groups, such as those with malabsorption syndromes, restricted eating patterns, osteoporosis, pernicious anaemia, and age-related macular degeneration, and those with long-term use of metformin or proton-pump inhibitors also may benefit from dietary supplements, she said.

Dr Manson also noted that the Physicians’ Health Study II, a large-scale randomised clinical trial of multivitamins in men, demonstrated that these supplements may modestly reduce the incidence of cancer. This finding is being explored further in an ongoing Cocoa Supplement and Multivitamin Outcomes Study (COSMOS) trial, which is testing whether multivitamins, with or without cocoa flavanols, can reduce the risk for cancer and CVD in older men and women. Initial results of the PHS II trial indicated that neither vitamin C nor vitamin E supplementation is associated with a

reduction in major cardiovascular outcomes, as compared with placebo, although vitamin E may be associated with a slightly higher incidence of haemorrhagic stroke, compared with placebo. Results from the COSMOS trial are expected in 2 years, so stay tuned.

Text 2: Questions 15-22

15. Dr Joonseok Kim uses the expression 'settle the controversy' to suggest there is

- Ⓐ A lack of agreement about the efficacy of MVM supplement use.
- Ⓑ Confusion over the terms used to define cardiovascular disease.
- Ⓒ A dispute over the findings from his latest published research.
- Ⓓ Misunderstanding about the methods used during his study

16. What are Dr Joann Manson's views on multivitamins?

- Ⓐ Far too many people take them.
- Ⓑ They help fight certain illnesses.
- Ⓒ We still don't know enough about them.
- Ⓓ They shouldn't replace a nutritional diet.

17. The writer explains the systematic review and meta-analysis of the studies reviewed by Dr Kim to highlight

- Ⓐ The amount of categories it covered.
- Ⓑ How many people it included.
- Ⓒ The age of the participants.
- Ⓓ Its global scope.

18. What does the word **they** in the fourth paragraph refer to?

- Ⓐ Multivitamins.
- Ⓑ Some vitamins.
- Ⓒ Health and well-being.
- Ⓓ Cardiovascular events.

19. How do Dr Haslam and Dr Prasad feel about multivitamin use for cardiovascular disease

- Ⓐ They are positive about their future application.
- Ⓑ They believe that some help and some don't.
- Ⓒ Although they appear to be promising, ultimately they don't work.
- Ⓓ Scientifically speaking they are still one of the best options available.

20. What particularly concerns Dr Manson?

- Ⓐ The way some people take multivitamins instead of their prescribed medication.
- Ⓑ The lack of understanding about the prolonged use of multivitamins.
- Ⓒ The ineffectiveness of multivitamins in many patients.
- Ⓓ The belief that multivitamins do no harm.

21. In the fifth paragraph, Dr Manson explains that supplements

- Ⓐ Should be regulated for use with particular diseases.
- Ⓑ Do have their place as a method of treatment.
- Ⓒ Work better in some people than others.
- Ⓓ Are helpful for vegetarians.

22. The writer mentions the Physicians' Health Study II to highlight that supplements

- Ⓐ Are ineffective in most situations.
- Ⓑ May work better depending on a person's gender.
- Ⓒ May work best when combined with other vitamins.
- Ⓓ Are still being investigated as a way of treating serious illness.

READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT remove OET material from the test room.

TEXT A

Growth Monitoring

The monitoring of children's growth (and weight in particular) is an important role of all health professionals. At every consultation (or at least yearly), health professionals need to have a conversation with families and carers around children achieving a healthy weight.

"Ask and Assess – use percentile charts to monitor growth"

Growth status in children and adolescents (age 0-18 years old) needs to be assessed using age- and sex-specific reference values, as the appropriate ratio of weight to height varies during development.

Reference values for assessing and monitoring weight, length/height and BMI have been developed by the World Health Organisation (WHO) and Centres for Disease Control and Prevention (CDC) in the form of the childhood growth charts. The choice of chart depends on the age and gender of the child. For children aged less than two years the WHO growth charts should be used. For children between 2 and 18 year either the WHO or the CDC growth charts can be used. However, it is important to ensure that the same chart is used over time.

TEXT B

Steps for discussing weight with families or carers

1. Make the family feel welcome and at ease.
2. Ask permission to discuss growth and development with the family.
3. Normalise the discussion of growth and development. Reassure the child and family that it is standard practice in primary care.
4. Explain what is measured and why? (Weight, Height, BMI).
5. Encourage involvement of parents/carers when measuring weight and height.
6. Explain how the child's growth tracks against the growth of all children. Engage the family in plotting and interpreting the growth chart, using the growth charts as a tool.
7. Advise parents or carers of child's BMI. Be sensitive and nonjudgmental but don't sugar-coat it. For example: "Tom's BMI is... which places him above the healthy weight range."
8. Use positive terminology to reinforce key concepts and advice:
 - Use "healthy eating", "eating" or "eating plan" rather than "diet" or "dieting".
 - Do not use 'good' or 'bad' to describe food or drinks – these terms reinforce a dieting mentality and create blame and guilt.
 - Let the family do most of the talking rather than you.
 - Be realistic - make sure everyone in the room understands that long term changes have to be made if the child is above the healthy weight range and that it will take time.
 - Avoid inappropriate words, such as "Tom is obese."
9. For adolescents, if appropriate, consider speaking to them without parent or carer present.
10. Agree on and develop an action plan

TEXT C

More than 25% of children and adolescents are overweight or obese. For obese children, complications are more likely to develop because they are obese longer. Risk factors for obesity in infants are low birth weight and maternal obesity, diabetes, and smoking. After puberty, food intake increases; in boys, the extra calories are used to increase protein deposition, but in girls, fat storage is increased. For obese children, psychologic complications (e.g., poor self-esteem, social difficulties, and depression) and musculoskeletal complications can develop early. Some musculoskeletal complications, such as slipped capital femoral epiphyses, occur only in children. Other early complications may include obstructive sleep apnea, insulin resistance, hyperlipidaemia, and non-alcoholic steatohepatitis. Risk of cardiovascular, respiratory, metabolic, hepatic and other obesity-related complications increases when these children become adults.

TEXT D

The possibility of obesity persisting into adulthood depends partly on when obesity first develops:

- During infancy: Low possibility
- Between 6 months and 5 yrs.: 25%
- After 6 yrs.: > 50%
- During adolescence if a parent is obese: > 80%

In children, preventing further weight gain, rather than losing weight, is a reasonable goal. Diet should be modified, and physical activity increased. Increasing general activities and play is more likely to be effective than a structured exercise program. Participating in physical activities during childhood may promote a lifelong physically active lifestyle. Limiting sedentary activities (e.g., watching TV, using the computer or handheld devices) can also help. Drugs and surgery are usually avoided but, if complications of obesity are life threatening, may be warranted.

READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** or the **Text Booklet** until you are told to do so.

Write your answers on the spaces provided on this **Question Paper**.

You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this Question Paper and the Text Booklet.

DO NOT remove **OET** material from the test room.

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate Text **Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

Assessing pain: Questions

Questions 1-7

For each of the questions, 1-7, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1. how to approach giving advice about overweight children? -----
2. how often to discuss a child's weight? -----
3. possible causes for obesity in infants and adolescent females? -----
4. the standards used to measure the development of children? -----
5. other conditions which are associated with obesity in children? -----
6. strategies to help stop children's weight increasing? -----

Questions 7-13

Answer each of the questions, **7-13**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

7. What influences the type of chart used to calculate a child's growth status?

8. What does the WHO and CDC assesses and monitor when forming childhood growth charts?

9. What increases in children once they reach pubescence?

10. If it is already in the family, what is the likelihood that a teenager will remain obese?

11. What needs to be agreed upon and developed with the family of an obese child?

12. What psychological symptoms are known to appear early in obese children?

13. If the situation is serious enough, what might be required in some children?

Questions 14-20

Complete each of the sentences, **14-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

14. If a mother has -----, smokes, or is herself obese it increases the risk of obesity in her child.
15. You should discuss healthy weight in children with parents -----
----- at a minimum.
16. Make sure to clarify the types of areas that are -----
----- and the reasons why.
17. The family should be encouraged to participate in the use of the -----
-----.
18. Convey information about the -----of children in an understanding way
19. If a child is obese for a length period of time, then they are more likely to experience -----
20. The family needs to appreciate that -----
changes must occur in children whose weight exceeds what is healthy.

READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this Question Paper until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this Question Paper.

DO NOT remove **OET** material from the test room.

HOW TO ANSWER THE QUESTIONS

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:



In this part of the test, there are six short extracts relating to the work of health professionals. For questions 1-6, choose the answer (**A, B or C**) which you think fits best according to the text.

1. The instructions explain that when making emergency phone referrals

- Ⓐ GRACE must always be contacted first.
- Ⓑ A referral must always be faxed right away.
- Ⓒ The on-call registrar must always grant approval.

Emergency referrals

Phone the relevant On-call Registrar via the Department of Emergency Medicine at your nearest facility. All urgent cases need to be authorised by the On-call Registrar. Urgent cases accepted via phone must be accompanied with a written referral and a copy faxed as soon as possible. If you have a patient who lives in The Prince Charles Hospital (TPCH) area and you are considering sending them to TPCH Emergency Department, please call the GP Rapid Access to Constatative Expertise (GRACE) hotline first. GRACE will provide access to internal medicine expertise to help with assessment, management and treatment.

2. To eliminate the risk of contamination from infectious agents or certain types of fluids, aprons and gowns should be

- Ⓐ Worn over all clothing.
- Ⓑ Changed when they become soiled.
- Ⓒ Discarded if they are damaged.

Extract from guidelines: Aprons and Gowns

International guidelines recommend that protective clothing (apron or gown) be worn by all healthcare workers when:

- Close contact with the patient, materials or equipment may lead to contamination of skin, uniforms or other clothing with infectious agents; or
- There is a risk of contamination with blood, body substances, secretions or excretions (except sweat).

The type of apron or gown required depends on the degree of risk, including the anticipated degree of contact with infectious material and the potential for blood and body substances to penetrate through to clothes or skin. Gowns and aprons must be changed between patients.

Clinical and laboratory coats or jackets worn over personal clothing for comfort and/or purposes of identity are not considered to be personal protective equipment. These items of clothing need to be changed dependant on activity and the extent of exposure to potential pathogens.

3. The purpose of the email about oral health clinical indicators is to

- Ⓐ Explain why the reports were introduced.
- Ⓑ Highlight what the reports have achieved.
- Ⓒ Provide future details about upcoming reports.

Update from the Dental Board of Australia: Oral health clinical indicators

Dear Practitioner,

The performance of public oral health clinics in Queensland is improving thanks to a range of oral health clinical indicators.

Since the first clinical indicator report was released in January 2015, there has been a 19 per cent increase in the rate of timely completion of root canal treatment, as clinics were prompted to review their local policies based on report findings.

The development of these reports has enabled Hospital and Health Boards to be better informed about the patient outcomes of oral health services, as well as how they compare to other services. The findings inform evidence-based quality improvement activities, and when generated over time; help determine if these activities are working.

4. The policy informs us that doctors treating athletes at sporting events in the UK

- Ⓐ Must be appropriately licensed.
- Ⓑ Are expected to have adequate training.
- Ⓒ Cannot treat anyone from another country.

Doctors working at sports events in the UK

Each year, athletes from all over the world come to the UK to take part in sports events and many bring their own doctors with them. There is a specific registration process for doctors who are travelling to the UK with their country's teams for sports events.

Doctors must be registered with a licence to practise medicine in the UK. This includes all doctors visiting from overseas and from Europe on a temporary basis. All team doctors accompanying their teams for sporting events must hold temporary registration with a licence to practise in order to be permitted to treat, diagnose and prescribe for their team. Please note that this type of registration does not permit doctors to treat UK nationals.

5. The purpose of the guidelines on remote consultations is to

- Ⓐ Help practitioners decide an appropriate course of action.
- Ⓑ Advise practitioners about a growing trend in modern medicine.
- Ⓒ Recommend a series of questions for practitioners to ask their patients.

Remote consultations

Remote consultations (over the phone, via video link or online) are on the increase. They can save doctors' time, benefit patients and help meet public demand for faster access to medical advice. But there are potential patient safety risks to consulting remotely. It is therefore important to identify and manage those risks, and to recognise that remote consultations are not always the right choice.

When deciding whether or not to perform a remote consultation, ask yourself:

- What do I know about the patient's needs and wishes?
- What are limitations and risks of communicating with the patient by phone/video link/online?
- Will I need to carry out a physical examination or other assessment?
- Do I have access to the patient's medical records and do I need them?
- Will the patient need follow up or ongoing management?

The answers should help you decide when a remote consultation is appropriate, and when a face-to-face consultation is required.

6. According to the extract, prescription medicines

- Ⓐ Are often unavailable due to stock shortages.
- Ⓑ Can be purchased as special orders from manufacturers if required.
- Ⓒ May show as being unavailable even though they can still be bought.

Medicine shortages information initiative

The Medicine Shortages Information Initiative provides information about a temporary or permanent disruption to the supply of a prescription medicine. This information is provided to assist health professionals and consumers when there is a temporary or permanent disruption (discontinuation) to the supply of a medicine in Australia.

The information provided refers to the supply of prescription medicines to the overall Australian market. This means that stock of a particular medicine may still be available at a pharmacy to fill a prescription until the specific pharmacy runs out of that medicine.

This information and further information on particular medicine shortages are published as alerts. These are updated when new information becomes available, including when the shortage has been resolved.

In this part of the test, there are two texts about different aspects of healthcare. For questions 7-22, choose the answer (**A, B, C or D**) which you think fits best according to the text.

Text 1: Stress and Burnout

Intensive care medicine (ICM) is an evolving high stakes specialty. Emerging evidence raises questions about the welfare and sustainability of the ICM workforce. Clinician burnout is a phenomenon resulting in consequences for both intensive caregivers and patients. While resident doctors, fellows and new consultants across many specialties display high levels of stress and burnout relative to the general population, ICM clinicians are disproportionately affected, rating higher on stress, burnout and compassion fatigue indices. Paediatric intensivists have markedly higher burnout rates than general paediatricians. An Australian ICM study described an 80% rate of psychological stress and discomfort in a practising ICM specialist population, with many reporting burnout symptoms.

Burnout is a state of psychological distress related to chronic stress. Prevalence estimation using different tools, alternate metrics and cut-off points have made epidemiological studies of ICM clinician burnout challenging. The Maslach Burnout Inventory is the most commonly used burnout instrument in this area and scores three major characteristics of burnout: emotional exhaustion, depersonalisation and low levels of personal effectiveness. Intensive care unit (ICU) physicians and nursing staff have similar rates of burnout symptomatology, with ICU nurses reporting higher emotional exhaustion rates and ICU physicians reporting higher rates of depersonalisation and reduction in professional achievement. These trends indicate specific risk factors within the ICM environment.

The burnout syndrome has been described in Australian emergency medicine clinicians. Some protective influences appear to be ongoing professional development, dedicated non-clinical time, and a feeling of teamwork. Burnout does not necessarily correlate with job satisfaction, with predisposing factors in this group including younger age, workplace conflict, a lack of exercise, and excessive alcohol consumption. In addition to many of these stressors, intensive care clinicians are repeatedly exposed to high stakes, ethically challenging decision-making processes.

The high” density” of ethical decision making in 1CM contributes to moral distress and may be exacerbated by the provision of” disproportionate care’ where there is a perceived inappropriate or harmful mismatch between the level of care provision and a patient’s needs.

Care of the health care provider and quality of patient care are interconnected. Physician burnout has been associated with lower patient satisfaction, reduced health outcomes and medical error. Burnout symptoms reduce potential 1CM workforce capacity through increased sick leave and decreased staff retention. The consequences of burnt out clinicians may ripple through an entire organisation, compromising interactions between individuals and teams.

Evolving trends in Australia may further exacerbate the problem of burnout. These changes include greater Intensivist coverage and shift work, an increasingly fractionalised workforce with unequal gender balance, and an evolving external and ward 1CM responsibility. The move towards physically larger Australian ICUs has coincided with enhanced public expectations of clinical outcome and an increase in interventional medicine. The prevention and remediation of burnout requires consideration of both individual and systemic factors. At an individual level, a holistic approach to the 1CM clinician, not just as a service provider, is required. A balance must be facilitated between work, life, clinical and non-clinical duties and career progression. Stress prevention and resilience strategies include mindfulness and cognitive techniques, coaching, mentoring and, perhaps most importantly, peer discussion. Leadership from clinicians will be important to drive change at an institutional level. Compassionate staffing, flexible rostering, ensured leave and ongoing employee assistance programs should be broadly available. Clinicians themselves will need to foster an acceptance of their own vulnerability and cultivate an environment where open dialogue about stressors is respected.

The College of Intensive Care Medicine and the Australian and New Zealand Intensive Care Society have roles to play in the development of performance indicators for workplace stress and burnout, with complementary advocacy for a safe, sustainable workplace. The 1CM training model should encompass self-assessment and resilience skills, supported by commensurate training of trainee supervisors and senior staff. A broader societal discussion about the antecedents of

moral distress and disproportionate care is required. Shared health goal setting before crises and preparing for realistic, appropriate decisions at the end of life continue to be of great importance. Such projects may be supported at government level, with direct expert input from 1CM clinicians.

While there is increasing evidence of the physical and emotional effects of the unique ICU environment on inter-professional practitioners, there remains a **paucity** of coordinated interventions aimed at understanding and addressing 1CM clinician burnout. It has therefore been suggested that a multilevel response is required in order to improve the welfare and sustainability of the Australian 1CM workforce.

Text 1: Questions 7-14

7. What concern does the writer express about ICM clinicians?

- Ⓐ There are not enough of them to treat the rising number of sick people.
- Ⓑ The numbers of those who work with children has increased.
- Ⓒ Many may not be performing effectively because of a pressured work environment.
- Ⓓ They usually work longer hours than is recommended.

8. In the second paragraph, the writer suggests that burnout

- Ⓐ Poses problems for the intensive care setting.
- Ⓑ Has become more prominent in recent years.
- Ⓒ Isn't currently able to be measured.
- Ⓓ Is very similar to depression.

9. What does the writer say about burnout syndrome in Australian emergency medicine clinicians?

- Ⓐ Not enough emphasis is placed on ways to relieve pressure.
- Ⓑ Training will help ICU staff better understand how they're feeling.
- Ⓒ The situation is made worse by of the level of responsibility placed on ICU staff.
- Ⓓ The impact of the condition is dependent on a range of personal choices.

10. The writer uses the term 'interconnected' to indicate that burnout can

- Ⓐ Make it difficult for staff to work well together.
- Ⓑ Impact on medical staff and patients.
- Ⓒ Cause some healthcare professionals to leave the industry.
- Ⓓ Affect clinicians for many years after they are first diagnosed.

11. What does the writer suggest about the problem of burnout in the fifth paragraph?

- Ⓐ It is being made worse by a reduction in healthcare professionals.
- Ⓑ It is developing in line with broader societal issues.
- Ⓒ It isn't going to be solved any time soon.
- Ⓓ It affects men more than women.

12. What point does the writer make in the sixth paragraph?

- Ⓐ It is difficult for staff to change their current behaviour.
- Ⓑ The existing approach within hospitals is there for a reason.
- Ⓒ Management should do more to improve working conditions.
- Ⓓ There needs to be a multifaceted response to workforce welfare.

13. The writer explains that the College of Intensive Care Medicine and the Australian and New Zealand Intensive Care Society

- Ⓐ Are integral in measuring the health of practitioners.
- Ⓑ Should take action before the situation worsens for practitioners.
- Ⓒ Require assistance from other sectors in order to help practitioners reduce their workload.
- Ⓓ Need to educate patients about how their actions affect practitioners.

14. In the final paragraph, the writer uses the word 'paucity' to suggest that

- Ⓐ Not enough is being done to help practitioners.
- Ⓑ Practitioners need to do more to help themselves.
- Ⓒ The ICU may not be a suitable place for all practitioners to work.
- Ⓓ It is natural that some practitioners handle stress better than others.

Text 2: Fluoride

If your toothpaste is fluoride-free, brushing and flossing alone aren't enough to keep cavities away, according to a recent study. The fluoride is what helps you avoid cavities, says lead researcher Philippe Hujoel, PhD, a Seattle periodontist and professor of oral health sciences at the University of Washington. "It's not [simply] keeping the teeth cleaner;' for years, dental professionals have debated the importance of the "clean tooth" hypothesis versus the "sound tooth" hypothesis in preventing cavities. Those in the first camp say good oral hygiene will remove the sticky film of acid-producing plaque that breaks down the enamel and allows cavity-causing bacteria to invade the teeth. Others argue that brushing and flossing, no matter how intense, isn't enough to prevent cavities. "The plaque is inaccessible and you can't get to it," Hujoel says. While oral hygiene may help a bit, it's the fluoride that makes the difference in getting to the plaque and breaking it down, he says. Cavities begin in tiny cracks and crevices in the enamel. Exactly how fluoride helps prevent cavities is not certain, according to Hujoel. "There is some evidence it may inhibit the enzymes that break down the tooth;' he says. In general, experts believe fluoride helps restore minerals to the enamel, helps strengthen the tooth, and even helps reverse the early cavity process.

While fluoride has been recommended for years, Hujoel's team wanted to focus on the intensity of oral hygiene to see if it made a difference in cavity prevention. They searched the published medical literature from 1950 to 2017 and found three randomised clinical trials, including 743 preteens and teens that were sound enough to include and analyse. Two were conducted in the U.S. and one in the UK. None were funded by commercial companies. In the studies, researchers assigned children to an intense oral hygiene group or to a usual or less intense hygiene group of brushing and flossing. In the intense group, the children had supervision of their oral hygiene, with plaque removal, at school, but no fluoride toothpaste was used at school. In the U.K. study, all used toothpaste with fluoride at home. Some in the U.S. studies used fluoride toothpaste and some did not at home, Hujoel says. While the design was not ideal, the key point was to compare intense hygiene with less intense hygiene, he says. Two studies were done in communities with non-fluoridated water supplies. "There was no significant difference in cavities between the groups," Hujoel

says. “These intensive oral hygiene interventions, which were successful in removing the biofilm, did not have an impact on the cavities.”

The study “supports what the dental association has said for years, that brushing with fluoride is good,” says Matthew Messina, DDS, a dentist in Columbus, Ohio, who is also a consumer advisor for the American Dental Association and assistant professor of dentistry at The Ohio State University in Columbus. To earn the ADA Seal of Acceptance, a toothpaste must contain fluoride, he says. The ADA recommends brushing twice a day with a fluoride-containing toothpaste, flossing once a day, eating a healthy diet, and seeing a dentist on a regular basis. “We know that works,” he says. That’s lifelong advice, he says, as cavities don’t just affect children. “We are seeing an increase in the rate of cavities in the older population,” he says. That’s partially due to improved dental techniques allowing people to keep their natural teeth longer, he says. But dry mouth, a side effect of numerous medications taken by older adults, can also make teeth more prone to decay, he says.

Fluoride offers one way to strengthen teeth and does have potential benefits, says Darryl Bosshardt, a spokesman for Redmond, which makes a fluoride-free toothpaste, Earthpaste. But tooth decay is not caused by fluoride deficiency, he says, and fluoride supplementation can’t reverse active cavities. “It can also have some potential negative aspects that some consumers would like to avoid if possible,” he says. As one of many examples, he cited a study finding fluoride-containing toothpaste ingestion as a main source of fluoride toxicity, according to the Association of Poison Control, especially in young children. “We are also quick to point out that a non-fluoride toothpaste may not be the best option for everyone. However, we similarly acknowledge that mandatory fluoride supplementation in all water supplies and in all oral care products may also not be the ideal solution for everyone.” He encourages people to weigh the pros and cons with their dentist.

While toothpaste with fluoride “is the best choice for oral health, we also recognise that not all the people that choose our brand want fluoride in their toothpaste, and we offer a fluoride-free alternative,” says Rob Robinson, a spokesman for Tom’s of Maine. The fluoride-free toothpaste from Tom’s does not carry an anti-cavity claim.

For those who do not want to use fluoride-containing toothpastes, another option is going on a very low-carb diet, generally less than 50 grams a day, Hujoel says. Doing so cuts back drastically on the sugars that can attack the teeth and lead to cavities, he says. That's the path he takes, but, he acknowledges, few can follow such a strict diet and so should use fluoride-containing toothpaste as part of their oral hygiene routine.

Text 2: Questions 15-22

15. The phrase 'the first camp' refers to dental professionals who

- Ⓐ Cannot agree on the effectiveness of fluoride.
- Ⓑ Believe keeping teeth clean prevents cavities.
- Ⓒ Feel that no amount of brushing reduced plaque.
- Ⓓ Are certain about how the entire process works.

16. Philippe Hujoel's comments in the first paragraph indicate that he believes

- Ⓐ No one knows how to prevent plaque.
- Ⓑ Everyone has plaque on their teeth.
- Ⓒ Only fluoride can remove plaque.
- Ⓓ Plaque is impossible to remove.

17. What does the word 'it' in the second paragraph refer to?

- Ⓐ Fluoride.
- Ⓑ Hujoel's team.
- Ⓒ Cavity prevention.
- Ⓓ The intensity of oral hygiene.

18. By examining clinical trials from the US and UK, Philippe Hujoel says his team established that

- Ⓐ Fluoride should be placed in all water supplies.
- Ⓑ Regular brushing is the best way to reduce cavities.
- Ⓒ Children need to be encouraged to brush their teeth.
- Ⓓ Intense oral hygiene is effective at cleaning teeth only.

19. Matthew Messina from the American Dental Association suggests that

- Ⓐ Scientific advances have reduced tooth decay.
- Ⓑ Fluoride isn't as important as many people believe.
- Ⓒ It takes a wide-ranging approach to prevent cavities.
- Ⓓ Increased life expectancy results in more dental disease.

20. Darryl Bosshardt's comments indicate that he believes

- Ⓐ It is up to individuals to make their own choice about fluoride.
- Ⓑ Fluoride is still the best way to protect teeth.
- Ⓒ There are no alternatives to fluoride.
- Ⓓ Fluoride is a dangerous product.

21. The writer includes the statement that fluoride-free toothpaste from Tom's of Maine does not carry an anti-cavity claim to suggest that

- Ⓐ This is the best toothpaste for people to buy.
- Ⓑ Some types of toothpaste may not prevent decay.
- Ⓒ They don't like offering this type of toothpaste.
- Ⓓ It is difficult to decide which toothpaste works best.

22. In the final paragraph, the writer uses Philippe Hujoel's words to highlight that

- Ⓐ Not everyone takes the same approach to oral hygiene.
- Ⓑ Some food can contribute to the problem.
- Ⓒ Fluoride is the best option.
- Ⓓ Not all diets are practical.

READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

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CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT remove OET material from the test room.

TEXT A

Snake bite is uncommon in Victoria and envenomation (systemic poisoning from the bite) is rare. The bite site may be evidenced by fang marks, one or multiple scratches. The bite site may be painful, swollen or bruised, but usually is not for snakes in Victoria.

There are no sea snakes in Victoria; however land-based snakes can swim.

Major venomous snakes in Victoria and effects of envenomation:

Snake	Coagulopathy	Neurotoxicity	Myotoxcity	Systematic symptoms	Cardiovascular effects	TMA
Brown	VICC	Rare and mild	-	50%	Collapse (35%) Cardiac arrest	10%
TIGER	VICC	30%	20%	Common	Rare	5%
Red-bellied black	Mild increase in aPITT and INR with normal fibrinogen, usually no significant bleeding	-	uncommon	Common often significant bite site pain and limb swelling	-	-

TMA: thrombotic microangiopathy. Haemolysis with fragmented red blood cells on blood film, thrombocytopenia and a rising creatinine.

Myotoxicity muscle pain, tenderness, rhabdomyolysis

Systemic Symptoms see history and examination.

VICC: Venom-induced consumptive coagulopathy (abnormal INR, high aPTT, fibrinogen very low, D-dimer high).

TEXT B

Assessment

Focus on evidence of envenomation.

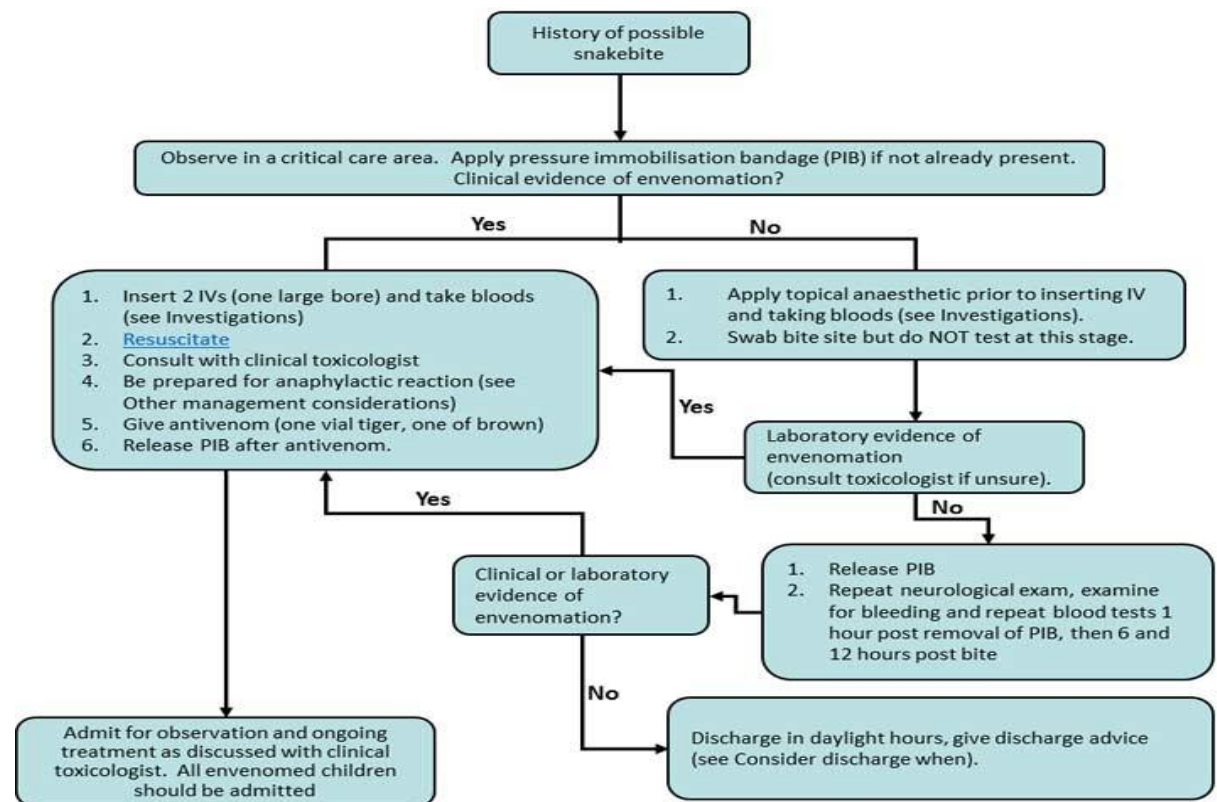
- Once the possibility of snakebite has been raised, it is important to determine whether a child has been envenomed to establish the need for antivenom.
- This is usually done taking into consideration the combination of circumstances, symptoms, examination and laboratory test results.
- Most people bitten by snakes in Australia do not become significantly envenomed.

History and Examination

Circumstances	Symptoms	Examinations
• Confirmed or witnessed bite versus suspicion that bite might have occurred • Were there multiple bites? • When? • Where? • First aid? • Past history? • Medications? • Allergies?	• Headache • Diaphoresis • Nausea or vomiting • Abdominal pain • Diarrhoea • Blurred or double vision • Slurring of speech • Muscle weakness • Respiratory distress • Bleeding from the bite site or elsewhere • Passing dark or red urine • Local pain or swelling at bite site • Muscle pain • Pain in lymph nodes draining the bite area • Loss of consciousness / collapse and/or convulsions	• Evidence of a bite / multiple bites. • Evidence of venom movement (e.g. swollen or tender draining lymph nodes) • Neurotoxic paralysis (ptosis, ophthalmoplegia, diplopia, dysarthria, limb weakness, respiratory muscle weakness) • Coagulopathy (bleeding gums, prolonged bleeding from venepuncture sites or other wounds, including the bite site) • Muscle damage (muscle tenderness, pain on movement, weakness, dark or red urine indicating myoglobinuria)

TEXT C

Snakebite management flowchart



TEXT D

Giving Antivenom

- Antivenom is indicated in all children where there is evidence of envenomation.
- Giving antivenom should occur in consultation with a clinical toxicologist.
- Dilute one vial in 100mls of 0.9% saline and give IV over 15-30 min.
- If the child is in cardiac arrest and this is thought to be due to envenomation, then give undiluted antivenom via rapid IV push.
- There is no weight based calculation for antivenom (the snake delivers the same amount of venom regardless of the size of the child). One vial of antivenom is enough to neutralise the venom that can be delivered by one snake. Clinical recovery takes time after antivenom administration and multiple vials do not speed recovery.

At discharge, ensure that the family is given advice on how to recognise serum sickness:

- Occurs in about 30% of children given antivenom.
 - Tends to occur 4 – 14 days following antivenom administration.
 - Consists of flu-like symptoms, fever, myalgia, arthralgia and rash.
- A letter should also be written to the child's GP regarding this

READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** or the **Text Booklet** until you are told to do so.

Write your answers on the spaces provided on this **Question Paper**.

You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this Question Paper and the Text Booklet.

DO NOT remove **OET** material from the test room.

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate Text **Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

Snakebite in children : Questions

Questions 1-7

For each of the questions, 1-7, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1. the factors to consider when evaluating snakebite? -----
2. what a snakebite can look like? -----
3. possible types of snakebite and their reactions? -----
4. signs that a child may be bitten by a poisonous snake? -----
5. when to release a pressure immobilisation bandage? -----
6. when it is safe to discharge a child who has not been envenomed? -----
7. what to tell parents to look for in a child having a response to serum? -----

Questions 8-15

Answer each of the questions, **8-15**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8. What **TWO** signs should you look for if snakebite is suspected?

9. When assessing a child, what might indicate the presence of venom movement?

10. In cases where neurotoxicity is rare or not severe, what type of snakebite could it indicate?

11. After removing the PIB, when should you conduct another blood test if there is no evidence of poisoning from snakebite?

12. What TWO anti-venoms should you give to a child with signs of envenomation?

13. What quantity and strength of saline should each vial of anti-venom be diluted in?

14. What time of day should discharge occur?

15. After anti-venom is administered, how long does it usually take for serum sickness to develop?

Questions 16-20

Complete each of the sentences, **16-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

16. If you suspect a snakebite has occurred, you must decide whether or not the child has been -----

17. Substantial pain and swelling is a likely sign that the child has been bitten by a -----

18. If clinical evidence warrants administration of antivenom, you should ensure it is done in conjunction with a -----

19. You will need to affix a ----- if it hasn't been done.

20. When deciding how much antivenom to administer, -----
----- is considered sufficient for each child irrespective of their body weight.

READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this Question Paper until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this Question Paper.

DO NOT remove **OET** material from the test room.

HOW TO ANSWER THE QUESTIONS

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:



In this part of the test, there are six short extracts relating to the work of health professionals. For questions 1-6, choose the answer (**A, B or C**) which you think fits best according to the text.

1. The purpose of the email to hospitals about prescribing of antibiotics is to

- Ⓐ Encourage them to implement appropriate procedures.
- Ⓑ Remind them of the seriousness of the issue.
- Ⓒ Advise them of upcoming changes.

Email from Health Service District Management: Control of Resistance

Antibiotic prescribing habits of clinicians and general practitioners are largely responsible for the emergence of resistant pathogens. The unnecessary use of antibiotics acts as a strong selective tool for the emergence of resistance microorganisms, and restriction of use should lead to the opposite effect.

Reducing antibiotic prescribing is far from easy, and a combined effort is mandatory. Adherence to prescribing guidelines (for hospital and community prescribing) (attached) and restriction policies that reduce the use of certain antibiotics (for hospital prescribing) may lead to the reduction in antibiotic overuse.

In addition to reducing antibiotic prescribing, judicious usage of antibiotics include considerations regarding choosing the right antibiotic class, taking into account factors like post-antibiotic effect and the tendency of certain antibiotic classes to induce resistance. All children hospitals should therefore develop an antimicrobial stewardship programme.

2. According to the memo, when keeping records staff should ensure

- Ⓐ They reflect everything that occurred.
- Ⓑ Important information is written down first.
- Ⓒ Their colleagues will be able to understand them.

Memo to all medical staff: Guidance for taking notes and keeping records

From the Medical Director

A number of problems are occurring throughout the hospital because some staff are keeping inaccurate and insufficient records of observations, assessments and treatments. I therefore want to draw the attention of all staff to the importance of good record-keeping and ask everyone to read these guidelines.

In summary they highlight that:

- Records primarily serve the interests of the patient.
- Records should be an accurate chronology of events — recording all consultations, assessments, observations, decisions and outcomes.
- Records are an essential means of communicating amongst staff.
- Records show that everyone has fulfilled his/her duty of care.

3. Why is the Queensland Bedside Audit conducted each year?

- Ⓐ To improve the quality of health service supply.
- Ⓑ To find out how parties feel about their treatment.
- Ⓒ To allow facilities to make advancements in technology.

Memo: Queensland Bedside Audit

The Queensland Bedside Audit (QBA) is a major clinical patient safety audit undertaken within Queensland every year. The data collected during the audit are used by Hospital and Health Services as evidence in meeting National Safety and Quality Health Service (NSQHS) Standards and other key safety and quality indicators.

The information is collected at the bedside and the results help to identify areas for improvement and establish a safety and quality framework that enables the delivery of the best possible care to patients.

The Queensland Bedside Audit consists of:

- A review of clinical documentation for all eligible patients
- A physical examination of consenting patients
- Asking patients questions on elements of their healthcare
- A review of the bed area.

By participating each year in the QBA, a facility can compare key outcome and process measures over time to assess the impact of their improvement initiatives.

4. The policy extract tells us that

- Ⓐ All haemophilia patients must be treated in a registered HTC.
- Ⓑ Only haemophilia patients with a factor level above 30% need to be treated in a HTC.
- Ⓒ The risks of being treated in a non-HTC facility have to be conveyed to relevant haemophilia patients.

Management of Haemophilia and Related Bleeding Disorders

In the interests of patient safety, patients with:

- Factor VIII <30%
- Factor IX <30%
- Factor XI <30%
- Von Willebrand Factor activity < 30%; and
- other rare bleeding disorders

should have elective surgery performed in a designated Haemophilia Treatment Centre (HTC) since they are “high risk patients” with an increased tendency to bleed, both early and late, and they require management by clinicians experienced in managing haemophilia and related bleeding disorders.

If a patient has a factor level that is over 30% and there is a specific reason for a procedure to be conducted at a hospital that is not a designated HTC, the patient/parent(s) or carer MUST be made aware of the potential risks attached to having surgery in a hospital that is not a designated or affiliated HTC and the Haemophilia Advisory Council Clinical Committee MUST be advised.

5. The guidelines inform us that GPs leaving general practice for more than three months

- Ⓐ Have a choice to make about whether or not to continue being recognised.
- Ⓑ Don't need to fulfil QI & CPD requirements if medicare still recognises them.
- Ⓒ Can decide whether or not to be recognised by medicare if they are a fellow.

GPs temporarily absent from general practice

A temporary absence from general practice may occur for many reasons (e.g. illness, parental leave, study leave or extended travel). GPs planning to leave general practice for more than three months need to temporarily remove their name from the list of recognised GPs.

General practitioners on the Royal Australian College of General Practitioners (RACGP) Fellows List may choose to remove or not remove their name from Medicare Australia's list of Recognised GP5 if they are planning to leave general practice for more than three months. However, requirements of the Quality Improvement and Continuing Professional Development (QI&CPD) Program need to be satisfied while they are on the list.

6. The hospital policy explains that

- Ⓐ It is generally preferable to only use items once.
- Ⓑ way in which items are reused is condition specific.
- Ⓒ Staffs are responsible for organising items into the correct group.

Reprocessing of Reusable Instruments and Equipment

Any instrument or piece of equipment that is to be reused requires reprocessing — cleaning, disinfection and/or sterilisation. The minimum level of reprocessing required for reusable instruments and equipment depends on the individual situation (i.e. the body site and the nature by which the instrument will be used).

The system used is based on instruments and items for patient care being categorised into critical, semi-critical and non-critical, according to the degree of risk for infection involved in use of the items.

Semi-critical items are defined as those that come into contact with mucous membranes or non-intact skin, and should be single use or sterilised after each use. If this is not possible, high-level disinfection is the minimum level of reprocessing that is acceptable.

In this part of the test, there are two texts about different aspects of healthcare. For questions 7-22, choose the answer (**A, B, C or D**) which you think fits best according to the text.

Text 1: Alcohol Consumption

Alcohol consumption is responsible for 2.8 million deaths per year across the globe, with cancer the leading cause of alcohol-related death among people aged 50 years and older, warn researchers, who also emphasise that there is no safe level of alcohol consumption. The findings come from the latest version of the Global Burden of Diseases, Injuries, and Risk Factors Study (GBD), which analysed data on 28 million people from 195 countries to estimate the prevalence of alcohol consumption, the amounts consumed, and the associated harms. The analysis found that among individuals aged 15 to 49 years, alcohol accounted for around 4% of deaths in women and 12% in men. Tuberculosis and road injuries were the leading causes of death related to alcohol. For those aged 50 years and older, alcohol was linked to 27% of deaths in women and 19% of deaths in men, with cancer the leading cause of alcohol-related death. Overall, consuming just one drink a day increased the risk of developing alcohol-related health problems by 0.5% vs abstaining; drinking five drinks a day led to a 37% increase in risk.

Lead author Max G. Griswold, MA, Institute for Health Metrics and Evaluation, University of Washington, Seattle, said in a release that although previous studies have suggested that alcohol is protective against some conditions, “we found that the combined health risks associated with alcohol increase with any amount of alcohol. “In particular, the strong association between alcohol consumption and the risk of cancer, injuries, and infectious diseases offset the protective effects for ischemic heart disease in women in our study;” he said. Although the health risks associated with alcohol start off being small with one drink a day, **they** then rise rapidly as people drink more,” he added. The new findings echo those from the 2014 World Cancer Report, which found a dose/response relationship between alcohol consumption and certain cancers. Griswold calls for public health policies to focus on “reducing alcohol consumption to the lowest level” and to revise the “widely held view of the benefits of alcohol.”

Co-author Emmanuela Gakidou, PhD, also from Institute for Health Metrics and Evaluation, went further, declaring: “Alcohol poses dire ramifications for future population health in the absence of policy action today. “Our results indicate that alcohol use and its harmful effects on health could become a growing challenge as countries become more developed, and enacting or maintaining strong alcohol control policies will be vital,” she said. She suggested that countries look at measures such as excise taxes and controlling the availability and advertising of alcohol. “Any of these policy actions would contribute to reductions in population-level consumption, a vital step toward decreasing the health loss associated with alcohol use.”

In an accompanying comment, Robyn Burton, PhD, Institute of Psychiatry, Psychology and Neuroscience, King’s College London, and Nick Sheron, MD, Division of Infection, Inflammation and Immunity, University of Southampton, United Kingdom, say that the results are “clear and unambiguous:” “Alcohol is a colossal global health problem and small reductions in health-related harms at low levels of alcohol intake are outweighed by the increased risk of other health-related harms, including cancer,” they write.

Burton and Sheron echo the call for public health policies to reduce population-level consumption to be prioritised. “These diseases of unhealthy behaviours, facilitated by unhealthy environments and fuelled by commercial interests putting shareholder value ahead of the tragic human consequences, are the dominant health issue of the 21st century:” they say. “The solutions are straightforward: increasing taxation creates income for hard-pressed health ministries, and reducing the exposure of children to alcohol marketing has no downsides.”

As if to support the views of Burton and Sheron, a recent study that showed that the websites and literature of 26 organisations related to the alcohol industry contained significant omissions or misrepresentations of the evidence linking alcohol to increased risk for many cancers. Despite all this, Griswold said he believes a ban on alcohol would be “a little step too far:” “There is a lot of cultural relevancy to alcohol, and there’s a long history in human society,” he said. “From a policy standpoint on a population level, we should work on reducing the amount of consumption, but I don’t

think taking a step of a ban would necessarily be the right direction, right away at least," he said. "I think that's a little severe."

Text 1: Questions 7-14

7. The writer uses the percentages in the first paragraph to highlight

- Ⓐ The size of the study.
- Ⓑ The illnesses identified by the study.
- Ⓒ The significance of the results of the study.
- Ⓓ The differences between certain groups who took part in the study.

8. What does Max Griswold, the lead author of the study, say concerns him?

- Ⓐ The fact that alcohol is consumed in larger quantities now than in the past.
- Ⓑ The misconception that in some instances alcohol may be good for you.
- Ⓒ The lack of education available about the effects of alcohol.
- Ⓓ The amount of ways alcohol has to make people sick.

9. What does the word 'they' in the second paragraph refer to?

- Ⓐ Types of threats.
- Ⓑ New discoveries.
- Ⓒ Number of drinks.
- Ⓓ Previous outcomes.

10. The writer uses the expression 'went further' to indicate Emmanuela Gakidou

- Ⓐ Would have liked the study to look at more cases.
- Ⓑ Wants more people to reduce the amount they are drinking.
- Ⓒ Believes the suggestions made by Max Griswold aren't enough.
- Ⓓ Isn't happy about the lack of action since the 2014 world cancer report.

11. Study co-author Emmanuela Gakidou suggests alcohol consumption

- Ⓐ Should not be of concern if current levels are monitored.
- Ⓑ Must be addressed by governments immediately.
- Ⓒ Has been increasing at an alarming rate.
- Ⓓ Is an excellent source of revenue.

12. What do Robyn Burton and Nick Sheron say about alcohol?

- Ⓐ More research is required before any action is taken.
- Ⓑ There are other issues that are more worrying.
- Ⓒ Curbing consumption decreases overall risk.
- Ⓓ Even drinking a small amount is dangerous.

13. In the fifth paragraph, Burton and Sheron suggest that

- Ⓐ There are currently not as many answers as there need to be.
- Ⓑ Financial considerations are being placed ahead of people's health.
- Ⓒ Most people will find it difficult to stop drinking even if it is bad for them.
- Ⓓ It would be very expensive for governments to reduce alcohol consumption.

14. In the final paragraph, Max Griswold expresses the belief that

- Ⓐ Alcohol has its place in the world.
- Ⓑ Too many companies lie about alcohol.
- Ⓒ Online advertising of alcohol should be prohibited.
- Ⓓ Action needs to be taken to rid communities of alcohol.

Text 2: Empathy

Every day, doctors, nurses and other health professionals are presented with situations that demand empathy and compassion. Whether telling a 40-year-old man with cancer he doesn't have long to live, or comforting an elderly woman who is feeling anxious, the health professional needs to be skilled in understanding what the other person is going through, and respond appropriately. With more demand on doctors and nurses and a push for quicker consultations, clinical empathy is being dwarfed by the need for efficiency. But this doesn't mean patients have stopped wanting to be treated in a caring and empathetic manner. And there is a growing body of evidence that this need is often not being met.

In the Novel To Kill a Mockingbird, Atticus Finch tells his daughter Scout that "you never really understand a person until you consider things from his point of view ... until you climb into his skin and walk around in it" This is empathy — where one identifies with another's feelings. It involves compassion and the ability to understand and respond to the feelings of others. Often, an empathetic response leads to a caring response. Empathy is different to sympathy which is described as feeling sorry for another person. This does not require us to understand the other person's point of view, but is an automatic, emotional response. In health care, feeling sympathy for another person can overwhelm us with sorrow and often preclude us from helping.

In recent times, poor communication, including lack of empathetic and caring behaviours, has resulted in an increasing number of complaints against health professionals in Australia. Shocking cases of maltreatment at a United Kingdom public hospital between 2005 and 2009 reveal the extreme consequences of negligence, poor communication and lack of empathy in health care. Incidents ranged from patients being forced to drink from flower vases to lying in their own excrement. More than 300 deaths were directly linked to this neglect. At the crux of the recommendations made in a report of the inquiry into the incidents was the need for improved communication between health care workers and patients. Empathy is fundamental to effective communication. For doctors and nurses, this means placing the patient at the centre of care. This skill leads to increased levels of satisfaction not

only in patients but also the doctors and nurses. Importantly, it is also associated with improved patient outcomes.

Technology has greatly contributed to health professionals 'diminishing levels of empathy. It has come at the cost of changing the way doctors and nurses interact with their patients. Because there are fewer opportunities for direct patient contact, it hinders the ability to develop a rapport with patients, monitor their non-verbal communication and elicit feedback on the interaction. For example, touch has historically been a large part of the work of a nurse. When nurses hold a patient's hand or arm to take their pulse, it contributes to the kind of connection that has been shown to release the feel-good hormone oxytocin. But taking a patient's pulse manually is now more often than not replaced by a probe attached to a patient's finger. Computers on wheels create a physical barrier for nurses when they use them to administer medications and access documents; and smart phones that support patient interviews have replaced the opportunity for a nurse to physically be present and develop a rapport with a patient. Meanwhile, virtual reality games and experiences are often used to distract patients undergoing painful procedures, when in the past a nurse may have held the patient's hand.

Learning often takes place using simulation technology, where students interact not with actual human beings but with computerized mannequins. It is understandably difficult to respond to a mannequin as a patient with emotional needs. Students subsequently find it difficult, in a real clinical setting, to integrate desired communication skills — in particular, empathy. University programs are often content-heavy, with graduates required to meet many competencies before they can be registered with professional bodies. The result can be that students in health professional courses tend to focus on clinical and technical skills at the expense of good communication.

The disruptiveness of technology may also be a factor affecting the ability of nurses and doctors to be empathetic and compassionate. Technology encourages multitasking, which is good for efficiency, but can distract health care professionals from important interpersonal interaction with patients. Funding constraints in the university sector, decreasing clinical placement opportunities, the increasing complexity of patients, and a heightened awareness of ensuring patient safety and

the associated legal responsibilities, all contribute to the increasing use of the controlled learning environment laboratories offer. Learning in laboratories using technology is being developed to maximise experiences that develop empathy. Good communication needs to be role-modelled, taught and assessed in university programs and throughout clinical practice. We need a better understanding of empathy development in health professions and more research on how to improve the situation with changing technologies. Most importantly, though, we always need to listen to our patients.

Text 2: Questions 15-22

15. What does the word 'this' in the first paragraph refer to?

- Ⓐ The patient's expectations of health professionals.
- Ⓑ The amount of work a health professional has to do.
- Ⓒ The lack of training modern health professionals receive.
- Ⓓ The lack of compassion shown by some health professionals.

16. The writer includes the quote from *To Kill a Mockingbird* to show that

- Ⓐ It's important to put yourself in someone else's position.
- Ⓑ It is difficult to know what another person is thinking.
- Ⓒ It's impossible to learn anything unless you listen.
- Ⓓ It's easier to understand people as you get older.

17. What concern does the writer express in the second paragraph?

- Ⓐ Empathy isn't fully understood by some health professionals.
- Ⓑ Sympathy can impede on professionalism in a clinical setting.
- Ⓒ Health professionals don't equally display empathy and sympathy.
- Ⓓ Empathy and sympathy are often confused by health professionals.

18. What point does the writer make in the third paragraph?

- Ⓐ The situation is worse in the UK than in Australia.
- Ⓑ The need for increased patient safety monitoring is clear.
- Ⓒ The clinical skill of healthcare professionals has decreased.
- Ⓓ The situation will improve for everyone if the patient is the focus.

19. What does the word 'it' in the fourth paragraph refer to?

- Ⓐ Touch.
- Ⓑ Oxytocin.
- Ⓒ The work of a nurse.
- Ⓓ Non-verbal communication.

20. The writer suggests technology has led to reduced levels of empathy because

- Ⓐ It means people don't always need to be in the same room.
- Ⓑ It sometimes results in people becoming preoccupied.
- Ⓒ It causes an extra level of stress for some people.
- Ⓓ It can create greater distance between people.

21. The writer explains that when students are learning in health professional courses

- Ⓐ There is a large written component.
- Ⓑ Achieving high results is the main goal.
- Ⓒ The emphasis is on being a good clinician.
- Ⓓ Working with mannequins can be confusing.

22. In the final paragraph, the writer suggests the best way to improve empathy is to

- Ⓐ Fund research into the issue.
- Ⓑ Utilise new lab-focussed initiatives.
- Ⓒ Pay greater attention to the patient.
- Ⓓ Undertake communication-based study

READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT remove OET material from the test room.

TEXT A

Osteoarthritis (OA), the most common joint disorder, often becomes symptomatic in the 40s and 50s and is nearly universal (although not always symptomatic) by age 80. Only half of patients with pathologic changes of OA have symptoms. Below age 40, most large-joint OA occurs in men and often results from trauma or anatomic variation (eg, hip dysplasias). Women predominate from age 40 to 70, after which men and women are equally affected.

Classification

OA is classified as primary (idiopathic) or secondary to some known cause.

Primary OA may be localized to certain joints (eg, chondromalacia patellae is a mild OA that occurs in young people). Primary OA is usually subdivided by the site of involvement (eg, hands and feet, knee, hip). If primary OA involves multiple joints, it is classified as primary generalized OA.

Secondary OA results from conditions that change the microenvironment of the cartilage. These conditions include significant trauma, congenital joint abnormalities, metabolic defects (eg, hemochromatosis, Wilson disease), infections (causing postinfectious arthritis), endocrine and neuropathic diseases, and disorders that alter the normal structure and function of hyaline cartilage (eg, RA, gout, chondrocalcinosis).

TEXT B

Diagnosis of knee osteoarthritis

Physical Examination

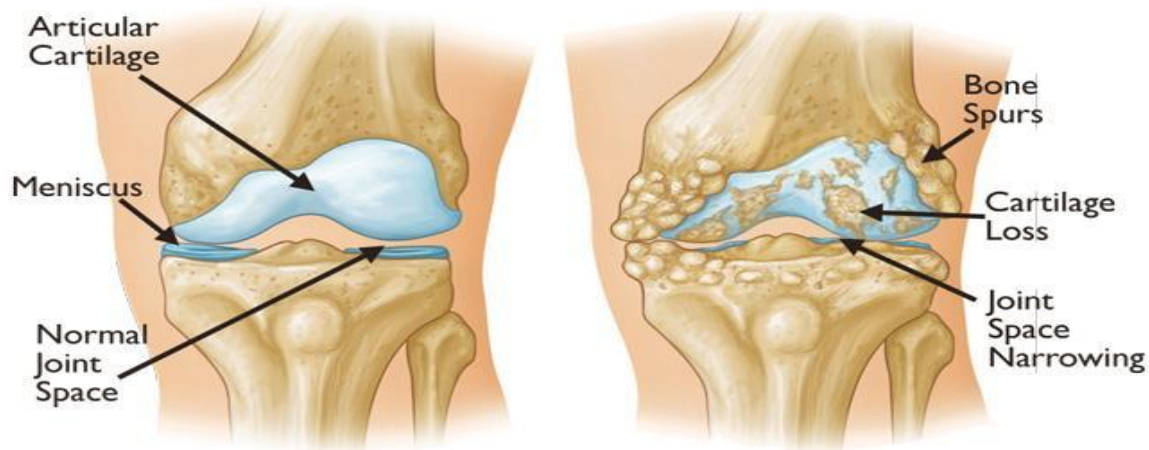
- Joint swelling, warmth, or redness
- Tenderness about the knee
- Range of passive (assisted) and active (self-directed) motion
- Instability of the joint
- Crepitus (a grating sensation inside the joint) with movement
- Pain when weight is placed on the knee
- Problems with gait
- Any signs of injury to the muscles, tendons, and ligaments surrounding the knee
- Involvement of other joints (an indication of rheumatoid arthritis)

Imaging Tests

- X-rays. These imaging tests create detailed pictures of dense structures, like bone. They can help distinguish among various forms of arthritis. X-rays of an arthritic knee may show a narrowing of the joint space, changes in the bone and the formation of bone spurs.
- Other tests. Occasionally, a magnetic resonance imaging (MRI) scan, a computed Tomography (CT) scan, or a bone scan may be needed to determine the condition of the bone and soft tissues of the knee.

TEXT C

Knee OA causes cartilage to be lost. The ligaments become lax and the joint becomes less stable, with local pain arising from the ligaments and tendons..



Osteoarthritis often results in bone rubbing on bone. Bone spurs are a common feature of this form of arthritis

TEXT D

Treating Osteoarthritis of the Knee

The primary goals of treating osteoarthritis of the knee are to relieve the pain and return mobility. The treatment plan will typically include a combination of the following:

- **Weight loss:** Losing even a small amount of weight, if needed, can significantly decrease knee pain from osteoarthritis.
- **Exercise:** Strengthening the muscles around the knee makes the joint more stable and decreases pain. Stretching exercises help keep the knee joint mobile and flexible.
- **Pain relievers and anti-inflammatory drugs:** This includes over-the-counter choices such as acetaminophen (Tylenol), ibuprofen (Advil, Motrin), or naproxen sodium (Aleve). Patients shouldn't take over-the-counter medications for more than 10 days without checking with their doctor. Taking them for longer increases the chance of side effects. If over-the-counter medications don't provide relief, patients may require an anti-inflammatory drug or other medication to help ease the pain.
- **Injections of corticosteroids or hyaluronic acid into the knee.** Steroids are powerful anti-inflammatory drugs. Hyaluronic acid is normally present in joints as a type of lubricating fluid.
- **Using devices such as braces:** There are two types of braces: "unloader" braces, which take the weight away from the side of the knee affected by arthritis; and "support" braces, which provide support for the entire knee.
- **Physical and occupational therapy.** Physical therapists assist patients with ways to strengthen muscles and increase flexibility in the joint. Occupational therapists teach ways to perform regular, daily activities, such as housework, with less pain.

Surgery: When other treatments don't work, surgery may be a good option. After any type of surgery for arthritis of the knee, there is a period of recovery. Recovery time and rehabilitation depends on the type of surgery performed.

READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

INSTRUCTIONS TO CANDIDATES:

DO NOT open this **Question Paper** or the **Text Booklet** until you are told to do so.

Write your answers on the spaces provided on this **Question Paper**.

You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this Question Paper and the Text Booklet.

DO NOT remove **OET** material from the test room.

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate Text **Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

Osteoarthritis : Questions

Questions 1-7

For each of the questions, 1-7, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1. The different types of osteoarthritis?
2. A comparison between a healthy knee and a knee afflicted by osteoarthritis?
3. The causes of osteoarthritis of the knee?
4. Evidence of osteoarthritis of the knee?
5. When osteoarthritis of the knee is likely to occur?
6. How to detect the condition of the bone within the knee?
7. Medication options to assist with knee pain?

Questions 8-13

Answer each of the questions, **8-13**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8. From what age will you generally start to see osteoarthritis in women?
.....

9. What will best show any alterations to the bone or joint space within the knee?

10. What is lost from the knees of patients suffering from knee osteoarthritis?

11. What common trait is found in patients with osteoarthritis of the knee?

12. What should you recommend to a patient to increase the joint mobility of their knee?

13. If a patient experiences pain in the joints beyond the knee, what might it indicate?

Questions 14-20

Complete each of the sentences, **14-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

14. The rate of occurrence of knee osteoarthritis is the same in men and women beyond the age of -----
15. If a patient is suffering with -----, an abrasive feeling within the joint, it is a sign of knee osteoarthritis.
16. A patient with osteoarthritis of the knee will experience localised pain from the -----
17. The main aims when managing knee osteoarthritis are to relieve pain, as well as -----
18. You may have to perform an MRI, CT scan or ----- in order to establish the state of the soft tissues of a patient's knee.
19. Recommending ----- for use on a patient's knee may be one way of providing much needed structural support.
20. If regular pain relief or anti-inflammatory medication doesn't relieve a patient's pain, you may need to suggest they consider ----- into the knee.

READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this Question Paper until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this Question Paper.

DO NOT remove **OET** material from the test room.

HOW TO ANSWER THE QUESTIONS

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:



In this part of the test, there are six short extracts relating to the work of health professionals. For questions 1-6, choose the answer (**A, B or C**) which you think fits best according to the text.

1. What message is being conveyed in this guideline extract?

- Ⓐ There is insufficient understanding of these systems by most employees.
- Ⓑ Employees are the only people capable of operating these systems.
- Ⓒ These systems are less effective if employees aren't included

Extract from guidelines: The core elements of safety and health management systems

Most successful safety and health management systems have six “core elements” all interrelated, each necessary to the success of the overall system. One of those is employee participation.

Employees have unique knowledge of the workplace and must be involved in all aspects of the safety and health management system—for example, setting goals, identifying and reporting hazards, investigating incidents, and tracking progress.

All employees understand their roles and responsibilities under the safety and health management system and what they need to do to carry them out effectively. Employees are encouraged to communicate openly with management and report safety and health concerns. Barriers to participation (e.g., language, lack of information, or disincentives) are removed.

2. The directive informs us that employees

- Ⓐ Don't always have to pay an administration fee.
- Ⓑ Split of the cost of the administration fee with their employer.
- Ⓒ Who change their packing arrangements must pay an administration fee.

Policy Directive: Procedures for salary packaging

Administration fees

Each health agency will charge each participating employee an annual fee for administering their Package which will be deducted on a pro rata basis for each period. The cost of the administration fee, including GST, will be shared equally by the agency and the employee.

Health agencies may elect to outsource the administration of the salary packaging scheme. An external salary packaging provider must be selected through a tender process. In all instances, the Health agency will require the employee to package salary for the administration fee liability. In other words, the employer recovers 50% of the administration fee from the employee.

An employee may review and alter their salary packaging arrangements at the end of each package year (usually 31 March) at no extra cost. If an employee elects to change their packaging arrangements during the packaging year, they may incur an alteration fee.

3. What is being described in this section of the guidelines?

- Ⓐ Different tactics to use with pregnant women who use drugs or alcohol.
- Ⓑ The problems mothers using drugs or alcohol cause their unborn babies.
- Ⓒ Why birth control information should be given to women with drug or alcohol issues.

Guidelines for the Management of Substance Use during Pregnancy, Birth and the Postnatal Period

Exposure to drugs and alcohol may have a serious effect on the foetus in the very early stages of pregnancy, particularly before the first missed period. Therefore, all women with problematic drug or alcohol use should be provided with advice on contraception. Long acting forms of contraception are generally preferred (e.g. progesterone implants, IUDs). This approach will facilitate planned rather than unplanned pregnancies, and reduce harm to the unborn child.

4. Food-borne botulism is known to

- Ⓐ take a long time to develop in most patients.
- Ⓑ require a specific course of treatment.
- Ⓒ cause death in severe cases.

Manual extract: food-borne botulism

Food-borne botulism is due to absorption of a preformed toxin from the gastrointestinal tract. The timing of symptoms is dependent on the dose of the toxin absorbed. Typically, however, onset is 12-26 hours (median 24 hours) post-ingestion, but can occur from 6 hours to 8 days after eating contaminating food. Food-borne botulism can be associated with symptoms such as nausea, vomiting, diarrhoea, and abdominal cramps, followed by constipation. Diagnosis is by detection of the botulism toxin or by isolation of the organism. A defined equine-derived antitoxin must be given as soon as possible — do not wait for diagnostic tests. Repeat doses may be given within 24 hours if deterioration continues. There is no specific antibiotic for this condition. Supportive care including ventilation, hydration and nutrition will be required during recovery.

5. The standard explains that in the case of living donors

- Ⓐ Collection can only take place once their history has been taken.
- Ⓑ A therapeutic goods order supersedes all other collection requirements.
- Ⓒ Interviews must outline everything that occurred at least one month prior to collection.

Requirements in relation to the medical and social history of prospective donors

In the case of a living donor, the blood, blood components, cells or tissues must only be collected from a living donor with whom an interview to obtain the donor's medical and social history has been conducted and recorded in accordance with the following requirements:

- a) The interview must be conducted with the donor or guardian/next of kin by a trained interviewer and should be a face-to-face interview.
- b) The interview must be conducted no more than 30 days prior to or 30 days after collection of the blood, blood components, cells or tissues, and must occur prior to any release of the relevant product from quarantine, unless otherwise specified in another, current Therapeutic Goods Order that applies to the relevant product.

6. The purpose of the announcement about the video call platform is to

- Ⓐ Advise users of changes that are taking place.
- Ⓑ Outline a range of useful features for users.
- Ⓒ Warn users about potential problems.

Important Announcement The tele health platform will be upgrading

On Thursday 13th September, Health direct Australia will be releasing version 3.9 of the video call platform. This upgrade will not affect the operation of the video call platform and you will be able continue to perform tele health consultations during the upgrade.

Some benefits of the new console include:

- Mobile-friendly — works well on tablets and smartphones, as well as desktops
- Simpler navigation, fewer clicks
- Improved functionality
- Quicker page response times after initial sign-in

If you experience any issues, please feel free to contact the tele health coordinator.

In this part of the test, there are two texts about different aspects of healthcare. For questions 7-22, choose the answer (**A, B, C or D**) which you think fits best according to the text.

Text 1: Is Australia prepared for the next pandemic?

Infectious diseases continue to threaten global health security, despite decades of advances in hygiene, vaccination and antimicrobial therapies. Population growth, widespread international travel and trade, political instability and climate change have caused rapid changes in human populations, wildlife and agriculture, in turn increasing the risk of infection transmission within and between countries and from animal species. New human pathogens have emerged, and previously "controlled" diseases have re-emerged or expanded their range. In the past decade alone, the global community has experienced infection outbreaks of pandemic influenza, Ebola and Zika viruses and Middle East respiratory syndrome (MERS). Planning for an effective response to the next pandemic is complex and requires extensive engagement between public health experts, clinicians, diagnostic laboratory staff, general and at-risk communities and jurisdictional and federal agencies. An effective response also requires access to real-time data, management of uncertainty, clear and rapid communication, coordination and, importantly, strong leadership. Are all these pieces of the plan currently in place in Australia?

The 2009 influenza pandemic tested Australia's capacity to respond to a highly transmissible emerging infectious disease. Public health units and frontline practitioners around the country were affected in different ways. The pandemic reached our states and territories at different times, leading to staggered and varied responses and pointing to clear gaps and challenges in logistics and governance. Although higher than usual rates of hospitalisation and intensive care admission, particularly among Aboriginal and Torres Strait Islander people and pregnant women, were observed early in the pandemic, most cases were mild. As the spectrum of disease became apparent, the existing plan, which had been based on a Spanish influenza-like worst-case scenario, was modified. The focus shifted from containing or limiting the spread of disease at a whole-of-population level to mitigation strategies targeted at key risk groups.

In keeping with similar exercises globally, the Review of Australia's health sector response to pandemic (H1N1) 2009 identified a need for greater flexibility in implementation plans to achieve an optimal response. Improved data sharing and synthesis within and between jurisdictions and internationally was defined as a key priority, to enable better understanding of the situation and evolving needs, to advise evidence-based practice and to inform clear, consistent messaging. The review also recommended development of a set of overarching ethical principles, to guide resource allocation in alignment with community expectations and values and to identify feasible interventions that are not disproportionately disruptive to society (disruptive interventions include social distancing measures such as school and workplace closures or travel restrictions). Failure to engage key populations at risk, including Aboriginal and Torres Strait Islander peoples, in preparedness activities before the 2009 pandemic was recognised as a critical deficiency.

Much has happened since 2009. The Australian health management plan for pandemic influenza, redrafted in 2014, is a nationally agreed plan for flexible and scalable responses in the health sector. It was developed in consultation with key stakeholders, including state and territory health departments and practitioner groups involved in implementing responses. The plan emphasises engaging with existing committees and practitioners to provide input to decision making under the leadership of the Australian Health Protection Principal Committee (AHPPC), the key decision maker in a national health emergency. The plan's recommendations on the use of infection control measures and pharmaceuticals, including antivirals and vaccines, are based on a wealth of national and international evidence emerging from the 2009 experience. Corresponding efforts have gone into strengthening the National Medical Stockpile and ensuring onshore vaccine manufacturing capacity to safeguard against the emergence of novel influenza strains.

However, influenza is not the only threat to Australia's health security. Recent outbreaks of MERS and Ebola and Zika virus infections have provided opportunities for the AHPPC and key stakeholders to practise and refine coordination and communication strategies to prevent, prepare for and respond to threats posed to Australians. These new threats highlighted the need to develop response plans that are agile, can be adapted to known and unknown pathogens and syndromes and are

well coordinated with international responses. The CDPLAN: Emergency response plan for communicable disease incidents of national significance, released in September 2016, provides a generic national framework for a primary response to outbreaks for which there is no pre-existing disease-specific plan. This plan is supported by the National Framework for Communicable Disease Control, a **roadmap** to improve national information sharing and facilitate a coordinated response to events of public health importance.

We are unable to predict when the next pandemic will occur or which new pathogen may appear, emphasising that every country must be well prepared. Australia has many pieces of the plan in place, but we must continue to fill gaps, test and refine existing systems and continually review what works to make sure we are as ready as possible for the next emerging infectious disease challenge. Louis Pasteur once said, "Gentlemen, it is the microbes who will have the last word' We need to ensure that he was wrong!

Text 1: Questions 7-14

7. The writer places the word “controlled” in inverted commas to indicate that viruses
- Ⓐ Have been regulated
 - Ⓑ Have started spreading.
 - Ⓒ Have never truly kept in check.
 - Ⓓ Were better understood in the past.
8. In the first paragraph, the writer suggests effective responses to disease outbreaks
- Ⓐ Require the development of new vaccines.
 - Ⓑ Need to be wide-ranging in scope.
 - Ⓒ Aren't presently well organised.
 - Ⓓ May be impossible to achieve.
9. What does the writer indicate about the 2009 influenza pandemic in Australia?
- Ⓐ It was worse than initially predicted.
 - Ⓑ It was similar to previous outbreaks overseas.
 - Ⓒ It demonstrated how unprepared various systems were.
 - Ⓓ It resulted in changes being made to the method of approach.

10. What problem did the review into the 2009 pandemic identify?

- Ⓐ Too many essential services were shut down during the incident.
- Ⓑ Media services fuelled a certain level of anxiety in the community.
- Ⓒ There wasn't enough consultation with the people who needed it most.
- Ⓓ Some individuals acted in a way that demonstrated a lack of principles.

11. The writer explains that the Australian health management plan for pandemic influenza

- Ⓐ Was created by health professionals to assist the general public.
- Ⓑ Is used as a means of developing future inoculations.
- Ⓒ Hasn't yet been used in a live response situation.
- Ⓓ Is largely used to improve communications.

12. What point does the writer make in the fifth paragraph?

- Ⓐ Any plans must be able to be rapidly modified to suit the issue.
- Ⓑ There are too many diseases to be able to combat them all.
- Ⓒ Australia has never been seriously affected by a pandemic.
- Ⓓ A lot of work still needs to be done on this global problem.

13. The writer uses the word 'roadmap' to illustrate that the National Framework for Communicable Disease Control

- Ⓐ is a guide.
- Ⓑ is significant.
- Ⓒ is widely used.
- Ⓓ is extremely effective.

14. The writer quotes Louis Pasteur in the final paragraph to say that

- Ⓐ Ultimately diseases will bring an end to humankind.
- Ⓑ We need to do everything we can to control disease.
- Ⓒ It is difficult to tell where a disease outbreak will occur.
- Ⓓ We are never as prepared as we need to be to fight disease.

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Text 2: Alzheimer's burden to double by 2060

Alzheimer's disease is a neurological disorder in which the death of brain cells causes memory loss and cognitive decline. At first, symptoms are mild, but they become more severe over time. Alzheimer's disease also accounts for 60 to 80 percent of cases of dementia in the United States. In 2013, 6.8 million people in the U.S. had been diagnosed with dementia. Of these, 5 million had a diagnosis of Alzheimer's.

Now, a report newly published by the Centers for Disease Control and Prevention estimates that the **burden** of Alzheimer's disease and related forms of dementia in the United States will double by the year 2060. This neurodegenerative disease is one of the leading causes of disability and the sixth-leading cause of mortality in the U.S. With annual healthcare costs of more than \$250 billion, the disease also puts a significant strain on the nation's healthcare system. Additionally, unpaid caregivers spend over 18 billion hours tending to those living with Alzheimer's.

Age is the most significant risk factor for Alzheimer's disease. Thus, as the population of the United States — along with that of the world — increases, it is important to ask: how many people will develop this form of dementia in the coming decades? Researchers from the Centers for Disease Control and Prevention (CDC) set out to investigate, and they published their findings in *Alzheimer's & Dementia: The Journal of the Alzheimer's Association*. Researcher Kevin Matthews, who currently works at the CDC's National Center for Chronic Disease Prevention and Health Promotion in Atlanta, GA, is the first author of the paper. He and his colleagues also looked at race and ethnicity, which are two "important demographic risk factors" for Alzheimer's. This made the study the first to predict Alzheimer's prevalence based on race and ethnicity.

Matthews and his colleagues used population projections obtained from the U.S. Census Bureau to calculate the projected number of seniors with Alzheimer's in the year 2060. To calculate the number of people living with Alzheimer's disease, researchers accessed data from the Centers for Medicare & Medicaid Services; specifically, they examined the number of Medicare Fee-for-Service beneficiaries aged 65 and above. The study revealed that compared with 2014, when the number

of people with Alzheimer's disease and other forms of dementia was 5 million, in 2060, this number will grow to 13.9 million. In terms of the population percentage, it represents an increase from 1.6 percent of the entire U.S. population in 2014 to 3.3 percent of the projected U.S. population in 2060. "Alzheimer's disease and other dementias burden will double to 3.3 percent by 2060 when 13.9 million Americans are projected to have the disease," write the study authors.

Also, the authors caution that in 2060, 3.2 million Hispanic people and 2.2 million African American people aged 65 and above will be living with the condition. African American people have the highest risk of developing Alzheimer's and other dementias; 13.8 percent of African American people who are aged 65 and over have the condition. Hispanic people fall second, with 12.2 percent, and non-Hispanic white people come third, with 10.3 percent. American Indian people and Alaska Natives fall fourth in the line-up, with 9.1 percent, and Asian and Pacific Islanders come fifth, with 8.4 percent. "These estimates conclude the authors, "can be used to guide planning and interventions related to caring for the Alzheimer's disease and related dementias population and supporting caregivers."

Dr Robert R. Redfield, the director of the CDC, commented on the findings, saying, "Early diagnosis is key to helping people and their families cope with loss of memory, navigate the healthcare system, and plan for their care in the future. This study shows that as the U.S. population increases, the number of people affected by Alzheimer's disease and related dementias will rise, especially among minority populations."

Text 2: Questions 15-22

15. The figures mentioned in the first paragraph indicate

- Ⓐ The significant rate of Alzheimer's disease in the U.S.
- Ⓑ The high death rate from Alzheimer's disease in the U.S.
- Ⓒ The increase in Alzheimer's disease rates in the U.S over time.
- Ⓓ The prediction for the rate of Alzheimer's disease in the U.S in the future.

16. The writer uses the word "burden" to show that Alzheimer's disease

- Ⓐ Is expensive to treat.
- Ⓑ Can be difficult to diagnose.
- Ⓒ Affects large numbers of people.
- Ⓓ Impacts on many areas of society.

17. The writer suggests that study by researcher Kevin Matthews is significant because

- Ⓐ it allows the U.S to plan for the future.
- Ⓑ it establishes the main cause of the disease.
- Ⓒ it focusses on areas that haven't previously been investigated.
- Ⓓ it is the most comprehensive review of its kind ever performed.

18. What point does the writer make in the fourth paragraph?

- Ⓐ Researchers found it difficult to access reliable data about Alzheimer's disease.
- Ⓑ Data revealed it is rare for people aged under 65 to develop Alzheimer's disease.
- Ⓒ The U.S. Census Bureau stored data on the amount of people with Alzheimer's disease.
- Ⓓ Medicare beneficiaries aged over 65 identify the amount of people with Alzheimer's disease.

19. What does the word "it" in the fourth paragraph refer to?

- Ⓐ The percentage increase of the population with Alzheimer's disease.
- Ⓑ The comparison between Alzheimer's disease rates in 2014 and 2060.
- Ⓒ The total number of people forecast to be suffering from Alzheimer's disease.
- Ⓓ The year the amount of people with Alzheimer's disease will reach 13.9 million.

20. The authors explain that the estimates outlined in the fifth paragraph

- Ⓐ Are heavily biased against some sections of the population.
- Ⓑ Require further investigation before being implemented.
- Ⓒ Might be able to assist decision makers in the future.
- Ⓓ Focus on providing treatment options to patients.

21. What concerns Dr Robert R. Redfield about the results of the study?

- Ⓐ The disproportionate effect on certain groups of people.
- Ⓑ The amount of time it can take to make a diagnosis.
- Ⓒ The lack of resources available to treat patients.
- Ⓓ The rate at which people are being affected.

22. In the final paragraph, the quote from Kevin Matthews is used to suggest that

- Ⓐ services require greater funding.
- Ⓑ the timing of intervention is significant.
- Ⓒ there aren't currently enough trained carers.
- Ⓓ people don't take responsibility for their health.

READING SUB-TEST – TEXT BOOKLET: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

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INSTRUCTIONS TO CANDIDATES:

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TEXT A

- Osteoporosis (OP) is defined as a disease characterized by low bone mass and deterioration of bone tissue, leading to bone fragility and an increase in fracture risk.
- Osteoporotic fractures usually result from a combination of decreased bone strength and injurious falls.
- Vertebral (spinal) fractures are the hallmark feature of OP and occur with a higher incidence and earlier in life than any other types of minimal trauma fracture.
- Non-vertebral minimal trauma fractures include hip, distal forearm, humerus, shoulder, ankle, pelvis and tibia.
- The lifetime risk of minimal trauma fracture is approximately 43% for women over 50 years of age and 56% among women over 60 years of age.
- Irrespective of fracture site, adults who sustain a fracture are at substantially greater risk (2 - 4 folds) of sustaining another fracture of a different type.

TEXT B

Diagnosis

- Diagnosis of OP is based on
 - ✓ Medical history and identification of risk factors
 - ✓ Clinical examination and
 - ✓ Confirmation by a dual energy x-ray absorptiometry (DXA) bone density measurement on 2 sites, preferably anteroposterior spine and hip
- Part of this diagnostic process is ensuring that other causes of bone fragility such as Metastatic cancers and endocrine disorders are excluded.
- The result of DXA scans are expressed as a 'T-score,' and will be in the range of
 - ✓ Normal (-1 or higher)
 - ✓ Osteopenia (low bone mineral density) (-1 to -2.5)
 - ✓ OP (-2.5 or lower)
- Applicable laboratory tests and radiographs of the thoracic and lumbar spine should also be considered.

Management

1. Factors complicating management

- In managing OP the following needs to be addressed
 - ✓ Cognitive impairment
 - ✓ Depression,
 - ✓ Anxiety disorders,
 - ✓ Falls risk
 - ✓ Medications (especially polypharmacy and potential drug interactions)

2. Support client self management

- Provide the client with OP resources.
- Maximize independent living using behaviour modification and exercise training.
- Utilize community support services (e.g. Home and Community Care) to enhance safety, reduce risk and support the client to stay in their own home.
- Ensure the home is safe from hazards such as trips, slips and burns.
- Encourage the client to identify barriers to adequate lifestyle modification and clinical.
- Adherence and develop goals to overcome those barriers based on their capacity and understanding.

TEXT C

Adequate vitamin D level

- Vitamin D deficiency is associated with low bone density and increased risk of falls
- Adequate vitamin D serum levels is critical for calcium absorption and is also important for bone growth and mineralization and muscle function
- People at risk of osteoporosis should maintain a serum vitamin D concentration of 75 nmol/L or more
- It is important to measure vitamin D serum levels and provide supplements for those at risk of vitamin D deficiency, including those
 - ✓ with limited sun exposure
 - ✓ with naturally occurring dark skin
 - ✓ who cover their skin (cultural or habitual clothing)
 - ✓ conditions affecting vitamin D metabolism
 - ✓ people in residential care or housebound, particularly the elderly
 - ✓ disabled, chronically ill or obese people
 - ✓ indoor workers
 - ✓ people with medical conditions or taking medications that interfere with vitamin D Metabolism

Medications:

- For those at risk of developing osteoporosis, vitamin D supplements should continue at doses that will maintain serum 25(OH)D levels greater than 60 nmol/L to prevent the onset of the disease
- Once this level has been achieved maintenance doses of vitamin D supplements
- should continue at daily doses of 800IU9
- Higher doses of 2000 - 4000 IU (50 - 100 micrograms) per day may be required in some individuals e.g. obese

	Maintenance dose (serum 25 (OH) D concentration >60nmol/L)	Mild vitamin D deficiency (serum 25 (OH) D concentration 30- 49 nmol/L)	Moderate to severe vitamin D deficiency (serum 25 (OH) D concentration < 29 nmol/L)
Colecalciferol	25 micrograms (1000 IU) orally, daily	25 – 50 micrograms (1000 – 2000 IU) orally, daily	75 – 125 micrograms (300- 5000 IU) orally, daily for at least 6-12 weeks Or 1250 micrograms (50,000 IU) orally, once a month for 3-6 months
		Check serum 25 (OH) D levels after 3 months	Check serum 25 (OH) D levels after 3 months When levels are 30 – 49 nmol/L, change to mild vitamin D deficiency therapy.

TEXT D

Table 5. Osteoporosis care plan

Action	Dx	Frequency
Height	✓	Once
BMI	✓	6 mthly
Weight	✓	6 mthly
Waist circumference	✓	6 mthly
BP	✓	6 mthly
Serum 25(OH)D levels	✓	3 mthly if being treated, otherwise annually
Carer education and support	✓	Each visit
Smoking cessation	✓	Ongoing support as required
Physical activity	✓	Exercise program as determined by physiotherapist
Diet and nutrition	✓	Each visit
Social emotional wellbeing	✓	Each visit
Influenza vaccine	Recommended - see the current edition of the <i>Australian Immunisation Handbook</i> for schedule	
Pneumococcal vaccine		
Lifestyle modification	✓	Each visit
Medication review	✓	3 - 6 mthly
Bone mass density testing	✓	Every 2 years
HW/RN R/V	✓	Each visit
MO/NP R/V	✓	6 mthly
Occupational therapist	✓	Referral as required
Physiotherapist	✓	Referral as required
Specialist review	✓	Referral as required
HACC	✓	Referral as required
Falls risk assessment	✓	As determined by allied health
Balance and strength exercise program	✓	As determined by allied health

READING SUB-TEST – QUESTION PAPER: PART A

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

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INSTRUCTIONS TO CANDIDATES:

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You must answer the questions within the 15-minute time limit.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the 15 minutes, hand in this Question Paper and the Text Booklet.

DO NOT remove **OET** material from the test room.

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate Text **Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should be correctly spelt.

Osteoporosis: Questions

Questions 1-7

For each of the questions, 1-7, decide which text (**A, B, C or D**) the information comes from. You may use any letter more than once.

In which text can you find information about

1. providing safe management to patient -----
2. overview of osteoporosis -----
3. the plan to be followed when treating a osteoporosis patient -----
4. Who to continue vitamin D supplements -----
5. The proportion of risks for women over 60 years of age -----
6. Elements muddling administration process -----
7. What to note while diagnosing an Osteoporosis patient -----

Questions 7-14

Answer each of the questions, **7-14**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

8. Sufficient vitamin D serum levels are basic for calcium assimilation and are additionally vital for ----- and mineralization and muscle work.

9. Osteoporosis fractures are resulting from a mixture of -----
----- and falls.
10. More than 1200 mg of colecalciferol is the required intake for the patient, when
his condition moves from moderate to ----- .
11. Some individuals need to take large amounts of vitamin D supplements, are
facing with ----- problems
12. Observe the patient thoroughly and suggest -----
review after 3 months.
13. Utilize ----- to improve safety, reduce hazard
and bolster the customer to remain in their own particular home.
14. If patient condition is -----, take serum
levels annually.

Questions 15-20

Complete each of the sentences, **15-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

15. Which two vaccines are used to treat a patient attacked with Osteoporosis?

16. Which supplements does a person need to escape from the prone of
Osteoporosis

17. What signs does a patient will get, if he is suffering from Osteoporosis

18. Whom the affected people need to contact for physical activity?

19. Which two body parts, tests and radiographs are also considered while diagnosis,

20. How much the quantity does the patient have to take, when his serum
concentration level is < 60 nmol/L?

READING SUB-TEST – QUESTION PAPER: PARTS B & C

CANDIDATE NUMBER:

LAST NAME:

FIRST NAME:

OTHER NAME: your details and photo will be printed here.

PROFESSION:

VENUE:

TEST DATE:

CANDIDATE SIGNATURE:

TIME: 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

DO NOT open this Question Paper until you are told to do so.

One mark will be granted for each correct answer.

Answer **ALL** questions. Marks are **NOT** deducted for incorrect answers.

At the end of the test, hand in this Question Paper.

DO NOT remove **OET** material from the test room.

HOW TO ANSWER THE QUESTIONS

Mark your answers on this **Question Paper** by filling in the circle using a 2B pencil.

Example:



In this part of the test, there are six short extracts relating to the work of health professionals. For questions 1-6, choose the answer (**A, B or C**) which you think fits best according to the text.

1. What was the reason for recent changes in healthcare?

- Ⓐ Reluctant to act friendly by doctors with patients.
- Ⓑ Nurse practitioners are acquiring high-positions, equal to doctors, because for extra qualifications.
- Ⓒ Lack of sufficient doctors in primary Healthcare hospitals.

Changing the healthcare Landscape?

Most of us have memories of visiting the family doctor when we were sick as children. This friendly and familiar figure checked your sore ears, listened to your heartbeat and gave you jellybeans at the end of your visit.

Unfortunately, the doctor shortage means it is getting harder to see a doctor, any doctor, when you really need them, especially if you live in a rural or regional area. But recent Medicare changes could bring about a change in our healthcare landscape with growing numbers of nurse practitioners likely to be working in primary and community care.

Unlike the practice nurse at your local GP surgery or a registered nurse that you may come across in hospital, nurse practitioners have extra qualifications allowing them to provide some of the care that previously only doctors could offer.

2. The information in these notes is intended to

- Ⓐ assist in the development of suitable procedures to this end.
- Ⓑ be conscious in operating with electrical equipment.
- Ⓒ project the lack of strict guidelines on operational handlings.

Hazards of Medical Electrical Equipment

Medical electrical equipment can present a range of hazards to the patient, the user, or to service personnel. Many such hazards are common to many or all types of medical electrical equipment, whilst others are peculiar to particular categories of equipment.

The hazard presented by electricity exists in all cases where medical electrical equipment is used, and there is therefore both a moral and legal obligation to take measures to minimize the risk. Because there is currently very little official guidance on precisely what measures should be in place in order to achieve this in respect to medical equipment, user organisations have developed procedures based on their own experience and risk assessments.

3. What does this extract from a handbook tell us about Microvascular Bleeding?

- Ⓐ Transfusion of blood components performed at the time of an operation.
- Ⓑ Hemostatic function can minimize the process of excessive bleeding.
- Ⓒ Use of potent platelet inhibitors to stop blood transfusion.

Microvascular Bleeding (MVB)

Patients undergoing cardiac surgery with cardiopulmonary bypass are at increased risk for microvascular bleeding that requires perioperative transfusion of blood components.

Platelet-related defects have been shown to be the most important hemostatic abnormality in this setting. The exact association between preoperative use of potent platelet inhibitors and either bleeding or transfusion in patients undergoing cardiac surgical procedures is currently being defined.

Laboratory evaluation of platelets and coagulation factors can facilitate the optimal administration of pharmacologic and transfusion-based therapy. However, their turnaround time makes laboratory-based methods impractical for concurrent management of surgical patients, which has led many investigators to study the role of point-of-care coagulation tests in this setting. Use of point-of-care tests of hemostatic function can optimize the management of excessive bleeding and reduce transfusion.

4. Why Clinical Medication Review gained prominence in recent times?

- Ⓐ Inability to accommodate patients in hospitals.
- Ⓑ To provide patient safety and for better health outcomes.
- Ⓒ Inappropriate medications are reflecting in hospital admissions.

Manual extract: Clinical Medication Review

Medication is by far the most common form of medical intervention. Four out of five people over 75 years take a prescription medicine and 36% are taking four or more drugs. However, we also know that up to 50% of drugs are not taken as prescribed; 2, 3, many drugs in common use can cause problems and that adverse reactions to medicines are implicated in 5-17% of hospital admissions. This leads to difficult decisions, particularly with the frail elderly, whether to initiate or discontinue medication.

Medication review is increasingly recognized as a cornerstone of medicines management, preventing unnecessary ill health and avoiding waste. Involving patients in prescribing decisions and supporting them in taking their medicines is a key part of improving patient safety, health outcomes and satisfaction with clinical care.

5. Why Are Case-Control Studies Used?

- Ⓐ To evaluate a conceivable relationship between an introduction and result.
- Ⓑ If the result of intrigue is uncommon or sets aside a long opportunity to happen.
- Ⓒ To alleviate recall and observation bias.

Case-Control Studies

Case-control studies are time-efficient and less costly than RCTs, particularly when the outcome of interest is rare or takes a long time to occur, because the cases are identified at study onset and the outcomes have already occurred with no need for a long-term follow-up. The case-control design is useful in exploratory studies to assess a possible association between an exposure and outcome. Nested case-control studies are less expensive than full cohort studies because the exposure is only assessed for the cases and for the selected controls, not for the full cohort.

Case-control studies are retrospective and data quality must be carefully evaluated to avoid bias. For instance, because individuals included in the study and evaluators need to consider exposures and outcomes that happened in the past, these studies may be subject to recall bias and observer bias.

6. Why does a patient cannot find the one, who have checked his case file?

- Ⓐ Healthcare is not having good security approach.
- Ⓑ Ethical privacy will make its prominence here.
- Ⓒ It is the policy of the Digital Health Research Centre.

Lax security culture in hospitals

A patient can look up My Health Record to check a log of which healthcare providers have opened their record, but won't be able to identify the individual health practitioner. When asked who records which individual doctors have accessed it, the ADHA declined to disclose this for "security reasons".

"When you have logins and you don't change them, and you have shared passwords, then yes ... it's difficult to tell who did what because your audit logs are going to have whoever was supposedly logged on," said Professor Trish Williams, Co-director of Flinders Digital Health Research Centre.

She said lax practices develop in hospitals due to time pressures and suggested the solution was to make logging on and off easier in the hospital environment. "One of the reasons why healthcare has been so bad at security has been the workf low," Professor Williams said.

In this part of the test, there are two texts about different aspects of healthcare. For questions 7-22, choose the answer (**A, B, C or D**) which you think fits best according to the text.

Text 1:

It was an ordinary day: me and my sister watching TV. Between endless series of horrifying news, we see one about the increasing number of both men and women who seek medical assistance and medication for depression. The same report informed my sister and I about the seriousness of the consequences of untreated depression, among these is suicide.

A couple years ago was the moment when I first saw news about depression that triggered my attention. I have experienced quite a few moments when I felt sad and needed to be alone. The constant invasion in the media about depression and how far things can get if not treated, taking into consideration my moments of weakness, have made me to even wonder myself: "What if my moments of sadness are signs of depression? Shall I look for help?"

'Mental states characterized by feelings of sadness, hopelessness, and loss of interest'. This is how depression is defined in the Oxford Dictionary of Sociology. What is more interesting is the fact that depression is considered to be evolved from melancholia. People feel melancholic because they are homesick or miss a friend.

It is normal to have moments when we miss someone so much that it hurts and we are sad because we cannot be with that person at that very precise moment, so we might wish to have some time for us, alone, to recover. But from experiencing this state of sadness, for the moment to give it a name, depression, there is only one small step in the eyes of the specialists. When I got in contact with the university life and found out more about the society, as well as read Mills' book "The Sociological Imagination", I further realized that the problems an individual experiences are issues with which the society confronts to and the dimension is much greater than believed. Therefore, my occasional sadness would probably be called, by specialists, mild depression, but this problem I am confronting sometimes has reached within the society a dimension that challenges me to further investigate the issue of depression.

The pharmaceutical industry has played an important role in the treatment of depression because these companies came up with an entire range of treatments meant to treat depression. However, this story with the pharmaceutical companies as the saviors emotional well-being of the people is quite an ambiguous one because it is hard to tell whether at first people experienced depression and then the drugs were invented, or the pharmaceutical industry made the drugs for the emotional recovering from sadness and renamed the state of sadness as depression, and then people started to use them.

In this journal I have chosen to focus on the subject of depression because I feel it is a personal topic. Experiencing minor episodes of depression myself, I would very much like to seek the history of depression and reveal whether depression is socially constructed or not, and acknowledge the true influence of the pharmaceutical industry in the treatment of depression.

For a long period of time, the concepts of illness and social reality were regarded as separate. In the 1960s, Szasz argued that the psychiatric perceptions about disease are actually social attributes to deviant behaviors because they are not built on an 'organic base'. In 1970, two perspectives were brought. On the one hand, Eliot Freidson made a distinction between the social constructed illness and the biological constructed illness and observed how particular problems or conditions of the human beings come to be defined as illnesses and bring a supplementary gain to the medical institutions and representatives. On the other hand, Foucault stated that people's behaviors, personal experiences and shape of identity can be influenced by the medical discourse. A few years after Friedson and Foucault's appreciations, Eisenberg claimed that there should be a differentiation between cultural and biological illness.

In the current society, medical sociologists include some forms of behavior and experiences of the people as 'medical conditions'. This is why the illness is shaped by a wide range of phenomena such as culture, knowledge, social contact and power; culture has an important meaning because it determines the way in which the illness is experienced, the reaction of the society towards illness, as well as the measures taken to cope with the illness. A very controversial and well known topic of

the present society has been through a complicated process in which culture has played an important role is depression.

Text 1: Questions 7-14

7. What made the author to think “Shall I look for help” in the second paragraph?

- Ⓐ He has lost someone, who is very lovable with him
- Ⓑ Sudden outbreak of news in Media about depression
- Ⓒ Author’s perception about his state of mental condition
- Ⓓ While seeing a article in a newspaper regarding suicidal cases increased abruptly in the last few years

8. What led the author to investigate about depression eagerly?

- Ⓐ Melancholia is considered to be a source of depression
- Ⓑ Because of his incidental sadness confronts within the society.
- Ⓒ To find out illness is made by a wide range of phenomena
- Ⓓ To disprove focault statements on depression.

9. The author suggests that problems as individual facing issues are confronting with society has

- Ⓐ Wide range of dimensions to believe
- Ⓑ Has perspectives that built on an organic base
- Ⓒ Acknowledged the true influence of the pharmaceutical industry
- Ⓓ Supplementary gain to the medical institutions

10. The word **Ambiguous** in the fifth paragraph implies that the role played by pharmaceutical companies as the saviors is

- Ⓐ Underpinned
- Ⓑ Explicit
- Ⓒ Dishonest
- Ⓓ Obscure

11. What made the author to feel depression as personal topic?

- Ⓐ The role played by the Pharmaceutical companies as the saviors.
- Ⓑ He himself has faced mild signs of depression
- Ⓒ To deter the opinion of differentiation between cultural and biological illness.
- Ⓓ An inspiration brought by reading the Mills Book

12. Authors view on Mental illness is

- Ⓐ the concepts of illness and social reality were regarded as same
- Ⓑ it is shaped by a wide range of phenomena
- Ⓒ culture alone influences a person mental condition
- Ⓓ there will be no evidence of social construction

13. How Szasz observations are different from others?

- Ⓐ Latter found them to be false
- Ⓑ Former observations are not Organic based
- Ⓒ Focused mainly on cultural observations
- Ⓓ Confined to behavioural features

14. What does the word 'this' in the final paragraph referring?

- Ⓐ Cultural process
- Ⓑ Behaviours and experiences
- Ⓒ Mental illness
- Ⓓ Medical conditions

Text 2: Alternative menopause therapies not best choice?

Too many Australian women are using treatments for menopause symptoms that don't work, the authors of a new study say. It's estimated nearly 500,000 women a month are using these medicines to control so-called vasomotor symptoms like night sweats, vaginal dryness and hot flushes says Dr Roisin Worsley, from Monash University's School of Public Health and Preventive Medicine. While some complementary therapies for menopause problems have not been as well researched as others, black cohosh and phytoestrogens at least has been the subject of multiple high quality studies known as randomised controlled trials and Meta -analyses, Worsley says. "There really was no evidence of any benefit."

Most alternative menopause therapies may also cause shorter term side effects including nausea, headache and upset stomach. Some known side effects of ginseng include hypertension, diarrhoea and sleeplessness. "It will reduce hot flushes by 80 per cent in most people," for instance, Worsley says. "It's really amazing how quickly it works as well." But women and doctors alike were scared off HRT after research findings released in 2002 suggested it increased the risk of breast cancer. The fear was understandable because "it was very scary evidence at the time". But the original analysis of study data was misleading because it focused on older women (average age 69) and those taking hormones for longer periods. This is because the original study set out to investigate a different question: whether oestrogen therapy could help prevent heart disease and dementia in older women. While the analysis showed HRT was linked with a raised risk of breast cancer, blood clots and strokes, "these were older women, who had already developed some forms of disease anyway".

Now the data has been reanalysed to work out the effect of the hormones on women who "actually want to use hormone therapy for their hot flushes". These are younger women (usually in their early 50s) who use hormones for a shorter period of time — and the conclusions are **offbeat**. "The reanalysis of the old data suggests the benefits of hormone therapy [for menopause symptoms] outweigh the risks for short-term use in healthy women." Current guidelines say women should take the lowest dose of HRT for the shortest amount of time possible, but can use it for up to five

years. However, all women should discuss their individual risk and personal preference with their doctor.

Phytoestrogens are compounds from plants that mimic the action of the human hormone oestrogen. Taken either as food supplements or in concentrated tablet form, they are the most commonly used complementary and alternative medicine for menopausal symptoms. 'We always thought they would help with hot flushes but unfortunately that hasn't worked out,' Worsley says. What's more phytoestrogens may pose a health risk because studies have shown when they are applied to isolated breast cancer cells in a laboratory dish, the cells multiply. Because of this, "we actually recommend if women have had breast cancer they shouldn't take these substances" Whether phytoestrogens might increase the risk of breast cancer in healthy women isn't known. "That's another point women don't realise: we don't have the long-term safety data on a lot of these remedies. They are a bit of an unknown quantity."

But treatments other than hormone therapy do exist and if women want to try them, Worsley thinks that's "completely reasonable". They include low-dose antidepressants and anticonvulsants. The key is to get good advice about options, something that can be tricky as it is very hard for GPs to stay up to date. "It's a really complicated topic and it's been changing rapidly over the last decade."

At present, "women with very severe debilitating symptoms have to navigate this really complex pathway. They try all different types of practitioners; they try every kind of diet and detox and various exercise things. And they're trying all kinds of supplements. I think a lot of women are not getting high quality information on which to make a decision". She suggests seeking out a "really good GP who's got an interest in women's health" or ask for a referral to a specialist who deals with menopausal symptoms. These are often gynaecologists or hormone specialists. There are also some lifestyle measures that can help. While menopause is a natural process, it "can be really disabling" for some women. "You can see why women are trying everything they possibly can to try and deal with it."

Text 2: Questions 15-22

15. The writer suggests that the potential harm to women was?

- Ⓐ Approaching artificial menopause therapies.
- Ⓑ Failing to take medication appropriately
- Ⓒ Looking for traditional therapies for longer benefits
- Ⓓ Modern lifestyle adaptations

16. When commenting on the Alternative menopause therapies, Dr Roisin Worsley shows his?

- Ⓐ Frustration on women depending alternative menopause therapies
- Ⓑ Reluctance of using those medicines that causes side effects
- Ⓒ Surprise that how most people are using these medicines
- Ⓓ Concern over the approaching of traditional therapies

17. The author used the words “it was very scary evidence at the time’ in the second paragraph to denote?

- Ⓐ A situation, when alternative therapies ending with cancer in people.
- Ⓑ The HRT research results feared off patients and doctors alike
- Ⓒ The fear of attacking cancer to the people, who have undergone regular therapies
- Ⓓ Mistaken view of the people, who had HRT research.

18. The meaning of the word **offbeat** in the third paragraph is

- Ⓐ Different
- Ⓑ Alike
- Ⓒ Confusing
- Ⓓ Uncommon

19. After reanalyzing the data, the effect of hormonal therapy on women is?

- Ⓐ Minimal
- Ⓑ Severe
- Ⓒ Negligible
- Ⓓ Outweighed

20. What drawback does the author mention in the fourth paragraph?

- Ⓐ Phytoestrogens are not suitable to consume as food supplements
- Ⓑ Phytoestrogens may cause breast cell multiply, which leads to cancer
- Ⓒ Oestrogen is taken as complementary food supplement
- Ⓓ Author recommending to take them in conc. tablet form

21. Worsley used the expression **“completely reasonable”** in fifth paragraph, it says

- Ⓐ He wants people to undergo continual alternative menopause therapies
- Ⓑ Suggesting other hormonal therapies
- Ⓒ To provide them better advice on treatment
- Ⓓ Very hard to cope with regular therapies

22. What does the word **‘they’** in the final paragraph refer to?

- Ⓐ Women
- Ⓑ Practitioners
- Ⓒ Gynaecologists
- Ⓓ Symptoms